

LET Reviewer Licensure Examination for Teachers 2017 Edition

- 1500 Test Question/Rationalization
- General & Professional Educational
- NCBTS Oriented

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MODELS OF COMMUNICATION

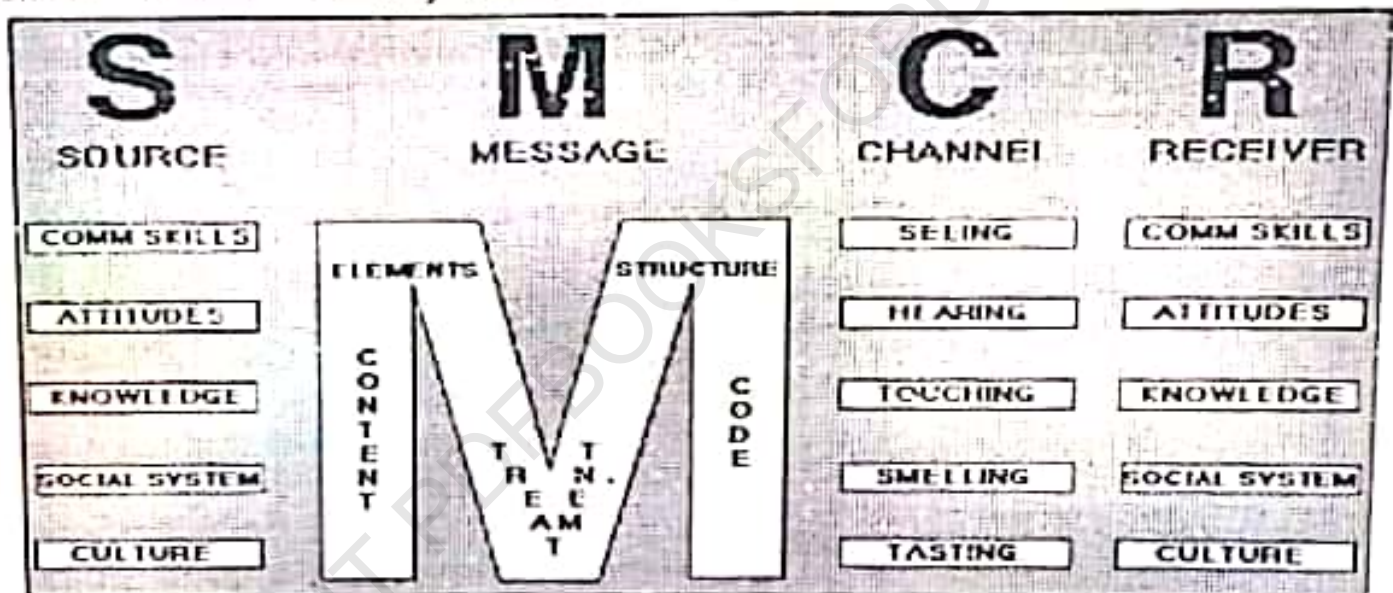
The communication process is indeed a paradox. We always communicate with each other, yet defining the process itself seems to be a very difficult task. Language experts of different historical periods, in their dire attempts to bridge the gap, were able to formulate their own theories to explain the process of communication. From the linear representation of Aristotle to the complex depiction of modern linguists, we can say that this endeavor has come a long way.

Renowned linguists have different points of view about the communication process, and this diversity proves to be the fuel that inspires linguists of today to take steps en route to the better understanding of this process.

Berio's representation, being the most widely cited and extensively used model, will be discussed to explain the basic components of the communication cycle.

Berio's Model

David Berlo's paradigm of the communication process is considered one of the most recognizable representations. It has four major components: source, receiver, message, and the channel. That is why at times this model is also called SMRC.



SMRC's linearity is oftentimes criticized, but Berlo's elucidations regarding the significance of each component redeem it all. According to him, the interlocutors' or conversation participants (source and receiver) communication skills, knowledge, socio-cultural system, and attitude are important for successful communication to take place (Berlo, 1960). For example, if the source has a higher level of language proficiency compared to the receiver, or vice versa, then problems may arise. Refer to the sample conversation that follows and then try to figure out what caused the lapse in communication.

Conversation Sample

Setting: JFK International Airport Immigration Counter

Situation: The immigration officer is interviewing a tourist about the latter's whereabouts.

Specifications: The immigration officer is a native speaker, while the tourist is a beginner learner of the English language.

Immigration Officer: Good morning, Miss! May I know your travel itinerary?

Tourist: Uhhh...sorry. What is that again?

Immigration Officer: Oh, I mean...where do you plan to go?

Tourist: Oh, I see. Actually I plan to go to the Metropolitan Museum and to the Museum of Modern Art, and maybe stay for another two days in New York City before flying to Florida.

Immigration Officer: Okay. Welcome to America. Enjoy your vacation!

In the sample conversation above, the difference between the proficiencies of the interlocutors led to lapses in communication. It is also noticeable that the success of the process relies largely on the one who has higher proficiency. Since the tourist has limited vocabulary (being a beginner), the immigration officer paraphrased the statement to fit the level of the tourist.

Discrepancy between the knowledge of the interlocutors also poses great threat. Read the sample conversation below and try to cite some reasons for the communication breakdown.

Conversation Sample

Setting: Computer Repair Shop

Situation: One client wants to have her laptop fixed.

Specifications: Both the client and the technician have the same language proficiency.

Client: Good morning! May I ask why my laptop shuts down automatically just right after turning it on?

Technician: Okay, let me see it.

Client: So what do you think?

Technician: I think your laptop has been infected by Trojans and worms...

Client: Huh? Trojans? Worms?

Technician: Yes. By the way, do you do defragmentation every once in a while? Doing so greatly helps the performance of your computer.

Client: Huh? What is that again? Defragment... what?

Technician: Defragmentation. Anyway, to fix your laptop, I will just reboot and reformat everything. Is that ok with you?

Client: I am not really familiar with what you are saying. So just fix it and send me the bill, OK?

In the conversation above, the technician uses jargon exclusive to those who are adept with matters pertaining to computers and information technology. For someone who is not really into the technicalities of computer software and hardware nomenclature, understanding what the technician is saying poses great difficulty that can later on lead to communication breakdown.

The Socio-cultural system is also a factor that affects the success of the communication process. A great example is the use of different gestures. In some cultures, nodding or shaking of the head has diverse meanings. In our culture, when one nods his head, it means approval, and shaking one's head from left to right means divergence. However, in some countries like India, shaking one's head from left to right signifies agreement. This means that an Indian shaking his head from right to left will be misinterpreted by a Filipino.

Another consideration is the attitudes of both the source and the receiver because if the interlocutors are at odds with each other, communication is expected to be less successful. If the source and the receiver have negative self-concepts, are averse to the subject matter, or to each other, then the message may not be sent or received in the way it is intended.

Aside from the interlocutors (source and the receiver), Berlo also explains the importance of the message. For him, the message is the physical product of the ideas of the source (1960). He also poses that the message has three factors: code, content, and treatment. Code pertains to the linguistic features, grammatical syntax, and morphological and phonological formations that are used by the source to express his message. Content is considered the main message, the idea to be expressed. Finally, treatment pertains to the way the code and the content are delivered.

For the message to be perceived by the receiver, the channel or the medium is very important. According to Berlo, the channel is the carrier of the message (code, content, and treatment) for it to be properly decoded by the receiver (1960). Since the primary aim of communication is for the source to achieve his desired response from the receiver, the former must choose the right channels for the latter to give the desired feedback/response. The use of graphic organizers in a business meeting is an example of this—the speaker explains the data, while the members of the board listen and at the same time, look at the visual aid.

On the whole, Berlo's model's main focus is to achieve high fidelity in terms of sending and receiving messages (1960). Fidelity is achieved, according to Berlo, when the source is able to receive the response/feedback that he desires. This is only possible if the source and receiver have similar communication skills, knowledge, socio-cultural upbringing, and right/positive attitude; if the message is well-coded, the content well thought of, and treatment carefully chosen; and finally, if proper channels are used.

English Language Learning

Learning the English language revolves around the topics mentioned below:

CUEING SYSTEMS	COMMUNICATIVE COMPETENCIES	MACRO SKILLS
PHONOLOGY	GRAMMATICAL	LISTENING
MORPHOLOGY	DISCOURSE	SPEAKING
SEMANTICS	SOCIOLINGUISTIC	READING
SYNTAX	STRATEGIC	WRITING
		VIEWING

THE FOUR CUEING SYSTEMS

1. Phonics - pertains to the sounds (phones) of the language
2. Phonology - pertains to the sounds of the language and their meaning
3. Morphology - pertains to formation of words

- types of morphemes
 - bound morpheme (dependent)
 - free morpheme (independent)
- 4. Semantics
 - pertains to meaning derivation
- 5. Syntax
 - grammar, structure, and form of the language

In teaching the English language, students are required to have at least a threshold competency for learning to take place, and according to CANALE & SWAIN (1983), there are four major communicative competencies needed, namely: Grammatical, Discourse, Sociolinguistic, and Strategic.

Grammatical competence refers to the ability of the student to use the specific forms of the language may it be a language's sounds (phonology), words (morphology), or sentence structure (syntax). Discourse competence pertains to the ability of the student to understand a particular language beyond sentential level—stories, letters, and conversations are some samples which can also be closely associated with the Sociolinguistic competence that in turn deals with the contexts (including culture, mode, tenor, and mood) where the language is used. Meanwhile, strategic competence, deals with the ability of the student to find ways on how to solve language gaps by creating a new form. An example of this is when a student lost for words utters "Lady Gaga-ish" when what he/she wants to say is outrageous or flamboyant.

The grammatical competencies are the most basic things that a student must possess for language learning to take place. Most experts would say that listening is the very first skill that students acquire. It can also be considered as the foundation of the other MACRO SKILLS in English. May it be reading, speaking, or even writing, awareness of the sounds of a particular language has always been the beginning; and that is why studying about the sounds of the language (phonology) is very important.

After being aware of the "sounds" of the language, the student must (it is so natural that learners don't need to really think about it anymore) now relate the sounds to written symbols (letters). A student must verbalize with automaticity the sounds of each letter and further blend different symbols to produce a meaningful utterance, thus bringing us to the study of words and word-formation (morphology). Knowledge about word formation is very much significant but is not the end of the endeavor, for words in isolation can be meaningless; Hence, they must be grouped together for them to take form.

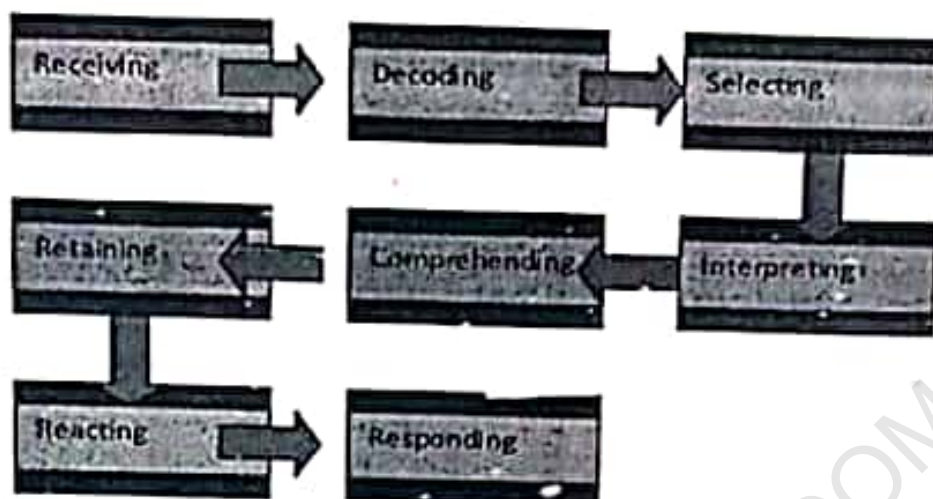
For example, the word "she" won't make any sense if not with other words. However, it is also a fact that one cannot just combine and group words. For example, the group of words "is pretty a she teacher" is just simply an utterance with no clear and sound meaning, and in this case, syntax is very important. For a group of words to be valid, meaningful, and sound, it must follow certain rules of grammar. Mastery of the skills above will be a great help in developing the next competencies which generally deal with how language is used in different contexts and discourses.

LISTENING

The Listening Process

Experts say that listening is the first language skill that students learn. Upon hearing things, then one can speak—just like how we learned our first utterances. Parents keep on repeating words until the babies imitate and repeat the utterances successfully (behaviorist). But is listening as easy as it may sound? In the next page is a diagram about the process of Listening.

The Listening Process



The receiving of the message from the source signals is the start of the listening process. The listener, then, will decode the message and select important parts of the message that will help the listener in interpreting it. Based on the interpretation, the receiver now comprehends the message and retains information needed for him to react and respond.

Kinds of Listening

Interactional/Empathic Listening

This type of listening is very common and at times done in an informal setting. This happens when persons listen to each other for the purpose of communicating and empathizing. At times, the aim of this type of listening is to accomplish tasks or to come up with something.

In a classroom setting, this is when students listen to each other for them to be able to accomplish given tasks and activities.

Transactional/Critical Listening

In this type of listening, the listener is concerned with acquiring new knowledge for the purpose of improving his skill in generalizing and decision making.

In a classroom setting, this is when students listen to their teacher to gain more knowledge.

Three Modes of Listening

Joan Morely in her essay "Aural Comprehension Instruction: Principles and Practices," made mention of the three modes of listening, namely: Unidirectional, Bidirectional, and Autodirectional.

- | | |
|------------------------|--|
| Unidirectional | - If the listener cannot respond to the things he/she heard |
| Bidirectional | - If the listening process is reciprocal—meaning, two way
(indicated by the prefix "bi") |
| Autodirectional | - If the listening process is reflexive—meaning, the speaker
This is intrapersonal listening. |

SPEAKING

Language and language learning developed as fast as civilization. In the olden times, man used language merely for survival—now, language functions as one important medium of change and innovation.

But the question still remains...what is language? Language was defined by Webster as the expression and communication of emotions or ideas between human beings by means of speech and hearing that is systematized and confirmed by usage among a given people over a period of time. If we are to look at Webster's definition, we can infer that there are only two functions of language.

The first one is to express how one feels (emotions); and the second one is to express how one thinks (ideas). He also mentioned two important factors of language: speaking (speech) and listening (hearing). The word systematized in Webster's definition can also be related to grammar—meaning it follows certain rules and systems, and lastly, he also mentioned "period of time," which can be interpreted as the changes language goes through with time.

Webster's definition is in line with the structuralist's. For them, language follows a system—meaning, one cannot just mix letters to form words. X, Y, K, and D cannot be combined to form a meaningful word—well, at least in English. In the sentence "She is pretty," it will be erroneous if one will change "is" to "are". These instances are the bases of the structuralists in saying that language is a system. They also believe that language is a means of communication and with that, no arguments were raised. That language is primarily vocal, is another claim of the structuralists. For them, and for Webster, speaking and listening are important language skills. They also adhere to the concept that language is arbitrary. For them, language constantly goes through series of innovations, developments, and changes. The words "thee", "thou", "growest", and "hath" are no longer used today. This is one proof that language, just as other things, constantly changes.

The cognitivists, however, believe that language is a mental phenomenon. For them, language learning is innate in the individual. According to them, there is one part in our brain called LAD (language acquisition device) that is responsible for language acquisition. If we are to analyze their definition, we can infer that language (for them) is not learned but acquired. The difference of learning from acquiring is that learning is voluntary, while acquiring is involuntary. We choose to learn, but we have no choice what to acquire. We acquire traits from our parents, but we learn how to solve math problems.

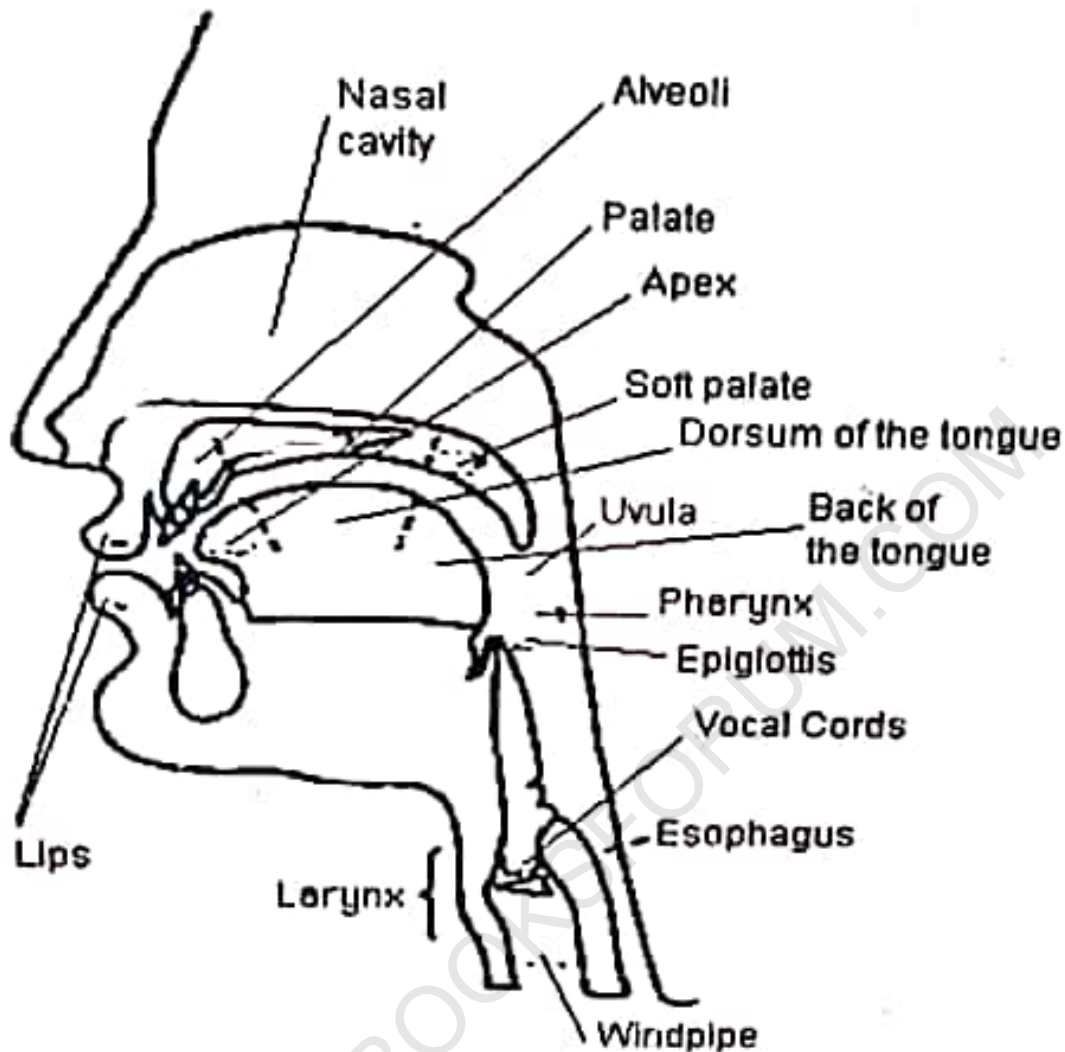
The functionalists have a rather functional definition about language. For them, language performs specific functions, such as to express, to persuade, to give or ask information, and to make someone to do something. Most people will agree with this definition, because practically, these functions are what language is really for.

The behaviorists also have their own definition of language. For them, language is learned through imitation, repetition, and reinforcement. If a teacher teaching grade one pupils will ask her students to say a word the way she does, then the pupils are expected to learn (imitation). If the teacher will ask the same pupils to repeat saying the word over and over again, then the pupils learned (repetition). If one pupil failed to say the word exactly how the teacher said it, then the pupil will be punished. On the other hand, if the pupil were able to say it correctly, then he/she will receive a prize.

THE SEGMENTALS

According to the structuralists, language is primarily vocal, but the question is—how should people speak in order to be considered proficient? Is it merely opening and closing the mouth? Or is there a complex process that guides speech production? In this section, you will know how the speech mechanism works.

The Speech Mechanism Model



Most likely, to produce sounds, one must open his/her mouth. Logically, the speech mechanism works when certain parts of the upper region touch certain parts of the lower region—like in pronouncing "P," where our upper lip touches the lower lip (bilabial). This is true especially in producing consonant sounds, but in producing vowel sounds, the mouth is always open (A, E, I, O, U).

The Family of Consonant Sounds

Consonant sounds can be classified according to three dimensions: the place of articulation (where the sound is made), the manner of articulation (how the sound is made), and the voicing (voiceless or voiced).

Manner of Articulation

Stops/Plosives

These are sounds produced when the air stream is blocked completely before it is abruptly released. The sounds P, T, K, B, D, G, are the members of this family.

Fricatives

These are sounds produced when the air stream is compressed and passes through a small opening creating friction. The sounds F, V, S, Z, H, TH (voiced) and (voiceless), Sh, Zh, are the members of this group.

Affricates

These are sounds produced when a plosive is followed by a fricative. Ts (Ch), and Dz (J) are the members of this group.

Nasals

Nasal sounds are produced when the oral cavity is blocked, and so the air passes through the nose. N, M, and Ng are nasal sounds.

Liquid/Lateral

Sounds are produced when air stream moves around the tongue in an unobstructed manner. Sounds like L and R are examples.

Glides

Glides are sounds that are close to vowel sounds, like W and Y.

Point of Articulation

- Bilabial - two lips touching each other
- Labio Dental - lower lip touches the upper teeth
- Dental - tip of the tongue and the inner edge of the upper teeth
- Alveolar - tip of the tongue and the alveolar ridge
- Palatal - the tongue and hard palate
- Velar - dorsal tongue and soft palate
- Glottal - throat passage

Voiced or Voiceless

A sound is voiced if the vocal cords vibrate, whereas a sound is voiceless if the vocal cords are not vibrating upon the production of sounds.

Consonant Chart

MANNER	POINT OF ARTICULATION						
	Bilabial	Labio dental	Dental	Alveolar	Palatal	Velar	Glottal
Stop	P B			T D		K G	
Fricative		F V	Th (voiced and voiceless)	S Z	SH ZH		H
Affricate					TS DZ		

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Affricate					TS DZ		
Glides							

VOWELS

Vowel sounds are produced with one's mouth open. It is also a fact that all vowel sounds are voiced sounds. Vowels sounds can be classified according to the height of the tongue and jaw, and the way the mouth opens.

A. VOWEL SOUNDS

	FRONT	CENTRAL	BACK
HIGH	/i/ /ɪ/		/u/ /ʊ/
MID	/e/ /æ/	/ʌ/ - accented schwa/ /ə/ - unaccented schwa/ /ɪ/ - accented or schwa sound /ʊ/ unaccented or schwa sound	/o/ /ɔ/
LOW	/a/		/ɑ/ /ɒ/

Vowel sounds can also be classified as SPREAD, ROUND, OR NEUTRAL.

SPREAD

- /i/ long /i/ sound as in "phoenix" and "beat"
- /ɪ/ short /ɪ/ sound as in "bit", "captain", "pin", and "marriage"
- /e/ soft /e/ sound as in "hate", "mate", "grape", and "grate"
- /ɛ/ hard /ɛ/ sound as in "let", "set", "any", and "jeopardy"
- /æ/ combination of /a/ and /e/ as in "family", "man", "anger", and "hamburger"

NEUTRAL

- /ʊ/ unaccented schwa sound as in "towel", and "America"
- /ʌ/ accented schwa as in "judge", and "cup"

ROUND

- /u/ long /u/ sound as in "balloon", "soon", and "pool"
- /ʊ/ short /ʊ/ sound as in "pull", "would", "push", and "cook"
- /o/ complete /o/ sound as in "boat", "phone", and "own"
- /ɔ/ incomplete /ɔ/ sound or the inverted "C" as in "hawk", "paw", "jaw" and "saw"
- /ɑ/ the /a/ sound as in "dark", "sergeant", and "psalm"

Source: Functional Oral Communication by Villamarzo, Relis, and Cruz (2003)

THE SUPRA SEGMENTALS

Juncture

Juncture is defined as the pauses and rests in a given speaking discourse. It is commonly represented by a single slash (/) for short pauses, double slashes (//) for long pauses, and three slashes (///) for rests at the end of the paragraph.

- Pitch** - This is the highness or lowness of sound.
- Volume** - This is the loudness or softness of sound.
- Intonation** - This is the rising and falling of sound.
- Stress** - This is the placement of emphasis or force on certain words or syllables.

READING

Jean Chall, proposes skills that are essential for real reading to take place. She said that there are five skills that a student must have:

- 1. Phonemic Awareness** - being familiar with the sounds of the language
- 2. Phonics** - drawing out the relationship between the symbol and the sound (graphophonemic relationship)
- 3. Fluency** - the ability to blend and mix the sounds to form a meaningful utterance
- 4. Vocabulary** - the ability to attach meanings to words
- 5. Comprehension** - the ability to create and decode meanings from a group of words

Chall also constructed a matrix of stages of reading development. She states that there are six stages, namely:

Stage 0 Pre-Reading Stage

In this stage, the students are being exposed to the different sounds of the language. This is the reason why songs, nursery rhymes, and poetry are being read to students of reading. Automaticity in recognizing a particular language is the key in the next stages. The teacher can determine if the students have phonemic awareness if they can name from language the words they hear are from.

Stage 1 Initial Reading Stage

In this stage, the students will start to realize the correspondence between the symbol and the sound. For example, upon seeing the symbol M, the students will know that the sound is /m/ as in "mother."

Stage 2 Confirmation Stage

This stage is deemed to be very crucial in reading development and according to Chall, most will mistake fluency for comprehension. She reiterates that mere verbalization of written text cannot be considered complete reading.

Stage 3 Reading to Learn Stage (Academic Reading)

The first three stages are considered "learning to read stages", while stage 3 up until stage 5 are "reading to learn stages." In this particular stage, students must already be beyond fluency. They must be able to comprehend what they read because if not, they will have difficulty in learning.

Stage 4 - Multiple Viewpoint Stage and Stage 5 - Construction and Judgment Stage

After reading a text, if students were able to give their opinions then they belong to the fourth stage. On the other hand, if students were able to criticize a particular text, and in turn were able to write their own, then they belong to fifth stage.

Purposes of Reading

- Skimming - reading to get the main idea
Scanning - reading to get specific information

Other Purposes of Reading

1. Information searching
2. General comprehending
3. Learning new information
4. Information evaluating and synthesizing

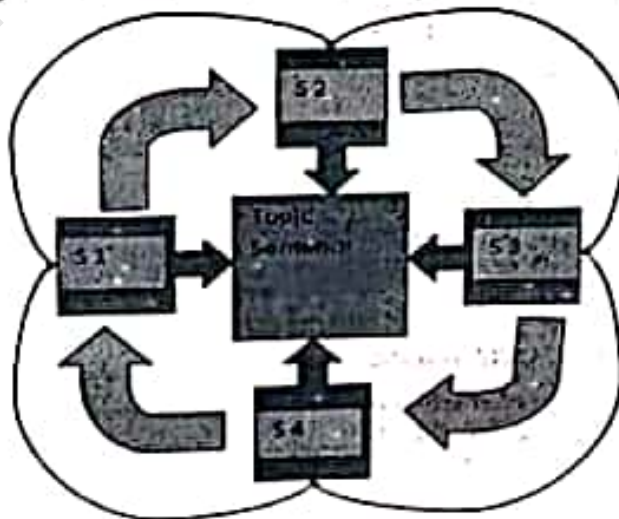
WRITING**The Paragraph**

The paragraph is defined as a group of unified, coherent, well-developed sentences that are properly and logically organized to support one specific idea or topic.

Based on the given definition, we can infer that it is important for a paragraph to have Unity, Coherence, logical Organization, support or Development, proper Emphasis, and one governing and limiting topic sentence—"UCODE TS".

The Paragraph Paradigm

Based on the diagram below, we can say that a paragraph is unified if all the sentences are about the topic sentence. It has coherence if all the sentences are connected with each other properly and in logical and correct order.



GRAMMAR
Parts of Speech
NOUNS

A noun names something – person, place, thing, events, phenomena, emotions, etc.

Kinds of Nouns

- | | |
|---------------------------|---|
| 1. Proper nouns | - names of specific persons, things, or places |
| 2. Common nouns | - refer to any one of a class of persons, places, or things |
| 3. Count nouns | - nouns that can be counted and quantified using numbers |
| 4. Non-count / Mass nouns | - nouns that cannot be counted and quantified using measurements
- mass nouns, abstract nouns, events, occasions and phenomena are under non-count nouns |
| 5. Collective nouns | - nouns that name a group of persons or things |
| 6. Compound nouns | - nouns that are made up of two words acting as a single unit
- Compound nouns may be written as separate words, hyphenated words, or combined words |
| 7. Concrete | - nouns that can be perceived by the senses |
| 8. Abstract | - nouns that cannot be perceived by the senses, such as concepts and emotions |

PRONOUNS

Pronouns are words that stand in the place of nouns.

Kinds of Pronouns

1. **Personal pronouns** - these are pronouns that refer to the person speaking, the person spoken to, or the person or thing spoken about.

PERSONAL PRONOUNS		
	SINGULAR	PLURAL
FIRST PERSON	I (Subjective) ME (Objective)	WE (Subjective) US (Objective)
SECOND PERSON	YOU	YOU
THIRD PERSON	SHE, HE, HER, HIM, IT	THEY (Subjective) THEM (Objective)

2. Possessive pronouns - These are pronouns that show possession.

POSSESSIVE ADJECTIVES (Subjective)	POSSESSIVE PRONOUNS (Objective)
MY OUR YOUR THEIR HIS HER ITS	MINE OURS YOURS THEIRS HIS HERS

Examples:

Possessive Adjective This is my book. ("my" qualifies / tells who owns the book)

Possessive Pronoun This book is mine. ("mine" represents the owner of the book)

3. Reflexive and Intensive Pronouns

	SINGULAR	PLURAL
FIRST PERSON	MYSELF	OURSELVES
SECOND PERSON	YOURSELF	YOURSELVES
THIRD PERSON	HIMSELF HERSELF ITSELF ONESELF	THEMSELVES

Examples:

REFLEXIVE	INTENSIVE
The pronoun refers to the subject	The pronoun shows emphasis
a. I see myself in his actions. b. Did you teach yourself? c. He cut the paper himself. d. They were shocked about themselves.	a. I, myself is the culprit. b. You, yourself teach. c. He, himself cut the paper d. They themselves were shocked.

4. Indefinite Pronouns - do not refer to a specific noun

SINGULAR	PLURAL	BOTH SINGULAR AND PLURAL
ANY EVERYONE ANYBODY NO ONE EVERYBODY ANYONE	BOTH SEVERAL FEW MANY	NONE ALL SOME

5. Interrogative pronouns - used in asking questions

INTERROGATIVE PRONOUNS		
WHO	HOW	HOW MUCH
WHAT	WHOM	HOW MANY
WHICH	WHOSE	
WHY	WHEN	

6. Demonstrative pronouns - Used in pointing nouns

DEMONSTRATIVE PRONOUNS		Near	Far
SINGULAR	THIS	THAT	
PLURAL	THESE	THOSE	

ADJECTIVES

An adjective qualifies and tells something about a NOUN or PRONOUN through descriptions.

It answers the questions:

- What kind?
- Which one?
- How many?
- How much?

Degrees of Comparison of Adjectives

POSITIVE	COMPARATIVE	SUPERLATIVE
BIG	BIGGER	BIGGEST
USEFUL	MORE USEFUL	MOST USEFUL
ACTIVE	LESS ACTIVE	LEAST ACTIVE
BAD	WORSE	WORST

Order of Adjectives

determiner	observation	Descriptive Adjectives				origin	material	qualifier	noun
		size	shape	age	color				
An	Expensive				Red	European		sports	Car
An	Extra-ordinary						Silver		Necklace
Str			Long-stemmed		White	American			Daisies
Her	Shiny	Short			Black				Hair
My		Big		Old		German			Dog
Those		1-2					Ceramic	Jewelry	Roses
That	Smooth	Big	round				Gold		Locket
Few		Tiny				African			Insects
Some	Exotic					Indian			Film

VERBS

Oftentimes, verbs are defined as action words. But in reality, verbs are far more than that. Verbs also link ideas in a sentence, help other verbs, and state conditions. We can classify verbs into (1) verbs of actions, (2) linking/be verbs, (3) helping verbs, (4) emphatic verbs, and of course, (5) modals.

Action Verbs

These are verbs that express action or movement. A majority of verbs are of this nature. Words like run, walk, talk, sing, dance, etc. are common examples of action verbs. We can further classify actions verbs into two more sub-classifications—Regular and Irregular verbs.

Regular verbs

An action verb is considered regular if its past form is derived by adding -d or -ed. Let us take the word walk for an example: the past form of "walk" is "walked".

BASE FORM	PAST FORM
Watch	Watched
Bake	Baked
Pick	Picked
Save	Saved

Irregular verbs

An action verb is considered irregular if its past form is derived not by adding d or ed, but through a change or through retention in spelling.

BASE FORM	PAST FORM
Pay	Paid
Put	Put
Say	Said
Write	Wrote

Linking Verbs**Linking Verbs as Copula**

The copula is defined by Celce-Murcia as the link between the subject and nonverbal predicates (nouns, adjectives, and some adverbials). The copula also carries the tense and will determine subject-verb agreement.

Examples:

She is beautiful. — She is a pronoun, and beautiful is an adjective.

She is beautiful. — She is singular, that is why we used "is."

Linking Verbs as Perception Copulas

These are verbs that express no action, but at the same time, are not the conventional (is, are, was, were) linking verbs. They are called Perception copulas because they are perceived through the senses (mental or sensory).

Examples:

Appear	Seem	Feel	Look
Smell	Sound	Taste	

Linking Verbs as State Copulas

State copulas are verbs that are not locomotive. They are more of a condition than an action.

Examples:

Lie	Remain
Rest	Stand

Linking Verbs as Change-of-State Copulas

These linking verbs do not express instant locomotion or movement. Mostly, these verbs express changes from one state to another.

Examples:

Become	Come	Fall
Get	Go	Grow
Turn		

Auxiliary Verbs

These are also called helping verbs because they always appear with an action verb in a sentence (main verb). Linking verbs such as is, was, are, were, are considered auxiliary verbs if they appear together with a verb in progressive form. Other helping verbs are has, have, and had.

The verb "has" is used for singular subjects in the present tense. The verb "have" is used for plural subjects in the present tense, and "had" is used for both singular and plural subjects in the past tense. Has, have, and had, are also considered Auxiliary verbs if they appear in a sentence with another verb (main verb) in the past participial form.

Example:

She has taken a bath already. -the verb "has" functions as an auxiliary

Emphatic Verbs

Emphatic verbs are used to give certain emphasis. These verbs are "do", "does", and "did". DO is used for plural subjects in the present tense. DOES is used for singular subjects in the present tense, and DID for both plural and singular in the past tense.

The verbs DO, DOES, and DID can also be used as main verbs. They are only considered emphatic verbs if they appear in a sentence with another verb in its base form.

Example:

She did not drink her milk. — DID is used as an emphatic verb

Modals

According to Celce-Murcia, modals are helping verbs that are used to give a proposition, a degree of probability, to express one's attitude, and to perform various social functions, such as expressing politeness or indirectness when making requests, giving advice, or granting permission. It is always coupled with a verb on its base form.

a. Stating Ability

I can do anything.

b. Expressing Regret

I should have loved you.

c. Giving Warning

You may be in danger.

d. Expressing Admission with Reservation

I might be wrong, but I know what I did.

e. Expressing Obligation

You must do this.

ADVERBS

Adverbs modify verbs, adjectives, or another adverb.

Adverb of Manner

This answers the question how a thing is done.

Example:

He did the job poorly.

Question: How did he do the job?

Answer: Poorly

Adverb of Time

This answers the question when the action happened.

Example:

He will go there tomorrow.

Question: When will he go there?

Answer: Tomorrow

Adverb of Frequency

This answers the question how often.

Example:

She is there every week.

Question: How frequent is she there?

Answer: every week

PREPOSITION

A preposition links words within a sentence. It also states how two separate things are related. A preposition may indicate a location, direction, possession, or cause.

Common Prepositions

In front of	About	At	In	Beneath
In regard to	Above	After	Near	During
Inside	Across	Around	Down	By
In spite of	Against	Before	Despite	By means of
Into	Among	Behind	From	Amid

CONJUNCTION

If prepositions link words within a sentence, conjunctions relate or join words within or across sentences

Coordinating Conjunctions – connect words or group of words

Examples: but, and, for, nor, or, so, yet

TIP: Coordinating conjunctions connect words of the same grammatical structures.

Correlative Conjunctions – connect words under the same grammatical structure, but they always appear in pairs.

Examples: either...or, neither...nor, not only...but also

Subordinating Conjunctions – connect two complete ideas to make one dependent to the other.

Examples: inasmuch as, whenever, unless

INTERJECTION

These are words that express strong feelings or emotions. They may function as an independent sentence with the speaker as the subject.

Examples: ouch, aha, alas, oh, hurray

Subject and Verb Agreement

Collective Nouns

These nouns can either take a singular or plural verb depending on how they are used in a sentence. On the one hand, if a collective noun is perceived as one entity, then it will take a singular verb. On the other hand, if a collective noun is perceived as individuals in the group, it will take a plural verb.

Examples: Our debate team has won the competition.
Our debate team have won all their battles.

Nouns ending in -s and -ics.

Nouns ending in s and ics are always paired with verbs in the singular form.

Examples: Mathematics is my favorite subject.
Measles is a dangerous disease.

Nouns in sets of twos.

Nouns of this nature take a singular verb if the word "pair" is present. If not, they take a plural verb.

Examples: A pair of scissors is on the table.
The scissors are on the table.

"A number" and "The number"

Sentences beginning with the phrase "a number" always take a plural verb, while sentences that begin with "the number" take a singular verb.

Examples: A number of students have been expelled.
The number of students being expelled is rising.

Fraction and Percent

Nouns in percent and in fraction take a singular verb if it paired with a non-count noun, and vice versa.

Examples: 50% of the Earth's water is still safe for drinking.
One-third of the students were expelled.

Compound Nouns

Compound subjects joined by a conjunction take plural verbs.

Example: Paul and Robin are brothers.

Neither...nor and Either...or

The verb must agree with the closest noun.

Example: Neither John nor his brothers are going to America.

As well and Together with

The verb must agree with the first noun (subject).

Example: The president, together with his advisers, is leaving tomorrow.

Tenses**Simple Present Tense**

Verbs in the present tense express habitual or factual actions.

Examples: She sings.

The sun shines.

Simple Past Tense

Verbs in the past tense express actions that happened in the past.

Example: She sang yesterday.

Simple Future Tense

Verbs in the future tense express actions that will happen in the future.

Example: She will sing.

Present Progressive Tense

Verbs in the present progressive tense, express actions that are happening at the moment.

Example: She is singing.

Past Progressive Tense

This expresses a continuing action that started and ended in the past.

Example: She was singing all afternoon yesterday.

Future Progressive Tense

This expresses a continuing action that will happen in the future.

Example: She will be singing in the competition.

Present Perfect Tense

This expresses an action that started in the past, but is still happening at present.

Example: I have sung a song.

Past Perfect Tense

This expresses two past actions, in which one happened before the other.

Example: She had sung before she danced.

Future Perfect Tense

This expresses two future actions, in which one will happen before the other.

Example: She will have sung before she dances.

Present Perfect-Progressive Tense

This expresses an action that started in the past and still happening at present and will most likely continue to happen in the future.

Example: She has been singing since this morning.

Past Perfect-Progressive Tense

This expresses two past actions, where the first one was still happening when the second one transpired.

Example: She had been singing before the bomb exploded.

Future Perfect-Progressive Tense

will/shall

This expresses two future actions, where the first future action is still happening when the second one transpires.

Example: She will have been singing before the bomb explodes.

Summary:

Present:

I dance.

Past:

I danced.

Future:

I will dance.

Present Progressive:

I am dancing.

Past Progressive:

I was dancing all afternoon yesterday.

Future Progressive:

I will be dancing.

Present Perfect:

I have danced.

Past Perfect:

I had danced before I sang.

Future Perfect:

I will have danced before I sing.

Present Perfect Progressive:

I have been dancing since then.

Past Perfect Progressive:

I had been dancing when the bomb exploded.

Future Perfect Progressive:

I will have been dancing when the party ends.



Faulty Modifiers

Dangling Modifier – absence of the word being modified

Example: Inside the store, shoes must be worn.

Who must wear shoes?

Customers must wear shoes inside the store.

Misplaced Modifier – the modifier modifies the wrong word

Example: Turning green, I watched the lights

I watched the lights turning green.

Fragment – group of words masquerading as a sentence

Example: The justice system in the olden times

Sentence

Four Kinds of Sentences according to Function

Declarative - states an idea and expresses facts and opinions

Example:

She is a pretty girl.

We must learn how to swim.

The earth is round.

Interrogative - used in asking questions

- ends with a question mark

Example:

Who are you?

What is the capital city of Hungary?

Imperative - used in asking someone to do something

- "you" is the implied subject

Example:

Go away.

Turn around.

Write your name

Exclamatory - used to express strong feeling or emotion
- ends with an exclamation point

Example:

Fire!

Bomb!

You won!

Four Kinds of Sentences according to Structure

Sentence Patterns with Transitive Verbs

S-AV-DO (Subject-Action Verb-Direct Object)

Beth read the book quickly.

I gave him a book.

S-AV-IO-DO (Subject-Action Verb-Indirect Object-Direct Object)

I gave the place a new coat of paint.

Mr. Padilla gave me the test results.

S-AV-DO-OC (Subject-Action Verb-Direct Object-Object Complement)

The judges considered him a champion.

The movie's ending made her happy.

Sentence Patterns with Linking Verbs

S-LV-PN (Subject-Linking Verb-Predicate Nominative)

That boy is the culprit.

One of the contestants is she.

S-LV-PA (Subject-Linking Verb-Predicate Adjective)

The show is cool.

The singer sounds bad.

Four Kinds of Sentences According to Structure

Simple Sentence - composed of one independent clause

Compound Sentence - composed of two or more independent clauses,

Complex Sentence - composed of one independent clause and two or more dependent clauses

Compound-Complex - composed of two or more independent clauses and one or more subordinate clauses

Examples:

Simple Sentence:	I tried to stop her.
Compound Sentence:	I tried to stop her, but she still went away.
Complex Sentence:	I tried to stop her when she left.
Compound Complex:	I tried to stop her when she left, but she still went away.

LITERATURE

Introduction to Fiction

FICTION

Fiction is a prose imaginative composition which may or may not be based on history or fact.

The different types of fiction are the following:

- **Short Story** – a relatively brief prose fictional composition based on a single main incident which is designed to produce a single dominant impression
- **Novel** – a prose fictional work of considerable length that deals with a series of complications involving characters in a particular setting
- **Drama** – a composition in prose or verse designed for stage performance through mime and dialogue
- **Allegory** – a symbolic fictional account conveying meaning/s beyond the literal

Elements of Fiction

Plot

The Plot is a series of events knit together following the principle of cause and effect. It is also deemed to be an arrangement of incidents, the narrative structure, the organization of a narrative, and the logical sequence of actions. A plot can be arranged and organized in two ways, the first one is through Chronology—which means that the events are arranged according to time and space, and the second one is through Climax—which means that the events are organized according to order of suspense.

There are also two types of plot. The first one is called organic, which means the story sprouted from just one conflict; and the second one is episodic, which means there are two or more sources of conflicts.

Conflict

Conflict is considered as the soul of the plot, and it is the tension between opposing forces in the story. It can be external, which means the conflict is from outside forces; or internal, which means the conflict resides with the main character.

Here are the types of conflicts:

- physical - man vs. nature
- social - man vs. man
- psychological - man vs. self
- cosmic - man vs. God

Character

Characters in the story are the moral agents of actions. They are the invented personages in fiction.

There are two types of characters, namely major and minor. Under major characters, we have the protagonist, who is the central character around whom the story revolves; and the antagonist, who prevents the protagonist from solving the conflict. Under minor characters, we have foil, who has the opposite traits of the main protagonist; the confidant, who serves as the friend of the protagonist; and the background characters, who are not closely related with the protagonist.

We also have two kinds of character. The first one is round, which means the character was able to undergo change, while the second one is called flat, which means there was no change in the outlook and action of the character.

Setting

The setting serves as the background of the story, may it be physical, mental, or spiritual. It serves as the backdrop and sets the mood of the characters. There are three elements of setting. The first one is time, which sets the duration of the events; next is place, which talks about the locality; and the third one is atmosphere, which is the emotion or the mood.

Theme

Theme is considered as the central message of the story. It is the universal truth expressed in the text.

Point of View

This pertains to the vantage point where the story is narrated. Below are the different types:

- **FIRST person** – a principal character in the story is the one narrating it.
- **SECOND person** – an indirect disclosure of the narrating self for characterization and analysis.
- **THIRD person (unlimited)** – also known as omniscient point of view, where the narrator is an all-knowing maker.
- **THIRD person (limited)** – also known as the central intelligence point of view; the author chooses a character from whose consciousness the entire story is told
- **CAMERA EYE** – presents the dialogues, and the incidents of a narrative like a mechanical recording device
- **REVOLVING** – characterized by a narrative shift from one point of view to another

COMPOSITE point of view – gives a comprehensive view of the events and incidents in the story through the different angles adapted by several narrating characters

Figurative Language

1. **Synecdoche** - an association of some important part with the whole it represents.
Example: The face who launched a thousand ships.
2. **Simile** - an indirect association using "like" or "as"
Example: She is like a flower.
3. **Personification** - giving human attributes to an inanimate object (animal, idea, etc.)
Example: The sun is looking down on me.
4. **Oxymoron** - a self-contrasting statement
Example: Loud silence
5. **Metonymy** - an association in which the name of something is substituted by something that represents it
Example: Toothpaste is sometimes called Colgate.
6. **Metaphor** - a direct comparison that disregards the use of "like" or "as"
Example: You are the sunshine of my life.
7. **Irony** - The contrast between what was expected and what actually happened.
Example: No smoking sign during a cigarette break
8. **Hyperbole** - An exaggeration
Example: Cry me a river.
9. **Euphemism** - Creating a positive connotation out of something negative
Example: Loved child (illegitimate child)
10. **Ellipsis** - omission of words in a sentence
Example: She walked away and so the world turns...
11. **Asyndeton** - Not putting any connectors (conjunctions or prepositions)
Example: No retreat, no surrender.
12. **Apostrophe** - A direct address to an abstract things or a person who passed away already
Example: Love, please come and take me!

WORLD LITERATURES

AUTHORS	WORKS
LEQI-UNNINNI, Scribe (700 BCE)	The Epic of Gilgamesh
HOMER (800 BCE)	Iliad Odyssey
CONFUCIUS (551-479 BCE)	The Analects
AESCHYLUS (525-456 BCE)	The Oresteia Agamemnon
SOPHOCLES (496-406 BCE)	Theban Plays: Oedipus Rex Oedipus at Colonus Antigone
EURIPIDES (484-406 BCE)	Alcestis Medea Hippolytus The Trojan Women Electra
HERODOTUS (484-425 BCE)	The Histories
THUCYDIDES (470-400 BCE)	The History of the Peloponnesian War
SUN-TZU (450-380 BCE)	The Art of War
ARISTOPHANES (448-388 BCE)	Lysistrata The Clouds The Birds
PLATO (428-348 BCE)	The Republic
ARISTOTLE (384-322 BCE)	Ethics Politics Poetics
MENCIUS (400-320 BCE)	The Book of Mencius
VALMIKI (300 BCE)	The Ramayana
VYASA (200 BCE)	The Mahabharata
Anonymous (200 BCE)	The Bhagavad Gita
SSU-MA CH'EN (145-86 BCE)	Records of the Grand Historian
LUCRETIIUS 1(00-50 BCE)	Of the Nature of Things
VIRGIL (70-19 BCE)	The Aeneid
AURELIUS, MARCUS (121-180)	Meditations
SAINT AUGUSTINE (354-430)	The Confessions



KALIDASA (400)	The Cloud Messenger Sakuntala/Shakuntala
MUHAMMAD (650)	The Koran
HUI-NENG (638-713)	The Platform Sutra of the Sixth Patriarch
FIRDAUSI (940-1020)	Shah Nameh
SEI SHONAGON (965-1035)	The Pillow Book
MURASAKI, Lady Shikibu (976-1015)	The Tale of Genji (First Novel in the world)
KHAYYAM, Omar (1048-?)	The Rubaiyat
ALIGHIERI, Dante (1265-1321)	The Divine Comedy
KUAN-CHUNG, Luo (1330-1400)	The Romance of the Three Kingdoms
CHAUCEER, Geoffrey (1342-1400)	The Canterbury Tales
Anonymous (1500)	1001 Nights/Arabian nights
MACHIAVELLI, Niccolo (1469-1527)	The Prince
RABELAIS, Francois (1483-1553)	Gargantua and Pantagruel
WU CH'ENG-EN (1500-1582)	Journey to the West
MONTAIGNE, Michel (1533-1592)	Essays—Apology for Raymond Sebond
SAAVEDRA, Miguel de Cervantes (1547-1616)	Don Quixote
SHAKESPEARE, William (1564-1616)	Romeo and Juliet Much Ado About Nothing Twelfth Night Hamlet Merchant of Venice
DONNE, John (1573-1631)	Devotions Sermons First and Second Anniversaries
GALILEI, Galileo (1574-1642)	Dialogue Concerning The Two Chief World Systems
HOBBS, Thomas (1588-1679)	Leviathan
DESCARTES, Rene (1596-1650)	Discourse on Method
MILTON, John (1608-1674)	Paradise Lost Lycidas Areopagitica
MOLIERE (1622-1673)	The School for Wives Tartuffe The Would-Be Gentleman
PASCAL, Blaise (1623-1662)	Thoughts
BUNYAN, John (1628-1688)	Pilgrim's Progress



DICKENS, Charles (1812-1870)	Pickwick Papers The Tale of Two Cities A Christmas Carol David Copperfield Great Expectations
TROLLOPE, Anthony (1815-1882)	The Warden
BRONTE, Charlotte (1816-1855)	Jane Eyre
BRONTE, Emily (1818-1848)	Wuthering Heights
THOREAU, Henry David (1817-1862)	Walden Civil Disobedience
TURGENEV, Ivan I (1818-1883)	Fathers and Sons
MARX, Karl (1818-1883)	The Communist Manifesto
MELVILLE, Herman (1819-1891)	Moby Dick
ELIOT, George (1819-1880)	The Mill on the Floss Middlemarch Silas Marner
WHITMAN, Walt (1819-1892)	Leaves of Grass
FLAUBERT, Gustave (1821-1880)	Madame Bovary
DOSTOYEVSKY, Fyodor I (1821-1881)	Crime and Punishment Brothers Karamazov
TOLSTOY, Leo (1828-1910)	War and Peace Anna Karenina
DICKENSON, Emily (1830-1886)	The Soul Selects Her Society A Dimple in the Tomb
TWAIN, Mark (1835-1910)	Huckleberry Finn Tom Sawyer
HARDY, Thomas (1840-1928)	The Mayor of Casterbridge
FREUD, Sigmund (1856-1939)	The Interpretation of Dreams
CHEKOV, Anton (1860-1904)	Uncle Vanya Three Sisters The Cherry Orchard
WHARTON, Edith (1862-1937)	The Age of Innocence The House of Mirth
FROST, Robert (1874-1963)	The Road Not Taken Stopping by the Woods on a Snowy Evening
FORSTER, E.M. (1879-1970)	A Passage to India
JOYCE, James (1882-1941)	Ulysses

WOOLF, Virginia (1882-1941)	Mrs. Dalloway To the Lighthouse Orlando
LAWRENCE, David Herbert (1885-1930)	Sons and Lovers Lady Chatterley's Lover The Fox
O'NEIL, Eugene (1888-1953)	A Long Day's Journey into the Night Mourning Becomes Electra
ELIOT, T.S (1888-1965)	Waste Land
HUXLEY, Aldous (1894-1963)	Brave New World
FAULKNER, William (1897-1962)	The Sound and the Fury A Rose for Emily
HEMINGWAY, Ernest (1899-1961)	Old Man and the Sea
ORWELL, George (1903-1950)	1984 Animal Farm
NARAYAN, R.K.	The English Teacher
BECKETT, Samuel (1906-1989)	Waiting for Godot
MARQUEZ, Gabriel Garcia (1928-present)	One Hundred Years of Solitude Love in the Time of Cholera
ACHEBE, Chinua (1930-present)	Things Fall Apart No Longer at Ease
MORRISON (1931-present)	Sula The Beloved Jazz Song of Solomon

FILIPINO AUTHORS and their WORKS

AUTHORS	WORKS
ALFON, Estrella	Magnificence and Other stories
ANGELES, Carlos	The Knifed Horizon A Stun of Jewels
ARGUILLA, Manuel	How My Brother Leon Brought Home a Wife and Other Stories
AYALA, Tita Lacambra	Sunflower Poems

BAUTISTA, Cirilo	The Archipelago Telex Moon Sunlight on Broken Stones
BENITEZ, Francisco	What is an Educated Filipino?
BENITEZ, Paz Marquez	Dead Stars Stepping Stones Half a Life
BRILLANTES, Gregorio	The Living and the Dead A Wind Over the Earth Distance to Andromeda
BULOSAN, Carlos	America is in the Heart (Autobiographical) The Laughter of My Father The Voice of Bataan The Power of the People
CORDERO-FERNANDO, Gilda	People in the War The Visitation of the Gods The Butcher, The Baker, and the Candlestick Maker
DAGUIO, Amador	The Wedding Dance
DEMETILLO, Ricardo	No Certain Weather Barter in Panay Daedalus and Other Poems Daedalus and Other Poems Masks and Signatures
ENRIQUEZ, Egmidio Alvarez	The Devil Flower House of Images
FERNANDEZ, Doreen	Palabas: Essays on Philippine Theater
FLORENTINO, Leona	Poems in Spanish and Ilocano
FORD, Aida Rivera	Now and at the Hour
GAMALINDA, Eric	Fire Poem/Rain Poem Popular Delusions Planet Waves
GLORIA, Angela Manalang	Poems

GONZALES, N.V.M.	The Winds of April A Season of Grace Seven Hills Away Children of the Ash-Covered Loam The Bamboo Dancers
GUERRERO, Amadis Ma.	Children of the City
HAGEDORN, Jessica	Dogeaters Gangster of Love
IGLORIA, Maria Luisa	Encanto Blood Sacrifice
JALANDONI, Magdalena	Juanita Crus Ang Dalaga sa Tindahan
JOSE, Francisco Sionil	Ermita Poon My Brother, My Executioner
JOAQUIN, Nick	The Woman Who Had Two Navels Summer Solstice May Day Eve
LATORENA, Paz	Small Key Desire Sunset
LOPEZ, Salvador	Literature and Society
LUMBERA, Bienvenido	Revaluation Abot-Tanaw
NAKPIL, Carmen Guerrero	My Humble Opinion Women Enough
POLOTAN-TUVERA, Kerima	The Virgin The Hand of the Enemy
RAMOS, Maximo	The Creatures of Philippine Lower Mythology
ROTOR, Arturo	Zita The Wound and the Scar
SANTOS, Bienvenido	The Volcano The Man Who (Thought He) Looked Like Robert Taylor The Day the Dancers Came Scent of Apples

SOTTO, Juan Crisostomo	Lidia
TIEMPO, Edith	His Native Coast The Tracks of Babylon Blade of Fern
TINIO, Rolando	Valediction sa Hillcrest Claudia and Her Mother
VILLA, Jose Garcia	Man Songs Footnote to Youth
ZUI UETA DA COSTA, Rafael	Like the molave
ZAFRA, Jessica	Twisted

IDIOMATIC EXPRESSIONS

EXPRESSION	MEANING
Come hell or high water	I am by your side come hell or high water. This means the speaker would not leave the one he/she is speaking with no matter what happens.
Put your finger in the pie	To finish the task at a much earlier time, everyone must put his/her finger in the pie. It means they must do their share in a particular task.
Be in the limelight	The soprano did her best to be in the limelight. It means to be at the center of everybody's attention.
Flogging a dead horse	Rallies and mass demonstrations against the RH law are like flogging a dead horse. It means that the work being done is futile.
Bring home the bacon	The speech coach cheered his contestants. He said "Bring home the bacon!" It means that he wants the team to win.
Between the devil and the deep blue sea	This situation is hopeless! It is like the devil and the deep blue sea. The situation is at its worst, and the resolution is seen to be unpleasant.
In your face	The debater was criticized straight in her face. To state something in an aggressive manner
Under the weather	She did not attend her class. She said that she feels under the weather. The person is sick.

Back to square one	Her efforts are wasted since she is back to square one. The person has to start again from the beginning.
Tickled pink	The news tickled her pink. Made one very happy
Hold your horses	The commandant reminds his team to hold their horses until the right time comes. Be patient.
When pigs fly	There is no chance of us being husband and wife, unless pigs fly. Impossible to happen
At the pink of health	She looks beautiful and well rested. I assume she is at the pink of health. Good health

Phrasal Verbs	Meaning
Add up	Add
Bring about	Cause to happen
Bring up	Raise
Call off	Cancel
Carry on	Continue
Back up	Support
Bring off	Accomplish
Carry out	Complete
Count in	Include
Cut down	Reduce
Fill out	Complete (printed form)
Fill up	Complete (container)
Hang up	Suspend
Hold up	Rob
Pay off	To complete payment
Touch up	Repair
Turn down	Refuse
Throw over	Reject
Save up	Accumulate
Put off	Postpone
Down play	Diminish
Figure out	Understand
Break down	To analyze

Multiple choice.

1. It is raining cats and dogs, Sophia; may I borrow _____ umbrella? I left _____ inside the car.

A. Your, Mine
B. Yours, My

C. You're, Mine
D. Your, My

Answer: A (your, mine). The speaker left her umbrella, so she is borrowing Sophia's. The speaker is the first person, while Sophia (being talked to) is the second person, so the pronoun is your.

2. In a classroom setting, which of the following kinds of listening do most students use?

A. To escape
B. To be informed

C. To be inspired
D. To enjoy

Answer: B (to be informed). In the classroom, where learning is supposed to take place, being informed is key to acquiring knowledge.

3. Graphic organizers aid in faster comprehension of ideas. If a writer wants to show how values (numerical) change over a period of time, what will be the best graphic organizer?

A. bar graph
B. line graph

C. pie chart
D. timeline

Answer: B (line graph). It shows movement (rise and fall) of values over time.

4. The President _____ the businessmen at the moment.

A. was meeting
B. meets

C. is meeting
D. will be meeting

Answer: C (is meeting). Because of the phrase "at the moment," we know it is in the present progressive tense.

5. In the statement "the moon is the brightest eye of the heavens above", what figure of speech is used?

A. Apostrophe
B. Simile
C. Metaphor
D. Onomatopoeia

Answer: C (metaphor). The two being compared are moon and eye, which are two different things originally. Since there is no use of the word "like" or "as," it is a direct comparison.

6. The Philippine Congress must pass a law that enforces the total ban _____ of cyanide in fishing.

- A. with the use
- B. on using
- C. on the use
- D. with using

Answer: C (on the use). The rest are ungrammatical.

7. If I _____ known her before, I would have hired her as my personal adviser.

- A. Had
- B. Have
- C. Could
- D. Will

Answer: A (had). The action can no longer take place because there is already an adviser. Hence, this shows a hindsight thought that talks about what could have been. Here, the best helping / auxiliary verb is "had."

8. The subject we were able to take last semester was too difficult for Jenny and _____.

- A. Me
- B. Myself
- C. I
- D. We

Answer: A (me). It is preceded by the preposition "for," making the word and "Jenny" the objects of the preposition. Being an object, the personal pronoun required is in the objective case. "Me" is the objective case of the personal pronoun "I."

9. She _____ finished writing her novel before the publishers arrive.

- A. Should
- B. Have
- C. Will have
- D. Have been

Answer: C (will have). The sentence is in the future perfect tense, which has two actions (one happening at the present and one being anticipated). The word "finished" signals the action at present, which must be accompanied by "will have."

10. What do we call a poem with fourteen lines and a specific rhyme scheme?

- A. Haiku
- B. Free Verse
- C. Sonnet
- D. Elegy

Answer: C (sonnet). It is based on the definition of a sonnet. Examples of sonnets are the Shakespearean and Petrarchan sonnets.

11. What do we call a traditional Japanese poem with three lines (with 17 syllables)?

- A. Haiku
- B. Free Verse
- C. Sonnet
- D. Elegy

Answer: A (haiku). A haiku is composed of three lines (with 5 syllables, then 7 syllables, and finally 5 syllables), and it usually discusses themes of nature and love.

12. Read the following lines from the Shakespearean play "Hamlet" and decide which of the following emotions is most reflected.

"To be or not to be, that is the question. Whether it is nobler in the mind to suffer the slings and arrows of outrageous fortune, or to take arms against a sea of troubles, and by opposing end them?"

- A. Sadness
- B. Wrath
- C. Indecision
- D. Doubt

Answer: C (Indecision). Words like "to be or not to be" and the constant presence of options shows a person who cannot make up his mind. Hence, there is indecision.

13. The invited resource speaker has a lot on his mind. He is such a **LOQUACIOUS** speaker. The capitalized word means:

- A. Reserved
- B. Lanky
- C. Verbose
- D. Limited

Answer: C (verbose). Loquacious means using a lot of words, or verbose.

14. Jacee was confused when she saw the printout of her speech because it had slashes after some words. What should she do when seeing a double slash?

- A. Stop the speech
- B. Wait for the audience's reaction
- C. Continue the speech
- D. Pause for a full second

Answer: D (pause for a full second). Slashes signal juncture or pausing in a speech. A double slash is for a full-second pause.

15. "The prodigal son, who is the black sheep of the family, has returned home." What figure of speech was used in the given statement?

- A. Metaphor
- B. Simile
- C. Irony
- D. Oxymoron

Answer: A (metaphor). The "prodigal son" was compared to the "black sheep" without the use of the word "like" or "as," making it a direct comparison.

16. "Two heads are better than one" means that it's better for two people to think of a solution than just one person. This is a classic example of this figure of speech.

- A. Simile
- B. Metaphor
- C. Synecdoche
- D. Metonymy

Answer: C (synecdoche). It is when a part of the whole (in this case, the head represents the whole person) represents the entire thing.

17. Novels of Nicholas Sparks are a big hit because they can make people feel romantic love—whether it is lost or requited. Selections like these can appeal to people successfully by achieving this literary standard.

- A. Style
- B. Spiritual value
- C. Emotional value
- D. Aesthetic value

Answer: C (emotional value). It is when a selection is able to make the readers feel what the author intends the reader to feel.

18. The planets of the solar system have mythological references. They are based on what genre of mythology.

- A. Greek mythology
- B. Roman mythology
- C. Norse mythology
- D. European mythology

Answer: B (Roman mythology). Jupiter, Venus, and Mars are examples of Roman gods and goddesses.

19. F. Sionil Jose is a national artist for literature famous for many selections, like *The God Stealer*. In the story, the protagonist, who hails from a tribal group, brought his foreigner boss to the community. The boss is an avid collector of unique items, and he desired the old *anito* of the protagonist's grandfather. Wanting to please his boss, he stole the *anito* and gave it to the foreigner. This caused the heartbreaking death of the grandfather. Afterward, grandson returned to his roots, wearing the tribal colors and shunning away his foreigner boss. Which of the following emotions best explains the grandson's reaction?

- A. fulfillment
- B. relief
- C. remorse
- D. sadness

Answer: C (remorse). Brought about by regret for what he did, the grandson returned to his roots out of remorse and the desire to make amends for his mistakes.

20. Who among the Filipino writers below focuses on the beliefs and traditions during the Filipino-Spanish period?

- A. Nick Joaquin
- B. Benvenido Santos
- C. NVM Gonzales
- D. F. Sionil Jose

Answer: A (Nick Joaquin). He was most famous for his Filipino-Hispanic-inspired writings.



21. The lines below are taken from the poem "The Spouse" by Luis Dato. Read the lines and decide which best describe the role of the wife.

"She holds no joys beyond the the day's tomorrow,
She finds no world beyond his arms' embrace
She looks upon the man behind the furrow
Who is her motion, time, and space"

- | | |
|-----------------|-------------------|
| A. Working Wife | C. Business Woman |
| B. Artist | D. House Wife |

Answer: D (house wife). The passage shows a woman who is subservient to the husband, who is her "motion, time, and space."

22. We often times see the phrase "Red Letter Day" on calendars. The quoted phrase is an example of:

- | | |
|--------------|-----------------|
| A. Hyperbole | C. Onomatopoeia |
| B. Idiom | D. Alliteration |

Answer: B (idiom). A red-letter day means a holiday, which in traditional calendars is signified by a red color. An idiom is a phrase not meant to be understood literally.

23. Sounds like /f/ and /v/ are caused by a friction between the lower lip and the teeth. They belong to this kind of consonant sound.

- | | |
|------------|--------------|
| A. palatal | C. stop |
| B. glottal | D. fricative |

Answer: D (fricative). It is caused by a friction between the teeth and the lower lip.

24. A poet admitted that for him to write a good piece, he must have read more than ten books. Indeed, the poet is a _____ reader.

- | | |
|--------------|----------------|
| A. Voracious | C. Industrious |
| B. Vehement | D. Indulgent |

Answer: A (voracious). Voracious is an adjective used to describe reader who reads a lot.

25. Venus, the Roman goddess of love, has always been the symbol of feminine pulchritude. The underlined word means:

- A. Homeliness
B. Plainness
C. Beauty
D. Grandiosity

Answer: C (beauty). Aphrodite is the goddess of beauty.

26. What is Jose Rizal's novel that centers on Crisostomo Ibarra, whose primary wish is to educate the youth?

- A. Noli Me Tangere
- B. La Liga Filipina
- C. El Filibusterismo
- D. For the Filipino Youth

Answer: A (Noli Me Tangere). This is the first of the two classic novels from Rizal.

27. An ample amount of water _____ needed for the city to sustain its communities.

- A. is
- B. were
- C. are
- D. was

Answer: A (is). Water is a mass / non-count noun. The subject in the sentence is "amount," which is also singular.

28. Who is the American writer behind the naturalistic novel "Red Badge of Courage"?

- A. Mark Twain
- B. Charles Dickens
- C. Stephen Crane
- D. George Orwell

Answer: C (Stephen Crane). This is a classic war novel.

29. What is considered as the very first masterpiece in Philippine Literature in English?

- A. The Small Key
- B. Footnote to Youth
- C. Dead Stars
- D. My Father Goes to Court

Answer: C (Dead Stars). This was written by Paz Marquez Benitez.

30. Had I studied very well, I _____ rewarded with a tour around El Nido.

- A. Was
- B. Would have been
- C. Will be
- D. Would be

Answer: B (would have been). This sentence shows regret about something that could have but did not happen. Of the choices, the one expressing regret over a missed opportunity is "would have been."

31. "Sturdy and strong, the Filipinos are like the molave" is an example of this figure of speech _____.

- A. Simile
- B. Irony
- C. Metaphor
- D. Hyperbole

Answer: A (simile). It is so because of the presence of the word "like," comparing Filipinos with the molave.



32. When preparing ingredients, she is always careful not to hurt _____.

- A. Ourselves
- B. Myself

- C. Himself
- D. Herself

Answer: D (herself). The subject is "she," and she is also the one preparing the ingredients. Hence, what is needed is the female reflexive pronoun "herself."

33. All of the books _____ returned prior to the holiday break.

- A. were
- B. was

- C. are
- D. is

Answer: A (were). The clues are the past-tense verb (returned) and the plural noun (books). Since "all" is an indefinite pronoun, it has to adjust to the clues in the sentence. "Books" is in the plural form, necessitating a plural verb. Likewise, it is in the past tense, so it needs a plural verb in the past tense—were.

34. The courage of the son who saved his father from a raging fire is worth _____.

- A. emulating
- B. mentoring

- C. narrating
- D. watching

Answer: A (emulating). Something that is admirable and commendable is worth imitating or learning from—emulating.

35. Neither the principal nor the coordinators _____ present in the event.

- A. were
- B. was

- C. am
- D. is

Answer: A (were). The sentence requires a plural verb since the nearer subject is "coordinators," which is plural.

36. "My concept of inner peace came from my mother's daily activities which I now recall with fondness and awe. She was a full-time housewife wholly dependent on my father's monthly salary. How she made both ends meet, guided us in our studies, and did small acts of charity on the side was beyond me."

Based on the recount, the mother's financial resources were _____.

- A. more than enough
- B. abundant
- C. nonexistent
- D. limited

Answer: D (limited). The resources were there, but they had to be managed well to "make both ends meet." Hence, the resources were limited.

37. "I am a retired public school teacher. As a teacher, I was branded as a terror in school. The pupils dreaded the day they would enter my class. Little did they know that behind my unpopular facade was a heart full of compassion. But how did I learn this moniker? I did not tolerate dirty pupils in my class. I wanted them to know that cleanliness of body was good for their health. I inspected their teeth, nails, footwear, handkerchiefs, clothes, ears, noses and hair."

Based on the recount, the teacher values _____.

- A. cleanliness
- B. study
- C. absence
- D. posture

Answer: A (cleanliness). This is shown by her inspection of teeth, nails, footwear, and hair, among others.

38. On your vacation in Palawan, if you chance upon Arturo, please give _____ my warm regards.

- A. Himself
- B. Myself
- C. Him
- D. Me

Answer: C (him). Arturo is a male and he is the one to receive the warm regards. Hence, you need the objective case of the male pronoun—"him."

39. Fables are stories using animals and objects as characters that are given human-like qualities. If this is the case, what figure of speech will be most prominent?

- A. assonance
- B. hyperbole
- C. onomatopoeia
- D. personification

Answer: D (personification). It happens when non-humans are given human-like qualities.

40. If a story is made out of an exchange of letters, it will be called _____ literature.

- A. Fiction
- B. Epistolary
- C. Episodic
- D. Classic

Answer: B (epistolary). Another term for letter is "epistle," as in the Epistles of St. Paul. An adjective form of epistle is epistolary.

41. My brother and I ran into a common friend of _____ in the park.

- A. ours
- B. us
- C. mine
- D. ourselves

Answer: A (ours). Since the friend is a common friend of the two, the answer should be in the plural form.

42. Payment is due _____ September 25.

- A. at
- B. on
- C. in
- D. during

Answer: B (on). A specific date (month and date and/or year) requires "on."

43. Several factors must be considered in order to make a sound _____.

- A. query
- B. decision
- C. change
- D. problem

Answer: B (decision). When making a decision or choice, one must consider several factors. The word "sound" collocates or is often found alongside the word "decision."

44. The Holy Bible contains a lot of life lessons. Specifically, it contains those that teach values. How do you call a story that is based on the Bible, particularly the Gospels?

- A. Tale
- B. Legend
- C. anecdote
- D. parable

Answer: D (parable). Examples include the Parable of the Prodigal Son.

45. We admire _____ when Hannah speaks Mandarin fluently in front of _____.

- A. Him, them
- B. Him, we
- C. Her, us
- D. Him, me

Answer: C (her, us). Hannah is a female. Therefore, the first word should be for females (her), while the second word should refer to the speakers (us).

46. "Because of PROFANITY, the program was suspended from television." The capitalized words best relates to this word.

- A. Obscenity
- B. History
- C. Decency
- D. Metaphor

Answer: A (obscurity). Profane is something negative because it is foul, shocks people, and does not respect decency.

47. The Philippines _____ the ASEAN integration agreement in 2008 before Timor Leste _____ for inclusion in the region some time in 2011.

- A. had signed, applied
- B. has signed, has applied
- C. will have signed, applied
- D. had signed, had applied

Answer: A (had signed; applied). The sentence is in the past perfect tense since both actions are already done.

English

48. Comprehend the following text: "Filipinos love eating. Regardless of the type of food, they enjoy it, such as Korean cuisines, Chinese delicacies, American staple food, even those from Israel."

What are the adjectives in the sentence?

- A. Korean, Chinese, American, Israel
- B. Korean, American, Israel
- C. Korean, Chinese, Israel
- D. Korean, Chinese, American

Answer: D (Korean, Chinese, American). Israel was used as a noun.

49. Though nothing can bring back
The hour of splendour in the grass,
Of glory in the flower,
We will grieve not,
Rather find strength in what remains behind.

This stanza expresses _____.

- A. Grief
- B. Hope
- C. Loss
- D. Despair

Answer: B (hope). It is signaled by the line "rather find strength in what remains behind."

For questions 51-53, comprehend the following text:

"If you feel downcast because you have gained weight during this Christmas season, let the words of St. Thomas Aquinas, who was himself known for being fat, console you. In his commentary on the 4th Book of Aristotle, he wrote: "Small and thin individuals can be called 'pretty' because of the appropriate proportion of their size in relation to their limbs; but they cannot be called beautiful because of the lack of magnitude that can sufficiently exude a splendor of order that catches the eye. Beauty can only be found in a large body."

50. Who was defending those who had a large body?

- A. The large people
- B. Saint Thomas Aquinas
- C. Aristotle
- D. Small and thin individuals

Answer: B (Saint Thomas Aquinas). It was from him that the quote was taken.

51. Based on the text, who can be called pretty?

- A. People with large bodies
- B. Aristotle
- C. Small and thin individuals
- D. Those who have splendor in their bodies

Answer: C (small and thin individuals). Their limbs are in proportion.

52. According to the quote, what do small and thin people lack to be called beautiful?

- A. Muscles
- B. Magnanimity
- C. Limbs
- D. Magnitude

Answer: D (magnitude). This, according to Saint Thomas Aquinas, is necessary.

53. A new show _____ on ABS-CBN two days ago.

- A. launched
- B. was launched
- C. will be launched
- D. is launching

Answer: B (was launched). The action took place in the past (two days ago), so it is in the simple past tense.

54. I watched the _____ news on television.

- A. late
- B. latest
- C. later
- D. lately

Answer: B (latest). News shows the most recent (latest) events. Likewise, latest is the only grammatically correct option.

55. The word "baduy" means:

- A. Uneducated
- B. Low profile
- C. Awkward looking
- D. Smart

Answer: C (awkward-looking). "Baduy" is a Tagalog term for things and attires that look awkward or out-of-style.

56. The Philippines _____ the ASEAN integration agreement in 2008 before Timor Leste _____ for inclusion some time in 2011.

- A. had signed, applied
- B. has signed, had applied
- C. signed, had applied
- D. signed, applied

Answer: A (had signed, applied). Both actions are in the past (2008 and 2011), so this is in the past perfect tense. In the past perfect tense, if the word "before" connects the two actions, the first action will take the "had + past participle" form, while the second action will take the simple past form.

57. Professionals _____ the concerns of clients regularly.

- A. handle
- B. will handle
- C. have handled
- D. handles

Answer: A (handle). The sentence is in the simple present tense, as signed by "regularly." Since the subject (professionals) is in the plural form, it requires a plural verb.

58. Orix Rental _____ a fleet of 32 cars on Saturday.

- A. buy
- B. bought
- C. buys
- D. will buy

Answer: D (will buy). The action has not yet taken place but will take place in the future (Saturday). Therefore, it needs the simple future tense.

59. Rolls Royce _____ the Philippine market a few years ago.

- A. has entered
- B. had entered
- C. entered
- D. will enter

Answer: C (entered). The action took place a few years ago, so it requires the simple past tense.

60. Comprehend the following text: "Learning is seen when there is a change in behavior, as caused by experience. Experience can come in two forms: personal and vicarious. A personal experience is something that the person concerned directly felt, saw, heard, tasted, or smelled, while a vicarious experience is something the person observed in others."

How many ways of learning are present?

- A. 1
- B. 2
- C. 3
- D. 4

Answer: B (2). There are two forms of learning through experience: personal and vicarious.

61. In the courtroom, the judge said that the findings have no bearing because they are _____.

- A. Inaccurate
- B. Inadmissible
- C. Confusing
- D. Inedible

Answer: B (inadmissible). Inadmissible evidence means it cannot be used in court.

62. The newly elected president demonstrated great **APLOMB** in dealing with the nasty and foul questions thrown by reporters. The capitalized word means:

- A. Nervousness
- B. Hesitation
- C. Composure
- D. Anger

Answer: C (composure). Aplomb in the face of nasty reporters is a sign that you can control yourself. Hence, the answer is composure.

63. The victim's request to reopen the case _____ very reasonable.

- A. were
- B. are
- C. is
- D. will be

Answer: C (is). Since there is already an existing request, we know that it is in the present, and since there is only one, it requires a singular verb.

64. Nostradamus' **APOCALYPTIC** view of the future is one of the most alarming in history. The capitalized word means:

- A. Destructive
- B. Scary
- C. Encouraging
- D. Exact

Answer: A (destructive). Apocalypse is known to cause destruction and chaos.

65. I cannot find _____ book. May I photocopy _____, Paul?

- A. my, you
- B. mine, your
- C. my, yours
- D. mine, yours

Answer: C (my, yours). Since the speaker lost his book, he relies on others for reference.

66. In Philippine Literature, who is the writer known as "Huseng Sisw"?

- A. Jose Rizal
- B. Francisco Baltazar
- C. Jose dela Cruz
- D. Andres Bonifacio

Answer: C (Jose dela Cruz). This is the nickname given to him.

67. "The goddess Aphrodite is the phantom of delight." The sentence is an example of:
- A. Metaphor
 - B. Simile
 - C. Irony
 - D. Hyperbole

Answer: A (metaphor). Aphrodite was directly compared to a phantom, in this case, the phantom of delight.

68. "It was many a year ago in a kingdom by the sea, there lived whom you may know by the name of Annabel Lee". This line is from a poem written by _____.
- A. John Keats
 - B. William Wordsworth
 - C. Edgar Allan Poe
 - D. Robert Frost

Answer: C (Edgar Allan Poe). He is the writer behind "Annabel Lee."

69. In the poem Invictus, the line "My head is bloody, but unbowed" is an example of:
- A. Metaphor
 - B. Oxymoron
 - C. Irony
 - D. Hyperbole

Answer: C (irony). Despite being bloody (which can be logically assumed to be bowed or tilted), the head remains upright.

70. Based on the reports of the inspector, the edifice collapsed because of _____ joints.
- A. Loose
 - B. Losing
 - C. Lose
 - D. Lost

Answer: A (loose). Loose (opposite of tight) joints can cause a structure to collapse or fall.

71. Teachers always TRANSMUTE the grades of the students. The capitalized word means:
- A. Change
 - B. Degrade
 - C. Upgrade
 - D. Add

Answer: A (change). Transmute is synonymous to change.

72. Comprehend the following text. "Dictatorship is a form of government in which an individual or a group holds absolute power. The term dictator originated in ancient Rome. The Roman Senate often appointed an individual as "temporary dictator" to handle national emergencies without the approval of the Senate."

Based on the text, dictatorship has its roots in this era of history.

- A. Greek empire
- B. Spanish colonization
- C. Roman era
- D. Babylonian era

Answer: C (Roman era). The only era mentioned in the historical detail is Roman.

73. This Shakespearean play talks about the ill-fated love affair between two individuals from warring families.
- A. Romeo and Juliet
 - B. Midsummer Nights Dream
 - C. Merchant of Venice
 - D. The Twelfth Night

Answer: A (Romeo and Juliet). This is the most famous romantic tragedy written by Shakespeare.

74. The cost of a new computer is quite expensive for _____. Besides, we have more urgent things to prioritize.
- A. Us
 - B. Them
 - C. We
 - D. They

Answer: A (us). The clue can be found in the second sentence, which begins with the subject "we." Hence, the correct answer is the pronoun "us," since it is preceded by the preposition "for." If it is preceded by "for," the word will become the object of the preposition. Thus, we need the pronoun in the objective case—us.

75. Karenin, Anna's husband, was betrayed for a number of times and yet he bears no RANCOR in his heart. He is not _____.
- A. Embarrassed
 - B. Bitter
 - C. Shy
 - D. Insulted

Answer: B (bitter). Rancor is an emotion related to holding a grudge, which is a sign of bitterness against others.

76. The statements of the witness are ~~impertinent~~ to the case because they are _____.
- A. Changeable
 - B. Irrelevant
 - C. Illogical
 - D. Important

Answer: B (irrelevant). Impertinent is a synonym of irrelevant.

77. The participants look _____. They showed great enthusiasm for the lectures.
- A. Angry
 - B. Offended
 - C. Interested
 - D. Happy

Answer: C (interested). This is signaled by the phrase "great enthusiasm" in the second sentence.

78. The Greek literary character who launched a thousand ships is _____.

- A. Patroclus
- B. Achilles
- C. Helen
- D. Agamemnon

Answer: C (Helen). Helen of Troy, as she came to be known, was the reason why King Menelaus launched a thousand ships to rescue her from her "abductors" in Troy. In reality, she was taken to Troy to be given to the most handsome man on Earth, Paris, as part of the deal between Paris and Aphrodite, the goddess of beauty.

79. Who is the most famous proponent of the Arena Theatre in the Philippines?

- A. Montano
- B. Avellana
- C. Tinio
- D. Carpio

Answer: A (Montano). His name is Severino Montano, one of the titans of Philippine theater.

80. The chief chef instructed his students to be careful not to hurt _____ while slicing the onions.

- A. Them
- B. Themselves
- C. Ourselves
- D. Themselves

Answer: B (themselves). The chef was warning the students, not himself. Since the students were also going to use the kitchen utensils, they were being warned not to hurt themselves. "Themselves" is a reflexive pronoun.

81. The transport system in the Philippines _____ drastically in the last ten years.

- A. Did improve
- B. Had improved
- C. Has improved
- D. Has been improving

Answer: A (did improve). The sentence does not refer to an action that began in the past and continues to the present. It just talks about something that is already done, completed over years. The best option is "did improve," which will make it a sentence in the emphatic past tense.

82. Mr. Padilla finally found time to sleep after _____ for a number of hours.

- A. Worked
- B. Working
- C. Have worked
- D. Had worked

Answer: B (working). The only grammatically correct choice is "working," as the others do not fit.

83. My bread and butter _____ acting.

- A. is
- B. was

- C. are
- D. were

Answer: A (is). "Bread and butter" is meant here as an idiomatic expression to refer to one's job or source of income. Hence, it is taken as a singular subject.

84. When Sam failed his exam for the second time, he resorted to DUTCH COURAGE before he went home to break the news to his family. What does DUTCH COURAGE mean?

- A. A trial or practice performance
- B. Courage which one gets from drinking alcohol
- C. A fair chance to do something
- D. Partial payment paid

Answer: B (courage which one gets from drinking alcohol). The Dutch are known for their penchant for drinking alcohol, which can make people forget their fears.

85. Comprehend the text: "Culture is a term used by social scientists for a way of life. Every human society has a culture. Culture includes a society's arts, beliefs, customs, institutions, inventions, language, technology, and values. A culture produces similar behavior and thought among most people in a particular locality."

What did social scientists define?

- A. beliefs
- B. customs
- C. culture
- D. society

Answer: C (culture). Culture is the main subject of the paragraph. All details provided refer to culture.

86. Edgar Allan Poe wrote "The Raven" from which the line is taken:

And the silken sad uncertain rustling of each purple curtain

Which of the following figures of speech is used by the poet to establish a mood?

- A. Onomatopoeia
- B. Alliteration
- C. Allusion
- D. Metaphor

Answer: B (alliteration). The repetition of the sounds /s/ and /t/ show it is an alliteration.

87. What is the mood of the lines below?

Memory, all alone in the moonlight
I can smile at the old days.
I was beautiful then.
I remember the time
I knew what happiness was,
Let the memory live again...

- A. Fear
B. Eagerness
C. Nostalgia
D. Humor

Answer: C (nostalgia). Longing for the past and remembering it fondly are signs of nostalgia, which can be understood as "missing something or someone."

88. Walter de la Mare wrote "Silver" from which the lines below are taken:

Slowly, silently, moon the moon
Walks the night in her silver shoon;
This way, and that, she peers and sees
Silver fruit upon silver trees.

Which of the figures of speech is used?

- A. Assonance
B. Personification
C. Metaphor
D. Simile

Answer: B (personification). Words like "peers," "sees," and "walks" are human-like qualities given to the moon in the passage.

89. Comprehend the following text: "The nitrogen cycle is the circulation of nitrogen and its compounds by living organisms in nature. Nitrogen in the air passes into the soil, where it is oxidized, by a process called nitrogen fixation."

Where does nitrogen get oxidized?

- A. air
B. soil
C. sun
D. living organisms

Answer: B (soil). Answer: found in the text, with the phrase "where it is oxidized."

90. This is a set of verbal or written symbols used by humans to communicate.

- A. understanding
B. encyclopedia
C. language
D. dictionary

Answer: C (language). Verbal and written symbols used for communication make up a language.

91. The Department of Education _____ K-12 before the Commission on Higher Education _____ preparing for 2016.
- A. had implemented; began
B. had begun; implemented
C. implemented; had begun
D. has begun; implemented

Answer: A (had implemented; began). The sentence is in the past perfect tense, with the Department of Education doing the first action (had implemented). The simple past form (began) is the second action.

92. The fourth-year high school students were told not to visit the Enchanted Kingdom before graduation, but they _____ on what they wanted.

Which verb tense will best fit if the writer wants to create the effect of emphasis?

- A. simple past – insisted;
B. past perfect – had insisted
C. past progressive – were insisting
D. emphatic past – did insist

Answer: A (simple past – insisted). The sentences simply state the contradiction in actions—being told not to go, but going anyway.

93. The world _____ the United Nations Day every October.

- A. celebrate
B. celebrates
C. celebrated
D. will celebrate

Answer: B (celebrates). With the presence of the phrase "every October," we know this sentence is in the simple present tense, since it is a habitual / regular action. For this tense, we use the base form of the verb.

94. The Board of Trustees will be convening _____ ten minutes.

- A. on
B. at
C. in
D. beyond

Answer: C (in). Of the given prepositions, only "in" is appropriate to introduce "ten minutes." The rest are grammatically incorrect.

95. The students who made the commotion in the library are now _____.

- A. investigating
B. investigated
C. being investigated
D. investigates

Answer: C (being investigated). If students cause the commotion, it is illogical to think that they will be the ones to investigate. Instead, they will receive the action of "investigating." Hence, the sentence is in the passive voice. Of the choices, "being investigated" is the one in the passive voice and with the word "now," the progressive tense is signaled.

96. The statistics _____ that the drop out rate this year is high.

- A. shows
B. show

- C. had shown
D. will have shown

Answer: B (show). The subject "statistics" is taken in the plural form because it talks about indicators and values.

97. Coordinators regularly _____ teachers to determine their competence.

- A. are observing
B. have observed

- C. observe
D. observes

Answer: C (observe). This sentence is in the simple present tense that expresses a habitual action (signaled by the adverb of frequency "have observed"). In this case, the subject is "coordinators," and the verb is in the plural form "observe."

98. Speaker Pantaleon Alvarez _____ the position since 2016.

- A. has been
B. has been holding

- C. had been holding
D. will have been holding

Answer: B (has been holding). The sentence is in the present perfect progressive tense, as signaled by the phrase "since 2016."

99. The Philippine government, with all of its officials, _____ one of the largest institutions in Asia.

- A. am
B. is

- C. are
D. will be

Answer: B (is). The phrase "with all of its officials" is a parenthetical expression enclosed between commas. Hence, the subject is "Philippine government," which is singular. Thus, the verb should be singular.

100. Philippine, American, and Italian are examples of _____.

- A. nouns
B. adjectives

- C. adverbs
D. verbs

Answer: B (adjectives). "American" and "Italian" can also be nouns, but "Philippine" is exclusively an adjective. Hence, the group is a group of adjectives.

101. _____ here! Look! _____ new car is parked _____!

- A. their, there, they're
B. they're, their, there

- C. they're, there, their
D. there, their, they're

Answer: B (they're, their, there). "They're" is a contraction of "they are," while "their" is a possessive pronoun for plural nouns. Finally, "there" is an indicator of location.

102. Behaviorism or cognitivism _____ a philosophy of education that teachers can adopt

- A. is
B. are

- C. was
D. were

Answer: A (is). In the case, the subjects are "behaviorism" and "cognitivism." However, they are connected by "or," which makes the sentence require a singular verb.

103. Bart wanted to tell Lisa what happened in school today. However, he wanted to tell Lisa and make her imagine the events as if they were still on-going

What tense will be best for this?

- A. simple past
B. past progressive

- C. past perfect
D. past perfect-progressive

Answer: B (past progressive). It is used to recall an event and retell it as if it were still happening.

104. Which among the four is the highest level of language, particularly in terms of the amount of meaning conveyed?

- A. clause
B. sentence

- C. word
D. phrase

Answer: B (sentence). The arrangement is word, phrase, clause, and sentence, with the sentence having the complete thought / meaning.

105. Its flag is very colorful, and each color signifies something.

What is the purpose of "its"?

- A. to show relationship
B. to show cause

- C. to show a contraction of "it is"
D. to show possession

Answer: D (to show possession). "Its" is a possessive pronoun to indicate ownership.

106. If President Duterte does not sign or veto a bill after thirty days, it _____ a law.

- A. become
B. becomes

- C. became
D. will become

Answer: D (will become). In this case, the conditional clause uses the present tense (does not), so the main clause should use the simple future tense (will become).

107. This is the kind of verb that adds -d or -ed to create its past form.

- A. transitive
- B. intransitive
- C. regular
- D. irregular

Answer: C (regular). Regular verbs are action verbs that form their past tense by adding -d or -ed to the last letter of the word.

108. Economics _____ a required subject in school, but the economics of the country _____ not improving because sound principles are not applied.

- A. is, is
- B. is, are
- C. are, is
- D. are, are

Answer: B (is, are). The first use of "economics" is as an academic subject or discipline, while in the second use, it is taken in the plural form because it refers to economic indicators and metrics.

109. The defense panel did _____ all the papers _____ for the incomplete one.

- A. accept, expect
- B. accept, except
- C. except, accept
- D. expect, accept

Answer: B (accept, except). "Accept" means to take in, while "except" means to exclude. In this case, the "incomplete one" was excluded. It is logical to think so since incompleteness is a sign of failure.

110. The convention will gather _____ from the various publishing companies for a day of sharing of good practices.

- A. editor-in-chief
- B. editor-in-chiefs
- C. editor-in-chieves
- D. editors-in-chief

Answer: D (editors-in-chief). In an hyphenated noun like "editor-in-chief," the plural form is given to the first word (editor), since the position is "editor," not chief.

111. During the deliberations, it was agreed upon that the honor students _____ decided by Friday.

- A. will have
- B. should have
- C. will have been
- D. could have been

Answer: C (will have been). This is a sentence in the future perfect tense. It is signaled by the deadline (by Friday). The action verb is given (decided) and needs the "will have been" in order to complete the sentence.

112. Four theorists were _____ the famous behaviorists, while _____ Vygotsky and Bruner, there was the similar view of constructivism.

- A. among, between
- ☒ B. between, among
- C. within, beside
- D. aside from, in the middle of

Answer: A (among, between). "Among" is a preposition used for three or more (in this case, four) entities being talked about, while "between" is strictly for two (Vygotsky and Bruner).

113. The classes were all in the rooms of the school, and stacks of papers were piled up.
What were the collective nouns in the sentence?

- A. classes, stacks
- ☒ B. classes, school
- C. school, stacks
- D. classes, school, stacks

Answer: A (classes, stacks). The "classes" stood for students, while "stacks" were for the papers.

114. "Don't give that bracelet as a gift _____, there's still that pendant _____ the dresser."

- A. beside, besides
- B. because, under
- C. notwithstanding, away from
- ☒ D. besides, beside

Answer: D (besides, beside). "Besides" means "in addition," while "beside" is a preposition indicating location.

115. "Bring home the bacon!" is an example of a/an _____.

- A. request
- B. irony
- C. personification
- ☒ D. idiomatic expression

Answer: D (idiomatic expression). It is an expression that uses words in a way that's different from how they literally mean. Idiomatic expressions are not meant to be understood literally.

116. "Peter Piper picked a peck of pickled peppers" is a sentence that uses what figure of speech?

- A. apostrophe
- B. onomatopoeia
- C. alliteration
- ☒ D. allusion

Answer: C (alliteration). This is when the same consonant sound begins words in succession, such as the consonant sound "/p/" which began the words "Peter Piper picked a peck of pickled peppers."

17. "And after years of anticipation, they finally met. Tensions were high because of this much-hyped meeting. What will happen? Will they fight? Will they hug? Those questions will be answered now."

This paragraph is best fit for what part for the short story?

- A. exposition
- B. rising action
- C. climax
- D. falling action

Answer: C (climax). It is the highest point of emotions and anticipation in the story.

118. "And know the stars are holding you, holding you tonight!" is a line that uses what figure of speech?

- A. simile
- B. metaphor
- C. personification
- D. irony

Answer: C (personification). The line "stars are holding you tonight" gives the stars a human-like quality of holding, which it does not otherwise have.

119. "Crispini Basilio! My children!" is an example of _____.

- A. allusion
- B. paradox
- C. apostrophe
- D. onomatopoeia

Answer: C (apostrophe). It is the figure of speech that is used when calling on people who are not physically present.

120. Who was the author of the work that had Inferno, Purgatorio and Paradiso?

- A. Dante Alighieri
- B. Francesco Petrarch
- C. Francis Bacon
- D. Edgar Allan Poe

Answer: A (Dante Alighieri). The works are part of the *magnum opus* Divine Comedy, written by the Father of Italian Literature.

121. This element of the short story is the combination of all events, which are caused to move because of the conflict.

- A. setting
- B. climax
- C. plot
- D. story train

Answer: C (plot). It contains all the events of the story, from the exposition to the resolution.

22. A sonnet is a poem that has fourteen lines and a total of _____ syllables.

- A. 130
B. 140

- C. 150
D. 10

Answer: B (140). With fourteen lines and ten syllables each, the sum of syllables is 140.

23. When pupils are asked to imitate animal sounds for a puppet show, they are using what figure of speech?

- A. simile
B. hyperbole
C. onomatopoeia
D. personification

Answer: C (onomatopoeia). It is the figure of speech directly related to the reproduction of animal and nature sounds.

124. Of the four, this is the shortest literary work.

- A. anecdote
B. short story
C. novelette
D. novel

Answer: A (anecdote). It is usually a paragraph or two only and can be used to introduce a concept or principle. It is also used to start a speech.

125. In the epic Beowulf, Grendel was incensed at being disturbed by the noise of the people in the Palace. As such, Grendel attacked and killed all of them.

What kind of character was Grendel?

- A. protagonist
B. antagonist
C. cameo
D. supporting

Answer: B (antagonist). Grendel wreaked havoc and caused chaos, bringing the negative events of the story. Hence, it is the antagonist.

126. Shakespeare's works like the Hamlet, Romeo and Juliet, and Merchant of Venice, are examples of this literary genre.

- A. tragedy
B. drama

- C. comedy
D. suspense

Answer: A (tragedy). All of these have a sad ending, which is a classic definition of a tragedy. The opposite of tragedy is "comedy," which has a happy ending.

127. "His ambitions are as high as Mount Everest" uses what figures of speech?

- A. simile and hyperbole
- B. metaphor and hyperbole
- C. illusion and apostrophe
- D. synecdoche and irony

Answer: A (simile and hyperbole). It uses a simile because of the use of the word "as" to compare ambition and Mount Everest, and also a hyperbole through the exaggeration of comparing the ambition to the tallest mountain in the world.

128. What is the thing that William Shakespeare, Dante Alighieri, and Miguel de Cervantes have in common?

- A. They were tragedians.
- B. They were great writers of satirical novels.
- C. They were Fathers of their country's respective literature.
- D. They were great poem writers.

Answer: C (They were Fathers of their country's respective literature). William Shakespeare is the Father of English (England) Literature, Dante Alighieri of Italian, and Miguel de Cervantes of Spanish.

129. Robin is to Batman, as Sancho Panza is to _____.

- A. Sherlock Holmes
- B. Beowulf
- C. Harry Potter
- D. Don Quixote de la Mancha

Answer: D (Don Quixote de la Mancha). Looking at the pairs of words, this is an analogy. Robin is the sidekick of Batman. Therefore, Sancho Panza is also a sidekick, this time of Don Quixote de la Mancha from Spanish literature.

130. The Holy Bible of the Christians has been around for centuries, yet it remains relevant to the present-day Christians.

What literary standard is shown in this situation?

- A. universality
- B. intellectual value
- C. style
- D. permanence

Answer: D (permanence). This literary standard is seen when a selection remains relevant despite the passing of time. As an old description goes, it is "timely and timeless."

131. "What is essential is invisible to the eye" is an example of a _____.

- A. theme
- B. moral
- C. virtue
- D. speech

Answer: B (moral). Another term for "lesson," the moral of the story is usually in the form of a sentence expressing a value, life lesson, or principle.

132. The Philippines' largest drug store is named after the Roman gods' _____.

- A. defender
- B. messenger
- C. warrior
- D. housekeeper

Answer: B (messenger). Hermes (Greek) or Mercury (Roman) is the messenger of the gods.

133. How do you call the relationship of Zeus and Jupiter?

- A. enemies
- B. friends
- C. siblings
- D. counterparts

Answer: D (counterparts). Zeus is to Greek, while Jupiter is to Roman mythology.

134. The Greek, Roman, and Norse stories have been very famous despite the passing of time. What is the term for stories about gods and goddesses?

- A. legend
- B. myth
- C. fantasy
- D. short story

Answer: B (myth). Gods and goddesses are the main characters in myths, the most famous of which come from the Greeks and Romans.

135. The Harry Potter series, as well as the renditions of Disney classics, have brought together families (parents and children), friends and other people from different walks of life in order to enjoy the stories unfold.

What literary standard is shown here?

- A. permanence
- B. universality
- C. emotional value
- D. style

Answer: B (universality). This literary standard means that a literary selection can be appreciated by people of different faiths, beliefs, nationalities, and cultures.

136. Who is the author of Les Misérables?

- A. Edgar Allan Poe
- B. Lewis Carroll
- C. Victor Hugo
- D. Alexandre Dumas

Answer: C (Victor Hugo). Les Misérables is an award-winning and classic musical adapted from the original story by Hugo.

7. The Adventures of Sherlock Holmes by Sir Arthur Conan Doyle, through the investigations of Sherlock, make people think and guess about whether their hypotheses are correct.

What is the literary standard shown here?

- A. emotional value
- B. permanence
- C. aesthetic value
- D. intellectual value

Answer: D (intellectual value). The words "think" and "guess" refer to cognitive processes. Furthermore, since the Adventures of Sherlock Holmes is a detective story, the profession (detective) gives the clue that it is related to thinking or the intellect.

38. "Data from the participants revealed that 92% wanted to go to college."

What chapter in the research is best suited for this?

- A. results
- B. conclusion
- C. method – data analysis
- D. literature review

Answer: A (results). It does not analyze the data; it just presents what the study found, without an explanation yet.

39. This lists the sources used in the study.

- A. appendices
- B. bibliography
- C. biography
- D. index

Answer: B (bibliography). It is synonymous to "references" or "reference list."

140. "What we know is outside their walls, there is a garden of bones. Every time they turn away a traveler, the garden grows." – Jorah (Game of Thrones) What happens to the travelers?

- A. They die.
- B. They plant more flowers.
- C. They serve as slaves.
- D. They disappear without explanation.

Answer: A (They die). Dead people eventually become bones, and since the garden of bones grows every time a traveler is turned away, it can be logically assumed that it was because the traveler died.

141. This refers to the scientific process of updating, validating, and discovering knowledge.

- A. scientific method
- B. experimentation
- C. investigation
- D. research

Answer: D (research). It can take many forms, like experimentations guided by the scientific method, or other forms of investigation.

142. This research method gathers data and analyzes them using numerical tools and procedures.

- A. statistical
- B. quantitative

- C. qualitative
- D. mathematical

Answer: B (quantitative). There are three research methods: quantitative, qualitative, and mixed. Of the three, quantitative uses numerical tools and procedures.

143. Apart from citing his sources to avoid plagiarism, Hans tries to edit the words of the author, infuses his own way of saying it, but stays true to the original intent and essence of the text. In doing so, he performs this process.

- A. editing
- B. summarizing

- C. paraphrasing
- D. citing

Answer: C (paraphrasing). This replaces the author's words with one's own without altering the meaning.

144. There are several punctuation marks in the English language. Which of these is used to replace the coordinating conjunctions like and, but, or, for, yet, and so?

- A. comma - ,
- B. colon - :

- C. period - .
- D. semi-colon - ;

Answer: D (semi-colon). It takes the place of coordinating conjunctions and connects sentences.

145. Complete the sentence with the correct coordination conjunction. "The research was a success _____ all questions were answered."

- A. for
- B. so
- C. but
- D. and

Answer: A (for). This sentence shows cause and effect, with the effect being the success of the research. The coordinating conjunction "for" is used to signal the cause.

146. Complete the stem. "The steps for the experiment are _____."

- A. as follow
- B. as follows
- C. following
- D. followed

Answer: B (as follows). The rest are not grammatically correct if fit within the given stem.

English

147. Of the different graphic organizers, this is the best fit for showing how one budgets his/her money for a month.

- A. pie chart
B. bar graph

- C. line graph
D. Venn diagram

Answer: A (pie chart). A pie chart shows distribution, and its total value is 100%.

148. Elizabeth Barrett-Browning is famous for her sonnet dedicated to her husband. It begins with the lines:

"How do I love thee
Let me count the ways
I love thee to the depth and breadth and height
My soul can reach
When feeling out of sight
For the ends of being and ideal grace"

Which word best describes the persona in the poem?

- A. emotionally detached
B. hopeful

- C. devoted
D. interested

Answer: C (devoted). The lines are overflowing with emotion for the husband, showing the wife's devotion to him.

149. Phil made the mistake of scheduling two meetings at the same time in the same place. Upon arriving at the venue, he realized he was in a catch-22 because the two people to meet were there at the same time, and he didn't know what to do.

What does a catch-22 mean?

- A. a silly mistake
B. a tragedy

- C. a blessing
D. a difficult situation

Answer: D (a difficult situation). Catch-22 is an idiomatic expression used to describe a predicament that does not seem to have a solution.

150. English has been the Lingua Franca of the world since America gained its international power. Lingua Franca means:

- A. Language of France
B. Language of the Rich

- C. Language of the World
D. Language of the Powerful

Answer: C. English has been considered as the International Language—it means the different nations in the world use English for them to understand each other. Therefore, Lingua Franca means Language of the World.



**MATHEMATICS
(GENERAL EQUATION)**

FRACTION

Definition: A fraction is defined as ratio or quotient of 2 numbers.

Examples: $\frac{3}{5}$, $\frac{4}{11}$, $\frac{10}{3}$, $\frac{18}{18}$

This is an example of a fraction $\frac{a}{b}$, (a = numerator b = denominator).

If a is less than b (a < b), $\frac{a}{b}$ is proper fraction.

Examples: $\frac{5}{8}$, $\frac{7}{11}$, $\frac{30}{101}$

If a is greater than or equal to b (a > b), $\frac{a}{b}$ is an Improper fraction.

Examples: $\frac{8}{5}$, $\frac{100}{5}$, $\frac{11}{11}$

MIXED NUMBER

Definition: A mixed number is defined as the sum of whole number and a fraction.

Examples: $5\frac{3}{4} = 5 + \frac{3}{4}$ $11\frac{5}{7} = 11 + \frac{5}{7}$

Convert the following to mixed numbers (use Law of Division):

Examples: 1. $\frac{8}{5} = 1\frac{3}{5}$ 2. $\frac{11}{2} = 5\frac{1}{2}$ 3. $\frac{18}{7} = 2\frac{4}{7}$

Convert the following to Improper fractions:

Examples: 1. $4\frac{2}{3} = \frac{14}{3}$ 2. $8\frac{1}{4} = \frac{33}{4}$ 3. $10\frac{3}{5} = \frac{53}{5}$

LCM

Definition: The Least Common Multiple (LCM) of 2 numbers (a and b) is the smallest number that is divisible by a and divisible by b at the same time.

Example: Find the LCM of the following:

1. 18 and 24

Method 1: Prime Factorization

$$18 = 2 \cdot 3^2$$

$$24 = 2^3 \cdot 3$$

The LCM, therefore, will be the product of the greatest powers of each kind of factors.

$$\text{LCM} = 2^3 \cdot 3^2 = 8 \cdot 9 = 72$$

Method 2: Division Method

$$2 \overline{) 18, 24}$$

$$2 \overline{) 9, 12}$$

$$2 \overline{) 6}$$

$$3 \overline{) 9, 3}$$

$$3 \overline{) 3, 1}$$

$$1 \overline{) 1}$$

$$\text{LCM} = 2^3 \cdot 3^2 = 8 \cdot 9 = 72$$

2.) 15 and 36

Method 1: Prime Factorization

$$2 \overline{) 15, 36}$$

$$2 \overline{) 15, 18}$$

$$3 \overline{) 15, 9}$$

$$3 \overline{) 5, 3}$$

$$5 \overline{) 5, 1}$$

$$1 \overline{) 1, 1}$$

$$\text{LCM} = 2^3 \cdot 3^2 \cdot 5 = 180$$

3.) 8 and 18

$$\text{LCM} = 18$$

$$\text{LCM} = 11(8) = 88$$

$$15 = 3 \cdot 5$$

$$36 = 2^2 \cdot 3^2$$

$$\text{LCM} = 2^3 \cdot 3^2 \cdot 5 = 180$$

4. 11 and 8

$$\text{LCM} = 11(8) = 88$$

The LCM of 2 numbers that are relatively prime to each other (Example: 11 and 8) is the product of the given 2 numbers.

Addition of Fractions**Examples:**

$$1. \quad \frac{3}{4} + \frac{1}{4} = \frac{3+1}{4} = \frac{4}{4} = 1$$



$$2. \quad \frac{5}{11} + \frac{7}{11} + \frac{4}{11} = \frac{5+7+4}{11} = \frac{16}{11} = 1\frac{5}{11}$$

$$3. \quad 2\frac{3}{5} + 4\frac{1}{5} = (2+4) + \left(\frac{3}{5} + \frac{1}{5}\right) = 6 + \frac{4}{5} = 6\frac{4}{5}$$

$$4. \quad \frac{5}{8} + \frac{7}{12} + \frac{3}{4} = \frac{3(5) + 2(7) + 6(3)}{24} = \frac{47}{24} = 1\frac{23}{24}$$

$$5. \quad 4\frac{3}{4} + 5\frac{2}{5} = (4+5) + \left(\frac{3}{4} + \frac{2}{5}\right) = 9 + \frac{5(3) + 4(2)}{20} = 9 + \frac{23}{20} = 9 + 1\frac{3}{20} = 10\frac{3}{20}$$

Subtraction of Fractions

Examples:

$$1. \quad \frac{5}{7} - \frac{2}{7} = \frac{5-2}{7} = \frac{3}{7}$$

$$2. \quad 8\frac{6}{11} - 4\frac{3}{11} = (8-4) + \left(\frac{6}{11} - \frac{3}{11}\right) = 4 + \frac{3}{11} = 4\frac{3}{11}$$

$$3. \quad \frac{5}{8} - \frac{3}{5} = \frac{5(5) - 8(3)}{40} = \frac{1}{40}$$

$$4. \quad 8 - \frac{2}{7} = \frac{7(8) - 1(2)}{7} = \frac{54}{7} = 7\frac{5}{7}$$

$$5. \quad 6\frac{1}{3} - 3\frac{1}{8} = (6-3) + \left(\frac{1}{3} - \frac{1}{8}\right) = 3 + \frac{8(1) - 3(1)}{24} = 3 + \frac{5}{24} = 3\frac{5}{24}$$

GCF/GCD

Definition: The *Greatest Common Factor/Divisor* (GCF or GCD) of two numbers (*a* and *b*) is the largest factor or divisor common to both *a* and *b*.

Example: Find the GCF or GCD of the following:

1. 18 and 24

Method 1: List all the divisors of 18 and 24.

Divisors or Factors of 18 = {1, 2, 3, 6, 9, 18}

Divisors or Factors of 24 = {1, 2, 3, 4, 6, 8, 12, 24}

Common Divisors = {1, 2, 3, 6}

GCD/GCF = 6

2.) 15 and 36

Method 1: List all the divisors of 15 and 36.

Divisors or Factors of 15 = {1, 3, 5}

Divisors or Factors of 36 = {1, 2, 3, 4, 6, 9, 12, 18, 36}

Common Divisors = {1, 3, 5}

GCD/GCF = 3

Method 2: Prime Factorization

$$18 = 2 \cdot 3^2$$

$$24 = 2^3 \cdot 3$$

The GCD/GCF, therefore, will be the product of the least powers of each kind of factors.

$$\text{GCD/GCF} = 2 \cdot 3 = 6$$

$$15 = 3 \cdot 5$$

$$36 = 2^2 \cdot 3^2$$

$$\text{GCD/GCF} = 3$$

3.) 8 and 8

$$\text{GCD/GCF} = 8$$

2.) 12 and 13

$$\text{GCD/GCF} = 1$$

Note: The GCD/GCF of 2 numbers which are relatively prime to each other is 1, like 12 and 13.

Multiplication of Fractions

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$$

Examples:

$$1. \quad \frac{21}{8} \cdot \frac{16}{3} = \frac{336}{24} = 14 \quad \text{or} \quad \frac{21}{8} \cdot \frac{16}{3} = \frac{21}{3} \cdot \frac{16}{8} = 7(2) = 14$$

$$2. \quad \frac{18}{25} \cdot \frac{5}{24} = \frac{90}{600} = \frac{90 \div 30}{600 \div 30} = \frac{3}{20} \quad (30 \text{ being the GCD of } 90 \text{ and } 600)$$

or by cancellation process using the principle of GCD:

$$\frac{18}{25} \cdot \frac{5}{24} = \frac{3}{5} \cdot \frac{1}{4} = \frac{3}{20}$$



Division of Fractions

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c} = \frac{ad}{bc}$$

Examples:

$$1. \quad \frac{16}{27} \div \frac{8}{36} = \frac{16}{27} \cdot \frac{36}{8} = \frac{2}{3} \cdot \frac{4}{1} = \frac{8}{3} = 2\frac{2}{3}$$

$$2. \quad 1\frac{3}{4} \div 2 = \frac{7}{4} \cdot \frac{1}{2} = \frac{7}{8}$$

$$3. \quad 1\frac{7}{8} \div 8\frac{1}{4} = \frac{15}{8} \div \frac{33}{4} = \frac{15}{8} \cdot \frac{4}{33} = \frac{5}{2} \cdot \frac{1}{11} = \frac{5}{22}$$

$$4. \quad \left(\frac{8}{9} - \frac{1}{3}\right) \div \left(\frac{3}{4} - \frac{1}{6}\right) = \left(\frac{8-3}{9}\right) \div \left(\frac{9-2}{12}\right) = \frac{5}{9} \div \frac{7}{12} = \frac{5}{9} \cdot \frac{12}{7} = \frac{5}{3} \cdot \frac{4}{7} = \frac{20}{21}$$

Applications:

Examples:

1. In budgeting their income, a young couple allows $\frac{1}{6}$ for rent, $\frac{1}{2}$ for food and the rest for clothing. What fractional part do they provide for clothing?

$$\text{Solution: Clothing} = 1 - \left(\frac{1}{6} + \frac{1}{2}\right) = 1 - \frac{2}{3} = \frac{1}{3}$$

2. A landowner who owns $25\frac{1}{2}$ hectares of land sold $8\frac{3}{4}$, $4\frac{1}{3}$ and 6 hectares of land during the months of January, February and March, respectively. How many hectares of land does he own now?

Solution: Number of hectares the landowner owns now

$$\begin{aligned} &= 25\frac{1}{2} - \left(8\frac{3}{4} + 4\frac{1}{3} + 6\right) \\ &= 25\frac{1}{2} - 19\frac{1}{12} = 6\frac{5}{12} \end{aligned}$$

3. How many pieces of $7\frac{1}{2}$ inch wire can be cut from $31\frac{1}{4}$ ft. roll of wire?

$$\text{Solution: } 31\frac{1}{4} \text{ ft.} = \frac{125}{4} \times 12 \text{ inches} = 375 \text{ inches}$$

$$375 \text{ inches} \div 7.5 \text{ inches} = 50 \text{ pieces}$$

4. A painter estimates that it will require 75 hrs to paint a certain house. If he works 7 hrs. a day, how many days will it take him to paint the house?

$$\text{Solution: } 75 \div 7.5 = 10 \text{ days}$$

5. If a coal costs P3,550.50 per ton, find the cost of $4\frac{3}{4}$ tons.

$$\text{Solution: } P3,550.50 \times 4.75 = P16,864.875$$

6. A woman bought 9.5 yd. of dress material at P305.00 per yd. and $4\frac{1}{2}$ yd. of ribbons at P37.50 per yd. What was the total amount that she paid?

$$\text{Solution: } 9.5 \times P305.00 = P2,897.50$$

$$4.5 \times P37.50 = P168.75$$

$$\text{Total amount paid} = P3,066.25$$

RATIO AND PROPORTION

Definition: A *ratio* is a certain relation between 2 numbers, expressed in terms of a quotient.

Examples:

1. $3:5 = \text{"3 is to 5"}$

$$3:5 = 3 \div 5 = \frac{3}{5}$$

2. A certain organization has 36 men and 15 women. Find the men-women ratio.

$$\text{Solution: } 36:15 = \frac{36}{15} = \frac{12}{5} = 12:5$$

3. A rod, 72 inches long is to be cut into 2 pieces in the ratio of 4:5. What is the length of the shorter piece?

$$\text{Solution: Shorter piece} = \frac{4}{9}(72) = 32 \text{ inches}$$

Definition: A proportion is an equality between 2 ratios.

$a:b = c:d$ is a proportion if and only if $ad = bc$. That is the product of the extremes (a and d) is equal to the product of the means (b and c).

$2:3 = 6:9$ is a proportion because,

$$2(9) = 3(6) = 18$$

$3:4 = 9:8$ is not proportion because,

$$3(8) \neq 4(9), \text{ that is } 24 \neq 36.$$

$$a:b = c:d \text{ is equivalent to } \frac{a}{b} = \frac{c}{d}$$

Find the missing term in each of the following:

1. $42 : x = 6 : 7$

$$6x = 42(7)$$

$$6x = 294$$

$$x = 49$$

2. $\frac{5}{4} = \frac{10}{x}$

$$5x = 40$$

$$x = 8$$

3. $x : \frac{1}{2} = 2 : \frac{3}{4}$

$$\frac{3}{4}x = \frac{1}{2}(2)$$

$$\frac{3}{4}x = 1$$

$$x = \frac{4}{3}$$

4. $\frac{4}{9} = \frac{2x}{1}$

$$18x = 4$$

$$x = \frac{4}{18} = \frac{2}{9}$$

Applications

Examples:

1. An investment of P15,000 results in a gain of P3,000. How much must be invested to earn P5,000?

Solution:

$$\frac{P15,000}{P3,000} = \frac{x}{P5,000}$$

$$x = P25,000$$

2. A motorcycle covers 225 km. in 2.5 hrs. How far can it go in 1.75 hrs?

Solution: $\frac{225}{2.5} = \frac{x}{1.75}$

$$x = 157.5 \text{ km}$$

2. The interior angles of a triangle are in the ratio of 2:3:4. What are the measurements of the 3 angles?

Solution: The sum of the interior angles of any triangle is equal to 180° .

$$2x + 3x + 4x = 180^\circ$$

$$2x = 40^\circ \quad \frac{2x}{2} = \frac{180^\circ}{2}$$

$$x = 20^\circ$$

$$4x = 80^\circ$$

Therefore, the measurements of the 3 angles are 40° , 60° and 80° .

PERCENTAGE

Definition: Percent is defined as an expression which indicates the number of parts taken from a hundred. *Per* is a Latin word meaning "for each" or "for every," while *cent* means a hundred. The symbol used to express percent is %.

Conversion Techniques

A. Changing Percent to Decimal

To change a percent to a decimal, simply move the decimal point two places to the left and drop the percent sign.

Examples:

1. $55\% = 0.55$

2. $0.08\% = 0.0008$

3. $\frac{1}{4}\% = 0.25\% = 0.0025$

B. Changing Decimal to Percent

To change decimal to percent, move the decimal point two places to the right and add the percent sign.

Example:

1. $0.38 = 38\%$

2. $0.4123 = 41.23\%$

3. $0.0062 = 0.62\%$

C. Changing Percent to Fraction

To change a percent to fraction, drop the percent sign and replace it with 100 as denominator. If the percent is in decimal, move the decimal point two places to the left after dropping the percent sign and convert the decimal to its fractional equivalent.

Examples:

$$1. \quad 36\% = \frac{36}{100} = \frac{9}{25}$$

$$2. \quad 7.5\% = 0.075 = \frac{75}{1000} = \frac{3}{40}$$

$$3. \quad 0.84\% = 0.0084 = \frac{84}{10,000} = \frac{21}{2,500}$$

D. Changing Fraction to Percent

To change a fraction to a percent, perform the indicated division, and then move the decimal point of the quotient two places to the right and add the percent sign.

Examples:

$$1. \quad \frac{3}{4} = 0.75 = 75\%$$

$$2. \quad 3\frac{1}{2} = 3.5 = 350\%$$

$$3. \quad \frac{1}{8} = 0.125 = 12.5\%$$

Sample Problems:

1. 40% of P200 is how much?

$$\begin{aligned} \text{Solution: } 40\% [P200] &= N \\ 0.4 (P200) &= N \\ P80 &= N \end{aligned}$$

2. What percent of P150 is P7.50?

$$\begin{aligned} \text{Solution: } R (P150) &= P7.50 \\ R &= \frac{P7.50}{P150} = 0.05 = 5\% \end{aligned}$$

3. What part of 30 is 180?

Solution: $R(30) = 180$
 $R = 6 \times 600\%$

4. 15% of P150 is what percent of 30% of P3,000?

Solution: $15\% \text{ of } P150 = 0.15 (P150) = P22.50$
 $30\% \text{ of } P3,000 = 0.30 (3,000) = P900$
 $P22.50$ is what percent of $P900$
 $P22.50 = R (P900) ?$

$$\frac{P22.50}{P900} = R = 0.025 = 2.5\%$$

5. P200 increased by 10% is how much?

Solution: $N = P200 + 0.10 (P200) = P200 + P20$
 $N = P220$

6. The gross profit of Ana's *sari-sari* store last year was 40% of the total sales. How much was the gross profit if total sales were P480,500?

Solution: Gross profit = $40\% (P480,000) = P192,200$

7. Ann, cashier of ABC Department store, is given a 15% discount on cash purchase. How much will her discount be if she buys 2 dresses at P180 each and one pair of shoes costing P650?

Solution: $P180 \times 2 \text{ dresses} = P360$
 Pair of shoes = $\frac{650}{}$
 Total purchase = $P1,010$
 Total discount = $0.15 (P1,010) = P151.50$

8. Student population in a certain barangay high school is composed of 2,300 females and 1,900 males.

A. What percent of the population are males?

B. What percent are females?

Solution: Total population = $2,300 + 1,900 = 4,200$

$$\text{Male \%} = \frac{1,900}{4,200} = 45.24\%$$

$$\text{Female \%} = \frac{2,300}{4,200} = 54.76\%$$

Divisibility of Numbers

Rules of Divisibility of Numbers

1. A number is divisible by 2 if it ends in 0, 2, 4, 6, or 8. An even number is divisible by 2.

Examples: 324, 50,630, 4,356

2. A number is divisible by 3 if the sum of the digits is divisible by 3.

Examples: 459 is divisible by 3 because the sum of its digits is 18 and 18 is divisible by 3. 13,462 and 2,352 are also divisible by 3.

3. A number is divisible by 4 if the number formed by the last 2 digits is divisible by 4.

Examples: 43,236 is divisible by 4 because the number formed by the last two digits is 36 and 36 is divisible by 4. 48,324 and 53,000 are also divisible by 4.

4. A number is divisible by 5 if it ends in 0 or 5.

Examples: 5,035, 23,210

5. A number is divisible by 6 if it is divisible by 2 and divisible by 3.

Examples: 4,452, 32,478

6. To determine divisibility by 7, double the last digit and subtract it from the rest of the numbers. If the result is divisible by 7, then original number is divisible by 7.

Examples: 252 Double the last digit ($2 + 2 = 4$).
Subtract 4 from 25. The result is 21 and since 21 is divisible by 7. Therefore, 252 is divisible by 7.
2,282 is also divisible by 7.

$$\begin{array}{r} 7 \\ 1 \overline{) 25} = 4 + 4 \\ 28 \\ \hline 2 \\ 28 \\ \hline 0 \end{array}$$

7. A number is divisible by 8 if the number formed by the last 3 digits is divisible by 8.

Examples: 11,648 since 648 is divisible by 8
11,648 is therefore divisible by 8.

8. A number is divisible by 9 if the sum of the digits is divisible by 9.

Examples: 45,927 is divisible by 9 because the sum of its digits is 27, and 27 is divisible by 9.

Math

Exercises: Find the value of k which satisfies the given condition.

- The number $432k$ is divisible by 5 if k is _____.
A. 2 B. 3 C. 4 D. 0
Answer: d. 0
- The number $532k6$ is divisible by 6 if k is _____.
A. 4 B. 1 C. 2 D. 0
Answer: c. 2
- The number $7456k8$ is divisible by 4 if k is _____.
A. 5 B. 4 C. 3 D. 1
Answer: b. 4
- The number $k96$ is divisible by 7 if k is _____.
A. 2 B. 1 C. 4 D. 5
Answer: b. 1
- The number $48k$ is divisible by 8 if k is _____.
A. 1 B. 5 C. 0 D. 3
Answer: c. 0

NUMBER SERIES

Each of the items contains a series of numbers which appears according to a certain rule or order. Relate the series from suggested answer and determine which number is next in the series.

- 4, - 7, - 10, - 13, Answer: - 16
- 3, 6, 12, 24, 48, Answer: 96
- $\frac{3}{2}, \frac{1}{2}, \frac{1}{6}, \text{---}, \frac{1}{54}$ Answer: $+\frac{1}{18}$
- 95, 88, 79, 68, __, 40 Answer: 55
- A, E, I, M, Q Answer: U

Operations on Real Numbers

A. Addition of Real Numbers

- To add numbers with the same sign, first add their absolute values. The sign of the answer (either + or -) is the same as the sign of the two numbers.

Example: $-8 + (-4) = -(8 + 4) = -12$

$(+12) + (+6) = 12 + 6 = 18$

-12 and 18 are called sums.

2. To add two numbers with different signs, first subtract their absolute values. The answer is positive if the positive number has the larger absolute value. The answer is negative if the negative number has the larger absolute value.

Example: $-8 + (+4) = -4$

$$8 + (-4) = 4$$

B. Subtraction of Real Numbers

For all real numbers a and b , $a - b = a + (-b)$. That is, change the sign of the second number then add. The result of subtraction is called difference.

Example: $10 - 8 = 10 + (-8) = 2$

$$-12 - (-7) = -12 + (+7) = -5$$

C. Multiplication of Real Numbers

The product of 2 numbers with same sign is positive. The product of 2 numbers with different signs is negative.

Example: $(-8)(-5) = 40$

$$(-8)(5) = -40$$

D. Division of Real Numbers

The quotient of 2 numbers with same sign is positive. The quotient of 2 numbers with different signs is negative.

Example: $30 \div -6 = \frac{30}{-6} = -5$

$$-24 \div -4 = \frac{-24}{-4} = 6$$

Note: For all real numbers a and b where $b \neq 0$, $\frac{a}{b} = a\left(\frac{1}{b}\right)$. There is no

reciprocal for the number 0, so division by 0 is not defined. Example $\frac{5}{0}$ is not defined.

Polynomials

Important Terms

1. Algebraic expression is any collection of variables, constants, grouping symbols and symbols of operation.

Math

Example:

$$3x^2y, \quad -2x + y, \quad \pi y^2$$

$$\sqrt{x + 2y}, \quad \frac{x + y}{x - y}$$

2. **Term** is any constant, variable or indicated product, quotient or roots of constants and variables. Terms in algebraic expressions are separated by plus or minus.

Examples:

a) In $4x^2 + 3x + 5$, there are 3 terms, namely $4x^2$, $3x$ and 5 .

b) In $3y^2 - \frac{2t}{z}$, there are 2 terms namely, $3y^2$ and $-\frac{2t}{z}$.

3. A **coefficient** is a factor in a term.

Example:

In the algebraic expression, $5x^3 - 4x^2y^2 + 3y^3$, $-4x^2y^2$ is a term,

where -4 is the numerical coefficient of x^2y^2 , x^2 is the coefficient of $-4y^2$

and y^2 is the coefficient of $-4x^2$.

4. **Polynomial** is an algebraic expression consisting of one or more terms that have real number coefficients. All variables are raised to natural number exponents, and the operations involving variables are limited to addition, subtraction and multiplication.

Examples of Polynomials:

a. $5 - 3x$

b. $6x^2y^3 - 3xy^2 + 3$

c. $-2xy$

d. $\sqrt{2}xy - \frac{1}{2}x + 3$

Examples of Not-Polynomials:

a. $\frac{2}{x} - \frac{3}{y}$

b. $x^{-3} - 2x^{-2} - 3x^{-1} + 3$

5. **Degree of a term with one variable** is the exponent of the variable.

Examples:

a. 3 is the degree of $-2t^3$

b. 6 is the degree of $5y^6$

c. 0 is the degree of 3, (since 3 can be written as $3x^0$)

Note: Degree of a constant is zero.



Degree of a term with 2 or more variables is the sum of the exponents of the variables.

Examples:

- a. 5 is the degree of $3x^2y^3$
- b. 7 is the degree of $-2x^2y^2z^3$

6. Degree of a polynomial is the highest degree of any of the terms in the polynomial.

Examples:

- a. 3 is the degree of $5a^3 - 3a^2 + 2$
- b. $3x^2 - 2x + \frac{1}{x}$ has no degree, since the given algebraic expression is not a polynomial.

7. Like terms or similar terms have the same variables to the same power.

Examples:

- a. $-2x$, $5x$ and $\frac{1}{2}x$ and are like terms.
- b. $3xy^2$ and $-5xy^2$ and are like terms.
- c. $2x^2y$ and $5xy^2$ and are not like terms.

Addition of Polynomials

Add similar terms together.

Example:

1. Add $5x^2 + 3xy - 2y^2$, $3x^2 - 2xy + 5y^2$ and $-4x^2 + 2xy - 3y^2$.

Solution:

$$\begin{array}{r} 5x^2 + 3xy - 2y^2 \\ 3x^2 - 2xy + 5y^2 \\ -4x^2 + 2xy - 3y^2 \\ \hline 4x^2 + 3xy \text{ (sum)} \end{array}$$

2. Add $-3a^2b + 2a^2b^2 - 5b^3$, $4a^2b^2 - 3ab^3 + 5b^3$ and $5ab^3 - 3a^2b^2 + 4b^3$.

Solution:

$$\begin{array}{r} -3a^2b \quad \quad + 2a^2b^2 - 5b^3 \\ \quad - 3ab^3 + 4a^2b^2 + 5b^3 \\ \quad \quad 5ab^3 - 3a^2b^2 + 4b^3 \\ \hline -3a^2b + 2ab^2 + 3a^2b^2 + 4b^3 \text{ (sum)} \end{array}$$

Subtraction of Polynomials

Change the sign of the subtrahend and proceed as in addition.

Example:

Subtract $5x^2 - 3xy + 4y^2$ from $-10x^2 + 2xy - y^2$

$$-10x^2 + 2xy - y^2 \text{ (minuend)}$$

$$5x^2 - 3xy + 4y^2 \text{ (subtrahend)}$$

$$\begin{array}{r} (-) \quad (+) \quad (-) \\ \hline \end{array}$$

$$-15x^2 + 5xy - 5y^2 \text{ (difference)}$$

$$\text{Another method } (-10x^2 + 2xy - y^2) - (5x^2 - 3xy + 4y^2)$$

$$= -10x^2 + 2xy - y^2 - 5x^2 + 3xy - 4y^2$$

$$= -15x^2 + 5xy - 5y^2$$

Removal of Grouping Symbols

To remove a grouping symbol preceded by a

- Minus sign, change the sign of each of the terms.
- Plus sign, no change of sign to be made.
- Factor, use the distribution law.

Simplify:

$$1. -(x - y + 2) + (4x - y) - 2(x + 3y - 2)$$

$$= -x + y - 2 + 4x - y - 2x - 6y + 4$$

$$= x - 6y + 2$$

$$2. - \{ -2x - y - (x + y - 2) - 4 \}$$

$$= - \{ -2x - y - x - y + 2 - 4 \}$$

$$= - \{ -3x - 2y - 2 \}$$

$$= 3x + 2y + 2$$

Laws on Exponents

$$I. a^m \cdot a^n = a^m = a^{m+n}$$

$$\text{Example: } 1. x^4 \cdot x^3 = x^{4+3} = x^7$$

$$2. (2a^3b)(-3a^2b^3) = -6a^5b^4$$

$$II. (a^m)^n = a^{mn}$$

$$\text{Example: } 1. (x^4)^3 = x^{4(3)} = x^{12}$$

$$2. (a^7)^2 (a^1)^4 = a^{10} (a^{12}) = a^{10+12} = a^{22}$$



III. $(abc)^n = a^n b^n c^n$

Example: 1. $(2x)^4 = 2^4 x^4 = 16x^4$
2. $-2^2 (-x^2)^3 = -4 (-x^6) = 4x^6$

IV. $\frac{a^m}{a^n} = a^{m-n}$, if $m > n$

Example: $\frac{x^7}{x^1} = x^{7-1} = x^6$

$\frac{a^m}{a^n} = \frac{1}{a^{n-m}}$, if $n > m$

Example: $\frac{x^1}{x^7} = \frac{1}{x^{7-1}} = \frac{1}{x^6}$

$\frac{a^m}{a^n} = 1$, if $n = m$

Example: $\frac{x^1}{x^1} = 1$

V. $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$

Example: $\left(\frac{2x}{3y^1}\right)^3 = \frac{2^3 x^3}{3^3 y^{1 \cdot 3}} = \frac{32x^3}{243y^3}$

VI. $a^0 = 1$, $a \neq 0$

Example: 1. $5^0 = 1$
2. $-2^0 = -1$

VII. $a^{-n} = \frac{1}{a^n}$

Example: 1. $3^{-2} = \frac{1}{3^2} = \frac{1}{9}$
2. $-1^{-4} \cdot 2^{-1} = -1 \cdot \frac{1}{2} = -\frac{1}{2}$
3. $\frac{2^{-4} \cdot 3^{-1}}{2^{-1} \cdot 3^{-4}} = \frac{\frac{1}{16} \cdot \frac{1}{3}}{\frac{1}{2} \cdot \frac{1}{81}} = 1$

VIII. $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$

Example: 1. $\left(\frac{3}{4}\right)^2 = \left(\frac{4}{3}\right)^2 = \frac{16}{9}$

2. $\left(\frac{2a}{b}\right)^3 = \left(\frac{b}{2a}\right)^3 = \frac{b^3}{8a^3}$

3. $(x^{-3} - y^3) = \left(\frac{1}{x^3} - \frac{1}{y^3}\right)^3 = \frac{x^3 y^3}{y^3 - x^3}$

Simplify:

1. $\frac{x+x}{x \cdot x}$

Ans. $\frac{2}{x}$

2. $\frac{-9^{4k}}{81^{2k}}$

Ans. -1

3. $\left(\frac{4x^3}{32x}\right)^3 \left(\frac{4}{x^3}\right)^3$

Ans. $\frac{1}{8}$

4. $\frac{-x^3}{y^3} - \left(\frac{x^3 y}{y^3 x}\right)^3$

Ans. $\frac{-2x^3}{y^3}$

5. $\left(\frac{4x^2}{(6(y^3)^2 x)}\right)^2 \left(\frac{12y^6}{8x}\right)^2$

Ans. 1

6. $\frac{4^{-1} + 2^{-1}}{4^{-1} - 2^{-1}}$

Ans. -3

7. $-2^{-2} - 2^0 - 2^2$

Ans. $-7/4$



8. $\frac{x^{-1} + y^{-1}}{x^{-2} - y^{-2}}$

Ans. $\frac{-xy}{y-x}$

9. $-(x-1) - (2-(x+1))$

Ans. 0

10. $(2x-3y)(4x+5y)$

Ans. $8x^2 - 2xy - 15y^2$

Multiplication of Polynomials

To multiply two polynomials, use distributive law and laws of exponents.

Example: Find the product

1. $(3x-y)(5x+3y) = 3x(5x+3y) - y(5x+3y)$

$= 15x^2 + 9xy - 5xy - 3y^2$

$= 15x^2 + 4xy - 3y^2$

2. $(x^2-2x-4)(3x^2+x-2)$

$= x^2(3x^2+x-2) - 2x(3x^2+x-2) - 4(3x^2+x-2)$

$= 3x^4 + x^3 - 2x^2 - 6x^3 - 2x^2 + 4x - 12x^2 - 4x + 8$

$= 3x^4 - 5x^3 - 16x^2 + 8$

Division of Polynomials

Rule 1: To divide monomials, use laws of exponents in division.

Example: $\frac{6x^3y}{2xy^4} = \frac{3x^2}{y^3}$

Rule 2: To divide a polynomial by a monomial, use $\frac{a+b}{c} = \frac{a}{c} + \frac{b}{c}$, $c \neq 0$

Example: $\frac{8a^3b - 4ab^2}{2ab} = \frac{8a^3b}{2ab} - \frac{4ab^2}{2ab} = 4a^2 - 2ab$

Rule 3: The last rule is to divide a polynomial by a polynomial with at least 2 terms. This type of division is applied only when the degree of the polynomial in the numerator is greater than or equal to the degree of the polynomial in the denominator.

Step 1. Arrange the terms of both the dividend and the divisor in descending powers of the variable.

Step 2. Divide the first term of the dividend by the first term of the divisor, giving the first term of the quotient.

Step 3. Multiply each term of the divisor by the first term of the quotient and subtract the product from the dividend.

Step 4. Use the remainder obtained in Step 3 as a new dividend, and repeat Steps 2 and 3.

Step 5. Continue the process until the remainder is reached, whose degree should be less than the degree of the divisor.

For exact division with remainder 0

$$\frac{\text{dividend}}{\text{divisor}} = \text{quotient}$$

For remainder $\neq 0$

$$\frac{\text{dividend}}{\text{divisor}} = \text{quotient} + \frac{\text{remainder}}{\text{divisor}}$$

Example 1: $(x^3 + 5x^2 + 5x - 2) \div (x + 2)$

$$\begin{array}{r}
 x^2 + 3x - 1 \\
 (x + 2) \overline{) x^3 + 5x^2 + 5x - 2} \\
 \underline{-(x^3 + 2x^2)} \\
 3x^2 + 5x \\
 \underline{-(3x^2 + 6x)} \\
 -x - 2 \\
 \underline{-(-x - 2)} \\
 0 = \text{remainder}
 \end{array}$$

Thus, the quotient is $x^2 + 3x - 1$.

Example 2: $(5x^3 - 2x^2 - 18x + 5) \div (x + 2)$

$$\begin{array}{r}
 5x^2 - 12x + 6 \\
 x + 2 \overline{) 5x^3 - 2x^2 - 18x + 5} \\
 \underline{-(5x^3 + 10x^2)} \\
 -12x^2 - 18x \\
 \underline{-(-12x^2 - 24x)} \\
 6x + 5 \\
 \underline{-(6x + 12)} \\
 -7 = \text{remainder}
 \end{array}$$

Thus, the quotient is $5x^2 - 12x + 6 - \frac{7}{x+2}$.

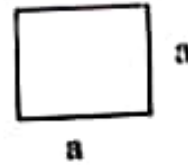


MENSURATION OF PLANE FIGURES

1. Square

$$\text{Area} = (\text{edge})^2$$

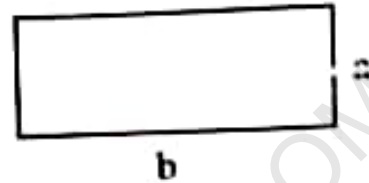
$$A = a^2$$



2. Rectangle

$$\text{Area} = \text{base} \cdot \text{altitude}$$

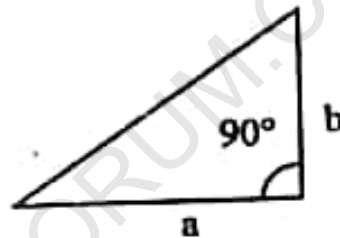
$$A = ba$$



3. Right triangle

$$\text{Area} = \frac{1}{2} \text{base} \cdot \text{altitude}$$

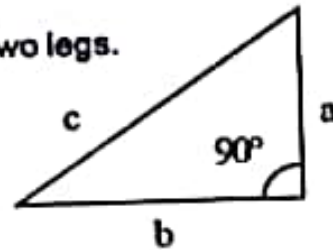
$$A = ba$$



Pythagorean Theorem

(Hypotenuse)² = sum of the squares of the two legs.

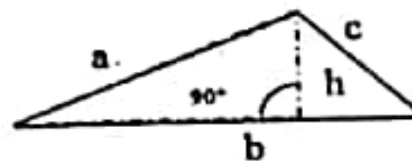
$$c^2 = a^2 + b^2$$



4. Oblique Triangle

$$\text{Area} = \frac{1}{2} \text{base} \cdot \text{altitude}$$

$$A = \frac{1}{2} bh$$

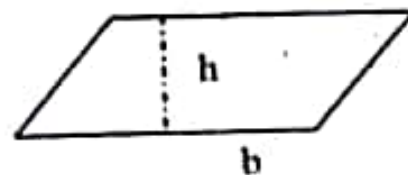


$$A = \sqrt{s(s-a)(s-b)(s-c)}, \text{ where } s = \frac{(a+b+c)}{2}$$

5. Parallelogram (opposite sides parallel)

$$\text{Area} = \text{base} \cdot \text{altitude}$$

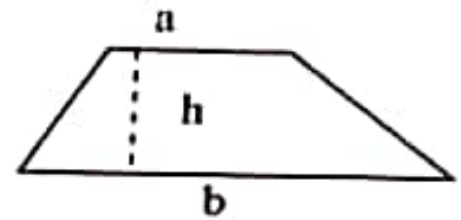
$$A = bh$$



6. Trapezoid (one pair of opposite sides parallel)

Area = $\frac{1}{2}$ sum of base • altitude

$$A = \frac{1}{2}(a + b)h$$



7. Circle

Circumference = $2\pi(\text{radius}) = \pi(\text{diameter})$

$$C = 2\pi r = \pi d$$

Area = $\pi(\text{radius})^2 = \frac{\pi}{4}(\text{diameter})^2$

$$A = \pi R^2 = \frac{\pi}{4} D^2$$



Examples:

1. A square has a side of 5 inches. What is its area and perimeter?

Solution: Let x = side of the square

$$A = x^2 = 5^2 = 25 \text{ sq. inches}$$

$$P = 4x = 4(5) = 20 \text{ inches}$$

2. A square has an area of 400 sq.cm. What is its perimeter?

Solution: Let x = side of the square

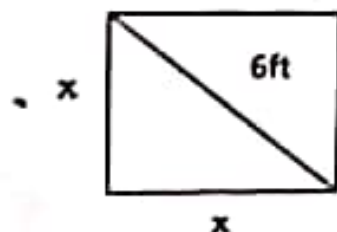
$$A = x^2 = 400$$

$$x = 20 \text{ cm}$$

$$P = 4x = 4(20) = 80 \text{ cm}$$

3. If the diagonal of a table with a square top is 6ft., what is the area of the table top?

Solution:



by Pythagorean theorem

$$x^2 + x^2 = 6^2$$

$$2x^2 = 36$$

$$x^2 = 18 \text{ sq ft} = \text{Area}$$

- Solution:** $A = LW = (500)(400) = 20,000$ sq. m.
 $A = 20,000$ sq. m. $\times$ 1 hectares = 2 hectares
 10,000 sq. m.

- Solution:** Let x = unknown side
 $2x + 2(20) = 60$
 $2x = 20$
 $x = 10$ inches

- Solution:** Area of square = 9 sq. units
Side of the square = 3 units
Perimeter of the square = $4(3) = 12$ units
Perimeter of the equilateral triangle = 12 units
Let x = side of the equilateral triangle
 $3x = 12$, therefore $x = 4$ units

- Solution:** Let x = length of the 3rd side
 $3x + 3x + x = 42$
 $7x = 42$
 $x = 6$ inches (3rd side)

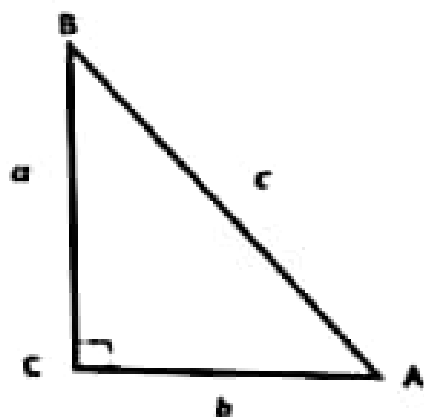
- Solution:**
 $A = \frac{1}{2}(6 + 12) 4 = 36$ sq. inches

- Solution:**
- Diameter = 4 ft Radius = 2 ft
- Circumference = $2\pi r = 2\pi(2) = 4\pi \text{ ft}$
- Area = $\pi r^2 = \pi(2)^2 = 4\pi \text{ sq. ft.}$

- Solution:** Let x = base of the right triangle
By Pythagorean theorem
$$x^2 + 4^2 = 5^2 \quad x = 3\text{ ft}$$
$$A = \frac{1}{2} (3) (4) = 6 \text{ sq. ft.}$$
$$P = 3 + 4 + 5 = 12 \text{ ft.}$$

PLANE TRIGONOMETRY

Six Trigonometric Functions



$$\text{Sine } A = \sin A = \frac{\text{Side Opposite } A}{\text{Hypotenuse}} = \frac{a}{c}$$

$$\text{Cosine } A = \cos A = \frac{\text{Side Adjacent } A}{\text{Hypotenuse}} = \frac{b}{c}$$

$$\text{Tangent } A = \tan A = \frac{\text{Side Opposite } A}{\text{Side Adjacent } A} = \frac{a}{b}$$

$$\text{Cosecant } A = \csc A = \frac{\text{Hypotenuse}}{\text{Side Opposite } A} = \frac{c}{a}$$

$$\text{Secant } A = \sec A = \frac{\text{Hypotenuse}}{\text{Side Adjacent } A} = \frac{c}{b}$$

$$\text{Cotangent } A = \cot A = \frac{\text{Side Adjacent } A}{\text{Side Opposite } A} = \frac{b}{a}$$

Pythagorean Theorem: The square of the hypotenuse is equal to the sum of the squares of the other 2 sides. As shown in the above figure, $c^2 = a^2 + b^2$.

Positive Functions in every Quadrant

QI = All 6 trigonometric functions

QII = Sine and Cosecant only

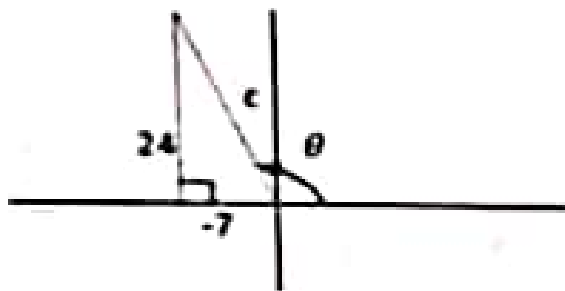
QIII = Tangent and Cotangent only

QII = Cosine and Secant only

Exercises

Find the value of the other circular functions given that of the following:

1) $\cot \theta = \frac{-7}{24}, \theta \in QII$



$$c = \sqrt{(24)^2 + (-7)^2} = 25$$

a. $\sin \theta = \frac{24}{25}$

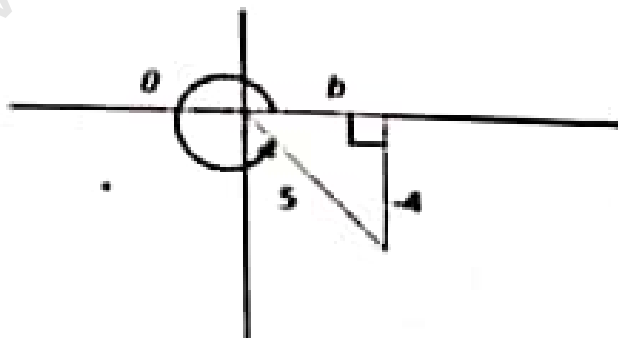
d. $\csc \theta = \frac{25}{24}$

b. $\cos \theta = -\frac{7}{25}$

e. $\sec \theta = -\frac{25}{7}$

c. $\tan \theta = -\frac{24}{7}$

2) $\sin \theta = \frac{-4}{5}, \cos \theta > 0$



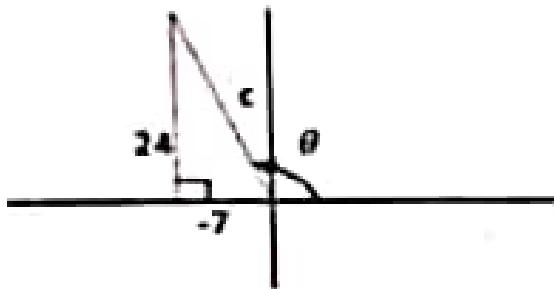
$$b = \sqrt{(5)^2 - (-4)^2} = 3$$

QIV = Cosine and Secant only

Exercises

Find the value of the other circular functions given that of the following:

1) $\cot \theta = \frac{-7}{24}, \theta \in QII$



$$c = \sqrt{(24)^2 + (-7)^2} = 25$$

a. $\sin \theta = \frac{24}{25}$

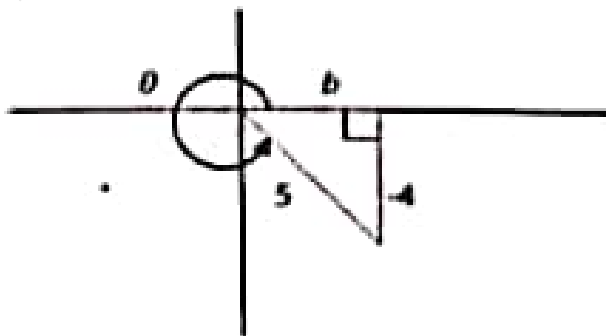
d. $\csc \theta = \frac{25}{24}$

b. $\cos \theta = -\frac{7}{25}$

e. $\sec \theta = -\frac{25}{7}$

c. $\tan \theta = -\frac{24}{7}$

2) $\sin \theta = \frac{-4}{5}, \cos \theta > 0$



$$b = \sqrt{(5)^2 - (-4)^2} = 3$$

a. $\cos \theta = \frac{3}{5}$

d. $\sec \theta = \frac{5}{3}$

b. $\tan \theta = -\frac{4}{3}$

e. $\cot \theta = -\frac{3}{4}$

c. $\csc \theta = -\frac{5}{4}$

Trigonometric Identities**Basic Identities**

$$\sin \theta = \frac{1}{\csc \theta}$$

$$\csc \theta = \frac{1}{\sin \theta}$$

$$\tan \theta = \frac{1}{\cot \theta}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cos \theta = \frac{1}{\sec \theta}$$

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\sec \theta = \frac{1}{\tan \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\cos^2 \theta + \sin^2 \theta = 1$$

$$\cos^2 \theta = 1 - \sin^2 \theta$$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$\tan^2 \theta = \sec^2 \theta - 1$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

$$\cot^2 \theta = \csc^2 \theta - 1$$

Double Number Identities

$$\sin 2\theta = 2\sin \theta \cos \theta$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

$$= 2\cos^2 \theta - 1$$

$$= 1 - 2\sin^2 \theta$$

$$\tan 2\theta = \frac{2\tan \theta}{1 - \tan^2 \theta}$$

Half Number Identities

$$\cos \frac{1}{2} \theta = \pm \sqrt{\frac{1 + \cos \theta}{2}}$$

$$\sin \frac{1}{2} \theta = \pm \sqrt{\frac{1 - \cos \theta}{2}}$$

$$\tan \frac{1}{2} \theta = \frac{1 - \cos \theta}{\sin \theta} = \frac{\sin \theta}{1 + \cos \theta} = \pm \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}}$$

INTRODUCTION TO PROBABILITY THEORY

Introduction

The study of probability is vital to us humans, for no day will pass without someone asking about probability or the chance that a certain event will occur. For example, before leaving home, one may ask about the chances that rain will occur. Another example is that of a close relative of a patient scheduled to undergo a risky operation asking about the survival chances of his kin. The study of probability, therefore, involves the success or failure of certain events and as a key to its understanding, the number of ways by which events can happen should be studied.

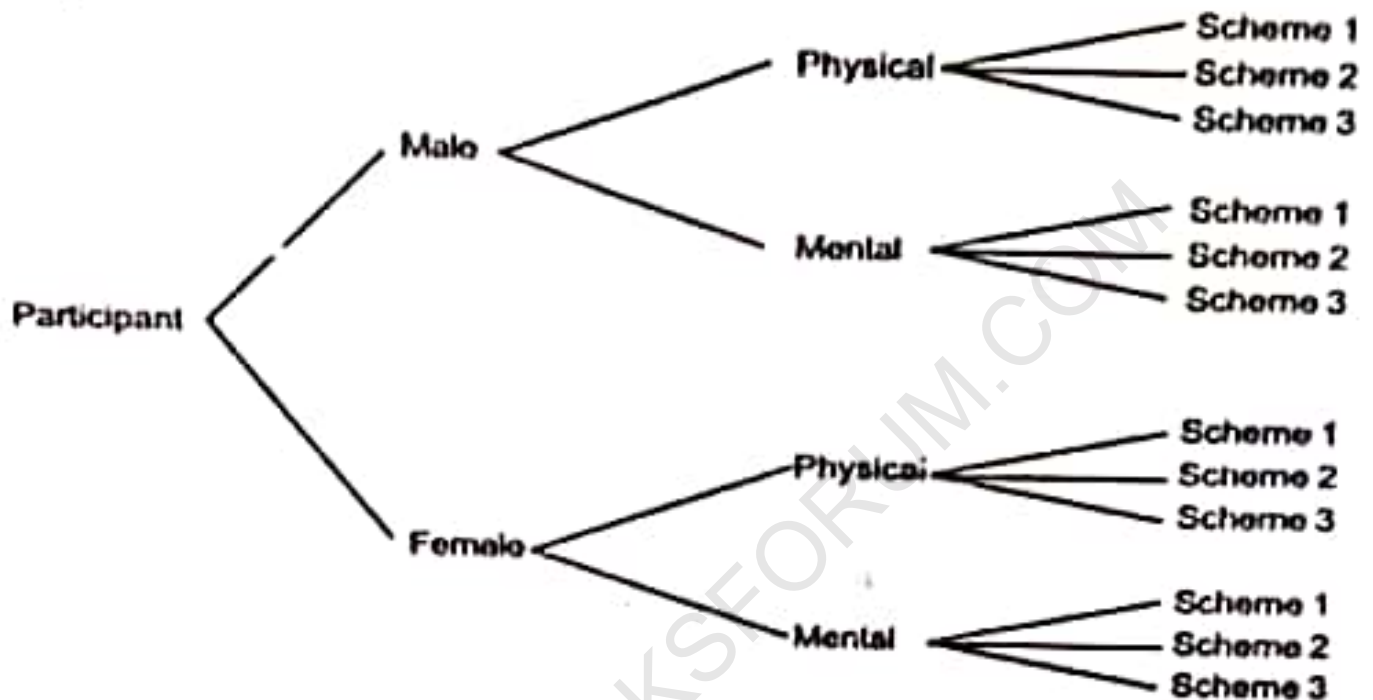
Techniques of Counting

There are different techniques for counting the number of ways that events can happen, namely, the tree diagram, the fundamental principle of counting, permutation, and combination. These techniques can be used singly or in combination.

The Fundamental Principle of Counting

Given two events, E_1 and E_2 , if E_1 can happen in n_1 ways and E_2 can happen in n_2 ways, then the number of ways that both events can happen in the specified order is $n_1 \cdot n_2$.

Example: In a planned longitudinal study on the effect of 3 different weights reducing schemes, the following factors were also considered: gender of participant (male and female) and type of daily activity (physical, mental). In how many ways may the participants be classified? Do a tree diagram.

Example

The participants in the weight-reducing scheme can be grouped into $2 \cdot 2 \cdot 3 = 12$ groups.

Permutation

Permutation is an arrangement of objects in which order is taken into account.

Factorial Notation

The expression (read as n factorial) is equal to $n(n-1)(n-2)\dots(n-r+1)$, where $r \leq n$.

Example: $5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$
 $5! = 120$

By definition:

Proof: If $r = n$, then $n - r = 0$. Thus, the last term in the expansion will be $(0 + 1) = 1$.

Permutation of n objects taken all together

$${}_nP_n = n!$$

Example:

1. ${}_6P_6 = 6!$

$$= 720$$

2. In how many different ways may four newly admitted patients be assigned to four available beds in the charity ward of a hospital?

$${}_4P_4 = 4! = 24$$

3. In how many different ways may seven victims of the cholera epidemic be assigned to a ward with seven beds if two of the victims who are sisters insist on being together?

$$\begin{aligned} {}_6P_6 \cdot P_2 &= 6! \cdot 2! \\ &= 720(2) \\ &= 1,440 \text{ ways} \end{aligned}$$

***Note:** The 2 sisters are considered as one group since they move together. However, they can interchange places by themselves.

Permutation of n objects r at a time

$${}_nP_r = \frac{n!}{(n-r)!}$$

Example:

1. ${}_6P_2 = \frac{6!}{4!}$

$$= \frac{6!}{(6-2)!}$$

$$= \frac{6 \cdot 5 \cdot 4!}{4!}$$

$$= 30$$

2. In how many different ways may the chief of a medical mission to a disaster-stricken area assign six out of nine doctors to head a team of health practitioners to six relief stations?

$${}_9P_6 = \frac{9!}{(9-6)!}$$

$$= \frac{162,880}{6}$$

$$= 60,480 \text{ ways}$$

Permutation of n objects not all distinct

When one cannot distinguish between objects of the same class or characteristics, the appropriate formula to use is:

Where
$$\frac{N!}{n_1!n_2!\dots n_k!}$$

N = total number of objects

n = number of objects belonging to a particular class

Example: In how many different ways can we arrange 10 experimental hamsters in 10 laboratory cages if 4 hamsters are white, 3 are brown, and 3 are gray?

$$\frac{10!}{4!3!3!} = \frac{3,628,800}{864} = 4,200 \text{ ways}$$

Circular Permutation

In circular permutation, one of the positions have to be fixed so that it will serve as the point of reference in counting the movement of the object in the circle. Thus, $(n-1)!$

Example: 1. ${}_{(n-1)}P_{(n-1)} = (n-1)!$

$$= 5!$$

$$= 120$$

In a culture and sensitivity test, a microbiologist has to arrange circular strips of different antibiotics in a circular fashion on a large Petri dish with culture media and swab it with the microorganism being tested. In how many ways may the microbiologist implant 7 of these antibiotic strips in a circle?



$$\begin{aligned} & (7-1)! \\ & = 6! \\ & = 720 \text{ ways} \end{aligned}$$

Combination

- Combination is an arrangement of objects where order or position is not important. The combination of n objects taken all together is 1.

$${}_nC_n = 1$$

Example: ${}_1C_1 = 1$

$${}_{100}C_{100} = 1$$

Combination of n object taken r at a time

$${}_nC_r = \frac{n!}{(n-r)!r!}$$

Example: 1. ${}_7C_3 = \frac{7!}{(7-3)!3!}$

$$\begin{aligned} & = \frac{7!}{4!3!} \\ & = \frac{7 \cdot 6 \cdot 5 \cdot 4!}{4!3!} = 35 \end{aligned}$$

2. On a survey on the effects of breastfeeding on infants, it was decided by the researchers that 40% of nursing mothers (those with infants being nursed) of every barangay in the study will be included in the sample. In a particular barangay, the researchers are to choose 4 out of 10 nursing mothers. In how many ways can this be done?

$$\begin{aligned} {}_{10}C_4 &= \frac{10!}{(10-4)!4!} \\ &= \frac{10!}{6!4!} \\ &= \frac{3,628,800}{(720)(24)} \\ &= 210 \text{ ways} \end{aligned}$$

Combination In a Series

$${}_nC_1 + {}nC_2 + \dots + {}nC_n = 2^n - 1$$

Example:

In how many ways may a rural health officer assign one or more of 6 available barangay health workers to do community work?

$$2^6 - 1 = 63 \text{ ways}$$

Classical Definition of Probability

The classical (a priori) concept of probability is given by:

$$\Pr(E) = \frac{n}{N}$$

Where E is any given event

n is the number of successes of the event

N is the total number of possible outcomes of the event.

Example: In a family of 2 children, what is the probability of having exactly one boy?

To answer this problem, the total number of outcomes must be listed first. This fundamental sample space (S).

$$S = \{B_1B_2, B_1G_2, G_1B_2, G_1G_2\}$$

There are 4 possible outcomes (N) of which only two contain exactly 1 boy. Therefore, if E_1 is the event of having exactly one boy, then:

$$\Pr(E_1) = \frac{2}{4} = \frac{1}{2}$$

Complement of an Event

The probability of non-occurrence of an event E (called failure) is represented by the complement of the event E . That is:

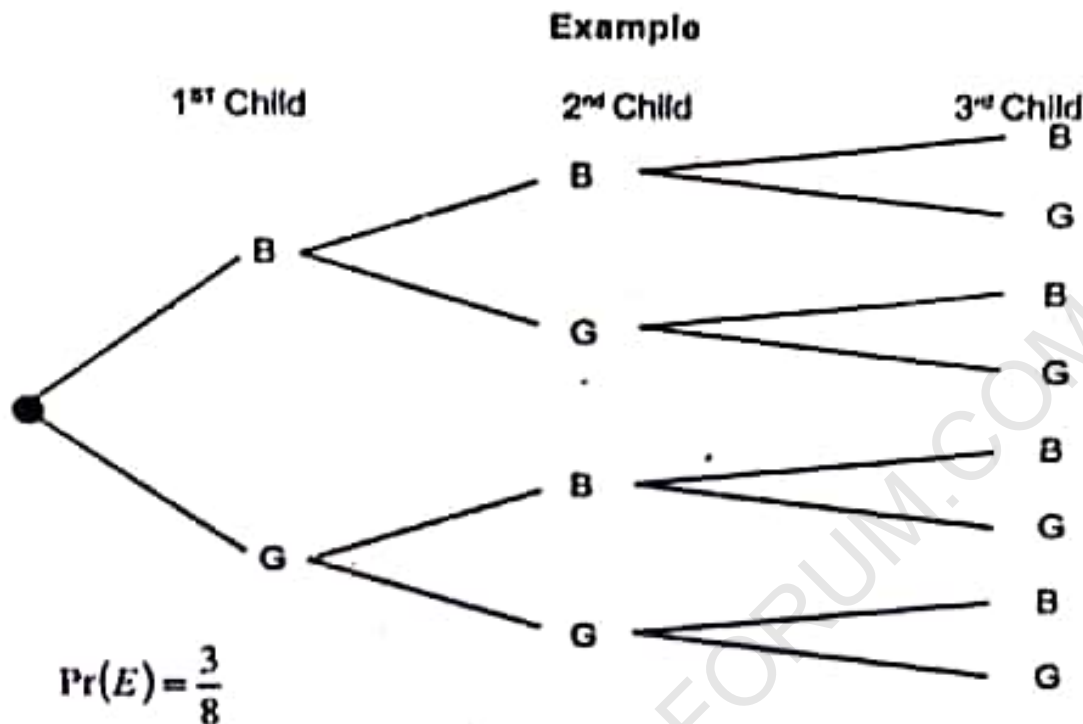
$$\Pr(\bar{E}) = 1 - \Pr(E)$$

Example:

Let E be the event of having 2 boys in a family of 3 children.

$$S = \{BBB, BGB, GBB, BBG, BGG, GBG, GGB, GGG\}$$

There are eight possible outcomes. These outcomes can easily be obtained using the tree diagram.



The complement of E which is $\bar{E}$ means not getting exactly 2 boys in a family of 3 children and this given by:

$$\Pr(\bar{E}) = 1 - \Pr(E)$$

$$= 1 - \frac{3}{8}$$

$$= \frac{5}{8}$$

Empirical Definition of Probability

- Empirical probability (a posterior) refers to the relative frequency of the occurrence of an event when the number of trials is large. Its value is based on the actual occurrence of the event.

Example:

- Given an event of getting a blood type O in a mass blood typing activity of the Philippine National Red Cross, suppose 600 of the 1,000 persons who had their blood examined came out as:

$$\Pr(E) = \frac{600}{1,000} = 0.6$$

Marginal and Joint Probability

- **Marginal probability** is the probability of occurrence of a single event or an event satisfying only one characteristic.
- **Joint probability** is the probability of two events occurring simultaneously in a single trial. Joint probability is also the probability of one event satisfying two or more characteristics. Let $Pr(E_1 \cap E_2)$ denote the probability of E_1 and E_2 .

Example:

- The following table shows the number of males and females who were examined for HIV during the past months.

SEX \ RESULTS	MALES	FEMALES	TOTAL
HIV +	2	3	5
HIV -	248	247	495
TOTAL	250	250	500

The marginal probability for HIV + cases is:

$$Pr(HIV+) = \frac{5}{500} = 0.01$$

The marginal probability for females is:

$$Pr(F) = \frac{250}{500} = 0.5$$

The joint probability for females, who are HIV + is:

$$Pr(E \cap HIV+) = \frac{3}{500} = 0.006$$

Mutually Exclusive Events

Two events E_1 and E_2 are mutually exclusive if the occurrence of E_1 precludes the occurrence of E_2 and vice and versa. The joint probability of E_1 and E_2 in this case is zero. If E_1 and E_2 are capable of occurring simultaneously in a single trial then they are said to be non-mutually exclusive.

The Addition Law

The probability that an event or an event but not both will occur in a single trial given by:

$$1. \quad Pr(E_1 \cup E_2) = Pr(E_1) + Pr(E_2)$$

For mutually exclusive events

$$2. \Pr(E_1 \cup E_2) = \Pr(E_1) + \Pr(E_2) - \Pr(E_1 \cap E_2)$$

For mutually exclusive events

Examples:

1. In a certain hospital, 5 infants are fed with infant formula A, 4 infants are fed with infant formula B, and 3 infants are fed with infant formula C. If one infant is selected at random, what is the probability he/she was fed with infant formula A or B?

$$\Pr(A \cup B) = P(A) + P(B)$$

$$= \frac{5}{12} + \frac{4}{12}$$

$$= \frac{9}{12}$$

$$= \frac{3}{4}$$

2. Of 10 medicines recommended for cough, 6 were found to be effective, 4 were found to be expensive, and 2 were found to be effective and expensive. Find the probability that one medicine chosen at random is either effective or expensive.

$$\Pr(E_1 \cup E_2) = \Pr(E_1) + \Pr(E_2) - \Pr(E_1 \cap E_2)$$

$$= \frac{6}{10} + \frac{4}{10} - \frac{2}{10}$$

$$= \frac{8}{10} = \frac{4}{5}$$

3. The probability of 0, 1, 2, 3 children will be attended to by 3 school dentists at a particular time are 0.08, 0.22, 0.45, and 0.25, respectively. Find the probability that in the same time frame.

- a. Exactly 2 children are attended to

Let X = number of school children attended to

$$\Pr(X = 2) = 0.45$$

- b. No child is being attended to

$$\Pr(X = 0) = 0.08$$

c. At least one child is being attended to

$$\Pr(x \geq 1) = 1 - \Pr(x = 0)$$

$$\begin{aligned} \text{or } \Pr(X \geq 1) &= \Pr(X = 1) + \Pr(X = 2) + \Pr(X = 3) \\ &= 0.22 + 0.45 + 0.25 \end{aligned}$$

Conditional Probability

The probability that an event, E_1 , will occur that some event E_2 , has already occurred is called conditional probability.

$$\Pr(E_1 | E_2) = \frac{\Pr(E_1 \cap E_2)}{\Pr(E_2)}$$

Example:

- The probability of randomly selecting a smoker (S) among 1,000 individuals is 0.50, and the probability of randomly selecting a person with lung cancer (C) among the same 1,000 individuals is 0.10. If the probability of selecting an individual who is both smoker and has lung cancer is 0.04, find the probability of selecting an individual who has lung cancer given that he is a smoker.

$$\begin{aligned} \Pr(C | S) &= \frac{\Pr(S \cap C)}{\Pr(S)} \\ &= \frac{0.04}{0.50} \\ &= 0.08 \end{aligned}$$

Dependent and Independent Events

Two events E_1 and E_2 are independent if the occurrence or non-occurrence of one does not affect the occurrence or non-occurrence of the other. Otherwise, the event is dependent.

The Multiplication Law

The probability that 2 events, E_1 and E_2 , will occur one after the other in the stated order is given by the product of their respective probabilities.

$$\begin{aligned} 1. \quad \Pr(E_1 \cap E_2) &= \Pr(E_1) \cdot \Pr(E_2) \\ &\dots \text{for independent events} \end{aligned}$$

$$\begin{aligned} 2. \quad \Pr(E_1 \cap E_2) &= \Pr(E_1) \Pr(E_2 | E_1) \\ &\dots \text{for dependent events} \end{aligned}$$

Example 1:

By way of experience of a certain hospital, it was found out that 60% of their triple bypass operations for patients with heart ailments are successful. Find the probability that for the next three patients operated on:

- a. All three patients survive.

$$\begin{aligned} P(All) &= (0.60)(0.60)(0.60) \\ &= 0.216 \end{aligned}$$

- b. Exactly 2 survive.

$$\begin{aligned} P(X = 2) &= (0.60)(0.60)(0.40) + (0.60)(0.40)(0.60) \\ &\quad + (0.40)(0.60)(0.60) \\ &= 0.432 \end{aligned}$$

Example 2:

In a sample of 10 school children, 4 were found to be malnourished. If the 2 children are selected one after another and sent to the school clinic for further examination, what is the probability that:

- a. Both are malnourished.

$$\begin{aligned} P(E_1 \cap E_2) &= P(E_1) \cdot P(E_2 | E_1) \\ &= \left(\frac{4}{10}\right)\left(\frac{3}{9}\right) \\ &= \frac{2}{15} \end{aligned}$$

- b. The first child is malnourished and the second child is not.

$$\begin{aligned} P(E_1 \cap \bar{E}_2) &= P(E_1) \cdot P(\bar{E}_2 | E_1) \\ &= \left(\frac{4}{10}\right)\left(\frac{6}{9}\right) \\ &= \frac{4}{15} \end{aligned}$$

Multiple choice

1. If the probability that it will rain today is $\frac{5}{8}$, what is the probability that it will not?

A. $\frac{1}{8}$ C. $\frac{1}{4}$
B. $\frac{3}{8}$ D. 100%

Answer: B. $\frac{3}{8}$ $P = 1 - \frac{5}{8} = \frac{3}{8}$

2. The thousandths digit of the number 1,234.5678, is _____.

A. 1 C. 8
B. 7 D. 5

Answer: B. 7 is the thousandths digit. 1 is the thousand digit.

3. The degree of the term $-4x^2y^1$ is _____.

A. 4th C. 2nd
B. 1st D. 3rd

Answer: D. The sum of the exponents of x and y is 3.

4. If x is an odd integer and y is an even integer, which of the following is even

A. $2x - y - 1$
 B. $x^2 + y$
C. $x^2 + 3y + 1$
 D. $4x - 1$

Answer: C. Let $x = 1$ (odd)

$y = 0$ (even)

$$x^2 + 3y + 1 = (1)^2 + 3(0) + 1 = 2 \text{ (even)}$$

5. The difference between gross profit and cost of goods sold is known as _____.

A. markdown
 B. discount
C. net profit
 D. invoice price

Answer: C. Net Profit

6. Julius has ₱ 15,000 in the bank. After a year, he expects it to earned ₱ 750. What is the rate of interest?

A. 2% C. 5%
B. 4% D. 6%

Answer: C. $r = \frac{I}{Pt} = \frac{750}{15000(1)} = .05 = 5\%$

7. The property of real numbers which describes $(5 + 6) - 3 = -3 + (5 + 6)$.

- A. Commutative
- B. Associative
- C. Distributive
- D. Reflexive

Answer: A, Commutative Property for Addition: $a + b = b + a$

8. Which is a proportion?

- A. $5 : 4 = 15 : 12$
- B. $8 : 2 = 16 : 9$
- C. $4 : 5 = 12 : 10$
- D. $12 : 7 = 5 : 2$

Answer: A, Product of extremes = product of means $5(12) = 4(15)$ $60 = 60$

9. After receiving a 30% discount, Jay paid ₱ 210.00 for an item. What is the regular price of the item?

- A. ₱ 300
- B. ₱ 350
- C. ₱ 380
- D. ₱ 400

Answer: A, $₱210 \div 70\% = ₱210 \div .7 = ₱300$

10. Peter borrowed ₱15,000 from the bank at the rate of 12% per year. How much will he owe the bank after 1.5 years?

- A. ₱16000
- B. ₱16700
- C. ₱17000
- D. ₱17700

Answer: D, $F = P(1 + rt) = ₱15000(1 + 0.12(1.5)) = ₱17700$.

11. $1/8$ in percent is _____.

- A. 10.5%
- B. 12.5%
- C. 14.5%
- D. 16.5%

Answer: B, $1/8 \times 100\% = 12.5\%$

12. Simplify this equation: $\left(\frac{8}{9} - \frac{1}{3}\right) + \left(\frac{3}{4} - \frac{1}{6}\right)$.

- A. $\frac{21}{20}$
- B. $\frac{20}{21}$
- C. $\frac{19}{20}$
- D. $\frac{20}{19}$

Answer: B, $\left(\frac{8}{9} - \frac{1}{3}\right) + \left(\frac{3}{4} - \frac{1}{6}\right) = \left(\frac{8-3}{9}\right) + \left(\frac{9-2}{12}\right) = \frac{5}{9} + \frac{7}{12} = \frac{5}{9} \cdot \frac{4}{4} + \frac{7}{12} = \frac{20}{36} + \frac{14}{36} = \frac{34}{36} = \frac{17}{18}$

13. The number of hours spent for review in Algebra, Chemistry and English is in the ratio of 4 : 3 : 1. How many hours does each student spend for Chemistry in a 24-hour review package?

- A. 3
- B. 6
- C. 9
- D. 12

Answer: C, $4 + 3 + 1 = 8$, $\frac{3}{8} (24) = 9$

$$\frac{3}{8} \times \frac{24}{1} = 9$$

14. What is the value in grams of 3.5 kilograms?

- A. 35
- B. 350
- C. 3500
- D. 35000

Answer: C, $3.5 \times 1000 = 3500$ grams

$$1\text{kg} = 1000\text{g.}$$

15. What is the value of 36° in radians?

- A. $\frac{\pi}{5}$
- B. $\frac{\pi}{4}$

- C. $\frac{\pi}{3}$
- D. $\frac{\pi}{2}$

Answer: A, $36^\circ \times \frac{\pi}{180^\circ} = \frac{\pi}{5}$ radian

16. What do you call a triangle with equal sides?

- A. scalene
- B. equilateral
- C. isosceles
- D. right

Answer: B, Equilateral triangle is a triangle with all sides equal.

17. Solve for the value of x to make the proportion $x : 8 = 4 : 16$ true.

- A. 1
- B. 2

- C. 3
- D. 4

Answer: B, $\frac{16x}{16} = \frac{32}{16}$ $x = 2$

18. How many centimeters are there in 100 millimeters?

- A. 0.1
- B. 1

- C. 10
- D. 10.5

Answer: B, $1\text{cm} = 10\text{mm}$ $100 \div 10 = 10\text{mm}$

19. A ball is drawn at random from a box containing 3 red balls, 4 white balls and 5 blue balls. What is the probability that it is not blue?

A. $\frac{1}{4}$

C. $\frac{7}{12}$

B. $\frac{1}{2}$

D. 1

Answer: C, $P(\text{not blue}) = \frac{3+4}{12} = \frac{7}{12}$

20. At 70 kph, Jay can reach his home within 50 minutes. At what rate should he drive his car so that he can reach home 15 minutes earlier?

A. 60 kph

C. 90 kph

B. 80 kph

D. 100kph

Answer: D, $x/70 = 50/35$ $x = 100\text{kph}$

21. What is the area of a rectangle of height h and base b ?

A. bh

C. $2bh$

B. $\frac{1}{2}bh$

D. $\frac{1}{3}bh$

Answer: A, Area of rectangle = base $\times$ height

22. Given the data below:

Staff	Female	Male	Total
Nurses	7	1	8
Physicians	3	2	5
Total	10	3	13

If a person staff is selected in a hospital, what is the probability that the subject is a nurse or male?

A. $\frac{5}{13}$

C. $\frac{10}{13}$

B. $\frac{8}{13}$

D. $\frac{11}{13}$

Answer: C, $P(\text{nurse or male}) = 8/13 + 3/13 - 1/13 = 10/13$

23. What is the circumference of a circle with an area of 4π square units?

A. 2π units

C. 6π units

B. 4π units

D. 8π units

Answer: B, $A = 4\pi = \pi r^2$ $r^2 = 4$ $r = 2$ $c = 2\pi r = 4\pi$

24. What is the probability of the FEU basketball team winning all in a series of 3 games, assuming the team are evenly matched?

A. $\frac{1}{8}$

B. $\frac{3}{8}$

C. $\frac{5}{8}$

D. 100%

Answer: A, $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$

25. What is the volume of the cube of edge 2 ft.?

A. 6 cubic ft.

B. 8 cubic ft.

C. 10 cubic ft.

D. 12 cubic ft.

Answer: B, $V = (\text{edge})^3 = 2^3 = 8$.

26. The probability that a man will be alive in 25 years is $\frac{3}{5}$ and that of his wife is $\frac{2}{3}$. What is the probability that both will be alive?

A. $\frac{1}{5}$

B. $\frac{2}{5}$

C. 50%

D. 100%

Answer: B, $P = \frac{3}{5} \cdot \frac{2}{3} = \frac{2}{5}$

27. The value of k for which 7456k8 is divisible by 4.

A. 5

B. 4

C. 3

D. 1

Answer: B, 745648 is divisible by 4 because the last two digits 48 is divisible by 4.

28. What is the value of $\cos^2 \theta + \sin^2 \theta$?

A. -2

B. -1

C. 1

D. 2

Answer: C, Pythagorean Identity $\cos^2 \theta + \sin^2 \theta = 1$

29. What is the total surface area of a box of dimensions 6 inches x 8 inches x 12 inches?

A. 216 sq. in.

B. 432 sq. in.

C. 576 sq. in.

D. 600 sq. in.

Answer: B, $S.A. = 2[(6)(8) + 6(12) + 8(12)]$
 $= 2(48 + 72 + 96)$
 $= 432 \text{ sq. in.}$



30. Simplify this: $8\frac{10}{11} + 5\frac{3}{11} - 4\frac{2}{11}$

A. $11\frac{1}{11}$

B. 11

C. $10\frac{1}{11}$

D. 10

Answer: D, $8\frac{10}{11} + 5\frac{3}{11} - 4\frac{2}{11} = 9\frac{11}{11} = 10$

31. 72 is $\frac{3}{4}$ of what number?

A. 76

B. 84

C. 86

D. 96

Answer: D,

Let x = the number

$$\frac{3}{4}x = 72$$

$$x = 96$$

32. 0.0125 expressed as fraction in lowest term

A. $\frac{1}{8}$

B. $\frac{1}{4}$

C. $\frac{1}{800}$

D. $\frac{1}{80}$

Answer: D,

$$0.0125 = \frac{125}{10000} = \frac{1}{80}$$

33. The value of $\frac{3}{1000} + \frac{4}{10} + \frac{5}{100} + \frac{6}{1}$ is _____

A. 6.453

B. 3.456

C. 4.536

D. 6.543

Answer: A, $\frac{3}{1000} + \frac{4}{10} + \frac{5}{100} + \frac{6}{1} = .003 + .400 + .05 + 6 = 6.453$

34. The value of the car has depreciated by ₱ 20,000. It is now worth ₱ 380,000. By how much percent did the value of the car decrease?

A. 5%

B. 10%

C. 20%

D. 25%

Answer: A, $\% \text{decrease} = \frac{20000}{400000} = 0.05 = 5\%$

35. If x is an integer and $90 > 3x$, then x cannot be _____.

- A. 29
B. 26

- C. 27
D. 30

Answer: D, $90 > 3x$, $30 > 30$ is False.

36. Simplify this $\frac{4^5}{8^4}$.

- A. $\frac{1}{2}$
B. $\frac{1}{4}$

- C. $\frac{1}{8}$
D. 4

Answer: B, $\frac{4^5}{8^4} = \frac{(2^2)^5}{(2^3)^4} = \frac{2^{10}}{2^{12}} = \frac{1}{4}$

37. If $x > \frac{5}{6}$ and $\frac{7}{4} > x$, then x can be _____.

- A. $\frac{1}{2}$
B. $\frac{2}{3}$

- C. $\frac{3}{2}$
D. 2

Answer: C, $\frac{7}{4} > x > \frac{5}{6}$ then $x = \frac{3}{2}$, $\frac{7}{4} > \frac{3}{2} > \frac{5}{6}$

38. The area of a square is 16 sq. units. How long is its diagonal?

- A. 6 units
B. 8 units

- C. 4 units
D. $4\sqrt{2}$ units

Answer: D,

Let x = side of the square

$a = x^2 = 16$, $x = 4$ units

Therefore, diagonal by Pythagorean Theorem

$$d^2 = 4^2 + 4^2$$

$$d^2 = 32, d = \sqrt{32} = \sqrt{(16(2))} = 4\sqrt{2}$$

39. How many times will the digit 7 appear between 1 to 100.

- A. 9
B. 20

- C. 19
D. 11

Answer: B, Count the numbers from 1 to 100, and there are 20 digit 7 between 1 to 100.

40. Find the value of $3a^2 - 2b^2$ if $a = -2$ and $b = 3$

- A. 6
B. -6

- C. 8
D. -8

Answer: B, $3a^2 - 2b^2 = 3(-2)^2 - 2(3)^2 = 3(4) - 2(9) = 12 - 18 = -6$

41. Lito cut $4\frac{1}{4}$ yd. of plastic cover from a bolt containing $35\frac{1}{8}$, how many yd of plastic cover were left in the bolt?

- A. $31\frac{1}{4}$
B. $30\frac{7}{8}$

- C. $31\frac{1}{8}$
D. $30\frac{1}{4}$

Answer: B, $35\frac{1}{8} - 4\frac{1}{4} = (35 - 4) + (\frac{1}{8} - \frac{1}{4}) = 31 - \frac{1}{8} = 30\frac{7}{8}$ yd

42. Which of the following numbers is prime?

- A. 91
B. 121

- C. 101
D. 119

Answer: C, 101 only has 2 factors 1 and 101.

43. The edges of the rectangular solid have measures of 1.5ft x 0.5ft x 3inches. What is the Volume in cubic inches?

- A. 324
B. 225

- C. 272
D. 27

Answer: A, 1.5ft = 18 inches, 0.5ft = 6 inches
volume = $18 \times 6 \times 3 = 324$ cubic inches

44. Simplify this $\frac{4x^2 - 9}{2x + 3}$

- A. $2x + 3$
B. $2x - 3$

- C. $3x + 2$
D. $3x - 2$

Answer: B, $\frac{4x^2 - 9}{2x + 3} = \frac{(2x+3)(2x-3)}{2x+3} = 2x - 3$

45. What is the average of $\frac{1}{2}$, $\frac{1}{4}$, and $\frac{1}{3}$?

- A. $\frac{13}{12}$
B. $\frac{13}{27}$

- C. $\frac{19}{29}$
D. $\frac{13}{36}$

Answer: D, $(\frac{1}{2} + \frac{1}{4} + \frac{1}{3}) \div 3 = \frac{13}{12} \times \frac{1}{3} = \frac{13}{36}$

**Math**

46. What is the exact interest of ₱15,000, invested at 5.5% simple interest rate for 108 days?

- A. ₱255.62
B. ₱265.62

- C. ₱260.62
D. ₱244.11

Answer: D, $I = 15000 \times 0.055 \times \frac{108}{365} = ₱244.11$

47. In how many ways can a photographer arrange 6 people in a row to pose for a photograph?

- A. 30
B. 600

- C. 120
D. 720

Answer: D, ${}_6P_6 = \text{permutation of 6 taken all at a time} = 6! = 720$

48. If a letter is chosen at random from the English alphabet, find the probability that the letter to be chosen will be after the letter G.

A. $\frac{5}{26}$

C. $\frac{7}{26}$

B. $\frac{19}{26}$

D. $\frac{10}{26}$

Answer: B, $\text{probability} = \frac{\text{no. of letter after G}}{\text{total number of letter}} = \frac{19}{26}$

49. If April 18, 2014 fell on Friday, in what day will April 18, 2016 fall?

- A. Monday
B. Tuesday

- C. Thursday
D. Sunday

Answer: A,

April 18, 2014 : Friday

April 18, 2015 : Saturday

April 18, 2016 : Monday

50. What is the single equivalent discount rate of the discount series 5%, 10%, and 25%?

- A. 40%
B. 35.9%

- C. 64.1%
D. 60%

Answer: B,

$$1 - 0.05 = .95 \quad 1 - 0.10 = .90 \quad 1 - 0.25 = .75$$

$$.95 \times .90 \times .75 = .64125$$

$$1 - .64125 = .35875 = 35.875\% = 35.9\%$$

51. The sum of 2 numbers is 10. Three times, the smaller exceeds the larger by 2. Find the smaller number.

- A. 1
B. 2

- C. 3
D. 4

Answer: C

Let x = larger number

$10 - x$ = smaller number

$$3(10 - x) - x = 2$$

$$30 - 3x - x = 2$$

$$28 = 4x \quad x = 7$$

$$10 - x = 3 \text{ (smaller number)}$$

52. What is the smallest positive number that is a multiple of both 15 and 18?

- A. 90
B. 180

- C. 75
D. 72

Answer: A, 90 the LCM of 15 and 18 is 90.

53. If 30% of x is 30, what is 50% of x ?

- A. 30
B. 40

- C. 50
D. 60

Answer: C

30% of x is 30, $x = 100$

Therefore, 50% of 100 is 50

54. A pair of fair dice is rolled. Find the probability that the sum 8 will appear.

- A. $\frac{5}{36}$

- C. $\frac{1}{6}$

- B. 1

- D. $\frac{2}{36}$

Answer: A, $P = \frac{(2,6),(3,5),(4,4),(5,3),(6,2)}{36} = \frac{5}{36}$

55. In how many ways can 6 persons form a committee of 3?

- A. 2
B. 25
C. 120
D. 20

Answer: D, Combination of 6 taken 3 at a time = ${}^6C_3 = 20$

56. How long had Bong been driving, if he had driven 186 km already and his car had moved at a constant rate of 62 km/hr?

- A. 3 minutes
B. 30 minutes

- C. 3 hours
D. 3 hours, 30 minutes

Answer: C. 3 hours

$$186 \text{ km} \div 62 \text{ km/hr} = 3 \text{ hours}$$

57. A department store had a summer sale of pairs of shoes at the following prices:

- A. P800 reduced to P550
B. P850 reduced to P600

- C. P950 reduced to P700
D. P1200 reduced to P850

which group of pairs of shoes was offered with the greatest rate of discount from its original price?

- A. a
B. b

- C. c
D. d

Answer: A. a P800 reduced to P550 with discount rate of $31.25\% = \frac{250}{800}$
 b with 29.41%
 c with 26.32%
 d with 20.83%

58. If 4 pipes can fill a swimming pool in 6 hours, how long will it take 12 pipes to fill the same pool?

- A. 2 hrs
B. 8 hrs

- C. 12 hrs
D. 18 hrs

Answer: A. 2hrs Indirect proportion $4 : 12 = x : 6$, $x = 2\text{hrs}$

59. In a graduating class with the same number of boys and girls, $\frac{1}{9}$ of the boys and $\frac{5}{6}$ of the girls are honor students. What part of the class are girls who are NOT honor students?

- A. $\frac{13}{48}$
B. $\frac{7}{48}$

- C. $\frac{1}{6}$
D. $\frac{11}{12}$

Answer: D. $\frac{1}{12}$

Let x = be the no. of boys

x = be the no. of girls

$x + x = 2x$ = total no. of students

$\frac{5}{6}x$ = total no. of girls who are honor students

$\frac{1}{6}x$ = total no. of girls who are NOT honor students

$$\frac{1}{6}x + 2x = \frac{1}{12}$$



60. If coconuts are twice as expensive as bananas, and bananas are one-third as expensive as mangoes, what is the ratio of the price of one coconut to one mango?

- A. 1 : 6
B. 2 : 3*

- C. 3 : 2
D. 6 : 1

Answer: B, 2 : 3

$$C = 2B$$

$$B = \frac{1}{3}M$$

$$C = 2\left(\frac{1}{3}M\right)$$

$$\frac{C}{M} = \frac{2}{3} = 2 : 3$$

61. A solution contains 28% chlorine. What fraction, in lowest terms, is this of the solution?

A. $\frac{28}{100}$

C. $\frac{3}{10}$

B. 0.28

D. $\frac{7}{25}$ *

Answer: D, 7/25

$$28\% = 28/100 = 7/25$$

62. What is the value of n in the proportion $\frac{3}{51} = \frac{4}{n}$?

A. 83

B. 68*

C. 53

D. 35

Answer: B, 68

$$\frac{3}{51} = \frac{4}{n} \quad \frac{1}{17} = \frac{4}{n} \quad n = 17(4) = 68$$

63. If the average of 5 consecutive even numbers is 8, which of the following is the smallest of the five numbers?

A. 2

B. 4*

C. 6

D. 8

Answer: B, 4

Let $x = 1^{\text{st}}$ even number

$$\frac{x + (x+2) + (x+4) + (x+6) + (x+8)}{5} = 8$$

$$\frac{5x + 20}{5} = 8 \quad x = 4$$

64. Which of the following is an irrational number?

A. $\sqrt{17} - 1$

C. $\sqrt{\frac{65}{5}}$

B. $\frac{7-7}{\sqrt{8}}$

D. $2\sqrt{36}$

Answer: C. $\sqrt{\frac{65}{5}} = \sqrt{13} = 3.60555... \text{ Irrational}$

A. $\sqrt{17} - 1 = \sqrt{16} = 4 \text{ rational}$

B. $\frac{7-7}{\sqrt{8}} = 0 \text{ rational}$

D. $2\sqrt{36} = 2(6) = 12 \text{ rational}$

65. What is the value of $\frac{abc^2 + bcd^2}{\frac{1}{2}abcd}$, if $a = 1$, $b = 2$, $c = -1$ and $d = -2$?

A. 2
B. -2

C. 3
D. -3

Answer: D, -3

$$\frac{(1)(2)(-1)^2 + (2)(-1)(-2)^2}{\frac{1}{2}(1)(2)(-1)(-2)} = \frac{-6}{2} = -3$$

66. The value of: -5^0

A. 1
B. -1

C. 0
D. undefined

Answer: B, -1

$$-5^0 = -(5^0) = -1 \quad x^0 = 1, x \neq 0$$

67. The value of: $(-5)^0$

A. 1
B. -1

C. 0
D. undefined

Answer: A, 1

$$(-5)^0 = 1 \quad x^0 = 1, x \neq 0$$

68. The value of: 0^0

A. 1
B. -1

C. undefined
D. indeterminate

Answer: D, indeterminate

$$0^0 \neq 0, 0^0 \neq 1 \quad 0^0 \text{ is indeterminate}$$

69. Simplify $2^{-3} + 2^3$

A. 0

B. -1

C. $\frac{65}{8}$

D. not define

Answer: C, $\frac{65}{8}$ $2^{-3} + 2^3 = \frac{1}{8} + 8 = 8\frac{1}{8} = \frac{65}{8}$

70. The absolute value of $x = |x| = x$ if

A. $x < 0$

B. $x > 0$

C. $x = 0$

D. $x \geq 0$

Answer: D, $|x| = x$ is true if $x \geq 0$

Ex. $|3| = 3$ $|0| = 0$

71. The absolute value of $x = |x| = -x$ if

A. $x < 0$

B. $x > 0$

C. $x = 0$

D. $x \geq 0$

Answer: A, $x < 0$ $|x| = -x$ if $x < 0$

Ex. $|-3| = -(-3)$

72. The value of x in $|x| = 4$.

A. 4

B. -4

C. 4, -4

D. $1/4$

Answer: C, 4, -4 $|4| = 4$ and $|-4| = 4$

73. The value of x in $|x| = -4$

A. 4

B. -4

C. 4, -4

D. no real number

Answer: D, no real number

The absolute value of a number is always positive or ≥ 0 .

74. Simplify: $-5 + 8 - 4 - 4 + 2$

A. 0

B. -2

C. 3

D. -3

Answer: D, -3 Addition of sign numbers

75. Simplify: $-3 + 4 \times 2 - 4$

- A. -2
- B. 2
- C. 1
- D. -1

Answer: C, 1 PMDAS

$$-3 + 4 \times 2 - 4 = -3 + 8 - 4 = 1$$

76. Simplify: $5 - 3(2 - 3) \times 2$

- A. 11
- B. -11
- C. 16
- D. -16

Answer: A, 11 PMDAS

$$5 - 3(2 - 3) \times 2 = 5 - 3(-1) \times 2 = 5 + 3 \times 2 = 5 + 6 = 11$$

77. The value of x in $|2x - 1| = 2$

- A. $\frac{3}{2}$
- B. -1
- C. $-\frac{1}{2}$
- D. $\frac{3}{2}$ & $-\frac{1}{2}$

Answer: D, $\frac{3}{2}$, $-\frac{1}{2}$

$$|2x - 1| = 2 \quad \text{at } x = \frac{3}{2}$$

$$|2(\frac{3}{2}) - 1| = 2 \quad |3 - 1| = 2 \quad |2| = 2$$

$$|2x - 1| = 2 \quad \text{at } x = -\frac{1}{2}$$

$$|2(-\frac{1}{2}) - 1| = 2 \quad |-1 - 1| = 2 \quad |-2| = 2$$

78. The value of x in the equation $\frac{1}{2}x + 2 = x - 2$,

- A. -8
- B. 8
- C. 2
- D. -2

Answer: B, 8

$$\frac{1}{2}x + 2 = x - 2$$

$$x + 4 = 2x - 4$$

$$4 + 4 = 2x - x, \text{ therefore } x = 8$$



79. Simplify: $\frac{2x-6}{3-x}$

- A. 3
- B. -3
- C. -2
- D. 2

Answer: C, -2

$$\frac{2x-6}{3-x} = \frac{2(x-3)}{-1(x-3)} = \frac{2}{-1} = -2$$

80. $7\frac{1}{2}\%$ in fraction simplified in form

A. $\frac{75}{1000}$

B. $\frac{3}{40}$

C. $\frac{5}{14}$

D. $\frac{15}{2}$

Answer: B, $\frac{3}{40}$

$$7\frac{1}{2}\% = 7.5\% = .075 = \frac{75}{1000} = \frac{3}{40}$$

81. What percent more than 25 is 30?

- A. 10%
- B. 15%
- C. 20%
- D. 25%

Answer: C, 20%

$$\frac{30-25}{25} = \frac{5}{25} = 20\%$$

82. What number decreased by 12.5% of itself gives 210?

- A. 125
- B. 240
- C. 250
- D. 300

Answer: B

Let x be the number

$$x - 0.125x = 210$$

$$0.875x = 210$$

$$x = \frac{210}{.875} = 240$$

83. John Lei bought a refrigerator on an installment plan. If he paid a downpayment of ₱1,188 which was 16.5% of the cost, what was the cost of the refrigerator?

A. ₱7,200
B. ₱7,000

C. ₱8,200
D. ₱8,000

Answer: A, ₱7,200

Let x be the cost of the refrigerator

$$₱1,188 = 0.165x$$

$$x = \frac{₱1,188}{.165} = ₱7,200$$

84. How many months will ₱1,000 accumulate to ₱1,020 at 8% simple interest rate?

A. 1
B. 2

C. 3
D. 4

Answer: C, 3 months

$$P = ₱1,000 \quad F = ₱1,020 \quad I = ₱20$$

$$r = 0.08 \quad t = \frac{I}{Pr} = \frac{20}{1000(0.08)} = .25 \text{ yr} = 3 \text{ months}$$

85. Lawrence invested ₱20,000 which accumulates to ₱22,500 in 10 months. Find the simple interest rate.

A. 5%
B. 10%

C. 15%
D. 20%

Answer: C, 15%

$$P = ₱20,000 \quad F = ₱22,500 \quad I = ₱2,500 \quad t = 10 \text{ months} = \frac{5}{6} \text{ year}$$

$$r = \frac{I}{Pt} = \frac{2500}{20000(\frac{5}{6})} = 0.15 = 15\%$$

86. In how many ways can 3 boys and 2 girls be seated in a row?

A. 5
B. 1

C. 12
D. 120

Answer: D, 120 ${}_5P_5 = 5! = 120$

87. In how many ways can 3 boys and 2 girls be seated in row alternately?

A. 5
B. 1
C. 12
D. 120

Answer: C, 12 $3! \cdot 2! = 6(2) = 12$



88. In how many ways can 5 persons form a group of 5?

- A. 1
B. 2

- C. 5
D. 120

Answer: A, ${}^5C_5 = 1$

89. The probability that John will pass the exam is $\frac{2}{3}$. The probability that Marsha will pass the same exam is $\frac{3}{4}$. What is the probability that both will NOT pass the said exam?

- A. $\frac{1}{2}$
B. $\frac{1}{12}$

- C. 100%
D. $\frac{1}{4}$

Answer: B, $\frac{1}{12}$

$$P = (1 - \frac{2}{3})(1 - \frac{3}{4}) = \frac{1}{3}(\frac{1}{4}) = \frac{1}{12}$$

90. A fair coin is tossed twice. Find the probability that at least one head will appear.

- A. 1
B. $\frac{3}{4}$

- C. $\frac{3}{4}$
D. 100%

Answer: C, $\frac{3}{4}$

$$n = (HT, TH, HH, TT) = 4$$

$$P = \frac{(HT, TH, HH)}{4} = \frac{3}{4}$$

91. Manny can type 5 pages in 6 minutes. How long will it take him to finish 12 pages?

- A. 12 min
B. 14.4 min

- C. 14 min
D. 12.5 min

Answer: B, 14.4 minutes

$$\text{Direct proportion } \frac{5}{12} = \frac{6}{x}$$

$$5x = 72$$

$$x = \frac{72}{5} = 14.4 \text{ minutes}$$

92. Find the value of x , $\frac{3.5}{x} = \frac{1}{2}$

- A. 4
B. 2

- C. 10
D. 14

Answer: D, 14



$$\frac{3.5}{x} = \frac{1}{2}$$

$$\frac{3.5}{x} = \frac{1}{4}$$

$$x = 3.5(4) = 14$$

93. If 5 men can do a job in 10 days, how many days will it take 10 men to complete this task, Assuming that they work at the same rate?

A. 3
B. 4

C. 5
D. 20

Answer: C, 5

Indirect Proportion $\frac{5}{10} = \frac{x}{10}$ $x = 5$ days

94. The ratio of the angles of a triangle is 2 : 3 : 4. What is the difference between the smallest angle and the biggest angle?

A. 300
B. 400

C. 600
D. 200

Answer: B, 40°

$$\text{biggest angle} = \frac{4}{9}(180^\circ) = 80^\circ$$

$$\text{smallest angle} = \frac{2}{9}(180^\circ) = 40^\circ \quad 80^\circ - 40^\circ = 40^\circ$$

95. There are 5 integers, A is their average. S is their sum. What is the ratio of the number S to the number A?

A. 1 : 5
B. 1 : 1
C. 2 : 1
D. 5 : 1

Answer: D, 5 : 1

$$A = \frac{S}{5} \quad \frac{S}{A} = \frac{S}{\frac{S}{5}} = 5 : 1$$

96. 20% of 30 is 6 Which is the appropriate term referring to 6 ?

A. Base
B. Rate
C. Percent
D. Percentage

Answer: D, Percentage

6 is 20% of 30.

P = Rate x Base

97. 15% of ₱300 is what percent of ₱180?

- A. 25%
B. 50%

- C. 20%
D. 60%

Answer: A. 25%

$$.15(₱300) = R(₱180)$$

$$R = \frac{.15(₱300)}{₱180} = .25 = 25\%$$

98. A class has 40 students. Of these, 25 are boys and the rest are girls. What percent of the students are girls?

- A. 62.5%
B. 25%

- C. 40%
D. 37.5%

Answer: D. 37.5%

$$\text{Girls} = 40 - 25 = 15$$

$$R = \frac{15}{40} = 0.375 = 37.5\%$$

99. A TV set which is listed at ₱12,000 is now being sold at ₱8000. What is the percentage decrease in the price of the TV set?

- A. $33\frac{1}{3}\%$
B. 33%

- C. 40%
D. $40\frac{1}{3}\%$

Answer: A. $33\frac{1}{3}\%$

$$R = \frac{12000 - 8000}{12000} = \frac{4000}{12000} = \frac{1}{3} = 33\frac{1}{3}\%$$

100. 0.8 is 50% of what number?

- A. 4
B. 4.4

- C. 1.6
D. 2.6

Answer: C. 1.6

$$0.8 = 0.50 B$$

$$B = \frac{0.8}{0.5} = 1.6$$

101. How much will ₱25000 amount to at 5% simple interest rate after 2 years?

- A. ₱2500
B. ₱27500

- C. ₱28500
D. ₱27000

Answer: B. ₱27500

$$P = ₱25000 \quad r = 5\% = 0.05 \quad t = 2 \text{ yrs.}$$

$$I = Prt = ₱25000(0.05)(2) = ₱2500$$

$$F = P + I = ₱27500$$



102. An agent gets 20% on the first ₱25000 and 10% on the amount over ₱25000 of his total sales. How much does he get from a total sales of ₱42000?

- A. ₱7700
- B. ₱6700
- C. ₱5700
- D. ₱8700

Answer: B, ₱6700

$$\begin{aligned}\text{Commission} &= 0.20(\text{₱}25000) + 0.10(\text{₱}17000) \\ &= \text{₱}5000 + \text{₱}1700 = \text{₱}6700\end{aligned}$$

103. John bought a gold bracelet worth ₱40000. How much should he sell it to realize an 18% profit?

- A. ₱47500
- B. ₱47200
- C. ₱46200
- D. ₱46500

Answer: B, ₱47200

$$\text{Amount} = \text{₱}40000 + 0.18(\text{₱}40000) = \text{₱}40000 + \text{₱}7200 = \text{₱}47200$$

104. If the probability that Jhen will pass the exam is $\frac{5}{8}$, what is the probability that she will not?

- A. 1
- B. $\frac{3}{8}$
- C. $\frac{5}{8}$
- D. $\frac{1}{8}$

Answer: B, $\frac{3}{8}$

$$P = 1 - \frac{5}{8} = \frac{3}{8}$$

105. A letter will be picked at random from the letters of the English alphabet. Find the probability that the letter is in the word DANCE or CANDY.

- A. $\frac{5}{26}$
- B. $\frac{7}{26}$
- C. $\frac{3}{13}$
- D. $\frac{5}{13}$

Answer: C, $\frac{3}{13}$

$$p = \frac{(\text{D,A,N,C,E,Y})}{26} = \frac{6}{26} = \frac{3}{13}$$



106. The least common multiple (LCM) of 36 and 24.

- A. 12
B. 36
C. 48
D. 72

Answer: D, 72

$$36 = 2^2 \cdot 3^2$$

$$24 = 2^3 \cdot 3$$

$$\text{LCM} = 2^3 \cdot 3^2 = 8(9) = 72$$

107. The greatest common factor (GCF) of 36 and 24.

- A. 12
B. 36
C. 48
D. 72

Answer: A, 12

$$36 = 2^2 \cdot 3^2$$

$$24 = 2^3 \cdot 3$$

$$\text{GCF} = 2^2 \cdot 3 = 4(3) = 12$$

108. The value of $(8 + 16) + 2^3 - 2 \times 3$

- A. 2
B. -2
C. -3
D. 3

Answer: C, -3

$$(8 + 16) + 2^3 - 2 \times 3 = 24 + 8 - 6 = 3 - 6 = -3$$

109. What are the 2 digits which when inserted in the blank spaces will make 524 _ _ divisible by 8?

- A. 2 and 0
B. 0 and 4
C. 1 and 2
D. 0 and 0

Answer: D, 0 and 0

52400 is divisible by 8

Divisibility by 8: A number is divisible by 8 if the last 3 digits is divisible by 8.

110. Complete the series: 95 88 79 68 _ _

- A. 65 , 60
B. 40 , 35
C. 50 , 45
D. 55 , 40

Answer: D, 55 , 40

$$95 \quad 88 \quad 79 \quad 68 \quad 55 \quad 40$$

$$7 \quad 9 \quad 11 \quad 13 \quad 15$$

111. The value of $\frac{25}{18} \cdot \frac{9}{15} \cdot \frac{6}{35}$

A. $\frac{8}{25}$

B. $\frac{1}{7}$

C. $\frac{4}{25}$

D. $\frac{3}{35}$

Answer: B, $\frac{1}{7}$

$$\frac{25}{18} \cdot \frac{9}{15} \cdot \frac{6}{35} = \frac{5}{2} \cdot \frac{1}{3} \cdot \frac{6}{35} = \frac{1}{1} \cdot \frac{1}{1} \cdot \frac{1}{7} = \frac{1}{7}$$

112. A can do a piece of work in 5 days and B in 8 days. How long will it take them, working together to complete the work?

A. 12 days

B. $3\frac{1}{13}$ days

C. 4 days

D. 13 days

Answer: B, $3\frac{1}{13}$ days

$$\text{together} = \frac{5(8)}{5+8} = \frac{40}{13} = 3\frac{1}{13} \text{ days}$$

113. Simplify: $\left(\frac{5}{8} - \frac{1}{4}\right) \div \left(\frac{11}{12} - \frac{5}{9}\right)$

A. $\frac{27}{26}$

B. $\frac{26}{27}$

C. $\frac{5}{12}$

D. $\frac{12}{5}$

Answer: A, $\frac{27}{26}$

$$\left(\frac{5}{8} - \frac{1}{4}\right) \div \left(\frac{11}{12} - \frac{5}{9}\right) = \left(\frac{5-2}{8}\right) \div \left(\frac{33-20}{36}\right) = \frac{3}{8} \cdot \frac{36}{13} = \frac{3}{2} \cdot \frac{9}{13} = \frac{27}{26}$$

114. In 4 years, Jean will be twice as old as she was 6 years ago. How old is Jean?

A. 14

B. 15

C. 16

D. 17

Answer: C, 16

Let x = present age of Jean

$$x + 4 = 2(x - 6)$$

$$x + 4 = 2x - 12$$

$$x = 16$$



115. The value of $\frac{27^3}{9^4}$.

A. 9

B. $\frac{1}{9}$

C. 3

D. $\frac{1}{3}$

Answer: C, 3

$$\frac{27^3}{9^4} = \frac{(3^3)^3}{(3^2)^4} = \frac{3^9}{3^8} = 3$$

116. A worm crawls on a branch 38 cm. long. How long is the branch in mm?

A. 380 mm

B. 3800 mm

C. 3.8 mm

D. 0.38 mm

Answer: A, 380 mm

$$1 \text{ cm} = 10 \text{ mm}$$

$$38 \text{ cm} = (38)(10) \text{ mm} = 380 \text{ mm}$$

117. The positive difference between the mean and the median of the numbers 25, 25, 27, 30, and 33 is

A. 0

B. 1

C. 2

D. 3

Answer: B, 1

$$\text{Mean} = \frac{140}{5} = 28$$

$$\text{Median} = 27$$

$$\text{Therefore} = 28 - 27 = 1$$

118. Which of the following is 7359.9966 rounded off to the nearest hundredths?

A. 7360.00

B. 7359.10

C. 7359.998

D. 7350.00

Answer: A, 7360.00

In 7359.9966, the hundredths digit is 9 and the thousandths digit is 6, add 0.01 to 7359.99, therefore the given number rounded off to the nearest hundredths is 7360.00.

119. The perimeter of a square of area 144 sq. meters.

- A. 24 m
B. 48 m

- C. 40 m
D. 60 m

Answer: B. 48 m

$$\text{Area square} = 144 \text{ sq. m} = s^2$$

$$s = \sqrt{144} = 12 \text{ m}$$

$$\text{perimeter } P = 4s = 4(12\text{m}) = 48 \text{ m}$$

120. The area of a rectangle is $x^2 - 3x - 4$. One of its sides is

- A. $x - 4$
B. $x + 4$

- C. $x - 1$
D. $x - 2$

Answer: A. $x - 4$

$$\text{Area} = x^2 - 3x - 4 = (x - 4)(x + 1)$$

The sides are $x - 4$ and $x + 1$

121. 24 is three-fourths of what number?

- A. 30
B. 32

- C. 24
D. 16

Answer: B. 32

$$24 = \frac{3}{4} B$$

$$B = (24(4))/3 = 32$$

122. 18% of ₱36,000 is what percent of ₱12,960?

- A. 50%
B. 25%

- C. 5%
D. 10%

Answer: A. 50%

$$0.18 (\text{₱}36,000) = R(\text{₱}12,960)$$

$$R = \frac{0.18 (\text{₱}36,000)}{(\text{₱}12,960)} = 0.50 = 50\%$$

123. After receiving 30% discount, Jay paid ₱210.00 for an item. What is the regular price of the item?

- A. ₱300
B. ₱350
C. ₱400
D. ₱380

Answer: A. ₱300

$$\text{Regular price} = \text{₱}210/(70\%) = \text{₱}210/.70 = \text{₱}300$$



124. A salesman gets 15% commission for the first ₱20,000 and 10% for the amount over ₱20,000 of his total sales. How much does he get for a total sales of ₱50,600?

- A. ₱8,060
- B. ₱6,060
- C. ₱7,060
- D. ₱5,060

Answer: B, ₱6,060

$$C = 0.15(₱20,000) + 0.10(₱30,600)$$

$$C = ₱3,000 + ₱3,060 = ₱6,060$$

125. Peter borrowed ₱15,000 from the bank at the rate of 12% per year. How much did he owe the bank after 18 months?

- A. ₱18,700
- B. ₱16,000
- C. ₱17,700
- D. ₱17,000

Answer: C, ₱17,700

$$F = P(1 + rt) = ₱15,000(1 + 0.12(1.5)) = ₱17,700$$

126. A car racer covers 250 km in 2 hours. How far can he go in $2\frac{3}{4}$ hours?

- A. $250\frac{1}{2}$ km
- B. 340 km

- C. $343\frac{3}{4}$ km
- D. 344 km

Answer: C, 343.75 km

$$\frac{250}{x} = \frac{2}{2.75}$$

$$x = \frac{250(2.75)}{2} = 343.75 \text{ km}$$

127. Five persons can finish painting a wall within 5 hours. If only 3 people are available, how many hours do they have to work to finish the same job?

- A. $7\frac{1}{2}$
- B. 8

C. 8

D. $8\frac{1}{3}$

Answer: D, $8\frac{1}{3}$

$$\frac{5}{3} = \frac{x}{5} \quad x = \frac{25}{3} = 8\frac{1}{3} \text{ hours}$$

Math

128. A father decided that his 30-hectare land be divided among his 3 children using 1 : 2 : 3 partition, the oldest getting the biggest share. How much did the second child receive?

- A. 5 ha
B. 15 ha

- C. 8 ha
D. 10 ha

Answer: D. 10 ha.

$$\frac{2}{6}(30) = 10 \text{ ha.}$$

129. A rectangular figure is 6 inches wide and 1 foot long. If the width is to be decreased by 2 inches, but the area remains the same, by how much should the length be increased?

- A. 6 inches
B. 8 inches

- C. 4 inches
D. 10 inches

Answer: A. 6 inches

$$\text{Area} = 6 \text{ inches}(12 \text{ inches}) = 72 \text{ in}^2$$

$$72 = 4 \text{ inches (Length)}$$

$$\text{Length} = 18 \text{ inches}$$

$$\text{Therefore, the length should be increased by } 18 \text{ inches} - 12 \text{ inches} = 6 \text{ inches}$$

130. The average of four numbers is 38. If three of the four numbers are 48, 36, and 24, what is the 4th number?

- A. 41
B. 44

- C. 40
D. 42

Answer: B. 44

Let x be the 4th number

$$\frac{48 + 36 + 24 + x}{4} = 38$$

$$x = 38(4) - 108 = 152 - 108 = 44$$

131. Thirty students had an average of 75, while 20 students had, an average of 90. What is the average of the entire group?

- A. 82.5
B. 81.5
C. 81
D. 82

Answer: C. 81

$$\text{average} = \frac{30(75) + 20(90)}{30 + 20} = 81$$



132. One-way fare from Manila to Cebu is ₱3,800. How many round-trip tickets can be purchased with the amount of ₱91,200?

- A. 8
- B. 10
- C. 14
- D. 12

Answer: D, 12

$$\text{Round trip tickets} = \frac{91200}{2(3800)} = 12$$

133. A certain number doubled and decreased by 12 is 8. What is the number?

- A. 14
- B. 12
- C. 10
- D. 8

Answer: C, 10

Let x be the number

$$2x - 12 = 8 \quad x = 10$$

$$\begin{aligned} 2x - 12 &= 8 \\ 2x &= 8 + 12 \\ 2x &= 20 \\ \frac{2x}{2} &= \frac{20}{2} \quad (x = 10) \\ x &= 10 \end{aligned}$$

134. What is the perimeter of a square board of area 36 square inches?

- A. 28 inches
- B. 24 inches
- C. 20 inches
- D. 18 inches

Answer: B, 24 Inches

$$\text{Area} = s^2 = 36 \text{ sq inches}$$

$$s = 6 \text{ inches}$$

$$\text{Perimeter} = 4s = 4(6 \text{ inches}) = 24 \text{ inches.}$$

135. The area of a rectangle is 40 square feet. The width is 5 feet. Find its perimeter.

- A. 30 ft.
- B. 6 ft.
- C. 26 ft.
- D. 21 ft.

Answer: C, 26 ft.

$$A = LW$$

$$40 = L(5)$$

$$L = 8 \text{ ft.}$$

$$P = 2L + 2W = 2(8) + 2(5) = 26 \text{ ft.}$$

Math

136. What is the height of a triangle of area 36 square units and base 4 units?

- A. 18 units
- B. 12 units
- C. 10 units
- D. 15 units

Answer: A, 18 units

$$\text{Area} = \frac{\text{base} \times \text{height}}{2}$$

$$36 = \frac{4 \times \text{height}}{2}$$

$$\text{height} = 18 \text{ units}$$

137. A swimming pool is an equilateral triangle in shape. One side is 18 meters. How many meters of rope are needed to enclose the pool?

- A. 54 meters
- B. 36 meters
- C. 48 meters
- D. 50 meters

Answer: A, 54 meters

An equilateral triangle has all sides equal.

$$\text{Perimeter} = 3(18\text{m}) = 54 \text{ m}$$

138. A square and an equilateral triangle have equal perimeter. If the side of the square is 3 units less than the side of the triangle, what is the area of the square?

- A. 64 square units
- B. 144 square units
- C. 81 square units
- D. 100 square units

Answer: C, 81 sq. units

Let x be the side of the equilateral triangle

$x - 3$ be the side of the square

Perimeter of square = perimeter of equilateral triangle

$$4(x - 3) = 3x$$

$$4x - 12 = 3x$$

$$x = 12 \text{ units} = \text{side of the equilateral triangle}$$

$$x - 3 = 12 - 3 = 9 \text{ units} = \text{side of the square}$$

$$\text{Therefore, Area of square} = 9^2 = 81 \text{ sq. units}$$



139. A ball is drawn at random from a box containing 4 red, 5 white, and 3 blue balls. Find the probability that the ball is NOT white.

A. $\frac{2}{3}$

C. $\frac{5}{12}$

B. $\frac{7}{12}$

D. $\frac{3}{4}$

Answer: B, $\frac{7}{12}$

$$P(\text{not white}) = \frac{12-5}{12} = \frac{7}{12}$$

140. A pair of dice is rolled. Find the probability that the sum "9" appears.

A. $\frac{2}{9}$

C. $\frac{1}{3}$

B. $\frac{1}{9}$

D. $\frac{4}{5}$

Answer: B, $\frac{1}{9}$

$$P = \frac{[(1,6), (4,5), (5,4), (6,3)]}{36} = \frac{4}{36} = \frac{1}{9}$$

141. A mother wants to have 2 children, both girls. Find the probability that both children will be girls.

A. $\frac{1}{2}$

C. 2

B. 100%

D. $\frac{1}{4}$

Answer: D, $\frac{1}{4}$

$$P(\text{both girls}) = P(\text{girl 1}) \cdot P(\text{girl 2}) \\ = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

142. The probability that the wife will survive in 10 years is $\frac{3}{5}$ and that of her husband is $\frac{2}{3}$. Find the probability that either one of them will survive.

A. $\frac{7}{15}$

C. $\frac{2}{5}$

B. $\frac{1}{2}$

D. 1

Answer: A, $\frac{7}{15}$

$$P = P(\text{wife will survive}) \cdot P(\text{husband will not survive}) + \\ P(\text{wife will not survive}) \cdot P(\text{husband will survive})$$

$$P = \left(\frac{1}{5}\right)\left(\frac{1}{3}\right) + \left(\frac{2}{5}\right)\left(\frac{2}{3}\right) = \frac{1}{15} + \frac{4}{15} = \frac{5}{15}$$

143. Simplify $\frac{2x-2}{4x^2-4}$

A. $2x+2$

E. $2x-2$

C. $\frac{1}{2x+2}$

D. $\frac{1}{2x-2}$

Answer: C, $\frac{1}{2x+2}$

$$\frac{2x-2}{4x^2-4} = \frac{2x-2}{(2x+2)(2x-2)} = \frac{1}{2x+2}$$

144. Simplify $\left(\frac{12x^2y^3}{8x^3y^2}\right) \div \left(\frac{20xy^2}{16xy}\right)$

A. $\frac{6y}{5}$

C. $\frac{6}{5y}$

B. $\frac{6x}{5}$

D. $\frac{6}{5x}$

Answer: D, $\frac{6}{5x}$

$$\left(\frac{12x^2y^3}{8x^3y^2}\right) \div \left(\frac{20xy^2}{16xy}\right) = \left(\frac{3y}{2x}\right) \div \left(\frac{5y}{4}\right) = \frac{3y}{2x} \cdot \frac{4}{5y} = \frac{6}{5x}$$

145. Simplify $(3x+2)\left(\frac{1}{12x^2-13x-14}\right)$

A. $\frac{1}{4x+7}$

C. $\frac{1}{4x-7}$

B. $4x+7$

D. $4x-7$

Answer: C, $\frac{1}{4x-7}$

$$\frac{3x+2}{12x^2-13x-14} = \frac{3x+2}{(3x+2)(4x-7)} = \frac{1}{4x-7}$$

146. Jack can finish a certain job in 5 days. Jill can finish the same job in 7 days. Working together, in how many days can they finish the job?

- A. 12
- B. $4\frac{1}{2}$
- C. 3
- D. $2\frac{11}{12}$

Answer: D, $2\frac{11}{12}$

$$\text{together} = \frac{5(7)}{5+7} = \frac{35}{12} = 2\frac{11}{12}$$

147. Jay is twice as old as John. Ten years ago, he was 4 times as old as John. Find Jay's present age.

- A. 20
- B. 25
- C. 30
- D. 40

Answer: C, 30

$$\begin{aligned} \text{Let } x &= \text{present age of John} \\ 2x &= \text{present age of Jay} \\ 2x - 10 &= 4(x - 10) \\ 2x - 10 &= 4x - 40 \\ 2x &= 30 \\ x &= 15 \text{ (age of John)} \\ 2x &= 30 \text{ (age of Jay)} \end{aligned}$$

148. How many gallons of 30% solution of HCl acid should be added to 2 gallons of 15% solution of the same acid to make a 20% solution?

- A. 2.5
- B. 1.5
- C. 1
- D. 2

Answer: C, 1

$$\begin{aligned} \text{Let } x &= \text{no. of gallons 30\% solution} \\ 0.30x + 0.15(2) &= 0.20(x + 2) \\ 30x + 15(2) &= 20(x + 2) \\ 30x - 20x &= 40 - 30 \\ 10x &= 10 \\ x &= 1 \end{aligned}$$

149. $x - 3p, -2x - 2p, -5x - p, \underline{\hspace{2cm}}$

- A. $-8x$
B. $8x$

- C. $8x - p$
D. $8x + p$

Answer: A, $-8x$

next term is obtained by adding $-3x + p$.

$$(-5x - p) + (-3x + p) = -8x + 0 = -8x$$

150. Value of n such that $324n0$ is divisible by 3

- A. 1
B. 2

- C. 3
D. 4

Answer: C, 3

$324n0 = 32430$ is divisible by 3

A no. is divisible by 3 if the sum of the digits is divisible by 3. In case of 32430 , the sum of the digits is 12 which is divisible by 3.

136. What is the height of a triangle of area 36 square units and base 4 units?

- A. 18 units
- B. 12 units
- C. 10 units
- D. 15 units

Answer: A. 18 units

$$\text{Area} = \frac{\text{base} \times \text{height}}{2}$$

$$36 = \frac{4 \times \text{height}}{2}$$

$$\text{height} = 18 \text{ units}$$

137. A swimming pool is an equilateral triangle in shape. One side is 18 meters. How many meters of rope are needed to enclose the pool?

- A. 54 meters
- B. 36 meters
- C. 48 meters
- D. 50 meters

Answer: A. 54 meters

An equilateral triangle has all sides equal.

$$\text{Perimeter} = 3(18\text{m}) = 54 \text{ m}$$

138. A square and an equilateral triangle have equal perimeter. If the side of the square is 3 units less than the side of the triangle, what is the area of the square?

- A. 64 square units
- B. 144 square units
- C. 81 square units
- D. 100 square units

Answer: C. 81 sq. units

Let x be the side of the equilateral triangle

$x - 3$ be the side of the square

Perimeter of square = perimeter of equilateral triangle

$$4(x - 3) = 3x$$

$$4x - 12 = 3x$$

$$x = 12 \text{ units} = \text{side of the equilateral triangle}$$

$$x - 3 = 12 - 3 = 9 \text{ units} = \text{side of the square}$$

$$\text{Therefore, Area of square} = 9^2 = 81 \text{ sq. units.}$$



139. A ball is drawn at random from a box containing 4 red, 5 white, and 3 blue balls. Find the probability that the ball is NOT white.

A. $\frac{2}{3}$

C. $\frac{5}{12}$

B. $\frac{7}{12}$

D. $\frac{3}{4}$

Answer: B. $\frac{7}{12}$

$$P(\text{not white}) = \frac{12-5}{12} = \frac{7}{12}$$

140. A pair of dice is rolled. Find the probability that the sum "9" appears.

A. $\frac{2}{9}$

C. $\frac{1}{3}$

B. $\frac{4}{9}$

D. $\frac{4}{5}$

Answer: B. $\frac{4}{9}$

$$P = \frac{[(3,6), (4,5), (5,4), (6,3)]}{36} = \frac{4}{36} = \frac{1}{9}$$

141. A mother wants to have 2 children, both girls. Find the probability that both children will be girls.

A. $\frac{1}{2}$

C. 2

B. 100%

D. $\frac{1}{4}$

Answer: D. $\frac{1}{4}$

$$\begin{aligned} P(\text{both girls}) &= P(\text{girl 1}) \cdot P(\text{girl 2}) \\ &= \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} \end{aligned}$$

142. The probability that the wife will survive in 10 years is $\frac{3}{5}$ and that of her husband is $\frac{2}{3}$. Find the probability that either one of them will survive.

A. $\frac{7}{15}$

C. $\frac{2}{5}$

B. $\frac{1}{2}$

D. 1

Answer: A. $\frac{7}{15}$

$$P = P(\text{wife will survive}) \cdot P(\text{husband will not survive}) + P(\text{wife will not survive}) \cdot P(\text{husband will survive})$$

$$P = \left(\frac{1}{5}\right)\left(\frac{1}{3}\right) + \left(\frac{2}{5}\right)\left(\frac{2}{3}\right) = \frac{1}{15} + \frac{4}{15} = \frac{5}{15}$$

143. Simplify $\frac{2x-2}{4x^2-4}$

A. $2x+2$

B. $2x-2$

C. $\frac{1}{2x+2}$

D. $\frac{1}{2x-2}$

Answer: C, $\frac{1}{2x+2}$

$$\frac{2x-2}{4x^2-4} = \frac{2x-2}{(2x+2)(2x-2)} = \frac{1}{2x+2}$$

144. Simplify $\left(\frac{12x^2y^3}{8x^3y^2}\right) \div \left(\frac{20xy^2}{16xy}\right)$

A. $\frac{6y}{5}$

B. $\frac{6x}{5}$

C. $\frac{6}{5y}$

D. $\frac{6}{5x}$

Answer: D, $\frac{6}{5x}$

$$\left(\frac{12x^2y^3}{8x^3y^2}\right) \div \left(\frac{20xy^2}{16xy}\right) = \left(\frac{3y}{2x}\right) \div \left(\frac{5y}{4}\right) = \frac{3y}{2x} \cdot \frac{4}{5y} = \frac{6}{5x}$$

145. Simplify $(3x+2)\left(\frac{1}{12x^2-13x-14}\right)$

A. $\frac{1}{4x+7}$

B. $4x+7$

C. $\frac{1}{4x-7}$

D. $4x-7$

Answer: C, $\frac{1}{4x-7}$

$$\frac{3x+2}{12x^2-13x-14} = \frac{3x+2}{(3x+2)(4x-7)} = \frac{1}{4x-7}$$



146. Jack can finish a certain job in 5 days. Jill can finish the same job in 7 days. Working together, in how many days can they finish the job?

A. 12

B. $4\frac{1}{2}$

C. 3

D. $2\frac{11}{12}$

Answer: D, $2\frac{11}{12}$

$$\text{together} = \frac{5(7)}{5+7} = \frac{35}{12} = 2\frac{11}{12}$$

147. Jay is twice as old as John. Ten years ago, he was 4 times as old as John. Find Jay's present age.

A. 20

B. 25

C. 30

D. 40

Answer: C, 30

Let x — present age of John

$2x$ — present age of Jay

$$2x - 10 = 4(x - 10)$$

$$2x - 10 = 4x - 40$$

$$2x = 30$$

$$x = 15 \text{ (age of John)}$$

$$2x = 30 \text{ (age of Jay)}$$

148. How many gallons of 30% solution of HCl acid should be added to 2 gallons of 15% solution of the same acid to make a 20% solution?

A. 2.5

B. 1.5

C. 1

D. 2

Answer: C, 1

Let x — no. of gallons 30% solution

$$0.30x + 0.15(2) = 0.20(x + 2)$$

$$30x + 15(2) = 20(x + 2)$$

$$30x - 20x = 40 - 30$$

$$10x = 10$$

$$x = 1$$



Math

149. $x - 3p, -2x - 2p, -5x - p, \underline{\hspace{2cm}}$

A. $-8x$

B. $8x$

C. $8x - p$

D. $8x + p$

Answer: A, $-8x$

next term is obtained by adding $-3x + p$.

$$(-5x - p) + (-3x + p) = -8x + 0 = -8x$$

150. Value of n such that $324n0$ is divisible by 3

A. 1

B. 2

C. 3

D. 4

Answer: C, 3

$324n0 = 32430$ is divisible by 3

A no. is divisible by 3 if the sum of the digits is divisible by 3. In case of 32430 , the sum of the digits is 12 which is divisible by 3.

Philippine History**Pre-historic**

- Origin of the Philippines
- The Land Bridge Theory
- Part of a Lost Continent
- Volcanic in Origin

The Land Bridge Theory was considered by some earlier scientists and scholars as the most plausible. This theory states that the Philippine archipelago was once a part of mainland Asia. The Philippines was landmass bridging the Asian continent to Indonesia, New Guinea up to the Australian Continent. About 250,000 years ago, during the Post-Glacial Period, the rise of the sea levels submerged lower regions, which included the land bridges linking main land China and the Philippines. What were not submerged became the archipelagoes of the Philippines and Indonesia.

Some American geographers and geologists theorized that the Philippines and Indonesia were remnants of a Pacific continent called Mu or Lemuria, which had sunk during the Prehistoric Period. This opinion was influenced by the legend of the lost continent of Atlantis, which was popularized by earlier European writers. This theory, however, is slowly becoming a myth to modern geologists who consider it to have no scientific basis.

The Earliest People

Many theories have been written on the origin of humans in the Philippines, but no definite evidence can show how and when the first people arrived in these parts.

Biblical theory states that humankind descended from Adam and Eve, while scientists stick to the evolutionary theory.

Filipino myth and folklore state that the first Filipino male was Malakas, and Maganda was his mate.

Migration Theory

Dr. H. O. Beyer's migration theory arranged the peopling of the Philippines with the coming of the Dawn Man, the pre-historic cousin of the Java man and Peking man of 250,000 years ago.

During the middle Pleistocene Period (Ice Age), roughly about 250,000 B.C., man first appeared in the Philippines via the land bridges, which connected mainland Asia and the Philippines.

The first known Filipino is the Tabon Cave Man, who lived in Palawan at about 22,000 B.C. (The skull cap of this Stone Age human was discovered by Dr. Robert Fox inside the Tabon Cave in 1962.)

Pre-Colonial

Prof. F. Landa Jocano, in his paper "The Vision of Our Future Must Be Rooted On Our Image Of The Past", strongly argued: "The Filipinos possessed an elaborate civilization in the past. This achievement did not become part of the Filipino consciousness, even today, because of the advent of colonization during which a systematic distortion of our perceptive about ourselves was carried out."

House

- Bahay Kubo (nipa hut) – It was suited to climate. Bamboo or wood stakes (haligi) held the hut above the ground.

Social Science

- **Batalan** – where washing and bathing were done
- **Badjaos** (sea gypsies) – Sea gypsies of Sulu lived in boat-houses. This was so, because these people mainly derived their living from the resources of the sea.

Mode of Dressing

- **Kanggan** – The male Filipinos wore a collarless short-sleeved jacket as the upper part of their clothing.
- **Bahag** – The lower part of their clothing, a strip of cloth wrapped around the waist and in between the legs
- **Putong** – The men's headgear
- **Baro or camisa** – The women wore a wide-sleeved jacket, the upper part of their clothing
- **Saya (Tagalog) and Patadyong (Visayans)** – The women's skirt or lower part
- **Tapis** – An additional cover was a piece of white or red cloth wrapped around the waist.

Ornaments

- **Kalumbiga** – Armlets, gold rings, gold necklaces, gold bracelets
- **Pintados** – The early Filipinos adorned their bodies and faces with tattoos.

The Visayan were the most tattooed people among the ancient Filipinos and for this reason, the Spanish missionaries called them Pintados.

Social Classes

The Pre - Spanish society was divided into three social classes:

- The nobles
- Free men
- Alipin

The chiefs, their families and relatives composed the upper class or nobility. They carried the title GAT or LAKAN.

The middle class or freemen who were usually free – born persons or freed slaves constituted the majority of the social classes.

The dependents or the ALIPIN belonged to the lowest class. There were two kinds of dependents:

- **Alipin namamahay** – owned houses and properties
- **Alipin sagigilid** – had NO right; could be sold anytime by his master(s)

Government

- **Barangay** – the pre-colonial political unit, composed of 30 to 100 families
It was a monarchy ruled by a chieftain called Datu, Hari, or Raja (in Muslim community).

Burial Mourning and Practices

- **Laraw/Larao** – mourning for a dead chieftain
- **Maglahe** – for a dead man
- **Morotal** – for a dead woman

Languages and dialects

More than a hundred languages and dialects were spoken by ancient Filipinos. The 8 major languages and dialects were:

- Tagalog
- Iloko
- Pangasinan
- Kapangpangan
- Sugbuhanon
- Hiligaynon
- Samaron
- Maguindanao

Malayo - Polynesian – Almost all of these languages and dialects originated from one linguistic family.

Literature

The pre-colonial Filipinos had both written and oral literature. Their oral literature consisted of the following:

- Awit (songs)
- Salawikain (proverbs)
- Hele (cradle songs)
- Bugtong (riddles)
- Ihiman (wedding songs)
- Kumintang (war songs)

Examples of their written literature were the myths and epics like:

- Hudhud and the Alim – Ifugao
- Biag ni Lam-ang – Ilocanos
- Handiong – Bicolanos
- Bantungan, Indirapatra, Sulayman, Parang Sabil – Muslims

Music and Dances

Musical instruments:

- Kudyapi – the Tagalog guitar,
- Tultogan – Visayan bamboo drum
- Silbay – Ilocano red flute
- Kutibeng – Ilocano five-stringed guitar

Dances:

- Balitaw and Dandansoy – Visayans
- Kumintang (love dance) – Tagalog
- Mahinhin (courtship dance) – Tagalog
- Kinnotan – Ilocanos ants dance

Asian Cultural Heritage

Indian

- Sri Vidjaya – An Indianized Malayo-Buddhist empire

- Majapahit – Hindu Malayan empire
- Orang Dampuan – Men of Champa who were vassals of Sri Vijaya Empire
- B'hattara – Great Lord
- Indra Battara – Most prominent deity
- Indra – sky god
- Agni – god of fire
- Surya – god of sun
- Mahabharata – Indian epic
- Putong – headgear
- Sarong – Indian sari

Chinese

- Ma-i – Old Chinese name for the Philippines
- Parian (now Binondo) – Chinese colony in Manila
- Chiensi – siyansi
- Baglai – bantay
- Sangle – sangla
- Inkong – ingkong
- Hebi – hibi
- A-chi – ate
- Bi-koe – biro
- Pin-to – pinto
- Pansit – pansit
- Ampau – ampaw

Arabic

- Mudum – The first who introduced Islam in the Malay Peninsula
 - Rajah Baginda – One of the Sumatra rulers of who went to Sulu and continued preaching the doctrines of Muhammad
 - Abu Bakar – Muslim missionary who arrived in the Philippines and continued propagating the Islamic faith
 - Paramisuli – Baginda's daughter
 - Lantaka – cannon
 - Vinta – warboat
 - Kuta – fort
 - Luwaran – code of laws
 - Darangan – epic poetry
- (Agoncillo and Guerrero, The History of the Filipino People)

Dates to be Remembered**The Age of Discoveries and Exploration**

- Ferdinand sighted the island of Samar. – March 16, 1521
- Magellan landed on Homonhon and named the islands "Archipelago of St. Lazarus." – March 17, 1521
- The first blood compact was held between Magellan and Rajah Kulambo of Limasawa. – March 29, 1521

- The first Christian Mass officiated by Father Pedro de Valderrama was celebrated in the island of Limasawa, Southern Leyte. – March 31, 1521
- Magellan was killed in the Battle of Mactan. – April 27, 1521

Spanish Period (Colonial Integration)

- Ruy Lopez de Villalobos named Samar and Leyte "Filipinas" in honor of Prince Philip II of Spain, the son King Carlos. The former, later became King Felipe II – Feb. 2, 1543
- General Miguel Lopez de Legazpi, another Spanish explorer, reached Cebu and claimed the island for Spain. – February 13, 1565
- Blood compact was contracted between Legazpi and Rajah Sikatuna of Bohol. – March 16, 1565
- Cebu, the first Spanish settlement established by Legazpi, was named La Villa de San Miguel. – April 27, 1565
- La Villa de San Miguel was renamed Ciudad del Santissimo Nombre de Jesus in honor of the Sto. Niño. It became the first Spanish permanent settlement in the Philippines. – January 1, 1571
- Manila was conquered by Martin de Goiti in the Battle of Bankusay. – June 3, 1571
- Manila was named "Distinguished and Ever Loyal City" by Legazpi, and made himself the first Governor – General in the Philippines. – June 24, 1571
- Martin de Goiti was killed by Limahong, a Chinese pirate. – November 30, 1574
- The Royal Audiencia, referred to as the Supreme Court, during the Spanish period was established to provide a check and balance of power among Spanish government officials. – May 16, 1584
- Defeat of the 6 Dutch fleet by the Spanish Navy led by Juan Ronquillo in the Battle of Playa Honda, Zambales – April 13 to 14, 1617
- Defeat of the Dutch fleet by the Spanish Navy in the Battle of Manila, hence the day was declared as the feast of La Naval de Manila. – October 3, 1646
- Manila was conquered by the British forces under General William Draper and Admiral Samuel Cornish. – October 5, 1762
- Britain surrendered the Philippines to Spain after losing the seven-year war. – May 17, 1764
- The truce signed between France and England ended the seven-year war and led to the recovery of Manila by Spain. – May 31, 1764
- The Dagohoy rebellion, the longest revolt in the Philippines, lasted for 85 years. – October 31, 1829
- Manila was opened to world trade, which led to an era of Commercial Revolution in the Philippines. – September 6, 1834
- Apolinario dela Cruz, known as "Hermano Pule," was executed for fighting against the Spanish government. – November 4, 1841

The Rise of Nationalism

- The birth of Dr. Jose P. Rizal in Calamba, Laguna – June 19, 1861
- A meeting in Cavite led by a Filipino Sergeant Lamadrid led to the take over the arsenal of Fort San Felipe, and the Filipinos killed their Spanish officers. – January 20, 1872
- The execution of GOMBURZA – February 17, 1872
- La Asociaaon Hispano Filipino, founded by Miguel Morayta aimed to introduce changes in the Philippines and to campaign for a Filipino Representative in the Spanish Cortes, the law-making body of Spain. – January 12, 1889

- The newspaper, *La Solidaridad* (SOL for short), founded by Graciano Lopez Jaena of Iloilo, served as the official organ of the Reform Movement. – February 15, 1889
- Rizal's second novel, *El Filibusterismo*, was published with the able financial assistance of Valentin Ventura. – February 18, 1891
- La Liga Filipina was founded by Rizal as a civic movement to unite the Filipinos in pushing for expressing reforms and freedom against Spanish colonial policies. – July 3, 1892
- Rizal began to be exiled in Dapitan, Zamboanga del Norte, Mindanao, and it lasted for four (4) years. – July 7, 1892
- The Kataas-taasan, Kagalang-galangang Katipunan ng mga anak ng Bayan (KKK) was secretly organized by Bonifacio, Arellano, Diwa, Plata and Diaz in Azcarraga Street (now C.M. Recto), in Tondo. – July 7, 1892
- The Katipunan was discovered by Father Mariano Gil through Teodoro Patino, resulting in the mass arrest and execution. – August 19, 1896
- Philippine Revolution – "Cry of Balintawak" – August 23, 1896
- The Battle of Pasong Tamo – August 25, 1896
- The Battle of Pinaglabanan – August 30, 1896
- General Emilio Aguinaldo led the uprising in Kawit, Cavite. – August 31, 1896
- The 13 men from Cavite, known as "Los Trece Martirez de Cavite," were executed. – September 12, 1896
- The Martyrdom of Dr. Jose Rizal in Luneta de Bagumbayan – December 30, 1896
- The Tejeros Convention resolved the conflict between the Magdalo and the Magdiwang factions of the Katipunan. – March 22, 1897
- The execution of Bonifacio brothers (Andres and Procopio) in Maragondon, Cavite – May 10, 1897
- The Biak-na-Bato, a revolutionary government, was established. – July 7, 1897
- The truce of Biak-na-Bato was signed between Pedro Paterno and Governor Primo de Rivera. – December 15, 1897

American Period

- The Spaniards blew up the American warship *Maine* in Havana, Cuba. – February 15, 1898
- The United States of America declared war against Spain. – April 25, 1898
- American Navy under Commodore George Dewey defeated the Spanish Armada led by Patricio Montojo in the famous Battle of Manila Bay. – May 1, 1898
- General Emilio Aguinaldo (in Barrio Alapan) waved the Philippine flag for the first time to celebrate its victory. – May 28, 1898
- Proclamation of Philippine Independence by Emilio Aguinaldo in Kawit, Cavite – June 12, 1898
- The Mock Battle of Manila Bay between Spain and America – August 13, 1898
- The military government was established by the American colonizers under General Wesley Merritt – August 14, 1898
- The Treaty of Paris was signed between Spain and America, ending the Spanish-American War. Spain surrendered the Philippines, Puerto Rico and Guam to the American government for the sum of \$20 million. – December 10, 1898

Period of Suppressed Nationalism

- The Philippines was considered America's Benevolent Assimilation. – December 21, 1898
- Inauguration of the First Philippine Republic or Malolos Republic with Emilio Aguinaldo as president – January 23, 1899
- The outbreak of Filipino – American War (Sta. Mesa, Manila) – February 4, 1899
- General Aguinaldo transferred the capital of the Republic from Malolos to San Fernando, Pampanga. – March 31, 1899
- Antonio Luna was assassinated by General Aguinaldo's guards in Cabanatuan, Nueva Ecija. – June 5, 1899
- General Gregorio del Pilar, the boy general, was killed by the Americans in the Battle of Tirad Pass. – December 2, 1899
- General Emilio Aguinaldo was captured at Palanan, Isabela by General Frederick Funston and his men. – March 23, 1901
- General Antonio Lucban and his brave men attacked the American garrison in Balangiga, Samar. – September 28, 1901
- The first Philippine Assembly was inaugurated at Manila Grand Opera, with Manuel Luis Quezon as the majority floor leader and Sergio Osmeña as the speaker. – October 16, 1907
- The University of the Philippines was established as the first public university. Murray Barlett became its first president, while Ignacio Villamor became its first Filipino president – June 18, 1908
- Francis Burton Harrison was the first American Governor – General of the Philippines. – October 6, 1913
- The signing of Underwood-Simons Tariff Law, allowing an open trade between the Philippines and United States – October 13, 1913
- The creation of bicameral legislatures, the House of Senate and the House of Representatives, as granted by the Jones Law – August 29, 1916
- The founding of the Communist Party of the Philippines (CPP) by Crisanto Evangelista – August 26, 1930
- The Right of Suffrage was granted to Filipinos by Governor Frank Murphy. – December 7, 1933
- The creation of Commonwealth Government as provided by Tydings-McDuffie Law and approved by the US President Franklin D. Roosevelt – March 24, 1934
- The signing of the American-sponsored Constitution of the Philippines by President Franklin D. Roosevelt. The 1934 Constitutional Convention (ConCon) was headed by Claro M. Recto. – March 23, 1935
- The creation of SAKDALISTA Party, a peasant organization against the commonwealth government, by Benigno Ramos. – May 12, 1935
- The Inauguration of Commonwealth Government with Manuel L. Quezon as President and Sergio Osmeña as Vice-President – November 15, 1935

The Japanese Occupation

- The bombing of Pearl Harbor, the US Naval Station in Asia-Pacific – December 8, 1941

- The Declaration of Manila as an Open City by General Douglas MacArthur to prevent the city from suffering further damage brought by Japanese Bombing – December 26, 1941
- The creation of HUKBALAHAP (Hukbong Bayan Laban sa Hapon) under the leadership of Luis Taruc – March 29, 1942
- The Fall of Bataan. Major General Edward P. King, the commander of US forces in Bataan surrendered the fortress to the Japanese forces. The Japanese led their captives on a cruel "Death March" from Bataan to San Fernando, Pampanga. – April 9, 1942
- The Fall of Corregidor Island. The last fortress of defense of the Filipino - American forces led by USAFFE Commander-in-Chief General Jonathan Wainwright surrendered to General Homma, Commander-in-Chief of the Japanese imperial forces – May 6, 1942
- The Inauguration of the Japanese-sponsored puppet Republic with Jose P. Laurel as president – October 14, 1943
- The Leyte Landing. General Douglas MacArthur landed at Red Beach, Palo, Leyte, together with President Sergio Osmeña and General Carlos P. Romulo. – October 20, 1944
- The Battle of Leyte Gulf – October 24 to 26, 1944
- The first atomic bomb was dropped in Hiroshima, Japan. – August 6, 1945
- Three days after, the second atomic bomb was dropped in Nagasaki, Japan. – August 9, 1945
- The Japanese Imperial forces surrendered to the Allied forces in Tokyo Bay aboard by SS Missouri. It ended the war in Asia-Pacific. – September 2, 1945

The Third Republic

- The approval of Philippine Rehabilitation Act by the US Congress, allowing the United States to dispose of, develop, and utilize (exploit) the Philippine natural resources – April 30, 1946
- The inauguration of the Third Republic and the signing of the Treaty of General Relations, which legalized the retention of US bases in the Philippines – July 4, 1946
- The signing of the formal agreement between US and the Philippines, allowing America to lease on 22 sites as US military Bases for 99 years. In return, the Philippines Army accepted US \$100 million worth of army equipment and supplies from the US army. – March 14, 1947
- President Manuel Roxas died of cardiac arrest at Clark Air Base in Pampanga. – April 4, 1947
- Vice-President Elpidio Quirino became the President. – April 4, 1947
- Ramon Magsaysay and Carlos P. Garcia were elected President and Vice- President, respectively. – November 10, 1953
- Luis Taruc, the commander of HUK guerilla surrendered to Benigno "Ninoy" Aquino., the emissary of President Magsaysay. May 17, 1954
- The SEATO (Southeast Asia Treaty Organization) was organized by the 9 countries to prevent spread of communism in Asia. – September 8 -10, 1954
- The Laurel – Langley Agreement was signed replacing the Bell Trade Act. – September 8, 1955
- President Ramon Magsaysay died in a plane crash in Mt. Manunggal, Cebu. – March 17, 1957
- Vice-President Carlos P. Garcia became the President, succeeding President

Magsaysay. – March 17, 1957

- Diosdado Macapagal, the poor boy from Lubao, Pampanga, was elected president. – November 11, 1957
- President Diosdado Macapagal changed the date of Philippine Independence Day from July 4, 1946 to June 12, 1898. – May 12, 1962
- President Diosdado Macapagal approved the Agriculture Land Reform Code, which abolished tenancy. – August 8, 1963
(Agoncillo and Guerrero, The History of the Filipino People)

Contemporary Period

- Congressman Ferdinand Marcos was elected president. – November 9, 1965
- The ASEAN (Association of Southeast Asian Nations) was created in a summit held in Manila. – August 8, 1967
- Gloria Diaz won the Miss Universe Title in Miami Beach, Florida, USA. – July 20, 1968
- The creation of CPP (Communist Party of the Philippines) by Jose Maria Sison – December 26, 1968
- President Ferdinand Marcos was reelected president. – Nov. 11, 1969
- 1971 Constitutional Convention – June 1, 1971
- Plaza Miranda Bombing – August 21, 1971
- Signing of Proclamation 1081, placing the entire Philippines under military rule (Martial Law) – September 21, 1972
- Two days after the signing, Marcos officially announced the Martial Law – September 23, 1972
- The creation of Batasang Pambansa – September 23, 1972
- Marcos lifted the Martial Law. – January 16, 1981
- Marcos won another six-year term. The opposition boycotted the election. – June 16, 1981
- Senator Ninoy Aquino was assassinated at the tarmac of Manila International Airport. – August 21, 1983
- Funeral processions for the burial of Ninoy were joined by millions of people – August 31, 1983
- Snap Election – February 7, 1986
- Marcos was proclaimed by Batasang Pambansa as President in spite of widespread protest on election fraud. – February 16, 1986
- Military mutiny headed by Defense Minister Juan Ponce Enrile and Vice Chief of Staff Fidel Ramos – February 22, 1986
- Archbishop Cardinal Jaime Sin appealed over Radio Veritas to protect military camps where the "rebels" were holed up. – February 23, 1986
- President Marcos fled into exile in Hawaii with the aid of US Air Force while Cory Aquino was sworn in as the first woman president of the Philippines by Senior Associate Justice of the Supreme Court Claudio Teehankee at Club Filipino in San Juan. – February 25, 1986
- The Mendiola Massacre – The Kilusang Magbubukid ng Pilipinas (KMP), a peasant movement, led a march to Malacañang Palace to present their sentiments and demands for a genuine land reforms. As the demonstrators marched across Mendiola Bridge (now Chino Roces Bridge), violence erupted between the peasants and the WPD policemen. – January 22, 1987
- Ratification of the new Constitution through referendum, replacing the National

- Assembly with a bicameral (2 houses) legislature – February 2, 1987
- Unsuccessful Coup d'état led by Lt. Col. Gregorio "Gringo" Honasan against the Cory Aquino Administration – August 28, 1987
- Another unsuccessful Coup d'état against President Aquino, this time by the combined forces of Reform the Armed Forces Movement and the Nationalist Army of the Philippines – December 1, 1987
- MV Doña Paz hit a tanker and sank, the worst sea tragedy in the history, leaving more than 5,000 casualties. – December 20, 1987
- The US military bases were extended for two more years – October 17, 1988
- A 7.6 intensity earthquake occurred in Central Luzon, especially Cabanatuan, Nueva Ecija, Baguio, and Dagupan. – July 16, 1990
- Eruption of Mt. Pinatubo – June 15, 1991
- The Philippine Senate rejected the possible extension of US Military Bases in the Philippines. The "Magnificent 12" senators who voted against the extension of US Military Bases were Senators Salonga, Pimentel, Tañada, Saguisag, Estrada, Ziga, Aquino, Laurel, Mercado, Maceda, Guingona and Enrile – September 16, 1991
- Secretary of Defense Fidel Ramos and Senator Joseph "ERAP" Estrada were elected President and Vice President, respectively. – 1992
- The remains of former president Marcos were brought back to his hometown in Batac, Ilocos Norte. – September 11, 1993
- The Philippines hosted the 4th Asia-Pacific Economic Cooperation (APEC). – November 24-25, 1996
- Vice-President Joseph Ejercito Estrada and Senator Gloria Macapagal-Arroyo were elected President and Vice-President, respectively. – May 11, 1998
- Centennial Celebration of Philippine Independence – June 12, 1998
- President Estrada was impeached by the House of Representatives, headed by speaker Manuel "Manny" Villar. Pres. Estrada was charged of plunder, graft and corruption, bribery, betrayal of public trust, and culpable violation of the Constitution. – November 13, 2000
- EDSA (People Power) II resulted when 11 out of 21 senator-Judge voted against the opening of the second envelope believed to contain important documents against President Estrada. – January 16, 2001
- Vice-President Gloria Macapagal-Arroyo was sworn in as the 14th President of the Philippines. – January 20, 2001
- The Philippine and US governments officially launched the "Balkatan 02-1, code named KALAYAAN-AGUILA 2002." – February 1, 2002
- The signing of Mutual Logistic Support Agreement (MLSA), which facilitates the exchange of logistic support, supplies, and services between the US Pacific Air Command and the Armed Forces of the Philippines for 5 years – November 21, 2002
- Manila hosted the 4th World Meeting of Families. – January 22 to 26, 2003
- President Arroyo signed the Overseas Absentee Voting Act of 2003 (RA 9189). – February 13, 2003
- Strengthening the Anti-Money Laundering Law (RA 9160) through RA 9194 - March 7, 2003
- Oakwood Mutiny, staged by enlisted military personnel known as the Magdalo Group – July 27, 2003
- The signing of RA 9225 (Citizenship Retention and Reacquisition Act), an act related to the citizenship of the Philippines for those who acquire Foreign Citizenship permanent – August 29, 2003

- Former Vice-President Salvador H. Laurel died. – January 28, 2004
- Signing of 9256, declaring August 21 as Ninoy Aquino Day – February 25, 2004
- President Arroyo and Senator Noli De Castro were elected President and Vice President, respectively. – May 11, 2004
- Fernando Poe Jr., one of the most influential and popular candidates for presidency in the May 11 election, died – December 14, 2004
- The Philippines was hit by two super typhoons in November and December (Typhoons Milenyo and Reming) – 2006
- The centennial year of the University of the Philippines – 2008
- Earth's Summit on climate change (Copenhagen, Denmark), – 2009
- National Elections on May 10, 2010
- First automated national elections in the Philippines – May 10, 2010
- Inauguration of President Benigno Simeon Cojuangco Aquino, III – June 30, 2010
- First State of the Nation Address (SONA) of P-NOY – July 26, 2010
- Second SONA of P-Noy – July 25, 2011
- Impeachment of Chief Justice Corona - May 29, 2012
- Pope Francis: Jorge Mario Bergoglio is the 266th and current Pope of the Catholic Church, elected on March 13, 2013. As such, he is Bishop of Rome, the head of the worldwide Catholic Church, and sovereign of the Vatican City State.

THE WORLD OF RIZAL'S TIME

Historical condition

- On the formation of national communities, Italy and Germany succeeded in their unification movements.
- Italy, under the able leadership and guidance of the triad of Count Cavour, Garibaldi, and King Victor Emmanuel, drove the last foreign enemies from Italian soil and unified Italy with Rome as capital city.
- The Prussians, led by their Iron Chancellor, Otto Von Bismarck, defeated the French in the Franco-Prussian War and set up the German Empire.
- Western imperialism, however, gained momentum during the century of Rizal's times.
- Great Britain during the reign of "Queen Victoria" emerged as the leading colonial power with the setting of a global colonial empire.
- The Chinese Manchu Dynasty was forced to cede Hong Kong to the British after their defeat during the First Opium Wars.
- India, Burma, and Malaya were consolidated to British Empire during the 19th century.
- France in 1858-1863, with the help of Filipino troops under Spanish officers, captured Vietnam; annexed Cambodia (1863) and Laos (1893) to French Indo-China.
- 19th century – dynamic period in European history; it was a century of rapid progress and radical changes.
- Social, political, and economic developments occurred, which directly and indirectly affected the Philippines.
- The modernization of living through industrialization and scientific progress

Political Condition

- The political changes in relationship between the government and the people brought about by:
 - Liberal ideas, democracy and the struggle for nationhood, were the landmarks of change during that period.
- By the 18th century, Europe supplied various products like woolen cloth, ship parts and accessories, hardwood, etc.
- The necessity to meet this demand led to the inventions of labor-saving machines.
- The use of machines in the manufacture of goods led to the shift from manual work to machine work, and from the domestic system to the factory system.

Industrialization

- Rapidly gained momentum in the 2nd half of the 19th century, this was accelerated by the discovery of oil and electricity as sources of power and industrial energy.
- The invention of the internal combustion engine heralded the coming of the motorized age.
- Railways and steamships were also constructed.
- The factory system employed thousands of workers, and a new group called capitalists also emerged.
- Poor workers sold their labor to the capitalists.
- Capitalists exploited the workers but with the spread of democracy, the laborers began to demand increased wages, shorter hours of work and other reforms.
- Competition for markets became stiff.

Scientific Developments

- With industrialization came scientific progress. These two developments almost rose at the same time for technology and science depended on each other.
- Factories multiplied in number.
- Vast quantities of coal and other minerals were needed to fill the requirements of such expansion.

Socialism

- There was a belief that government should own and control the means of production for the benefit of the many and not only for a few individuals, such as capitalists.
- Karl Marx in 1848 declared that only a violent revolution can improve the lives of the proletariat (workers).
- He espoused the abolition of private ownership of land and the centralization of economic planning.
- The workers, as a result of Marxist agitation, became more aware of their role and power in the economic system of their country.

Democracy

- The most significant achievements in the 19th century are the growing of European appreciation for the worth and dignity of individual and his right to participate in government.
- American President Abraham Lincoln defined democracy in his Gettysburg Address as "the government of the people, by the people, and for the people."



Spain during Rizal's time

- The 19th century was the twilight period of Spain as colonial power.

Why twilight?

- Spain lost one by one her rich colonies in South and Central America.
- She (Spain) lost Paraguay in 1811
 - ...Argentina in 1816
 - ...Chile in 1818
 - ...Colombia in 1819
 - ...Ecuador in 1822
 - ...Venezuela in 1821
 - ...Peru in 1821
 - ...Bolivia in 1825

The Philippines during Rizal's time

Social system

- Filipinos had their own culture, government, and religion.
- The Spanish colonizers forced them to accept foreign culture and Catholicism, which they never completely understood.

Limpieza de Sangre (purity of blood)

- Creating social ranking among various groups
- The social structure was pyramidal in form, with the Spanish Peninsulares occupying the APEX, while the INDIOS or natives were placed at the base of the pyramid.

Political system

- King
- Ministro De Ultramar (Ministry of Colonies)
- Gobernador-General (National government)
- Alcalde mayores/alcaldia (Provincial)
- Cabildo/Ayuntamiento (City)
- Gobernadorcillo (Town)
- Cabeza de Barangay (Barangay)

Frallocracia

"Rule of the friars"

Education

- Royal Decree of 1863 – Establishment of schools for elementary pupils
- Royal Decree of 1865 – Establishment of schools for secondary students
- Moret Decree of 1870 – Intended to secularize higher education in the Philippines; The friars vigorously OPPOSED that idea of the government controlling the educational system.

ANCESTRY

- Rizal was of mixed racial origin. He descended from a hard-working and intelligent Chinese merchant, Domingo Lam-co, who married Inez de la Costa, a Chinese mestiza from the Chinese community in Manila. The family migrated to Biñan and became tenants of the Dominican friars.
- Lam-co's son, Francisco, who was to be Rizal's great grandfather, was appointed Captain of Biñan in 1783. The family adopted the surname Mercado, complying with the Claveria Decree, which mandated that all Filipinos use Spanish surnames.
- Francisco Mercado and his wife, Bernarda Monicha, had two children: Juan and Clemente. Juan married Cirila Alejandra, also a Chinese mestiza. They bore 14 children, one of whom was Francisco, Jose's father.

MATERNAL

- The family of Teodora Alonzo, Jose's mother, was more prominent than that of the Mercado's. The Alonzos had engineers, lawyers, priests and government officials.

JOSE'S IMMEDIATE FAMILY

- Francisco Mercado studied Latin and Philosophy at the College of San Jose in Manila. Jose described him as "a model of fathers."
- Jose inherited from his father his sense of dignity, self-respect, and seriousness.
- Doña Teodora Alonzo was born in Santa Manila. She was a talented and remarkable woman. Jose admired and adored her.

"My mother is not a woman of ordinary culture. She knows literature and speaks Spanish better than I do...She even corrected my poems and gave me wise advice when I was studying rhetoric...she is a mathematician and she reads many books..."

Pepe's siblings:

- Saturnina
- Paciano
- Narcisa
- Olimpia
- Lucia
- Maria
- Jose
- Concepcion
- Josefa
- Trinidad
- Soledad

LOVE OF COUNTRY

Rizal's first essay "The Love of Country" was written when he was 21. It was published first in *Diariong Tagalog*, a nationalist newspaper, on August 20, 1882, under the name of Laong Laan. It was subsequently published in *La Solidaridad* on October 31, 1890.

TO THE YOUNG WOMEN OF MALOLOS

In the letter dated February 1889, Rizal congratulated the women of Malolos for their courage in petitioning Governor-General Valerino Weyler, that they should be permitted to

learn Spanish lessons. He was glad that at last, some Filipino women were asserting their rights to education.

SOBRE LA INDOLENCIA DE LOS FILIPINOS (THE INDOLENCE OF THE FILIPINOS)
Rizal's longest essay was published in five installments in the *La Solidaridad* from July 15 to September 15, 1890.

FILIPINAS DENTRO DE CIEN AÑOS (The Philippine a Century Hence)
This essay came out as a series of articles in *La Solidaridad* on 30 Sept. 1889, 31 October 1889, 15 December 1889, and 1 February 1890. Rizal envisioned the future for the Philippines in the essay. What will become of the Philippines within a century? Would it become independent? Will it become a province of Spain or become a colony of other nations?

BRINDIS

This was Rizal's memorable speech honoring the painters Juan Luna and Felix Resurreccion Hidalgo at the commemorative banquet on June 25, 1884. Luna won first prize for his painting "Spolarium", while Hidalgo won a silver medal for his "Virgins Exposed to the Populace" at the art exhibit in Madrid.

(Santa Romana and Romero, Rizal and the Development of National Consciousness)
(Maguigad and Muhi, Rizal...)

POLITICAL SCIENCE AND OTHER RELATED LAWS IN EDUCATION

Reduced to its simplest terms, political science is the systematic study of the state and government. The word "political" is derived from the Greek word polis, meaning a city. The word "science" comes from the Latin word scire, "to know."

- **Political Theory** – the entire body of doctrines relating to the origin, form, behavior, and purposes of the state is dealt with in the study of political theory.
- **Public Law** – The (a) organization of governments, (b) the limitations upon government authority, (c) the powers and duties of governmental offices and officers, and (d) the obligations of one state to another are tackled in the study of public law.
- **Public Administration** – In the study of public administration, attention is focused upon the methods and techniques used in the actual management of state affairs by the executive, legislative, and judicial branches of government.
- **Political Science and History** – "History is past politics, and politics present history."
- **Political Science and Economics** – Economic conditions affect the organization, development, and activities of states, which in turn modify or even prescribe economic conditions.
- **Political Science and Geography** – Geopolitics is a science dealt with the study of the influences of physical factors, such as population pressures, sources of raw materials, and geography.
- **Political Science and Sociology & Anthropology** – Knowing the origins and nature of governmental authority
- **Political Science and Psychology** – The political scientist as well as the psychologist promotes studies of the mental and emotional processes motivating the political behavior of individuals and groups.
- **Political Science and Philosophy** – The concepts and doctrines of Plato, Aristotle, and Locke are important to the specialist in academic philosophy and also to the political scientist.

State – a community of persons more or less numerous, permanently occupying a definite portion of territory, having a government of their own to which the great body of inhabitants renders obedience, and enjoying freedom from external control

ELEMENTS

- **People** – This refers to the mass of population living within the state. Without people, there can be no functionaries to govern and subjects to be governed.
- **Territory** – It includes not only the land over which the jurisdiction of the state extends, but also the rivers and lakes therein, a certain area of the sea which touches upon its coasts and the air space above it.
- **Government** – It refers to the body through which the will of the state is formulated, expressed, and carried out.
- **Sovereignty** – to have freedom from foreign control; the supreme power of the state to command and enforce obedience to its will from the people within its jurisdiction and corollarily

ORIGINS

- **Divine Right theory** – It holds that the state is of divine creation, and the ruler is ordained by God to govern the people.
- **Necessity or Force theory** – It maintains that states must have been created through force.
- **Paternalistic theory** – It attributes the origin of states to the enlargement of the family which remained under the authority of the father or mother.

Preamble

- Preambular – “to walk before”
- Sets down origin and purposes of the Constitution
- Tells us the authors of the Constitution and for whom it has been promulgated
- States the general purposes
- May serve as an aid in its interpretation

Sources

- The Filipino people
- Sovereign people

Purposes and Aims

- To build a just and humane society
- To establish a Government that shall:
 - embody our ideals and aspirations
 - promote the common good
 - conserve and develop our patrimony
 - secure to ourselves and our posterity the blessings of independence and democracy under the rule of law and a regime of truth, justice, freedom, love, equality, and peace

LEGISLATIVE DEPARTMENT

- **Legislative power** – is essentially the authority under the Constitution to make laws and subsequently, when the need arises, to alter and/or repeal them.
- **Laws** – refer to statutes which are the written enactment of the legislature governing the relations of the people among themselves or between them and the government and its agencies.

Advantages of Bicameralism

- A second chamber/house (Senate) is necessary to serve as a check to hasty and ill-considered legislation.
- Serves as a training ground for future leaders
- Provides a representation for both regional and national interests
- Less susceptible to bribery and control of big interest

Disadvantages

- The bicameral set-up has not worked out as an effective fiscalizing machinery.
- Although it affords a double consideration of bills, it is no assurance of better-considered and better-deliberated legislation.
- It produces duplication of efforts and serious deadlocks in the enactment of important measures with the Conference Committee of both Houses.
- All things being equal, it is more expensive to maintain than a unicameral legislature.
- The prohibitive costs of senatorial elections have made it possible for only wealthy individuals to make it to the Senate; and as to the claim that a Senate is needed to provide a training ground for future leaders.

EXECUTIVE DEPARTMENT

- **Executive Power** – the power to administer the laws, which means carrying them into practical operation and enforcing their due observance

Suspension of sentence – postponement of a sentence for an indefinite time

Commutation – the reduction of the sentence imposed to a lesser punishment, as from death to life imprisonment. It may be granted without the acceptance and even against the will of the convict.

Pardon – an act of grace proceeding from the power entrusted with the one in-charge of the execution of the laws (President of the Philippines) which exempts the individual on whom it is bestowed, from the punishment the law inflicts for a crime he has committed.

Object of Pardoning Power

Executive clemency exists to afford relief from undue harshness or evident mistake in the operation or enforcement of the criminal law.

Kinds of Pardon

- **Absolute** – when it is NOT subject to any condition whatsoever
- **Conditional** – When it is given subject to any condition or qualification the President may see fit
- **Remission** – prevents the collection of fines or the confiscation of forfeited property

- **Amnesty** – an act of the sovereign power granting oblivion or a general pardon for a past offense usually granted in favor of certain classes of persons who have committed crimes of a political character, such as treason, sedition, or rebellion

JUDICIAL DEPARTMENT

- **Judicial Power** – the power to apply the laws to contests or disputes concerning legally recognized rights or duties between the State and private persons, or between individual litigants in cases properly brought before the judicial tribunals.

Systems of Education

- Education service contracting scheme
- Five-year literacy program
- Reduction of entrance age
- Values education framework
- Parents education program
- Educational development projects
- National Secondary Assessment Test / National Elementary Admission Test

National Diagnostic Examination

- Study Now- Pay Later Plan
- Accelerated Learning Program for Elementary Schools
- Secondary Education Development Program
- Family Planning
- Program for Decentralized Educational Development
- Youth Entrepreneurship and Cooperatives in School
- Effective and Affordable Secondary Education
- Non-Formal Education Accreditation and Equivalency

Compulsory elementary education for all children of school age and the natural right and duty of Filipino parents to rear their children

• Establish and maintain a system of free public education in the elementary and high school levels. Without limiting the natural right of parents to rear their children, elementary education is compulsory for all children of school age."

- Section 2(2), Article XIV, 1987 Constitution

Indigenous Community

• The State recognizes and promotes the rights of indigenous cultural communities within the framework of national unity and development."

- Section 22, Article II, 1987 Constitution

TAXATION

Taxation is the act of laying a tax, i.e., the process or means by which the sovereign, through its law-making body, raises revenues to defray the necessary expenses of government. Expressed in another way, it is a method of apportioning the cost of government among those who in some measure are privileged to enjoy its benefits and must, therefore, bear its burdens. (Hector De Leon quoting Am.Jur.34; 1, Cooley 72 -73)

Importance and purpose of taxation

The purpose of taxation on the part of the government is to provide funds with which to promote the general welfare and protection of its citizens, and to enable it to finance its multifarious activities (Hector De Leon quoting Am.Jur. 34 -35)

TAXES

Taxes are the enforced proportional contributions from persons and property levied by the law-making body of the state by virtue of its sovereignty for the support of the government and all public needs (Hector de Leon quoting Cooley).

Characteristics

1. **It is an enforced contribution.** – A tax is NOT a voluntary payment or donation, and its imposition is in NO way dependent upon the will or assent of the persons taxed (Hector De Leon quoting AM. Jur. 38 -39)
2. **It is generally payable in money.** – Unless qualified by law, the terms "taxes" or "tax" are usually understood to be a pecuniary burden – an exaction to be discharged alone in money, which must be in legal tender.
3. **It is proportionate in character.** – A tax is laid by some rule of apportionment according to which the persons or property share the public burden. It is ordinarily based on the ability to pay.
4. **It is levied on persons or property.** – A tax be imposed on acts or transactions or contracts. In each case, however, it is only a person who pays the tax.
5. **It is levied by the state which has jurisdiction over the person or property.** – The persons and property must be subject to the jurisdiction of the taxing state. This is necessary in order that the tax can be enforced. The taxing power of a state necessarily stops at its boundary lines.
6. **It is levied by the law-making body of the State.** – The power to tax is a legislative power which only the legislative body (national or local) can exercise through the enactment of statutes or ordinances. The power to tax is also granted by the Constitution and by law to local government units (provinces, cities, municipalities and barangays), but the power of the latter is subject to such limitations as may be provided by law.
7. **It is levied for public purposes.** – Taxation involves and a tax constitutes, a charge or burden imposed to provide income for public purposes – the support of the government, the administration of the law, or the payment of public expenses.

ECONOMICS

Economics is a social science which deals with the efficient allocation of scarce resources to satisfy the unlimited human needs and wants.

Divisions of Economics

1. **Macroeconomics** – a branch of economics which deals with study of the economic activity as a whole, such as inflation, unemployment, business cycle, etc.
2. **Microeconomics** – a branch of economics which deals with the study of the individual behavior of the economy

Major Economic Resources

1. **Land** – refers to all natural resources needed to produce goods and services
2. **Labor** – refers to physical and mental efforts of men to produce goods and services
3. **Capital** – These are man-made goods used to produce another good
4. **Entrepreneur** – a person responsible to combine all the resources to produce goods and services

Major Economic Problems

1. What to produce?
2. How to produce?
3. For whom to produce?

Economic System

1. Traditional Economies
2. Command Economies
3. Market Economies

Demand – the ability and willingness of people to buy a given alternative prices for a common period of times

Supply – the ability and willingness of the people to produce goods and services at different prices in specified period of time

Market Equilibrium – point where demand is equal to supply

Surplus – situation when supply is greater than demand

Shortage – situation when demand is greater than supply

Unemployment – is when many of the labor force are willing and able to work but cannot find work.

Inflation – general increase in price level

AGRARIAN REFORM

Land reform refers to the full range of measures that may or should be taken to improve or remedy the defects in the relations among men (e.g., between the tiller and owner of land, employee and employer in a farm) with respect to their rights in land (Raup, The Contribution of Land Reforms to Agricultural Development, page 1).

Land reform thus involves the "transformation of agrarian structure" or what are sometimes called "structural reforms" (Progress in Land Reform, p.30, a UN publication).

MEANING OF AGRARIAN REFORM

The term comprises NOT only land reform (i.e. the reform of tenure, production and supporting services structures) but also the reform and development of complementary institutional framework, such as the administrative agencies of the national government, rural, educational and social welfare institutions, cooperatives, etc., and not limited simply to the question of the relationships of the farmers to the land.

In the Comprehensive Agrarian Reform Law of 1988 (R. A. # 6657), agrarian reform is defined to mean "the redistribution of land, regardless of crops or fruits produced, to farmers and regular farm workers who are landless, irrespective of tenurial arrangement, to include the totality of factors and support services designed to lift the economic status of the beneficiaries and all other arrangements alternative to the physical redistribution of lands, such as production or profit-sharing, labor administration, and the distribution of shares of stock, which will allow beneficiaries to receive a just share of the fruits of the lands they work." (Sec.3 (a) thereof).

In short, agrarian reform means solving the agrarian problem and thereby elevating the quality of the rural populace and making them active participants in the economic, social, and political affairs of their locality.

EXAMPLES OF AGRARIAN REFORM MEASURES

Agrarian reform, therefore, also covers the following:

- Public health programs;
- Family planning;
- Education and training of farmers;
- Reorganization of agrarian reform agencies;
- Application of labor laws to agriculture workers;
- Construction of infrastructure and facilities, such as feeder roads, irrigation systems, etc., and the establishment of rural electrification;
- Organization of various types of voluntary associations (farmers' associations, cooperatives, youth organization, etc.) as a means of insuring popular support or overcoming opposition to the reform;
- Providing employment opportunities to underemployed or surplus rural labor (through the development of cottage and small and medium - scale industries); and
- Other services of a community development nature.

COMPONENTS OF AGRARIAN REFORM

Agrarian reform has at least two (2) components.

- (1) **Land distribution** – The most important component of agrarian reform is land redistribution, which is the most effective way of providing security of tenure to tillers of the soil. This is the meat of agrarian reform – to make the tillers of the land the owners of the land they are tilling and to insure their right to a just share in the fruits of their labor.
- (2) **Companion measures** – It is NOT enough that the tillers become the owners of the land they till. They must also become successful owners of the land. In other words, after the proper redistribution of the land and after transferring ownership to the users of the land, the other problem must now be tackled and solved, namely: the increased maximized and modernized utilization of the land for the benefit of NOT only the new owner but that of society. This will mean the creation and development of new social institutions that can effectively carry out the other companion measures of agrarian reform. The farmers need to be provided with the necessary skills and know-how to enhance their productivity.

COMPANION MEASURES TO LAND DISTRIBUTION

- (1) **Credit** – The first companion measure required for a successful land redistribution program is the extension of credit to the tiller-owners on easy terms. When the farmer become's the owner of the land he tills, he has to shoulder the whole capitalization of his farming operations. He has to provide for the seeds, the transplanting, the fertilizers, his family's subsistence until the crop is harvested, etc.
- (2) **Modern and better methods of production** – This is the second most needed companion measure for a successful land redistribution program. There is need for improved soil fertility, etc. In this regard, government help is needed, but government efforts in technological matters will not succeed without the cooperation of the farmers. This principle applies in the case of agricultural production. The agricultural scientist or technician will have to depend not only on the information to be given by the farmers with respect to their farms and the results of the agricultural methods in them, but also on the free acts of the farmers themselves, which are essential for the implementation of many agricultural projects for mass benefit.
Therefore, the cooperation of the farmers is essential. They should participate in every process of experimentation and formulation of new methods. In this way, there will be no need to convince the farmers of methods later on because they participated in their formulation. However, all these can only be done if the farmers were organized.
- (3) **Marketing facilities, equitably pricing, Infrastructure** – Another important companion measure for a successful land redistribution program is the provision for marketing facilities and equitable price for the farmers' produce and the accompanying infrastructure requirements needed to transport and the produce from the barrio to the town or city.
- (4) **Cooperatives** – It is evident that all these companion measures cannot be done without the organized and conscious efforts of the farmers themselves. Thus, there is a necessity for cooperatives and other forms of social organizations. These cooperatives can, in turn, be the most effective channels for proper agricultural planning for both the micro and macro levels, to ensure food sufficiency in terms of adequate supply and proper distribution. Cooperatives and other forms of social organization can also serve as a vehicle for information and the effective molding of a responsive public opinion (Instructor's Manual, note 1 at pp 60-61).

CONSTITUTIONAL PROVISIONS RELEVANT TO AGRARIAN REFORM

The provisions of the 1987 Constitution relevant to agrarian reform are NOT confined in one article. They cannot be found solitary and alone considering that agrarian reform covers all aspects of institutional development, including tenure, production, and supporting services and structures, and related institutions such as local government units, public administration in rural areas, rural education and rural social welfare institutions, farmers' organizations, law enforcement agencies, etc. However, only the more pertinent ones will be considered (De Leon, Agrarian Reform and Taxation, 2000).

Article II, Sec. 9, 1987 Constitution – "The State shall promote a just and dynamic social order that will ensure the prosperity and independence of the nation and free the people from poverty through policies that provide adequate social services, promote full employment, a rising standard of living, and an improved quality of life for all."

Article II, Sec. 19, 1987 Constitution – "The State shall develop a self-reliant and independent national economy effectively controlled by Filipinos."

Section nineteen (19) states the guidelines in the development of the national economy: economic self-reliance, independent national economy, and effective Filipino control of the national economy.

Article II, Sec. 21, 1987 Constitution – "The State shall promote a comprehensive rural development and agrarian reform."

Article III, Sec. 4, 1987 Constitution – "NO LAW shall be passed abridging the freedom of speech, of expression, or of the press, or the right of the people peaceably to assemble and petition the government for redress of grievances."

Article III, Sec. 8, 1987 Constitution – "The right of the people, including those employed in the public and private sectors, to form unions, associations, or societies for purposes not contrary to law shall not be abridged."

Article III, Sec. 9, 1987 Constitution – "Private property shall NOT be taken for public use without JUST compensation."

Article XII, Sec. 1, 1987 Constitution – "The goals of the national economy are a more equitable distribution of opportunities, income, and wealth; a sustained increase in the amount of goods and services produced by the nation for the benefit of the people; and an expanding productivity as the key to raising the quality of life for all, especially the underprivileged."

The State shall promote industrialization and full employment based on sound agricultural development and agrarian reform, through industries that make full and efficient use of human and natural resources and which are competitive in both domestic and foreign markets. However, The State shall protect Filipino enterprises against unfair foreign competition and trade practices.

In the pursuit of these goals, all sectors of the economy and all regions of the country shall be given optimum opportunity to develop. Private enterprises, including corporations, cooperatives and similar collective organizations shall be encouraged to broaden the base of their ownership."

Article XII, Sec. 2, 1987 Constitution – "All lands of the public domain, waters, minerals, coal, petroleum, and other mineral oils, all forces of potential energy, fisheries, forests or timber, wildlife, flora and fauna, and other natural resources are owned by the State. With the exception of agricultural lands, all other natural resources shall not be alienated. The exploration, development, and utilization of natural resources shall be under the full control and supervision of the State."

Lands of the public domain are classified into:

- Agricultural
- Forest or timber
- Mineral Lands
- National parks

Adoption of Filipino First Policy

Article XII, Sec. 12, 1987 Constitution – "The State shall promote the preferential use of Filipino labor, domestic materials and locally produced goods, and adopt measures that help make them competitive."

AGENCY TO PROMOTE VIABILITY AND GROWTH OF COOPERATIVES

Article XII, Sec. 15, 1987 Constitution – "The Congress shall create an agency to promote the viability and growth of cooperatives as instruments for social justice and economic development."

The Constitution gives proper recognition to the important role that Cooperatives can play as "instruments for social justice and economic development." R.A. 6939 creates the Cooperatives Development Authority under the Office of the President to carry out the provisions of the Act.

Article XIII, Sec. 1, 1987 Constitution – "The Congress shall give highest priority to the enactment of measures that protect and enhance the right of all the people to human dignity, reduce social, economic, and political inequalities, and remove cultural inequalities by equitably diffusing wealth and political power for the common good."

"The State shall regulate the acquisition, ownership, use, and disposition of property and its increments."

Article XIII, Sec. 2, 1987 Constitution – "The promotion of social justice shall include the commitment to create economic opportunities based on freedom of initiative and self-reliance."

Article XIII, Sec. 3, 1987 Constitution – "The State shall regulate the relations between workers and employers, recognizing the right of labor to its just share in the fruits of production and the right of enterprises to reasonable returns on investments, and to expansion and growth."

Article XIII, Sec. 4, 1987 Constitution – "The State shall, by law, undertake an agrarian reform program founded on the right of farmers and regular farmworkers, who are **LANDLESS**, to own directly or collectively the lands they TILL or in the case of other farmworkers, to receive a just share of the fruits thereof. To this end, the State shall encourage and undertake the just distribution of all agricultural lands, subject to such priorities and reasonable retention limits as the Congress may prescribe, taking into account ecological, development, or equity considerations, and subject to the payment of **JUST COMPENSATION**. In determining retention limits the State shall respect the right of small landowner. The State shall further provide incentives for voluntary land-sharing."

THE CODE OF AGRARIAN REFORMS (R.A. 3844)

Beneficiaries under the Code

The Code of Agrarian Reforms of the Philippines was enacted specially to reach four underprivileged groups marking up the majority of the agricultural population, each of which represents a corresponding group of problems.

- **Tenant farmers** – The share crop or "kasama" system, a carry-over from the Spanish colonizers, which deprives the farmer of a just share of the produce, makes his tenure insecure; exposes him to usurious practices and to perpetual

indebtedness; and forces him to be idle after a few months spent on the farm. Under the Code, share tenancy is automatically converted to leasehold and landless farmers are given a chance to own the lands they till;

- **Agricultural wage-earners or farm workers** – The helpless position of these workers makes them unorganized and unprotected by laws on labor standards. Under the Code, they are given, in general, all the rights and opportunities now enjoyed by industrial workers;
- **Settlers including migrant workers** – The ineffective and haphazard settlement system which prevented those who moved from highly tenanted areas from taking up productive lives in public agricultural lands. Under the Code, they are given help in acquiring titles to public lands and solving their problems; and
- **Owner-cultivators of less than family-size farms** – The lack of credit, irrigation, marketing and information facilities among them, aggravated their lack of organization. Under the Code, they are to be given opportunities to purchase additional lands.

(Pres. Diosdado Macapagal's Message to Congress, March 19, 1963).

LANDS COVERED BY THE CODE

1. **Tenanted areas** – They are areas worked on by the tenant-farmer either under the shares cropping or lease hold systems. Under Presidential decree No. 27, only private agricultural lands primarily devoted to rice or corn are covered
2. **Landed estates** – They are private agricultural lands acquired by the Department of Agrarian reform through the Land Bank for redistribution to the tent-tillers both under the Code and previous laws, under the scheme of family-size farms.
3. **Old settlements** – They are the settlement projects developed by the Department of Agrarian Reform from the public domain and are still under its administration and management.
4. **Proposed settlements** – They are of the public domain either marked for settlement purposes or have potentials for such purposes. Landless farmers and former dissidents who have NOT acquired lands from tenanted agricultural lands are allocated farms lots in resettlement projects. In these areas, farms families are resettled in an orderly manner and their agricultural activities are initially supported by the government.

COOPERATIVE CODE OF THE PHILIPPINES (Republic Act # 6938)

The Cooperatives Development Authority (CDA), under the Office of the President, created by virtue of R.A. # 6939, is the new agency now charged with the authority and responsibility for implementing the program of cooperatives development. Since functions in connection with the development of cooperatives are centralized under the CDA, agrarian reform has to be actively supported by said agency in terms of services towards the organization of cooperatives. This agency creation is in pursuance of a conditional mandate:

Article XII, Sec. 15, 1987 Constitution – "Create an agency to promote the viability and growth of cooperatives as instruments for social justice and economic development."

AGRARIAN REFORM PROGRAM AND COOPERATIVES DEVELOPMENT PROGRAM COMPARED

1. **Objectives of agrarian reform** – The Agrarian Reform Program breaks up the ownership of the land and apportions this ownership among tenant-farmers who are tilling the soil.

2. **Objectives of cooperatives development** – The Cooperative Development Program seeks to UNITE interests, talents, and resources in special kinds of organization whereby our farmers can perform better not only but also farmers and citizens. They, therefore, become better human beings.
3. **Cooperatives must complement agrarian reform.** – The act of breaking up must, therefore, be accomplished by the act of consolidating. Agrarian reform must be accompanied by the development of cooperatives. The redistribution of land ownership in order to give farmers a more dignified existence must be followed with measures to consolidate the management of land, so that our farmers can increase productivity and efficiency at the same time. The breaking up of outmoded, paternalistic, and oppressive relationship between the landlords and tenants must be complemented by the creation of new relationship based on mutual assistance and equality – such as the loyalties of farmers to their organizations or to the good for their community. (De Leon, Agrarian Reform and Taxation, 2000).

SPECIFIC OBJECTIVES OF THE PROGRAM

- Prepare the farmers to become better producers;
- Enforce timely payment of land amortization;
- Enforce savings among farmers;
- Encourage farmers to perform economic activities collectively; and
- Develop a cooperative system, a unified and properly operating set of cooperative institutions and supportive agencies (Clemente Terso, Jr., "The Cooperative Development Program of the DLGCD, "in Seminar Papers on Agrarian Reform, published by the Agrarian Reform Institute, U.P., p 85.)

Meaning of Cooperative

A cooperative is a duly registered association of persons, with a common bond of interest, who have voluntarily joined together to achieve a lawful common social or economic end making equitable contributions to the capital requirements and accepting a fair share of the risks and benefits of the undertaking in accordance with universally accepted cooperative principle (Article 3, R.A. # 6938).

COOPERATIVE PRINCIPLES

- **Open and voluntary** – Membership in a cooperative shall be voluntary and available to all individual regardless of their social, political, racial or religious background or beliefs.
- **Democratic control** – "Create an agency to promote the viability and growth of cooperatives as instruments for social justice and economic development." Cooperatives are democratic organizations. Their affairs shall be administered by persons elected or appointed in a manner agreed upon by the members. Members of primary cooperatives shall have equal voting rights on a one-member, one-vote principle. However, in the case of secondary and tertiary cooperatives, the provisions of Article 37 of the Cooperative Code shall apply (Article 37 – Voting System).
- **Limited interest in capital** – Share capital shall receive a strictly limited rate of interest.
- **Division of NET surplus** – Net surplus arising out of the operations of a cooperative belong to its members and shall be equitably distributed for cooperative development, common services, indivisible reserve fund, and for limited interest on capital and / or patronage refund in the manner provided in the code and in the articles of Cooperation or BY-LAWS.

- **Cooperative Education** – All cooperatives shall make provisions for the education of their members, officers and employees and of the general public based on the principles of cooperation; and
- **Cooperation among cooperatives** – All cooperatives, in order to best serve the interest of their members and communities, shall actively cooperate with other cooperatives at local, national, and international levels (Article 4, R. A. 6938).

ORGANIZATION OF COOPERATIVES

A cooperative may be organized and registered by at least fifteen (15) persons for any or all of the following:

1. To encourage thrift and savings mobilization among the members;
2. To generate funds and extend credit to the members for productive and provident purposes;
3. To encourage among members, systemic production and marketing;
4. To provide goods and services and other requirements to the members;
5. To develop expertise and skills among its members;
6. To acquire lands and provide housing benefits for the members;
7. To insure against losses of the members;
8. To promote and advance the economic, social and educational status of the members;
9. To establish, own, lease or operate cooperative banks, cooperative wholesale and retail complexes, insurance and agricultural/industrial processing enterprises, and public markets;
10. To coordinate and facilitate the activities of cooperatives; and
11. To undertake any all other activities for effective and efficient implementation of the provisions of the Code. (Article 6, R.A. 6938).

Objectives of Cooperatives

1. The primary objective of every cooperative is to provide goods and services to its members and thus enable them to attain increased income and savings, investment, productivity and purchasing power and promote among them equitable distribution of net surplus through maximum utilization of economies of scale, cost-sharing and risk-sharing without, however, conducting the affairs of the cooperative charitable purposes.
2. A cooperative shall provide maximum economic benefits to its members, teach them efficient ways of doing things in a cooperative manner and propagate cooperative practices and new ideas in business and management and allow the lower income groups to increase their ownership on the wealth of the nations (Article 7, R.A. 6938).

Privileges of cooperatives

1. Cooperatives enjoy the privilege of depositing their sealed cash boxes or containers, documents or any valuable papers in the safes of the municipal or city treasurers and other government offices free of charge.
2. Cooperatives organized among government employees, notwithstanding any law or regulation to the contrary, enjoy the free use of any available space in their agency whether owned or rented by the government.
3. In areas where appropriate cooperatives exist, the preferential right to supply government institutions and agencies, rice, corn and other grains, fish and other marine products, meat, eggs, milk, vegetables, tobacco and other agricultural commodities produced by their members is granted to the cooperatives by the appropriate government agencies.

4. Preferential treatment in the allocation of fertilizers and in rice distribution is granted to cooperatives by the appropriate government agencies.
5. Cooperatives and their federations, such as market vendor cooperatives, have preferential rights in management of public markets and or/ lease of public market facilities stall or spaces.
6. Credit cooperatives and/or federations are entitled to loans, credit lines, rediscounting of their loan notes, and other eligible papers with the Development Bank of the Philippines, The Philippine National Bank, the Land Bank of the Philippines, and other financial institutions, except the Central Bank of the Philippines.

THE COMPREHENSIVE AGRARIAN REFORM LAW: LEGAL BASIS OF CARP (An Excerpt)

The Comprehensive Agrarian Reform Program was a response to the people's clamor and expectations of a more effective land reform program that will supposedly correct the many flaws that plagued the previous land reform programs.

Republic Act 6657, signed into law on June 10, 1988 by President Corazon Aquino, known as the Comprehensive Agrarian Reform Law of 1988 (CARL), is an act instituting a Comprehensive Agrarian Reform Program to promote social justice and industrialization, providing the mechanism for its implementation, and for other purposes.

The law's major features are the following:

- It provides for the coverage of all agricultural lands regardless of crops produced or tenurial status of the tiller.
- It recognizes as beneficiaries of the program all workers in the land given that they are landless and willing to till the land.
- It provides for the delivery of support services to program beneficiaries.
- It provides for arrangements that ensure the tenurial security of farmers and farmworkers such as the leasehold arrangement, stock distribution option and production and profit sharing.
- It creates an adjudication body that will resolve agrarian disputes.

Land Reform during the Cory Aquino Administration

The predecessor of CARP was the Accelerated Land Reform Program, in PD 27, which imposed a ceiling of seven hectares for all croplands, the distribution of large privately-owned farms, rice and corn lands, small farms, alienable as well as disposable lands, exempting areas such as ancestral tribal lands and those that are used for public service. Other features of the program include tenancy regulation and voluntary land sharing and corporate stock sharing as alternative schemes to land reform. It may have contrasting features compared to past land reforms; however it still had flaws in it. Hence, the Corazon Aquino government drafted Executive Order No. 229, which focused on the administrative procedures and not on the substance of an agrarian reform measure. It detailed the mechanics of land registration, private land acquisition and the compensation procedures to land owners.

It also specified the composition and functions of the governing entities, which will coordinate and supervise the implementation of the program. The land reform issues such as the retention limit and priority areas were left for the Congress to define. Both Houses produced their own agrarian bills. These two bills later mirrored the contrasting interests of both Houses. While the landlord-dominated Lower House reflected the interests of

landowners, the urban-based Senators emphasized the need for a land reform to attain economic development. These disputes and debates, and the fact that there is diversity among the landowners themselves when it comes to land reform measures, have paved the way for them to compromise. It is within this context that the CARL was put into law.

- The CARP is so far the most comprehensive agrarian reform program ever formulated. Unlike PD 27, which included only rice and corn lands, CARP covers all private and public agricultural lands regardless of commodity produced and tenurial status of the tiller, including other lands of the public domain suitable for agriculture.

CARP recognizes as beneficiaries of the agrarian reform program not only farmers but all workers in the land given that they are landless and willing to cultivate.

Land Reform during the Ramos Administration

The agrarian reform policies of the Ramos administration focused on accelerating the direct land transfer and non-land transfer programs through adopting more rational and simpler operating procedures and a fair, expeditious and inexpensive settlement of agrarian disputes. It focused in the adoption of a fair land valuation formula and prompt payment of just compensation to encourage landowners to cooperate and support agrarian reform.

The administration also encouraged the development of alternative schemes of landowner compensation to motivate them to invest in rural-based industries that have strong linkages with agriculture. It also adopted a progressive agricultural land tax to encourage smaller landholdings among large landowners, a land conversion tax to discourage land conversion, and idle land tax to encourage landowners to cultivate the land. These taxes were also needed to augment the Agrarian Reform Fund aside from mobilizing both local and foreign resources.

The administration also pursued for the amendment of Section 63 of the CARL making the ARF a revolving fund and increasing the fund to P100 Billion. It also planned to increase the composition of the DAR's Adjudication Board's full-time members from three to nine and upgrading their salaries. The budget of DAR therefore had to be increased to cover reorganization costs. The protection of ARBs whose lands were converted to commercial, industrial or residential use by making them shareholders or co-investors of the industrial/commercial venture was also one of CARP's major agenda.

Also, the CARP bureaucracy had to be motivated further for more successful results and its partnership at the provincial level with various government and non-government organizations, local government units, farmer-beneficiaries, landowners, legislature, media and the academe has to be enhanced.

The Ramos administration set a target of 3.4 million hectares of land to be distributed to farmer-beneficiaries in which it was able to accomplish 2.6 million hectares or 33.3 percent of the total CARP scope of 7.8 million hectares. It has brought the total accomplishment for land acquisition and distribution at the end of June 1998 to 4.7 million hectares or 60 percent of the scope.

Land Reform during the Estrada Administration

The Estrada administration focused on fast-tracking land acquisition and distribution (LAD). It aimed to reduce distortions and uncertainties in land market in the rural areas to be able to help increase farmers' productivity and the private sector investment as well.

Another major step was the intensification of the delivery of support services and social infrastructure to boost incomes of ARBs. It also prioritized the improvement and protection of the tenure status of stakeholders and the promotion of agri-industrialization in CARP areas through joint ventures, corporatives, contract farming and other types of production and marketing arrangements. It also aimed for the completion of land parcel mappings covered by collective Certificate of Land Ownership Awards (CLOAs).

The Estrada administration also focused on the strengthening of the databases of the implementing agencies, i.e., DAR and DENR on the location of lands to cover and on the potential beneficiaries of CARP. It also promoted the use of market-based instruments in land distribution such as progressive agricultural land tax and direct land transfer. Lastly, the Estrada administration pursued to accelerate the resolutions of agrarian-related cases.

The Estrada administration has promised to complete the distribution of the CARP scope of 7.8 million hectares by 2004. From July 1998 to September 2000, the total number of beneficiaries of CARP under the Estrada administration was 182,762.

Multiple choice.

1. The Philippine Legislature has two houses: Senate and House of Representatives. What term best describes this setup??

A. Bicameralism
B. Bipartisanship
C. Unicameralism
D. Co-legislative powers

Answer: A. A bicameral legislature is one in which the legislators are divided into two separate assemblies, chambers or houses

2. Which of the choices ensures that the citizenry will have the right to seek public office, vote for elective officials, and be active in government?

A. Political rights
B. Right of suffrage
C. Socio-civil rights
D. Right to due process

Answer: B. Political rights include natural justice (procedural fairness) in law, such as the rights of the accused, including the right to a fair trial; due process; the right to seek redress or a legal remedy; and rights of participation in civil society and politics such as freedom of association, the right to assemble, the right to petition, the right of self-defense, and the right to vote.

3. Social and environmental justice is interconnected. Which concept best showcases this?

A. Industrialized countries' toxic disposed to poorer countries
B. Temperature warming and rise of sea level
C. Desertification of verdant regions
D. Extinction of rare animal species

Answer: A. Social justice and environmental justice are inseparable

4. How do you call the tax imposed on all employed and practicing professionals?

A. Income tax
B. Real estate tax
C. Community tax
D. Inheritance tax

Answer: A. An income tax is a tax imposed on individuals or entities (taxpayers) that varies with the income or profits (taxable income) of the taxpayer

5. The Philippine government provides for preferential attention to the welfare of the less fortunate. What policy stipulates this?

A. Social Justice
B. Criminal Justice
C. Distributive Justice
D. Bill of rights

Answer: A. Social Justice is the fair and just relation between the individual and society. This is measured by the explicit and tacit terms for the distribution of wealth, opportunities for personal activity and social privileges

6. Who among the following is a famous sculptor?

- A. Edades
- B. Manansala
- C. Castrillo
- D. Cuevas

Answer: C. Eduardo Castrillo (October 31, 1942 – May 18, 2016) was an award-winning Filipino sculptor. He was born in Santa Ana.

7. This man served as the last Spanish Governor-General of the Philippines.

- A. Ramon Blanco
- B. Diego de los Rios
- C. Basilio Agustin
- D. Fermin Jaudenes

Answer: B. Diego de los Rios y Nicolau (9 April 1850 - 4 November 1911) was the last Spanish Governor-General of the Philippines

8. If the social system is based on meritocracy, which is a possible effect?

- A. Rule by the wealthy and powerful
- B. Leadership by the people of talent
- C. Culture of elitism
- D. Rule by those of noble birth

Answer: B. Meritocracy is a political philosophy holding that power should be vested in individuals almost exclusively based on ability and talent.

9. Whose view gives prominence to faith in understanding the existence of God?

- A. St. Benedict
- C. St. Augustine
- B. St. Peter
- D. St. John

Answer: C. When the Western Roman Empire began to disintegrate, Augustine developed the concept of the Church as a spiritual City of God (City of God).

10. This field deals with the study of how human beings behave.

- A. Philosophy
- C. Psychology
- B. Morality
- D. Ethics

Answer: C. Psychology is the study of behavior and mind, embracing all aspects of conscious and unconscious experience as well as thought.

11. If this principle is applied well, no branch of government will be too powerful or abusive.

- A. Impeachment
- C. Check and balance
- B. Ombudsman
- D. Laissez Faire

Answer: C. Checks and Balances refers to the system in Philippine Constitution to ensure that no one branch of government shall become too powerful.



12. Which of the following is considered the lowest form of learning?

- A. Perceiving
- B. Memorization
- C. Conditioning
- D. Teaching

Answer: B. According to Bloom's Taxonomy of Cognitive Development, memorization is the lowest form of learning

13. The Filipinos are deeply religious; however, sometimes this religiosity instills the **NEGATIVE** attitude of _____

- A. Social responsibility to participate in the development of society
- B. Resignation, which leads to doing nothing to improve one's condition
- C. Total surrender to God in times of helplessness is best
- D. The Lord, who is omnipotent and omniscient, will allow what is best

Answer: B. Allowing fate through religious rituals to predetermine future events

14. Eastern religion is best related to this concept.

- A. Optimism
- B. Secularism
- C. Mysticism
- D. Pessimism

Answer: C. It may also refer to the attainment of insight in ultimate or hidden truths, and to human transformation supported by various practices and experiences.

15. He declared the date of the Philippine Independence from July 4 to June 12.

- A. QUIRINO
- B. ROXAS
- C. MACAPAGAL
- D. MAGSAYSAY

Answer: C. He declared June 12th as the official celebration of Philippine Independence

16. Maslow is popular in this theory _____

- A. Multiple Intelligence
- B. Hierarchy of Needs
- C. Social theory
- D. Constructivism

Answer: B. Abraham Maslow's Hierarchy of Needs.

Some of which focus on describing the stages of growth in humans. Maslow used the terms "physiological", "safety", "belongingness" and "love", "esteem", "self-actualization", and "self-transcendence" to describe the pattern that human motivations generally move through.

17. Who is the Philippine President known as the "Father of Social Justice and of the National Language"?

- A. ELPIDIO QUIRINO
- B. MANUEL ROXAS

- C. MANUEL QUEZON
- D. RAMON MAGSAYSAY

Answer: C. The First President of the Commonwealth Government. He signed Laws protecting the interest of the Filipino laborers (Minimum Wage Act and 8-Hour Labor Law)

18. Which was the 1st labor union in the Philippine Island founded by Isabelo delos Reyes in 1904?

- A. Union Obrera Democratica
- B. Union Trábedadores de Filipinos
- C. Association of Philippine Labor
- D. Association de Campania Tabacalera

Answer: A. U-O-D-: was a trade union centre in the Philippines. The organization was the first modern trade union federation in the country, earlier labor groups had been more of mutual aid societies and guilds.

19. The Philippine archipelago lies in the _____, an area where many volcanoes are active.

- A. Volcanic rim
- B. Archipelagic fault line

- C. Wheel of fire
- D. Ring of fire

Answer: D. The Ring of Fire is a major area in the basin of the Pacific Ocean where a large number of earthquakes and volcanic eruptions occur.

20. What law set the date of Philippine Independence from America on July 4, 1946 after a 10 year transition period under a Commonwealth government of Manuel L. Quezon?

- A. Jones Law
- B. Tydings Mc Duffie Law

- C. Cooper Act
- D. Hare-Haw-Cutting

Answer: B. The Tydings-McDuffie Act- Is a United States federal law that established the process for the Philippines as an American colony to become an independent country after a ten-year transition period.

21. As stipulated in Article VI of 1987 Phil. Constitution, which best describes the division of the Legislature into the Senate and the House of Representatives?

- A. Co-legislative power
- B. Bicameralism

- C. Unicameralism
- D. Bipartisanship

Answer B. Bicameralism - A bicameral legislature is one in which the legislators are divided into two separate assemblies, chambers or houses

22. It is being recognized as man-made wonders of the Philippines.

- A. Manila Bay
- B. Taal Volcano

- C. Rice Terraces
- D. Mount Makiling

Answer C. Rice terraces - The Banaue Rice Terraces are 2,000-year-old terraces that were carved into the mountains of Ifugao in the Philippines by ancestors of the indigenous people. The Rice Terraces are commonly referred to as the "Eighth Wonder of the World".

23. On June 12, 1898 at Kawit Cavite this band played the Marcha Nacionalista Filipino during the declaration of Philippine Independence.

- A. Malabon Band
- B. San Francisco del Monte Band
- C. Kawit Cavite Band
- D. San Francisco de Malabon Band

Answer D. San Francisco de Malabon Band - On June 11, 1898, Aguinaldo then requested Felipe to teach the music to the members of the band of San Francisco de Malabon so it could be played the next day (June 12, 1898).

24. Participation in governance, including the right to vote and right to be elected in public office as well is secured within the citizenry's _____.

- A. Right to due process
- B. Right of suffrage
- C. Political rights
- D. Socio-civic rights

Answer B. Right of Suffrage - Suffrage, political franchise, or simply franchise is the right to vote in public, political elections (although the term is sometimes used for any right to vote). Suffrage, political, franchise, or simply franchise is the right to vote in public, political elections (although the term is sometimes used for any right to vote).

25. The right to search for privacy and information is invoked through the _____.

- A. writ of habeas corpus
- B. writ of certiorari
- C. Writ of habeas data
- D. writ of amparo

Answer C. Writ of Habeas Data - Habeas data is a writ and constitutional remedy available in certain nations. The writ that protects privacy of information.

26. Which situation is NOT included in the social justice mandate in the 1987 Constitution?

- A. Creation of economic opportunities for all
- B. Sense of individualism among citizens

- C. Elimination of cultural inequalities
- D. Equitable distribution of wealth

Answer B. Sense of individualism among citizens - The 1987 Constitution promotes social justice among the Filipino people regardless of religion, socio-economic status, gender and the like.

27. Which is the fundamental and highest law of the Philippines?

- A. The Philippines Constitution
- B. The Civil and Criminal Codes
- C. The Ten Commandments
- D. The Bill of Rights

Answer A. The Philippine Constitution - Constitution is the Fundamental and Supreme Law of the land

28. The biggest among the seven continents of the world is _____.

- A. Asia
- B. Australia
- C. Africa
- D. Europe

Answer A. Asia - Asia largest and most populous continent, located primarily in the eastern and northern hemispheres and sharing the continental landmass of with the continent of Europe.

29. The province in the Philippines which was called "Ma-I" by the Chinese in the 14th century and Mina de Oro by the Spaniards.

- A. Ilocos
- B. Mindoro
- C. Cavite
- D. Palawan

Answer B. Mindoro - Ma-I- known as the 'Mina de Oro' - Mindoro

30. In 1565, which religious missionary order first arrived in the Philippines?

- A. Augustinian
- B. Jesuit
- C. Dominican
- D. Franciscan

Answer A. Augustinian - The then Prior General of the Order, Fr. Tadeo de Perusa, issued the decree creating the new Augustinian Province in the Philippines to be known as the Province of the Most Holy Name of Jesus of the Philippines (Santisimo Nombre de Jesus de Filipinas).

31. Which power of the state enables it to impose charge of burden upon persons, property or property rights for the use and support of the government expenditures for social services and a way of revenue collection?

- A. Eminent Domain
- B. Expropriation
- C. Value added tax
- D. Taxation

Answer D. Taxation - The levying of taxes aims to raise revenue to fund governing and/or to alter prices in order to affect demand. States and their functional equivalents throughout history have used money provided by taxation to carry out many functions such as economic infrastructure (roads, public transportation, sanitation, legal systems, public safety, education, health-care systems), military, scientific research, culture and the arts, public works, distribution, data collection and dissemination, public insurance, and the operation of government.

32. Which best describes the division of the Legislature into the Senate and the House of Representatives?

- A. Unicameralism
- B. Bipartisanship
- C. Co-legislative powers
- D. Bicameralism

Answer D. Bicameralism - A bicameral legislature is one in which the legislators are divided into two separate assemblies, chambers or houses

33. What kind of cooperative that unites small scale producers for joint production of goods

- A. Service cooperative
- B. Producers cooperative
- C. Marketing cooperative
- D. Multi-purpose cooperation

Answer B. Producers cooperative - A kind of cooperative uniting small-scale producers for joint production of goods and rendering of services.

34. The economic theory that considers good action is one that helps the greatest number of people is called _____.

- A. mercantilism
- B. colonialism
- C. capitalism
- D. utilitarianism

Answer D. Utilitarianism - Utilitarianism is an ethical theory that states that the best action is the one that maximizes utility. "Utility" refers to resources, such as human beings and other animals

35. Which runs against the Filipino family value of sacrifice?

- A. Willingness to forgiveness
- B. Delayed satisfaction of desires
- C. Inability to forgive
- D. Tolerance of pain and mistakes

Answer C. Inability to forgive - The attitude of unforgiveness runs against the Filipino family value of sacrifice

36. Is it correct to say that "Bahalana" is a Filipino dysfunctional trait?

- A. Yes, it is tantamount to leaving one's fate to another
- B. It depends on the situation of the person saying it
- C. It depends on the intention of one's action
- D. No, "Bahalana" is an attitude of daring combined with genuine concern

Answer D. Bahalana - Bahalana- from the expression " Bathalana" (God is in control) to Bahalana- "Whatever will be will be"

37. What was the first term given by Marcelo H. del Pilar to the notorious invisible influence and domination by Spanish religious priests over the colonial government.

- A. Iassuertepartidas
- B. Pase region
- C. frailocracia
- D. complace

Answer C. Frailocracia - Aside from spreading the word of God, the missionaries help in pacifying the country. They created more than 300 towns; built churches, roads and bridges, and other infrastructures through FORCED LABOR. They also established schools, colleges and universities, hospitals, orphanages, and asylums across the country.

38. Those who advocate the revival of the classics in art literature are called _____.

- A. humanists
- B. rationalist
- C. reformists
- D. existentialist

Answer A. Humanism - to describe Renaissance humanism, the movement that flourished in the Italia Renaissance to revive classical learning.

39. Which law set a full free trade policy in 1946 abolishing the quota limitations on Philippine exports to the United States?

- A. Payne-Aldrich Act
- B. Underwood-Simmons Act
- C. Hare-Hawes Cutting Act
- D. Bell Trade

Answer D. Bell Trade -The Bell Trade Act of 1946, also known as the Philippine Trade Act, was an act passed by the United States Congress specifying policy governing trade between the Philippines and the United States following independence of the Philippines from the United States

40. What problem can result owing to bias among parents and the youth towards manual or technical occupations, e.g. masons, mechanics, plumbers, etc?

- A. Boss mentality
- B. White collar mania
- C. Blue collar mania
- D. Technocrat mind

Answer B. White collar mania - Belonging or pertaining to the ranks of office and professional workers whose jobs generally do not manual labor or wearing of a uniform .

41. When one controls the supply/production of goods, this implies

- A. inflation
- B. economic stability
- C. monopoly
- D. economic equilibrium

Answer C. A monopoly exists when a specific person or enterprise is the only supplier of a particular commodity

42. Which Spanish policy provided the resettlement of Filipino communities to establish town centers of cabaceras?

- A. Assimilation
- B. Encomienda system
- C. Reduccion
- D. Bandala

Answer C. This "reduccion" – a term the Spanish used to name their efforts of subduing, converting, and gathering Chamorros through the establishment of missions – enacts the cultural, political, geographic, and linguistic "reduccion" that has accrued from three centuries of colonialism.

43. What was the social relevance of Noli and Fili to the conditions among Filipinos during the late 19th century period?

- A. Unfulfilled secularization movement of Gom-Bur-Za
- B. Need for violent uprising against Hispanic imperial rule .
- C. Rule of friars over Hispanic colonial governments sir roger pls verify the answer
- D. Human civil oppression of native Filipinos

Answer C. Rule of friars over Hispanic Imperial rule. Noli Me Tangere and El Filibusterismo as social documents

44. Art is product of man's need to express himself with a high degree of sensitivity towards his environment. Who is the painter of fisher folks and "BayanihansaBukid" of his hometown in Rizal?

- A. Vicente Manansala
- B. Anita Magsaysay
- C. Carlos V. Francisco
- D. Fernando Amorsolo

Answer C. Carlos "Botong" Francisco Botong's great works include Blood Compact, First Mass at Limasawa, The Martyrdom of Rizal, Bayanihan, Magpupukot, Fiesta, Bayanihan sa Bukid, Sandugo, Portrait of Purita.

45. Art is the handmaid of religion. Why do some religious allows signs and symbols?

- A. People might worship images themselves
- B. Signs and symbols display hidden mysteries as iconic representation
- C. Human figures are difficult art forms
- D. They display more traditional themes

Answer B. Signs and symbols display hidden mysteries as iconic representation
Religious symbolism provides an additional means to communicate the underlying message of a religion via symbol

46. Taoism is another Chinese philosophy about the same period as Confucianism, but more spiritual rather than ethical interest as it stressed that nature contains a divine impulse that direct all life. Who is the philosopher behind Taoism known as TAO TE CHING?

- A. Confucius
- B. Lao-Tzu
- C. The Buddha
- D. Sidharta Gautama

Answer B. Lao Tzu was an ancient Chinese philosopher and writer. He is known as the reputed author of the Tao Te Ching[2] and the founder of philosophical Taoism, and as a deity in religious Taoism and traditional Chinese religions.

47. What document contained the concrete promise of the Americans of Philippine independence "as soon as stable government can be established"?

- A. Preamble of the Jones Law
- B. Fairfield Bill of 1924
- C. Preamble of the Commonwealth Government
- D. First 1919 Parliamentary Mission of the U.S.

Answer A. Preamble of Jones Law. Whereas it is, as it has always been, the purpose of the people of the United States to withdraw their sovereignty over Philippine Islands and to recognize their independence as soon as a stable government can be established therein.

48. A person's efforts to cope with stressful situations in life is called _____.

- A. reaction
- B. anger
- C. defense mechanism
- D. emotion

Answer C. Defense mechanism. A defence mechanism is an unconscious psychological mechanism that reduces anxiety arising from unacceptable or potentially harmful stimulus



49. Which characterizes psychologically healthy people?

- A. pre-occupied only with their interest
- B. concerned with goals beyond themselves
- ☒ C. feels superior over others
- D. can't accept others as they are

Answer B. Concerned with goals beyond themselves Psychological health concerns itself with how you cope, how you are doing in response and whether you find life to be interesting and enjoyable

50. The fundamental right invoked by filing the "writ of amparo" is _____.

- A. The right to self-defense
- B. The right to due process
- ☒ C. The right to life, liberty and security
- D. The right to be defended by a public attorney

Answer C. Life, Liberty and Security. Writ of Amparo -It is a remedy available to any person whose right to life, liberty, and security has been violated or is threatened with violation by an unlawful act or omission of a public official or employee, or of a private individual or entity. The writ covers extralegal killings and enforced disappearances or threats thereof.

51. What does a professional code of conduct prescribe?

- A. Civic conduct for all
- B. Professional traditions and mores
- ☒ C. Moral and ethical standards
- D. Stricter implementation of laws

Answer C. Moral and Ethical Standards. RA 6713: Code of Conduct and Ethical Standards for Government Officials and Employees

52. He established the tobacco monopoly during the Spanish era.

- | | |
|--|--------------------|
| ☒ A. Jose Basco | C. Basilio Agustin |
| B. Rafael Izquierdo | D. Francisco Rizzo |

Answer A. Spanish Governor General Jose Basco. The Tobacco Monopoly refers to the 1782 economic program of Spanish Governor General Jose V. Basco, in which tobacco production in the Philippines was under total control of the government.

53. Through the Galleon Trade (1565-1815), the Philippines engaged in trade with _____.

- | | |
|--|------------------|
| ☒ A. Mexico | C. Spice Islands |
| B. China | D. Spain |

Answer A. Mexico. Manila-Acapulco, Mexico

54. He was the last Spanish Governor-General of the Philippines.

- A. Basilio Agustin
- B. Ramon Blanco
- C. Diego de los Rios
- D. Fermin Jaudones

Answer C. Spanish Governor General Diego de los Rios. Diego de los Rios y Nicolau was the last Spanish Governor-General of the Philippines. He signed the Treaty of Paris on December 10, 1898 representing Spain.

55. The penname Agapito Bagumbayan was used by this Katipunan leader.

- A. Emilio Jacinto
- B. Pio Valenzuela
- C. Julian Felipe
- D. Andres Bonifacio

Answer D. Agap-ito Bagumbayan. Agap-ito Bagumpayan – the pseudonyms of Gat. Andres Bonifacio

56. The fundamental right involved in the "writ of habeas data" is _____.

- A. The right to information privacy
- B. Right to transmit data
- C. Right to gather information data
- D. The right to search for information

Answer A. The Rights to Information privacy. Writ of Habeas Data – Privacy of Information

57. The fundamental right invoked by filing the "writ of amparo" is _____.

- A. The right to due process
- B. The right of life, liberty and security
- C. The right to self-defense
- D. The right to be defended by a public attorney

Answer B. The Right of Life, Liberty and Security. Writ of Amparo -It is a remedy available to any person whose right to life, liberty, and security has been violated or is threatened with violation by an unlawful act or omission of a public official or employee, or of a private individual or entity. The writ covers extralegal killings and enforced disappearances or threats thereof

58. This characteristic of cognitive skills makes them difficult to observe.

- A. hidden
- B. psychological
- C. overt
- D. mental

Answer D. Mental. Occurring or experienced in the mind : INNER mental anguish

59. Political idealists advocate ideals in politics such as justice, equality and fairness. Political realists have a more realist viewpoint of politics, aptly stated by "Might is right." Who among the following is more of a political realist rather than political idealist?

☒ A. Julius Caesar
B. Mohatma Ghandi
C. Martin Luther King
D. Benigno Jr. Aquino

Answer A. Julius Caesar. He centralised the bureaucracy of the Republic and was eventually proclaimed "dictator in perpetuity", giving him absolute authority

60. This law, passed in August 1909 by the U.S Congress, established a partial free trade in the Philippines.

☒ A. Treaty of General Relations
☒ B. Payne Aldrich Act
C. Bell Trade
D. Underwood-Simmons Act

Answer B. Payne -Aldrich Act. The Payne-Aldrich Tariff Act of 1909 named for Representative Sereno E. Payne (R-NY) and Senator Nelson W. Aldrich (R-RI), began in the United States House of Representatives as a bill raising certain tariffs on goods entering the United States

61. He was the first editor of La Solidaridad and an orator of the Reform Movement.

A. Maximo Viola
B. Valentin Ventura
☒ C. Graciano Lopez Jaena
D. Marcelo H. del Pilar

Answer C. Graciano Lopez-Jaena. The First Editor-in-Chief of La Solidaridad: Graciano Lopez-Jaena. The Second Editor-in-Chief of La Solidaridad: Marcelo H. Del Pilar

62. A branch of government cannot become too powerful or abusive because of this constitutional safeguard.

A. Impeachment
☒ B. Check and balance
C. Laissez Faire
D. Ombudsman

Answer B. Check and Balance. All three branches have "checks and balances" over each other to maintain the balance of power and not to exceed the constitutional limits

63. He was elected Speaker of the Philippine Assembly in 1907.

A. Manuel Roxas
B. Claro M. Recto
C. Manuel Quezon
☒ D. Sergio Osmeña, Sr.

Answer D. Sergio Osmena, Sr. first Speaker of the Philippine House of Representatives from 1907 to 1922, and Senator from the 10th Senatorial District for thirteen years

64. This is the largest country in Asia.

- A. Philippines
- B. Indonesia
- C. India
- D. China

Answer D. China. Second largest country in Asia, after Russia. Excludes Taiwan, disputed territories with India, and disputed islands in the South China Sea (total 137,296 km²)

65. The Philippines lies in the _____, an area where many volcanoes are active.

- A. Wheel of fire
- B. Volcanic rim
- C. Ring of fire
- D. Archipelagic faultline

Answer C. Ring of Fire. The Ring of Fire is a major area in the basin of the Pacific Ocean where a large number of earthquakes and volcanic eruptions occur. In a 40,000 km (25,000 mi) horseshoe shape, it is associated with a nearly continuous series of oceanic trenches, volcanic arcs, and volcanic belts and/or plate movements

66. It refers to the full and fair equivalent of the property taken from its owner by the expropriator, and the gauge for computation is not the taker's gain but the owner's loss. In order for the payment to be "just," it must be real, substantial, full, and ample; made within a reasonable time from the taking of the property

- A. Property
- B. Public use
- C. Eminent domain
- D. Just compensation

Answer D. Just compensation: In general, a fair and reasonable amount of money to be paid for work performed or to make one "whole" after loss due to damages

67. In Northeast Luzon, the longest river in the country originates from the mountains of Quirino and Nueva Vizcaya and drains the whole Cagayan Valley existing through the town of Aparri and Cagayan. What is the name of this river?

- A. Agno River
- B. Abra River
- C. Pampanga River
- D. Rio Grande de Cagayan

Answer D. Rio Grande de Cagayan: The Cagayan River, also known as the Rio Grande de Cagayan, is the longest and largest river in the Philippine Archipelago. It is located in Cagayan Valley.

68. A sociological research method that approaches social phenomena through quantifiable evidence, and often relies on statistical analysis of many cases (or across intentionally designed treatments in an experiment) to create valid and reliable general claims
- A. Qualitative design
B. Quantitative design
C. Normative
D. Formative

Answer B. Quantitative design: As indicated above, quantitative research focuses on analysis of numeric data.

69. The "La Liga Filipina" was a concrete evidence of Rizal's desire to _____.
- I. Unite the entire Philippine archipelago
II. Fight violence and injustice
III. Revolt against the Spaniards
- A. I and II
B. II and III
C. I and IV
D. I and III

Answer A. Unite the entire archipelago and fight injustice in the country.

70. The culture of the Ilocanos can easily be felt through their ants dance known as _____.
- A. Balitaw
B. Dandansoy
C. Mahinhin
D. Kinnotan

Answer D. Kinnotan: One favorite Ilocano dance was the Kinnotan or ants' dance. ... Aside from the kinnotan, the Kinnalogong is another favorite dance of the Ilocanos

71. Cooperatives can eventually put up banks of their own. What are the evidences of their successes?
- A. Many banks in Metro Manila and Central Luzon started out as cooperatives.
B. Most of the rural banks started out as cooperatives.
C. Cooperatives are not heavily taxed.
D. Opening of rural bank is allowed on a cooperative concept.

Answer B. Most of the rural banks began as cooperatives, catering to the basic needs of the farmers (loans for fertilizers, seeds, etc.

72. When a combination of independent business organizations form to regulate production, pricing, and marketing of goods by the members, which term applies?
- A. Cartel
B. Monopoly
C. Depression
D. Inflation

Answer A. Cartel: A cartel is a formal (explicit) agreement among competing firms. It is a formal organization of producers and manufacturers that agree to fix prices.

73. What are considered major goals of multicultural education?

- I. Catering to diversity of learners
- II. Considering social class and ethnic groups
- III. Providing equal opportunities to education
- IV. Helping individuals and group survive despite differences in culture and beliefs.

- A. II and III
- B. I, II and IV

- C. III and IV
- D. I, II, III and IV

Answer: D. All of the above. "Multicultural education is a field of study and an emerging discipline whose major aim is to create equal educational opportunities for students from diverse ethnic groups."

74. What right and duty do Filipinos exercise when they participate in ratifying a proposed constitution?

- A. citizenship
- B. habeas corpus

- C. suffrage
- D. Bill of Rights

Answer C. Suffrage: The right to vote and right to be elected in public office as well.

75. Martial law remained in force until 1981 under Proclamation # _____

- A. 1081
- B. 1425

- C. 9994
- D. 7610

Answer A. Proclamation 1081 was in effect from September 21, 1972 to February 1981.

76. The process by which a person learns the requirements of the culture by which he or she is surrounded, and acquires values and behaviors that are appropriate or necessary in that culture is called _____.

- A. assimilation
- B. enculturation

- C. cultural relativism
- D. acculturation

Answer: B - Enculturation. Enculturation is the process by which people learn the requirements of their surrounding culture and acquire values and behaviors appropriate or necessary in a particular group of people.

77. The exchange of cultural features that results when groups of individuals from different cultures come into continuous first-hand contact. The original cultural patterns of either or both groups may be altered, but the groups remain distinct

- A. assimilation
- B. enculturation

- C. cultural relativism
- D. acculturation



Answer D. Acculturation: It explains the process of cultural and psychological change following the meeting between cultures.

78. The Commander of the American and Filipino forces in Bataan was _____.
- A. General Arthur MacArthur
B. General Douglas MacArthur
C. General Jonathan Wainwright
D. General Edward P. King

Answer D. General Edward P. King: When President Roosevelt ordered General MacArthur to leave Bataan for Australia on March 11, 1942, Wainwright also acquired command of all of the U.S. forces in the Philippines, and King became the Commanding General of the Philippine-American forces on Bataan. The Japanese launched their final assault on Bataan on April 3, 1942, and on April 5, they succeeded in capturing Mt. Samat, a major observation post and artillery station. Unable to continue fighting due to the insufficient nutrition and diseases which were plaguing his forces, General King, fearing that his men would be annihilated, disobeyed orders from Wainwright and MacArthur to counterattack the Japanese and surrendered Bataan on April 9, 1942. This was the largest surrender of a military force in U.S. military history.

79. The first American Civil Governor was _____.
- A. William Howard Taft
B. Wesley Merritt
C. Arthur Mac Arthur
D. Murray Bartlet

Answer A. William Howard Taft, who served as President of the Commission his title until he became civil governor in September 1901

80. This is a collection of traditional courtship song in the Mexican-Spanish tradition based on the *habanera* rhythm.
- A. Harana
B. Kundlman
C. Jazz
D. modern

Answer A. Harana: Harana includes traditional courtship songs in the Mexican-Spanish tradition based on the *habanera* rhythm.

81. It is characterized by shared experiences and mutual responsibilities.
- A. Gesellschaft
B. Gemeinchaft
C. Enculturation
D. Acculturation

Answer B. Gemeinchaft: Generally translated as "community"

82. It features the exchange of goods, money and services.

- A. Gesellschaft
B. Gemeinchaft
C. Enculturation
D. Acculturation

Answer A. Gesellschaft: "Society"

83. One of the advantages of presidential system of government is ____.
- ☒ A. the principle of the separation of powers and checks and balances is observed.
 - ☐ B. unity of authority
 - ☐ C. It cuts costs and redundancy in lawmaking.
 - ☐ D. It promotes strong political party.

Answer A. Article II: "Declaration of Principles and State Policies" and Article VII- The Executive Department"

The principle of the separation of powers and checks and balances shall be observed.

84. The fulfillment of the collective needs to the human species, including global society, is to pursue ____.
- ☒ A. public interests
 - ☐ B. common good
 - ☐ C. justice
 - ☐ D. political will

Answer A. Public Interests. The public interest refers to the "common well-being" or "general welfare."

85. This is the determination of the government to relate to what is perceived as beneficial to public interests and acts to implement its decision on such in order to achieve a better future for its citizens.
- ☐ A. Public interests
 - ☐ B. Common good
 - ☐ C. Justice
 - ☒ D. Political will

Answer D. Political Will: Political will is the exercise of an abstract feature of political authority to enforce certain acts for the benefit of its constituents.

86. Those provinces that were pacified and were already recognizing Spanish rule were called ____.
- ☒ A. Alcaldias
 - ☐ B. Alcalde mayor
 - ☐ C. Indulto de comercio
 - ☐ D. Corregimientos

Answer A. Alcaldias: The jurisdiction or office of an alcalde; also, the building or chamber in which he conducts the business of his office.

87. During the Spanish period, this refers to the privilege of the provincial Governor to engage in trade.
- ☐ A. Alcaldia
 - ☐ B. Alcalde mayor
 - ☒ C. Indulto de comercio
 - ☐ D. Corregimientos

Answer C. Indulto de Comercio: The Indulto de Comercio or license to trade, was issued to the Alcalde Mayor or Corregidor.



88. What form of government did we have when President Aguinaldo proclaimed the Philippine Independence on June 12, 1898?

- ☒ A. Dictatorial
- ☐ B. Revolutionary
- ☐ C. Freedom
- ☐ D. Presidential

Answer A. Dictatorial Government: The Dictatorial Government lasted for only a month, from May 24 to June 23, 1898. On June 23, Aguinaldo established the Revolutionary government.

89. The military provinces that were unpacified during the Spanish period were called _____.

- ☐ A. Alcadia
- ☐ B. Alcalde mayor
- ☐ C. Indulto de comercio
- ☒ D. Corregimientos

Answer D. Corregimientos: The Corregimientos were usually led by a Corregidor, helped by a lieutenant.

90. The longest revolt in Philippine History is called the _____.

- ☒ A. Dagohoy revolt
- ☐ B. Diego Silang revolt
- ☐ C. Gabriela Silang revolt
- ☐ D. Palaris revolt

Answer A. Dagohoy revolt: The Dagohoy Rebellion, also known as Dagohoy Revolution or Dagohoy Revolt lasted 85 years, from 1744-1829.

91. The most devastating typhoon in 2009, with a damage of \$1.09 billion and 747 fatalities was known as _____.

- ☐ A. Reming
- ☐ B. Milenyo
- ☐ C. Ondoy
- ☒ D. Habagat

Answer C. Ondoy: September 2009

92. Violation of human rights is a violation of the _____ of persons.

- ☒ A. dignity
- ☐ B. intelligence
- ☐ C. emotions
- ☐ D. freedom

Answer A. Dignity is a term used in moral, ethical, legal, and political discussions to signify that a being has an innate right to be valued and receive ethical treatment.

93. One of the new seven wonders of nature that can be found in Palawan is

- ☒ A. Puerto Princesa Underground River
- ☐ B. El Nido
- ☐ C. Palawan Underground River
- ☐ D. Puerto Princesa River

Social Science

Answer D. Puerto Princesa Underground River National Park was declared in 2011 as one of the new 7 Wonders of Nature.

94. The Act/Law that led to the granting of ten-year transition period is called _____.

- A. Hare-Hawes Cutting Bill
- B. Tydings-McDuffie Law
- C. Parity Rights
- D. Independence Law

Answer B. The Tydings-McDuffie Act (officially the Philippine Independence Act, Pub.L. 73-127, 48 Stat. 456, enacted March 24, 1934)

95. Republic Act 9994 is also known as the _____.

- A. Anti-Child Abuse Law
- B. K-12 Act
- C. Expanded Senior Citizens' Act
- D. Prison Modernization Law

Answer C. Republic Act 9994

96. What is the very foundation of genuine peace and reconciliation?

- A. Happiness
- B. Joy
- C. Confrontation
- D. Social Justice

Answer D. Social justice is defined as justice exercised within a society, particularly as it is exercised by and among the various social classes of that society.

97. This is the location of the island known as the "Isla de Pintados" by the Spaniards.

- A. Luzon
- B. Visayas
- C. Mindanao
- D. Isla Verde

Answer B. Panay: Visayas is home to Panay, an island in the Philippines. It is divided into four provinces: Aklan, Capiz, Antique and Iloilo.

98. What was our government under the 1935 Constitution?

- A. Revolutionary
- B. Monarchy
- C. Commonwealth
- D. Military

Answer C. The Commonwealth government was in exile from 1942 to 1945, when the Philippines was under Japanese occupation. In 1946, the Commonwealth ended.

99. In economics, _____ is a rise in the general level of prices of goods and services in an economy over a period of time.

- A. Inflation
- B. mercantilism
- C. Depression
- D. free trade

Answer A. In economics, inflation is a rise in the general level of prices of goods and services in an economy over a period of time.

100. What values are being promoted by the Philippine Government when it launched the "Juan Time" project in 2011?

- A. Punctuality
- ~~B.~~ Austerity
- C. Palabra de Honor
- D. Close family ties

Answer: A. Juan Time project is an advocacy that aims to teach Filipino respect for time to avoid being late.

101. A UNESCO World heritage Site in Ilocos Norte is _____.

- A. Bangui windmills
- B. Paoay Church
- C. Currimao Rock Formations
- D. Pagudpud

Answer: B. Paoay Church, became the Ilocanos, centerpiece for their cultural campaign in promoting the Ilocos's cultural icons.

102. An international convention where the Philippine can validate its claim to the Spratlys and other territories in the West Philippine Sea is known as the _____.

- A. United Nations Convention on the Law of the Sea (Unclos)
- ~~B.~~ Geneva Convention
- C. Rio De Janeiro Convention
- D. Kyoto Protocol

Answer: A. All conflicting claims should be resolved based on the rules-based regime of the UNCLOS.

103. The author of "Flora de Filipinas" circa 1837, the botanical illustrations of plant specimens that grew into the Philippines herbarium is _____.

- A. Gov. Polavieja
- B. Padre Emmanuel Gonzales
- C. Fr. Manuel Blanco
- D. Gov. Narciso Claveria

Answer: C. He conducted the initial illustrations and collections of plants that grew in the Philippines.

104. The Philippines was ruled by Spain through the _____ until 1814.

- A. Cuba
- B. Puerto Rico
- C. Viceroy of Mexico
- ~~D.~~ Guam

Answer: C. The Philippines was ruled by Spain through a viceroy in Nueva España or New Spain (Mexico), from the 16th to the 19th century.

105. This is a guarantee given by the Constitution to certain units of government where their approved annual appropriations shall be automatically and regularly released.

- A. General Appropriations Act
- B. Fiscal Autonomy
- C. Fiscal Discipline
- D. Internal Revenue Allocations

Answer: B. Fiscal autonomy is a guarantee given by the Constitution to certain units of the government. It is intended as a guarantee of separation of powers and of independent political agencies.

106. Which order of government correctly sequences what happened during the American regime?

- A. Civil, Military, Commonwealth
- B. Military, Civil, Commonwealth
- C. Civil, Commonwealth, Military
- D. Military, Commonwealth, Civil

Answer: B. After the capture of Manila in 1898, a military government was established. On July 1, 1901, a civil government was inaugurated. Before the granting of full independence from the United States of America, a ten-year transition government known as Commonwealth was created in 1935.

107. Which order of presidents correctly applies to Philippine history?

- I. Carlos P. Garcia
- II. Diosdado Macapagal
- III. Manuel Roxas
- IV. Ferdinand Marcos

- A. III, I, II, IV
- B. IV, II, I, III

- C. I, II, III, IV
- D. IV, III, II, I

Answer: A. Aguinaldo-Quezon-Laurel-Osmena-Roxas-Quirino-Magsaysay-Garcia-Macapagal-Marcos-Cory Aquino-Ramos-Estrada-Arroyo-Noynoy Aquino

108. In whose presidency was the famous MIRACLE RICE produced?

- A. Ferdinand Marcos
- B. Fidel Ramos
- C. Joseph Estrada
- D. Sergio Osmeña

Answer: A. The International Rice Research Institute (IRRI) was conceptualized and implemented during the term of President Ferdinand Marcos.

109. If you keep seeing your ways of doing things as the right way and everybody else's as the wrong way, you tend to have the attitude called

- A. Cultural relativism
- B. Ethical relativism
- C. Ethnocentrism
- D. Xenophobia

Answer: C. The attitude that your culture is the most superior.



110. After the Second World War, which emerged as new powers?

- A. People's Republic of China and United States of America
- B. South Korea and Soviet Union
- C. Germany and Soviet Union
- D. United States of America and Soviet Union (USSR)

Answer: D. Cold war, characterized by these two eventually engaged in the political conflict and military tension between USA and USSR.

111. Legislation granting special privileges to senior citizens helps neutralize prejudice and discrimination toward a social category based on:

- A. Age
- B. Ethnicity
- C. Gender
- D. Race

Answer: A. Republic Act 9994 or Expanded Senior Citizens Act of 2010: It protects and neutralizes the rights and privileges of the senior citizens, who are often discriminated based on age.

112. The employees of MET Review Center want to associate, pool their savings, and use the fund as soon as possible to grant loans to members and capitalized and operate the canteen of the said review center. What type of cooperative will they organize and register?

- A. Service cooperative
- B. Producer's cooperative
- C. Consumer cooperative
- D. Credit cooperative

Answer: D. Credit cooperative – generates thrift among its members and creates funds in order to grant loans; Consumer's cooperative – Procures and distributes commodities to members and non-members; Producer's cooperative- Undertakes joint production (agricultural or industrial); Service cooperative- Engages in medical and dental care, hospitalization, transportation, insurance, housing, labor, electric light and power, communication or other services.

113. One of the types of positive economics that collects and presents data that can describe the phenomena is _____.

- A. Descriptive economics
- B. Economic theory
- C. Growth
- D. Stability

Answer: A. Descriptive economics is simply the collection and presentation of data that describe phenomena and facts.

114. The complex set of relationships within the agricultural sector and between tenure structure, production structure, and the structure of supporting services is called _____.

- A. Land reform
- B. Agrarian structure
- C. Comprehensive Agrarian Reform Law
- D. Agrarian reform measures

Answer: B. Agrarian structure is defined as a complex set of relationships within the agricultural sector and between tenure structure, production structure, and the structure of supporting services.

115. The earliest mode of economic life in human history according to Karl Marx is called _____.

- A. Enlightened self-interest
- B. Reciprocity
- C. Primitive communism
- D. Agricultural

Answer: C. As a model, primitive communism is usually used to describe early hunter-gatherer societies, which had no hierarchical social class structures or capital accumulation.

116. It refers to any liability to pay money arising out of a contract, express or implied.

- A. Poll tax
- B. Tax
- C. Debt
- D. Debtors

Answer: C. Debt. Section 20, Article III, 1987 Philippine Constitution.

117. The state of entire subjection of one person to the will of another is called _____.

- A. Peonage
- B. Slavery
- C. Suppressed
- D. Depressed

Answer: B. Slavery. Section 18, Article III, 1987 Philippine Constitution

118. It means that when a person is charged with an offense and the case is terminated either by acquittal or conviction or in any other manner without the express consent of the accused, the latter cannot again be charged with the same or identical offense.

- A. Writ of Habeas Corpus
- B. Right against illegal detention
- C. Writ of Amparo
- D. Right against double jeopardy

Answer: D. Right against double jeopardy. Section 21, Article III, 1987 Philippine Constitution.

119. The families of victims of extrajudicial killings and enforced disappearances can invoke the writ when the right to life, liberty, or security of a person is violated or threatened with violation by an unlawful act or omission of a public official or employee or of a private individual or entity.

- A. Writ of Habeas Corpus
- B. Right against illegal detention
- C. Writ of Amparo
- ☒ D. Right against double jeopardy

Answer: C. Writ of Amparo. This special writ prohibits respondents from using the defense denial. They will have to produce documents or evidence to support claims that they did not violate the rights to life, liberty, and security of the aggrieved party.

120. This court shall have jurisdiction over civil and criminal cases involving graft and corrupt practices and such offenses committed by public officers and employees in relation to their offices as may be determined by law.

- A. Court of Appeals
- B. Regional Trial Courts
- ☒ C. Sandiganbayan
- D. Supreme Court

Answer: C. Sandiganbayan. Section 5, Article XI. 1987 Philippine Constitution.

121. These are the only officials removable by impeachment.

- ☒ A. The President, Vice-President, members of the Supreme Court, members of the Constitutional Commissions, and the Ombudsman.
- B. The President, Vice-President, members of the Supreme Court, members of the Constitutional Commissions and the Senators
- C. The President, Vice-President, members of the Supreme Court, members of the Constitutional Commissions and the Congressmen/women.
- D. The President, Vice-President, members of the Supreme Court, members of the Constitutional Commissions and the Commander-in-Chief of the Armed Forces.

Answer: A. The President, Vice-President, members of the Supreme Court members of the Constitutional Commissions, and the Ombudsman. Section 2, Article XI. 1987 Philippine Constitution.

122. This is the legal process whereby the registered voters of a local government unit may directly propose, enact, or amend any law or ordinance at polls called for the purpose independently of the regularly constituted local legislative body.

- A. Referendum
- B. Recall
- C. Order
- ☒ D. People's Initiative

Answer: D. People's Initiative. Section 3, Article X. 1987 Philippine Constitution.

123. He became governor, with new capital at Iloilo, 1898 until Treaty of the Paris was signed, December 10, 1898, handing sovereignty over the Philippines to the United States for \$20 million. His term as governor was from August 13, 1898-December 10, 1898, and Manila, on January 1, 1899

- ☒ A. Diego De Los Rios
- ☐ B. Basilio Agustin

- ☐ C. Miguel Lopez De Legazpi
- ☐ D. Narciso Claveria

Answer: A. Diego De Los Rios. He was the last Spanish Governor-General in the Philippines.

124. He was a native of Cataluña, and count of Caspe. He became governor in 1891, around the time when La Liga filipina (Philippine League) was founded in Manila by Rizal in 1892. He introduced many reforms; popular with natives but aroused the wrath of religious orders, who are said to have paid \$100,000 for his dismissal. His term as governor was from 1891-1893.

- ☒ A. Gov. Gen. Despujol
- ☐ B. Gov. Gen. Claveria

- ☐ C. Gov. Gen. De Los Rios
- ☐ D. Gov. Gen. Polavieja

Answer: A. Governor General Despujol.

125. The Republic Act that requires the compulsory teaching of Rizal Course and the lives of other Filipino patriots is _____.

- ☐ A. Republic Act 7610
- ☐ B. Republic Act 9262

- ☒ C. Republic Act 1425
- ☐ D. Republic Act 9994

Answer: C. Republic Act 1425 - mandates the teaching of the lives, works and writings of Rizal and other Filipino patriots

126. This Republic Act created the Technical Education and Skills Development Authority (TESDA).

- ☒ A. Republic Act 7796
- ☐ B. Republic Act 9262

- ☐ C. Republic Act 1425
- ☐ D. Republic Act 9994

Answer: A. Republic Act 7796

127. It prohibits hazing and other forms of Initiation rites in fraternities, sororities, and other organizations.

- ☐ A. Republic Act 7796
- ☐ B. Republic Act 9262
- ☐ C. Republic Act 1425
- ☒ D. Republic Act 8049

Answer: D. RA 8049 - An Act regulating hazing and provides penalties for the violators.



128. An Act Strengthening Teacher Education in the Philippines by establishing Centers of Excellence and creating a teacher Education Council.

- A. Republic Act 7784
- ☒ B. Republic Act 9262
- C. Republic Act 1425
- D. Republic Act 8049

Answer: A. Ra 7784 – Establishing the Centers of excellence in the country.

129. Rizal was eight years old when he wrote this poem. He realized the ultimate importance of having a native tongue.

- A. Remembrance to my Hometown
- B. Our Mother Tongue
- ☒ C. To Education
- D. Intimate alliance between religion and good education

Answer B. Our Mother Tongue was written in 1869

130. This poem, was written by Rizal when he was 15 years old. His memories of Calamba demonstrated his innate love of nature that ultimately developed into a greater love of nation.

- A. Remembrance to My Hometown.
- ☒ B. Our Mother Tongue
- C. To Education
- D. Intimate alliance between religion and good education

Answer A. Remembrance to My Hometown was written in 1876.

131. In this poem, Rizal compared education to the goddess of light, wisdom, hope, peace and truth.

- A. Remembrance to my hometown.
- B. Our Mother Tongue
- ☒ C. To Education
- D. Intimate alliance between religion and good education.

Answer C. To Education was a literary achievement during Rizal's Ateneo days. He compared education to an eternal lighthouse.

132. In this poem, Rizal noted the role of religion in attaining good education.

- A. Remembrance to my Hometown
- B. Our Mother Tongue
- C. To Education
- ☒ D. Intimate alliance between religion and good education

Answer D. Intimate alliance between religion and good education. According to Rizal, education is directionless with religion.

133. This was a prize winning-poem of Rizal when he was a student in UST. This poem proved that an Indio was not only equal but also more superior than the Spaniards in poetry writing.

- A. To the Filipino Youth
- B. They ask Me For Verses
- C. Hymn to Talisay
- D. Kundiman

Answer A. To the Filipino Youth. The ultimate challenge of Rizal in this poem is to excel and be an international figure.

134. Rizal's loneliness is profoundly expressed in this poem which has the lines "His lyre had long ago become so mute and broken; his muse stammers and no longer smiles at him." What is the title of this poem?

- A. To the Filipino Youth
- B. They ask Me For Verses
- C. Hymn to Talisay
- D. Kundiman

Answer B. They Ask Me For Verses. In this poem, Rizal shows his literary talents.

135. Written in 1895, the verses, were used against him during his trial in December 1896.

- A. To the Filipino Youth
- B. They ask Me For Verses
- C. Hymn to Talisay
- D. Kundiman

Answer C. Talisay, in the jurisdiction of Dapitan, was where Rizal with his money from lottery bought a house, built a school and set a farm plantation.

136. Originally written in Tagalog, the poem is one of the only two poems that Rizal wrote in his vernacular. He, however, denied authorship of such.

- A. To the Filipino Youth
- B. They ask Me For Verses
- C. Hymn to Talisay
- D. Kundiman

Answer D. Kundiman. During his trial, the poem was cited as evidence against him.

137. The highest mountain in the continent of Africa is _____.

- A. Mt. Everest
- B. Mt. Kilimanjaro
- C. Mt. Fuji
- D. Mt. Apo

Answer B. Kilimanjaro, with its three volcanic cones, Kibo, Mawenzi, and Shira, is a dormant volcanic mountain in Kilimanjaro National Park, Kilimanjaro Region, Tanzania.



138. He coined the term "survival of the fittest."

- A. August Comte
- B. Karl Marx
- C. Herbert Spencer
- D. Arnold Toynbee

Answer C. Spencer developed an all-embracing conception of evolution as the progressive development of the physical world, biological organisms, the human mind, and human culture and societies.

139. This is a multilateral agreement regulating international trade. According to its preamble, its purpose was the "substantial reduction of tariffs and other trade barriers and the elimination of preferences, on a reciprocal and mutually advantageous basis."

- A. APEC
- B. ASEAN
- C. NAFTA
- D. GATT

Answer D. The General Agreement on Tariffs and Trade (GATT) was a multilateral agreement regulating international trade. According to its preamble, its purpose was the "substantial reduction of tariffs and other trade barriers and the elimination of preferences, on a reciprocal and mutually advantageous basis."

140. The mountain that separates the continents of Asia and Europe is ____.

- | | |
|-------------------------|----------------|
| <u>A. Ural Mountain</u> | C. Mt. Everest |
| B. Mt. Kilimanjaro | D. Mt. Apo |

Answer A. Ural Mountain range forms a rugged spine in west-central Russia and the major part of the traditional physiographic boundary between Europe and Asia.

141. The "yellow race" is also known as:

- | | |
|----------------------|-------------|
| A. Caucasians | C. Negroids |
| <u>B. Mongoloids</u> | D. Malays |

Answer B. Mongoloids probably originated in what is now the north of Tibet. The Caucasoids who mingled with yellow race once extended halfway across Siberia.

142. The major rivers in China are ____.

- A. Tigris and Euphrates
- B. Hwang-Ho and Yangtze
- C. Nile and Indus
- D. Aegean and Mekong

Answer B. Hwang Ho and Yangtze. The Yellow River (or Huang He) is the second-longest river in Asia after the Yangtze.

143. Year 2014 has been declared by Pope Francis as Year of the _____.

- A. Laity
B. Hope

- C. Faith
D. Righteousness

Answer A. 2014 has been declared by Pope Francis as "Year of the Laity" - the people of a religious faith as distinguished from its clergy.

144. _____ is a major land mass on the northern coast of the Black Sea that is almost completely surrounded by water. The peninsula is located just south of the Ukrainian mainland and west of the Russian region of Kuban. It is surrounded by two seas: the Black Sea and the smaller Sea of Azov to the east.

- A. Gibraltar
B. Cape of Good Hope

- C. Falkland
D. Crimea

Answer D. Crimea.

145. The Russian President who wants to annex Crimea is named _____.

- A. Vladimir Putin
B. Michael Gorbachev

- C. Joseph Stalin
D. Nikita Khrushchev

Answer A. Vladimir Putin.

146. The Russian President who gave Crimea to Ukraine in 1954 is _____.

- A. Vladimir Putin
B. Michael Gorbachev

- C. Joseph Stalin
D. Nikita Khrushchev

Answer D. Nikita Khrushchev

147. The British Prime Minister who supported the European Union's measures against Russia on the issue of Crimea is _____.

- A. Margaret Thatcher
B. Winston Churchill

- C. David Cameron
D. Tony Blair

Answer C. David Cameron

148. Which constitute a quorum in each House of Congress?

- A. One-third of the members of each House
B. Two-thirds of the members of each House
C. A majority of each House
D. Three-fourths of the members of each House

Answer C. House of Representatives: Section 74: Quorum: A majority of all the members of the House shall constitute a quorum. Senate: Section 44: A majority of the Senators shall constitute a quorum.



149. The function by which schools help children to participate effectively in larger society is called:

- A. Socialization
- ☒ B. Education
- C. Enculturation
- D. Acculturation

Answer A. Socialization refers to the lifelong process of inheriting and disseminating norms, customs and ideologies, providing an individual with the skills and habits necessary for participating within his or her own society.

150. The principle of checks and balances is intended to maintain balance among executive, legislative and judicial departments of the government. Which one is a check on the executive department by the judiciary?

- A. Determining the salaries of the President and Vice President
- B. Declaring a legislative measure is unconstitutional
- ☒ C. Declaring an act of the President unconstitutional
- D. Impeaching the President

Answer C. Judicial department is the one who interprets the law
Declaring the Law Constitutional or Unconstitutional

SCIENCE

- From the Latin word "scientia," which means "knowledge"
- Any methodological activity, such as observational, experimental investigation, and theoretical explanation of natural phenomena
- Systematized knowledge based on facts

Divisions of Science

- Social Science
- Political Science
- History
- Mathematics
- Natural science – deals with nature and natural phenomena

Branches of Natural Science

- **Physical Science** – deals with non-living things
- **Biological Science** – deals with living things

Scientific Method

- An orderly, logical, and rational manner of solving problems
- Enables and leads scientists toward unveiling the truths about observable phenomena, helping them construct their clear representations

Steps In Solving Problems

- **The Problem** – question formulated from observations to which you want to find specific answers

- Identify, state, know, and understand the problem
- Identify the variables

Types of variables

- Independent variable – the one which is manipulated or changed
- Dependent variable – the one which is measured and observed; depends on the independent variable

Example: In the experiment, "the effect of sunlight on the growth of the plants"

Sunlight - Independent variable

Growth - dependent variable

Hypothesis

- It is an inference or a scientific guess.
- It can be a positive or a negative hypothesis.

Experimentation

- Designed and conducted to test the validity of the hypothesis
- A good experiment is one where the least time, effort, and logistics are used, but yields the best result.

- Steps in Performing an Experiment
 - Gather the materials to be used in experimentation
 - Set the materials
 - Observe
 - Gather the data
 - Interpret the data
- **Conclusion**
 - Draw the conclusion by stating the findings in the manner that allows the problem to be solved / answered.
- **Application**
 - This is an optional step. If the results of the experiment are good enough, then, there is a need to apply the results of the experiment in real-life situations.

Scientific Attitude

- This is the scientist's way of doing and thinking, especially when performing an experiment or any scientific investigation.
- Some of the scientific attitudes are the following:
 - **Curiosity** – pays particular interest and asks questions about observations
 - **Honesty** – reports results of observations truthfully
 - **Humility** – accepts that he does not have the monopoly of ideas about the phenomenon
 - **Unbiased** – ability to separate his own personal ideas from the actual findings
 - **Open-mindedness** – readiness to accept or consider opinions of others
 - **Patience and perseverance** – does not easily give up when investigation seems difficult

Law vs. Theory

- **Law** – generalizations formulated from observations that are proven correct and true.
Ex. Ohm's Law: that current is directly proportional to the voltage but inversely proportional to the resistance. This explains that an increase in voltage will increase the current if the resistance is constant and if the voltage is held constant, and increase in resistance will give a decrease in current.
- **Theory** – generally accepted explanations of observations, but are yet to be fully proven because some questions still need to be answered.
Ex. Theory of Evolution: results of scientific investigations explain changes in organisms, but questions that cannot undergo experimentations are left unanswered.

PHYSICAL SCIENCES

Branches of Physical Science

- **Physics** – deals with matter and energy and of the interactions between the two
- **Chemistry** – deals with the composition and properties of matter
- **Earth Science** – deals with the physical aspects of the Earth

Physics

Scientific Notation

- Short-hand in writing extremely large or small numbers using the power of ten notation
- Takes the form $A \times 10^n$ where A is an integer from 1 to 9.9999 and n is the exponent in the power of ten.
- The exponent n may be dealt also as the number of times the decimal point is moved, positive if moved to the left, and negative when moved to the right.

Ex. 361000 = 3.61×10^5 - decimal point is moved to the left 5 times so that the number to the left of the decimal point is less than ten.

Ex. 000361 = 3.61×10^{-4} - decimal point is moved to the right 4 times so that the number to the right of the decimal point is less than ten.

Measurement

- The process of comparing the quantity to be measured and the corresponding standard
- The quantity is the one being measured, not the object.

Physical Quantities

- There are two physical quantities, namely: the fundamental quantities and the derived quantities.
- The fundamental quantities are also called basic quantities. The following are the fundamental quantities, together with their corresponding standard units:
 - length meter
 - mass kilogram
 - time second
 - temperature Kelvin
 - luminous intensity candela
 - electric current ampere
- Derived quantities are quantities taken from fundamental quantities.
 Ex. Area = $l \times l$, area is taken from the fundamental quantity, length
 Density = mass/volume, and volume = $l \times l \times l$, density is derived from fundamental quantities length and mass.

Relationship between Quantities

- **Direct Proportionality** – When one quantity increases, the other quantity increases also in proportion to each other.
 - A is directly proportional to B
 - $A/B = K$
 - If A is plotted against B , the resulting graph line is straight line slanting to the right
- **Direct-square Proportionality** – When one quantity increases, the other quantity increases also, but faster and greater than the other.

- is directly proportional to B^2
- $A/B^2 = K$
- If A is plotted against B, the resulting graph line is a parabola.
- **Inverse Proportionality** – When one quantity increases, the other quantity decreases.
 - A is inversely proportional to B.
 - $AB = K$
 - If A is plotted against B, the resulting graph line is a hyperbola.
- Physical quantities are also classified as vector and scalar quantities.
 - Scalar quantities are quantities with magnitude only.
Ex. Area, density, distance, speed
 - Vector quantities are quantities with both magnitude and direction.
Ex. Displacement, velocity, force
 - Vector quantities can be represented by an arrow, the tail represents the magnitude and the arrowhead represents the direction.

Mechanics

- **Force**
 - Push or pull that changes or tends to change the motion of the body
 - It is a vector quantity.
 - Its unit is newton or kg-m/sec² in MKS.
 - Sets of forces
 - **Parallel forces** – forces whose lines of action are parallel to each other
 - **Concurrent forces** – forces whose lines of action meet at a common point
 - **Friction** – a force between two surfaces that opposes the motion of the body
 - **Laws of Equilibrium**
 - First law of equilibrium – for a body to be at rest, the sum of all the forces should be equal to zero.
 - Second law of equilibrium – for a body to be at rest, the sum of all the torques should be equal to zero.
- **Motion**
 - Continuous change in position of a body with respect to a reference point
 - **Speed** is the rate of motion.
 $v = d/t$, where v = speed, d = distance, and t = time
 - **Velocity** is also the rate of motion, only it is a vector.
 - **Uniform motion** is one where the speed or velocity remains constant.
 - **Acceleration** is the change in velocity at a given time interval.
 - **Rectilinear motion** – motion along a straight path. An example of this is the motion of a freely-falling body.
 - **Curvilinear motion** – motion along a curved path. Circular motion and projectile motion are examples of curvilinear motion.
 - **Newton's Three Laws of Motion**
 - **Law of Inertia** – a body at rest will remain at rest and a body in motion will

continue to move in uniform motion unless acted upon by an unbalanced force

- **Law of Acceleration** – When acted upon by a net force, the body will accelerate, and that the acceleration of the body is directly proportional to the force applied and inversely proportional to the mass of the body $F = ma$.
- **Law of Interaction** – for every action, there is an equal but opposite reaction.

• **Work**

- The product of the force applied and the displacement through which the force is directed.

$W = Fd$, measured in N.m or joule (J)

- Can be done easier and faster with the use of machines

- **Machine** is any device that helps in doing work.
- The **six simple machines** are: lever, pulley, wheel, and axle, inclined plane, screw and wedge.
- Mechanical advantage is the number of times a machine can do work compared to the work done without a machine.
- The efficiency of a machine is the ratio of work output to work input.
Efficiency = work output/work input x 100%

• **Power**

- Rate of doing work
 $P = W/t$, measured in joule/sec or watt

• **Energy**

- Ability of doing work
- **Forms of energy**
 - **Mechanical energy**
 - **Potential energy (PE)** – energy due to position of the body
 $PE = mgh$, where m = mass of the body, h = height and g = acceleration due to gravity which is 9.8 m/s^2 .
 - **Kinetic energy (KE)** – energy of the body which is in motion
 $KE = 1/2mv^2$
 - **Electrical energy** – energy of the moving electrons
 - **Chemical energy** – energy in the atoms and molecules of matter
 - **Nuclear energy** – energy resulting from the fission or fusion reactions in atoms
 - **Radiant energy** – energy in the form of electromagnetic waves
 - **Thermal energy** – internal energy of a body which is the total kinetic energy of the molecules of a body

Fluids

- **Density** is the mass of the substance at a given volume, $d = m/V$.
- If the density of the material or substance is greater than the density of the liquid, the material will sink in the liquid; if the density of the material or substance is less than the liquid, the material will float in the liquid.
- The greater the density of the liquid, the greater is the buoyant force.
- **Pressure** is the force exerted at a given unit area, $p = F/A$

- **Liquid pressure** depends on the density and height of the liquid. Liquid pressure, $p = dhg$ where p is the liquid pressure, d is the density of the liquid and h is the height of the liquid. g is a constant and does not affect the pressure.
- **Atmospheric pressure** or **air pressure** depends on the height. The higher the position, the thinner is the air, and the lesser is the air pressure.

Heat and Temperature

- **Heat** is the total kinetic energy of molecules of a body.
- **Temperature** is the average kinetic energy of molecules.
- **Temperature conversion**
 - $^{\circ}\text{C} = \frac{5}{9} (^{\circ}\text{F} - 32)$
 - $^{\circ}\text{F} = \frac{9}{5} \text{C} + 32$
 - $^{\circ}\text{K} = ^{\circ}\text{C} + 273$
- **Methods of Heat Transfer**
 - **Conduction** – heat transfer from one molecule to adjacent molecule
 - **Convection** – heat transfer by the actual movement of heated molecules. The circulation of the heated molecules is called **convection current**.
 - **Radiation** – transfer of heat across space in the form of electromagnetic waves
- **Change of Phase**
 - **Melting or Fusion** – change of phase from solid to liquid at its normal melting point
 - **Freezing or Solidification** – change of phase from liquid to solid at its normal freezing point
 - **Vaporization** – change of phase from liquid to gas at its normal boiling point
 - **Condensation** – change of phase from gas to liquid
 - **Sublimation** – change of phase from solid to gas without passing the liquid state

Wave and Sound

- A **wave** is a disturbance propagated through a medium. There are two kinds of waves: transverse wave and longitudinal wave.
- The properties of waves are reflection, refraction, diffraction, and interference.
- **Sound** is a longitudinal wave, coming from a vibrating source, transmitted through a medium and interpreted by the sense of hearing.

Light

- Theories about the nature of light:
 - **Corpuscular theory** – Light is made up of particles known as corpuscles.
 - **Wave theory** – Light is a wave.
 - **Electromagnetic Wave theory** – Light is composed of electric field and magnetic field.
- Since light behaves as a wave, it can be reflected, refracted, diffracted, and interfered.
- The image formed by a mirror is due to reflection of light. The kinds of mirror are plane and curved mirrors. Curved mirrors can be convex or concave.
- The image formed by a lens is due to refraction of light. Lenses are categorized as diverging and converging.

Electricity

- **Classifications of electricity:**
 - **Electrostatics** is electricity at rest. It only involves electric charges and their behaviour. Electric charges are of two kinds:
 - Negative charge – when there is an excess of electrons
 - Positive charge – when there is a deficiency of electrons
 - **Current electricity** – electricity due to moving charges.
- **Basic components of electric circuit:**

Quantity	Unit	Measuring Device
Current	Ampere	Ammeter
Voltage	Volt	Voltmeter
Resistance	Ohm	Ohmmeter
- Ohm's Law states that the current flowing is directly proportional to voltage and inversely proportional to resistance, $I = V/R$.

Electromagnetism

- **Magnetism** – derived from Magnesia, an island in the Aegean Sea
 - **Magnet** – an object that attracts magnetic objects like metals
 - A magnet has two poles, north and south.
 - Like poles repel; unlike poles attract.
- **Generator** – a device that changes mechanical energy to electrical energy
- **Motor** – a device that changes electrical energy to mechanical energy

Nuclear Energy

- **Atom** – The smallest and indivisible particle of matter it is composed of sub-particles, namely: electron, which carries negative charge, proton which is positively charged; and neutron, which is neutral.
- **Types of Nuclear Reaction:**
 - **Fission** – when a heavy nucleus splits together with the release of energy
 - **Fusion** – when two light nuclei combine with the release of energy

Chemistry

- Study of the composition and changes in matter

Matter

- Anything that has mass and has volume
 - Mass** – amount of the substance indicative of the inertia it possesses
 - Volume** – space occupied
- **Classification of Matter:**
 - **Pure substances** – made of definite kind of material
 - **Elements** – made of the same atoms
Ex. Gold, aluminium, oxygen, carbon, platinum, etc.
 - **Compounds** – made of 2 or more different atoms in definite composition or



ratio and which cannot be separated by ordinary physical means

Ex. Water – H_2O – composed of 2 atoms of hydrogen and 1 atom of oxygen

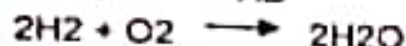
Glucose – $C_6H_{12}O_6$

- **Mixtures** – physical combination of 2 or more substances which can be separated by mechanical means
 - **Homogeneous mixture** – a mixture in which the molecules are thoroughly mixed; a mixture that is uniform throughout.
 - Solutions are homogeneous mixtures. The components of solution are **solute**, which is the dissolved particles, and the **solvent**, which is the dissolving particles.
 - **Heterogeneous mixture**
 - **Suspensions** – heterogeneous mixture where particles are too large that they settle at the bottom of the container
 - **Colloids** – heterogeneous mixtures whose particles are not large enough to settle nor small enough to be dissolved, like the Tyndall Effect, which is the scattering of light by the particles
- **Methods of Separating Mixtures**
 - **Filtration** – use of filter paper to separate liquid from solid components
 - Filtrate – liquid that passes through the filter paper
 - Residue – substance that did not pass through the filter paper
 - **Decantation** – pouring off a layer of liquid from a mixture
 - **Magnetism** – use of magnets to separate magnetic materials from the nonmagnetic ones
 - **Centrifugation** – Substance is subjected to circular or rotational motion in a centrifuge.
 - **Distillation** – Liquid is set to boiling. Vapour is collected and later cooled in order to condense.
 - **Chromatography** - passing mixture in solution or suspension or as a vapour (as in gas chromatography) through a medium in which the components move at different rates
- **Properties of Matter**
 - **Physical properties:**
 - **Extensive** – depends on the amount of the substance
Ex. mass, volume, weight, pressure
 - **Intensive** – does not depend on the amount of substance
Ex. density, taste, colour, smell, specific heat capacity, melting or boiling point, malleability, etc.
 - **Chemical Properties**
 - Property exhibited due to composition of the substance
Ex. combustion, corrosion, decomposition, etc.
- **Changes In Matter**
 - **Physical change** – change in form and appearance but not in the composition
 - **Change in phase:** melting, evaporation, condensation, sublimation, freezing

- Change in form or shape: cutting, breaking, folding, etc
- **Chemical change** – change in the composition resulting in the formation of a new substance
 - Observed when substances react with each other and form a different substance
 - Forming precipitates, evolution of gas, release or absorption of heat, change in color and taste, production of sound and light

- **Types of Chemical reactions**

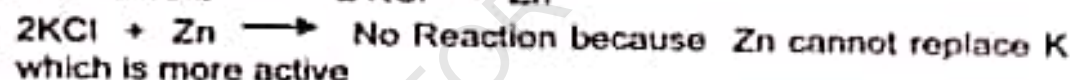
- **Combination or synthesis** – 2 simple substances combine to form one new substance



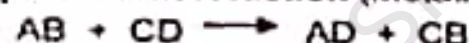
- **Decomposition Reaction** – a substance breaks down into simpler substances



- **Single Replacement Reaction** – a more active element replaces a less active atom in a compound



- **Double Replacement Reaction (Metathesis)** – replacement of the ions



- **Neutralization** – reaction between an acid and a base resulting in the formation of salt and water



- **Law of Conservation of Energy**

- During chemical reaction, the energy before and after the reaction is constant. Energy is not created nor destroyed, but only transformed.

- **Law of Conservation of Mass**

Total mass of the reactants before the reaction is equal to the total mass of the products after the reaction.

- **Phases of Matter**

- **Solid** – molecules are compact; small intermolecular spaces; strong molecular attraction; molecular motion limited to vibrations; definite shape, size and volume
- **Liquid** – molecules are farther apart than solids, larger intermolecular spaces; weaker intermolecular force of attraction; molecules move past one another, definite volume; takes the shape of container
- **Gas** – molecules are far apart; large intermolecular spaces; very weak intermolecular force of attraction; molecules are free to move; no definite volume; takes the shape of the container

Kinetic Molecular Theory of Gas

- **Kinetic Energy** – energy in motion
- **Postulates:**
 - Gas is a collection of particles that travel randomly along a straight-line path.
 - Molecules do not occupy a definite volume.
 - Molecules exhibit perfectly elastic collision.
 - Molecules do not exhibit attraction nor repulsion.

Gas Laws

- **Boyle's Law:**

At constant temperature, the pressure exerted by a gas is inversely proportional to its volume.

- The graph of pressure against volume is hyperbolic.
- When volume is doubled, the pressure is reduced to one-half.
- $P_1V_1 = P_2V_2$

- **Charles' Law**

- At constant pressure, the volume of a gas is directly proportional to its temperature expressed in Kelvin.
- The graph of volume against the temperature is a straight diagonal line. Increasing the temperature (K) twice will double the volume of the gas.
- $V_1T_2 = V_2T_1$

- **Gay-Lussac's Law**

- At constant volume, the pressure exerted by gas is directly proportional to the temperature in Kelvin.
 - The graph of pressure against the temperature is a straight diagonal line.
 - Increasing the temperature twice will double the pressure exerted.
 - $P_1T_2 = P_2T_1$

- **Combined Gas Law**

- $P_1V_1T_2 = P_2V_2T_1$

- **Avogadro's Law**

- The amount of gas is directly proportional to its volume.
- A mole of gas at standard temperature and pressure (STP), at 0°C and 1 atmosphere, occupies a volume of 22.4 liters (L).
- When the amount is doubled, the volume is also doubled.

- **Ideal Gas Law:**

- $PV = nRT$ where: P – pressure; V – volume; n – number of moles; T temperature in Kelvin; R is the gas constant: 8.3145 J/mol-K

Atomic Structure

• Atom

- Building blocks of matter
- Composed of 3 particles
 - **Protons** – positively charged; located in the nucleus
 - **Electrons** – negatively charged; orbiting around the nucleus
 - **Neutrons** – no charge particle located in the nucleus
- Has equal number of protons and electrons
- **Atomic number** is the same as the number of protons, while the **atomic mass** is the sum of the number of protons and neutrons
- Exists in different versions called **Isotopes**

Ex. Isotopes of hydrogen : protium (H-1) = 1 proton, 1 electron, 0 neutron

Deuterium (H-2) = 1 proton, 1 electron, 1 neutron

Tritium (H-3) = 1 proton, 1 electron, 2 neutron

• **Formula Mass** – mass of an ionic compound

Ex. $\text{Ba}(\text{OH})_2$: $\text{Ba} = (1 \text{ Ba atom})(137.3 \text{ g/mol atom}) = 137.3 \text{ g}$
 $\text{O} = (2 \text{ O atoms})(16 \text{ g/mol atom}) = 32.0 \text{ g}$
 $\text{H} = (2 \text{ H atoms})(1 \text{ g/mol atom}) = 2.0 \text{ g}$
 Formula mass = 171.3 g

• **Molar Mass** – mass of one mole of a substance which is equivalent to the formula mass expressed in grams.

The molar mass of the $\text{Ba}(\text{OH})_2$ in the above example is 171.3g/mol.

• **Percentage Composition of a Compound**

Ex. Find the % composition of $\text{Ba}(\text{OH})_2$

$\text{Ba} : (137.3/171.3)(100\%) = 80.2\%$

$\text{O} : (32.0/171.3)(100\%) = 18.7\%$

$\text{H} : (2.0/171.3)(100\%) = 1.2\%$

Percent composition = 100%

• **Chemical Bonds** – forces that join atoms together to form molecules

- **Ionic Bond** – formed between charged particles called ions due to electrons transfer
 Ionic bond forms the NaCl compound.
- **Covalent Bonds** – formed when atoms share electrons
- **Non-Polar Covalent** – bond between atoms of the same kind that share electrons equally.
- **Polar Covalent** – bonds between atoms of molecules that do not equally share the electrons
- **Metallic Bonds** – bonds between metals



EARTH SCIENCE

- Branch of physical science that deals with the physical aspects of the Earth
- The study of Earth Science encompasses the following topics:
 - Earth's origin and its relation to other celestial bodies
 - Earth's structure and composition
 - Earth's activities that bring about changes in the condition of the planet

Origin of the Earth

- The "Big-Bang Theory" is the closest in explaining how the Earth evolved.
- In this theory, the Earth started to exist at the time the solar system originated.
- The other theories that may explain the origin of the Earth are the following:
 - Planetesimal Theory
 - Binary Stars Theory
 - Dust-Cloud Theory

The Solar System

- Composed of several celestial bodies, namely:
 - The sun, which is the central figure of the system
 - The planets, which are generally characterized by its revolution around the sun
 - The moons or natural satellites that are revolving around its mother planet
 - The asteroids, meteors, meteorites, and comets
- The solar system is composed of inner planets, Mercury, Venus, Earth, and Mars. The outer planets are Jupiter, Saturn, Uranus, and Neptune.
- Pluto, which was once an outer planet, was temporarily excluded in the solar system because astronomers were not able to track the movement of the planet.

Our Moon: its Effects on the Earth

- Lone natural satellite of the Earth
- It makes one complete revolution in $27 \frac{1}{3}$ days but its apparent revolution, that is, completing its phases, takes $29 \frac{1}{2}$ days.
- One effect of the moon is the occurrence of eclipse. **Solar eclipse** occurs when the moon passes directly between the Earth and the sun at each new moon.
- **Lunar eclipse** occurs when the moon passes through the shadow of the earth, and this happens during full moon.
- Another effect of the moon is the occurrence of tides on the Earth. **Spring tides** occur when the moon and the sun are aligned every two weeks (new moon and full moon) resulting in high tides, which are higher than usual.
- **Neap tides** occur when the sun and the moon are at right angles to each other (first and last quarters), resulting in low tides, which are lower than usual.

The Motions of the Earth and their Effects

There are two principal motions of the Earth:

- **Rotation**
 - The Earth rotates on its axis every 23 hours, 56 minutes, and 4.09 seconds. This is the length of the day on Earth.
 - As the Earth rotates, half of the Earth is lighted by the sun. This part experiences daytime. The other side which does not receive light experiences night time.
- **Revolution**
 - The Earth revolves around the sun in 365 $\frac{1}{4}$ days. This is the length of the year.
 - The seasonal change is caused by the tilt of the Earth and the angle of sunlight where it hits the Earth.

The Four Spheres (Layers) of the Earth

- **Lithosphere**
 - It is the solid, rocky crust covering the entire planet. It is made up of the following layers that extend from the outermost to the innermost part of the Earth: crust, mantle, outer core, and inner core.
- **Hydrosphere**
 - It is composed of all the water on or near the surface of the Earth.
 - This includes the oceans, lakes, rivers, streams and even the moisture in the air.
- **Atmosphere**
 - It is the body of air that envelops the Earth.
 - The atmosphere is composed of 80% nitrogen and just around 16% oxygen. The small amount remaining, about 4%, is made up of other gases, such as carbon dioxide, helium, argon and neon.
- **Biosphere**
 - The biological sphere came from "bio," which means life.
 - It is composed of all living organisms – the plants, animals, and other organisms.

Composition of the Earth

The Earth is composed of the materials and substances found in the lithosphere, hydrosphere, and atmosphere, which are solid materials, water, and air, respectively. Of these three, the solid materials are predominantly visible and diversified. These are the minerals, rocks, and soil.

- **Minerals**
 - **General Characteristics**
 - Single elements or compounds found naturally in the Earth's crust
 - Inorganic, that is, not derived from living things
 - Solids
 - **Visible Characteristics**
 - **Color** – the most dominant color of the mineral



- **Streak** – the color of a thin layer of a mineral. This is sometimes seen as a line of color in a mineral.
- **Luster** – the ability of a mineral to reflect, refract, or absorb light falling on its surface
- **Crystal form** – the geometric pattern of the mineral due to the arrangement of atoms and molecules
- **Cleavage and fracture** – definite flat surfaces that yielded after the breaking of minerals
- **Distinguishing Properties**
 - **Hardness** – the ability of a mineral to scratch or cut another mineral
 - **Specific Gravity** – density of the substance compare to the density of water.
- **Classification of Minerals**
 - **Siliclous** – ex. Quartz, Agate, Hornblende
 - **Non-metals** – ex. Calcite, Sulfur, Gypsum
 - **Metal-ore** – ex. Gold, Silver, Copper
 - **Gems** – ex. Jade, Topaz, Opal
- **Rocks**

The study of rocks is called **Petrology**. They are classified according to their color, texture, composition, and origin. Based on their origin, rocks are classified as:

 - **Igneous rocks**
 - Formed as magma cools off and crystallizes
 - Those formed inside the Earth are called intrusive, while those outside are called extrusive rocks.
 - Examples are volcanic tuff, granite, and basalt.
 - **Sedimentary rocks**
 - Formed from sediments of weathered materials
 - Used to tell much of Earth's history because these rocks contain fossils
 - Examples of these rocks are limestone, gravel, and pebbles
 - **Metamorphic rocks**
 - These rocks undergo stages of development due to heat, pressure, and chemical reaction.
 - Examples of these rocks are slate, marble, and graphite.
- **Soil**

Soil is a natural resource formed out of weathered rocks with humus on the Earth's surface. It consists of visually and texturally distinct layers which are classified as:

 - **Topsoil** – the uppermost layer with decomposed organic matter, mixed with small amount of minerals. Plants mostly grow in topsoil because it is rich in humus.
 - **Subsoil** – This is also called mineral layers because it is composed of the soil and its parent mineral. This contains less humus and also acts as reservoir of water.

• **Soil Conservation**

The following are some of the ways to conserve soil:

- **Contour plowing** – plowing in furrows which follow the contours of the land
- **Strip cropping** – planting crops arranged in alternate bands of row crops and cover crops
- **Terracing** – construction of steplike ridges following the contours of the slopes of the land
- **Crop rotation** – alternate planting of row crops one year with cover crops the next year

Activities that Change the Condition of the Earth

- **Weathering** – the breaking down of rocks by physical or chemical means
- **Erosion** – the transport of weathered materials
- **Diastrophism** – the movement of the landforms
- **Continental drift** – a theory that explains the drifting of one continent to its present position
- **Plate tectonics** – This theory explains that the outermost rigid layer of the Earth, called plates is continuously moving. This also explains the cause of earthquakes, volcanoes, ocean trenches, and formation of mountains.
- **Rock deformation** – refers to the changes in volume and shape of rocks that may cause fractures and lead to the formation of faults.
- **Earthquake** – vibration resulting from the movement along the existing fault line. The point where the vibration originated is called **focus**. The point on the surface which is directly above the focus is known as **epicenter**.
- **Volcanism** – covers all kinds of volcanic activities which include formation of volcanoes, building, and expulsion of magma. Volcanoes may be classified according to cones, which are:
 - **Cinder cone** – with narrow base and steeper slope
 - **Shield cone** – with broad base and gentle slope
 - **Composite cone** – with a nearly perfect shape whose slope is gentle at the base and become steeper as one approaches the peak.

Weather and Climate

- **Meteorology** is the study of weather and climate.
- **Weather** refers to the atmospheric condition at a particular place and time.
- **Climate** is the pattern of weather in a bigger land area over a long period of time.
- **Tropical disturbances** are classified as tropical depression, storm, and typhoon.

These disturbances are measured by the strength of accompanying winds.

- The prevailing winds in the Philippines are the northeast monsoon and the southwest monsoon.



BIOLOGICAL SCIENCE

BIOLOGY

- Coined from the Greek words 'blos' meaning life, and 'logos' meaning study of.
- Study of the living matter both plants and animals and the plant-like and animal-like organisms.
- Branches and fields of study
 - **Anatomy** – macroscopic structure of multi-cellular organisms
 - **Biochemistry** – chemicals of life, like carbohydrates, proteins, amino acids
 - **Botany** – structure, characteristics, classification of plants
 - **Cytology** – properties and structure of the cell
 - **Embryology** – embryo or developing organism
 - **Entomology** – kinds, characteristics, and structure of insects
 - **Ecology** – interaction between the organism and its environment
 - **Genetics** – heredity or transmission of traits from parents to offspring
 - **Herpetology** – reptiles and amphibians
 - **Ichthyology** - fish
 - **Molecular biology** – structure and function of molecules in the living things
 - **Morphology** – form and structure of living organisms
 - **Microbiology** – structure of microorganisms and their effects on other organisms
 - **Marine biology** –marine organisms
 - **Ornithology** – birds
 - **Parasitology** – parasites
 - **Pathology** – disease-causing organisms
 - **Paleontology** – fossils
 - **Physiology** – physical function of living organisms
 - **Zoology** – animals

Living Things

- Composed of cells
 - Unicellular (1 cell) or multicellular (many cells)
- Exhibit levels of organization; complex and organized
 - Cell – tissue – organ – system – organism
 - Cell as the basic unit of structure
- Grow and develop
 - Increase in size (grow) develop (less to more complex organism) through cell division
 - Mitosis** – division in somatic or body cells
 - Meiosis** – division of gametes or sex cells
- Respond to stimulus/stimuli
 - Ability to react to any change in condition
 - Stimulus** – the condition to which an organism reacts/responds

- **Need and Transform energy**
 - Energy is required in order for the organism to grow, develop, and later reproduce.
- **Adapt or adjust**
 - Ability to adjust to conditions in its environment in order to survive
 - Tropism:** phototropism (light); hydrotropism (water); geotropism (gravity)
 - Mimicry, camouflage, growing thorns, producing chemicals that affect the other organisms
- **Reproduce – the ability to produce a new organism of its own kind**
 - **Asexual** – no sex cells involved; one parent only; binary fission, budding, spore formation, regeneration, vegetative propagation
 - **Sexual** – sex cells or gametes are involved; male and female parents

Microscope

- A tool necessary in the study of the organisms
- Composed of lenses used to produce image of the organism under study. The images are magnified many times so the eyes can see.
- Come in different types:
 - **Light Microscope** – refracts light using lenses to magnify the image formed. Its usual magnification: 40x, 100x, 400x
 - **Stereoscope** – 3D view, 10-20x magnification, viewing with 2 eyes
 - **Scanning Electron Microscope** – uses beam of electrons to magnify an specimen two million times
 - **Transmission Electron Microscope** – uses beam of electrons that pass through the specimen allowing to view the inner parts of the specimen

Cell Theory

- Describes or explains what a cell is.
 - Cell is the basic component of a living organism.
 - New cells are produced from existing cells.
 - Cell is the building block of life.
- **Kinds:**
 - **Prokaryotic** – has no nucleus; ex. Bacteria
 - **Eukaryotic** – has true nucleus; ex. Mammals
- **Parts / Structures**
 - **Cell Wall** – protects and supports the plant cell; allows entry of water and other substances into the cell
 - **Cell membrane** – between the cell and its environment which maintains homeostasis
 - **Nuclear membrane** – controls that substances entering and leaving the nucleus
 - **Nucleus** – carries the genetic material and controls cellular activities
 - **Cytoplasm** – Jelly-like part outside the nucleus that houses and protects the organelles
 - **Endoplasmic Reticulum** – network of tubes that carries materials through the cell



- **Ribosome** – synthesize proteins
- **Mitochondrion** – breaks down sugar and release energy; where aerobic respiration takes place
- **Vacuole** – food or water storage
- **Lysosome** – breaks large food molecules to smaller ones
- **Chloroplast** – where photosynthesis takes place
- **Nucleolus** – 'nucleus' inside the nucleus responsible for making ribosomes
- **Golgi Apparatus** – modify and export proteins
- **Centrioles** – separate chromosome pairs during mitosis

- Transport of Materials through Cell Membrane or Plasma Membrane

Whether or not a substance can pass through the plasma membrane is determined by the permeability of the membrane.

- **Permeable** – Almost all substances can pass through.
 - **Impermeable** – No substance can pass through.
 - **Selectively permeable** – Some substances can pass through, while others cannot.
 - **Active Transport** – Energy is required so substances can pass through.
 - **Passive Transport** – substances can pass through without energy use.
 - **Diffusion** – Substance moves or spreads from area of higher concentration to an area of lower concentration
 - **Osmosis** – Spreading of water molecules from region of higher concentration of solute to region of lower concentration.
- Osmotic Pressure** – force exerted by the water molecules that spread through the area
- **Hypotonic** – less solute concentration in the area, so water moves away
 - **Hypertonic** – greater solute concentration in the area, so water moves toward
 - **Isotonic** – equal concentration on both sides of membrane, so water does not diffuse
- **Endocytosis** – engulfing materials
 - **Phagocytosis** – Particles are engulfed in the vacuole.
 - **Pinocytosis** – Fluid is enclosed.

Energy: Requirement for Survival

- **Energy** is a requirement for an organism to survive. It enables one to do work. The energy needed by an organism is basically coming from the food it takes. The food also has taken its energy from the sun.
- **Classification of organisms based on energy**
 - **Producer / autotrophs** – makes its own food through the process of photosynthesis. During photosynthesis, carbon dioxide and water are converted to carbohydrates with the use of sunlight as energy source. The process releases oxygen.

$$6\text{CO}_2 + 6\text{H}_2\text{O} \xrightarrow{\text{light}} \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$$
 - **Consumers/Heterotrophs** - obtain energy from other organisms
 - **First-order consumer / herbivores**

- Second-order consumer
- Third-order consumer
- **Decomposers/Saprotrophs** – special type of consumers because they obtain energy by consuming both the producers and consumers when they are already dead.

Cell Division

- **Mitosis** – Body cells (somatic cells) are produced for growth and development.
- **Meiosis** – Gametes or sex cells are produced, and these are necessary for reproduction.
- **Cell Cycle** – exhibits the stages in the life of a cell

$$G1 \rightarrow S \rightarrow G2 \rightarrow M$$
 - **Synthesis / Replication (S)** – the making of the exact copy of the genetic material so the daughter cells will have the same genetic make-up as the parent cell
 - **Gaps (G1 and G2)** separate DNA Synthesis from Mitosis; mediation to switch in cellular activity
 - **Mitosis (M)** – separation of the copied genetic materials into the daughter cells
- **Stages of Mitosis**
 - **Interphase** – resting stage; includes the G1, S, and G2
 - **Prophase** – chromosomes become visible and condensed; nuclear membrane disappears; centrioles move toward opposite poles; spindle fibres from centrioles and attach to chromatids
 - **Metaphase** – centrioles reach the poles; chromosomes line up in the equator.
 - **Anaphase** – spindles shorten, pulling the sister chromatids apart.
 - **Telophase** – chromosomes become less condensed; nuclear envelope forms; cytokinesis happens.
- **Cytokinesis** – breaking or division of the cytoplasm to form the daughter cells.
- **Meiosis** – reduces the chromosome by one-half (diploid cell becomes haploid)
 - Haploid daughter cells become diploid after fertilization.
 - Half of the chromosomes come from the female gamete, while half comes from the male gamete.
- **Cancer** – develops when the cells divide uncontrollably



Multiple choice.

1. Which of the following statements does a scientist always consider?
- A. Experiments provide information that will always support predictions.
 - B. Accurate observations are necessary to organize problems.
 - C. Predictions do not point the way to possible solutions to problems.
 - D. Graphs are not useful in interpreting data.

Answer A. Experimentation is the most important step in scientific method. The results of the experiment or the information provided by the experiment will answer the problem and will prove or disprove the hypothesis.

2. Arrange the following steps in the most logical order of scientific investigation.
- I. Formulate the hypothesis
 - II. Perform the experiment
 - III. State the problem
 - IV. Draw the generalization

- A. III, II, I, IV
- B. III, IV, II, I

- C. III, I, II, IV
- D. III, II, IV, I

Answer C. In scientific method, the logical steps in solving problem start with the problem, followed by hypothesis, then experimentation and generalization. Generalization includes conclusion and application.

3. What is the importance of control set-up in experimentation?
- A. It will identify the variables in the experiment.
 - B. It will provide a basis of comparison between treated and untreated subjects.
 - C. It will provide a basis for measuring data.
 - D. It will eliminate mechanical errors in experimentation.

Answer B. Control set-up is one where all the factors in an experiment or any scientific investigation are removed or changed. For this matter, it serves as the basis of comparison between the treated and the untreated subjects.

4. The process of comparing one quantity with the corresponding standard is called _____.
- A. Measurement
 - B. Formulation
 - C. Interpolation
 - D. Extrapolation

Answer A. Measurement is comparing one quantity with the corresponding standard. For example, if we wish to measure the length of the table, we use a meterstick, which is supposed to be the standard device for measuring length. We then compare the length of the table with the length of the meterstick.

5. Which of the following relationships is correct?

- A. A meter is less than a kilometer.
- B. A second is more than a minute.
- C. A centigram is equal to a milligram.
- D. A decimeter is greater than a decameter.

Answer A. One kilometer is 1,000 meters. It is obvious that a meter is less than a kilometer.

6. The relationship between quantity A and B is expressed in the equation, $AB = K$. What graph line is formed if quantity A is plotted against quantity B?

- A. Straight line slanting to the right
- B. Horizontal line
- C. Parabola
- D. Hyperbola

Answer D. Quantity A is inversely proportional to B if the equation is $AB=K$; thus, if A is plotted against B, the graph line will be hyperbola.

7. The three states of matter are solid, liquid, and gas. What is the fourth state of matter

- A. Cell
- B. Ion
- C. Plasma
- D. Quarks

Answer C. The fourth state of matter is plasma, but this occurs only at very high temperatures and consists of ionized atoms, that is, at this temperature, the electrons are separated from the nuclei.

8. A student walks from home to school 3 blocks away and then returns home for lunch. After lunch, he goes back to school. His total displacement is _____.

- A. 0 block
- B. 3 blocks
- C. 6 blocks
- D. 9 blocks

Answer B. Displacement is a vector, so, the direction is also to be considered. When the student returns home, the 3 blocks displacement is cancelled and going back to school will give 3 blocks.

9. In the race between the man and the dog, the man appeared to move faster than the dog and finished the race ahead. How will you describe the motion of the two during the race?

- A. The man travelled a longer distance than the dog.
- B. The dog used a longer time to finish the race.
- C. The man used a shorter time to finish the race.
- D. The man and the dog travelled the same distance at the same time.

Answer C. The speed of a body depends on two factors: distance and time. The longer the distance travelled in a shorter time interval will result in greater speed.

Since it is a race, the man and the dog travelled the same distance. The man used a shorter time and so, he travelled faster and finished the race ahead.

10. Which of the following is true when a block of wood lays at rest on top of the table?
- A. There are no forces acting on the block.
 - B. The sum of the forces acting on the block is zero.
 - C. The block is in uniform motion.
 - D. The block is accelerated.

Answer B. For a body to be at rest, the sum of the forces acting on a certain body should be zero. There are forces acting on the block, but since there is no resultant force acting on the block, this block stays at rest on top of the table.

11. What property of matter is involved in the sign, "Fasten Your Seat Belt"?
- A. Impenetrability
 - B. Inertia
 - C. Acceleration
 - D. Interaction

Answer B. Inertia is the ability of a body to remain at rest when at rest and to continue moving in uniform motion when it is in motion unless acted upon by a net force. When a sudden stop is made in a car which you are riding, your body moves forward, thus, there is a need to fasten your belt.

12. Which laboratory equipment/s is/are used to find whether a ring is genuine or fake?
- I. Platform balance
 - II. Graduated cylinder
 - III. Beaker
- A. I only
 - B. II only
 - C. I and II only
 - D. I, II and III

Answer C. Density is a distinguishing property of a material, therefore, it can be used to find whether a ring is fake or genuine. Density is mass/volume, so, what is needed is the platform balance to measure the mass and the graduated cylinder to measure the volume by displacement.

13. If three-fourths of the block of wood floating in water is below the surface of the water, the density of this block of wood is _____.
- A. Greater than the density of water
 - B. More than half the density of water
 - C. Equal to the density of water
 - D. Less than half the density of water

Answer B. Whatever is the part of the floating object submerged in water, it is the density of the object. Three-fourths of the block is submerged in water, so its density is three-fourths, which is more than one-half the density of water.

14. A razor blade can float on the surface of water because of _____.

A. Surface tension
B. Capillarity

C. Buoyant force
D. Liquid pressure

Answer A. Surface tension is a property of liquids arising from the molecular cohesive forces at or near the surface. As a result, the surface tends to contract, resembling a stretched elastic membrane. That's why a razor blade can float in water despite its greater density than water.

15. Pressure is defined as force exerted at a given unit area. Its unit is newton per square meter and is equivalent to _____.

A. pascal
B. joule

C. watt
D. ampere

Answer A. The equivalent unit given to pressure is pascal. This is to honor Blaise Pascal who pioneered on the concept of hydraulics, such as hydraulic brakes, hydraulic lift, dentist chair, hydraulic jack, etc.

16. A rectangular block of iron has a dimension of 5 cm x 10 cm by 15 cm and weighs approximately 60 N. How should this block be placed on top of the table to exert the greatest pressure on the surface?

A. On the 5 cm x 10 cm side
B. On the 5 cm x 15 cm side

C. On the 10 cm x 15 cm side
D. Any of the sides

Answer A. The 5 cm x 10 cm side has the smallest area. Since pressure, $p = F/A$, is inversely proportional to area, the one with the smallest area will have the greatest pressure exerted.

17. Four circular holes of equal size arranged from top to bottom were cut 2 cm apart into a large plastic bottle. The holes were covered first with tape, and the bottle was filled with water. The tape was then removed simultaneously. Which of the following statements will best describe the result?

A. Water from each hole will squirt out the same distance from the bottle.
B. Water from the top hole will squirt out the farthest.
C. Water from the bottom hole will squirt out the farthest.
D. The sudden release of water will cause the bottle to collapse.

Answer C. The liquid pressure is proportional to the height of the liquid as expressed in the equation, $p = Dhg$, where D is the density of the liquid, h is the height, and g is a constant which is acceleration due to gravity. In other words, the higher the liquid, the greater is the pressure at the bottom.

18. A ball rolls down an inclined plane. Which of the following is not correct to describe the potential and kinetic energy of the ball?

A. The greatest potential energy is at the highest end of the inclined plane.
B. The greatest kinetic energy is at the lowest end of the inclined plane.



- C. Kinetic energy is transformed to potential energy as it rolls downward.
- D. Potential energy is transformed to kinetic energy as it rolls downward.

Answer C. A, B, and D are all correct, while C is not. The ball starts from the highest point; thus, the potential energy is the greatest at that point. As it rolls downward, the ball is accelerated and so the kinetic energy increases. Therefore, the transformation of energy is from potential to kinetic.

19. Which of the following does not describe the temperature of a given body?

- A. It is the total kinetic energy of the molecules of the body.
- B. It is the average kinetic energy of the molecules of the body.
- C. It is the measure of the hotness and coldness of the body.
- D. It is measured by a thermometer.

Answer A. The total kinetic energy of the molecules of the body is known as heat. B, C, and D are all pertaining to temperature.

20. Which of the following explains why a housewife is able to hold the handle of a heated pan?

- I. The handle is made up of good insulator of heat.
- II. The handle has a very low thermal conductivity.
- III. The handle expands when heated.

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II, and III

Answer A. I and II explain why the housewife is able to hold the handle of a heated pan. It is true that most solid expands when heated, but it is not a good reason to tell that expansion conducts heat.

21. Which statement best explains why ice is a good refrigerant?

- A. It cools by evaporation.
- B. It absorbs heat while melting.
- C. It conducts heat quickly.
- D. It does not melt inside the refrigerator.

Answer B. A substance becomes cold when heat is removed from that substance. For example, when ice is placed in water, the heat from the water is absorbed by the ice making the water cold due to the removal of heat.

22. Bats, which are nocturnal, can fly swiftly in the dark without bumping to a barrier. This is because _____.

- A. They have supersensitive eyes in the dark.
- B. They project fluorescence light in the dark.
- C. They have sonar at the tip of the nose that emits sound waves.
- D. They are made invisible at the dark.



Answer C. The sonar at the tip of the nose of the bat emits sound waves and these waves after hitting the barrier in front will be reflected and goes back to the sonar to tell that there is a barrier in front.

23. Which of the following statements is true for both sound and light waves?
- Both can be reflected.
 - Both travel in a medium.
 - Both have similar wavelengths.
 - Both travel as a longitudinal wave.

Answer A. Reflection is a general property of a wave. Since sound and light are both waves, both can be reflected. Sound cannot travel without a medium. Sound and light have different wavelengths. Sound is longitudinal, while light is a transverse wave.

24. Where does visible light fall on the electromagnetic spectrum?
- Between infrared and ultraviolet radiations
 - Between FM and AM radio waves
 - Between microwaves and infrared radiations
 - Between gamma rays and x-rays

Answer A. Infrared means "before red" while ultraviolet means "beyond violet". Thus, the visible light, which is responsible for seeing an object, is located between infrared and ultraviolet radiations.

25. What kind of mirror is best used as side mirrors in cars?
- Concave
 - Convex
 - Plane
 - Irregular

Answer B. The image formed by a convex mirror is always upright, virtual, and located near the aperture of the mirror whether the vehicles following the car is near or far. The image formed by the concave mirror is very deceiving because there are cases in which the image formed is inverted or even no image is formed. The image formed by the plane mirror is also deceiving because the size and distance of the image are always the same as the size and distance of the object.

26. An object becomes positively charge if it _____.
- Gains protons
 - Gains electrons
 - Loses protons
 - Loses electrons

Answer D. An object becomes positively charge when it loses electrons, so that the number of protons is greater than the number of electrons. The electrons are the only ones that can transfer to or from the atoms, not the protons.

27. Which of the following equations shows the relationship between the current and voltage when resistance remains constant?

A. $I = V/R$
B. $R = IV$

C. $I = VR$
D. $V = I/R$

Answer A. In Ohm's Law, the current in a circuit is directly proportional to voltage, thus, $I = V/R$.

28. If three light bulbs of different wattages are connected in series to a battery, which of the following will be true?

A. The resistance in each light bulb will be the same.
B. The current in each light bulb will be the same.
C. The light output of each light bulb will be the same.
D. The power consumed by each light bulb will be the same.

Answer B. In a series circuit, the current passing in each resistor (light bulb) is the same.

29. The force that holds the atoms and molecules together to form larger bodies is known as _____.

A. Gravitational force
B. Electromagnetic force

C. Weak nuclear force
D. Strong nuclear force

Answer D. The strong nuclear force is responsible for holding electrons together in an atom. This force is also the one that holds the atoms and molecules together to form matter.

30. A radioactive sample has a mass of 16 grams, and its half life is 10 days. What mass of the original sample remains after 20 days?

A. 1 gram
B. 2 grams

C. 4 grams
D. 8 grams

Answer C. The problem tells that the half life of the given radioactive sample is 10 days, so after 10 days, the mass of the sample becomes 8 grams. It follows, therefore, that after 20 days the mass of the sample becomes 4 grams.

31. Which warning label indicates a chemical property of the material being labelled?

A. Fragile
B. Handle with care

C. Shake well
D. Flammable

Answer D. When the material is labelled Fragile, Handle with care or shake well, the labels will show the physical property of the material because the composition of the material remains as is. Flammability, on the other hand, is a chemical property because after burning the composition of the material changes.

32. Which of the following statements about atomic structure is not true?

- A. Atoms are mostly empty space.
- B. All atoms of the same atomic number are atoms of the same element.
- C. Most of the mass of the atom is concentrated at the nucleus.
- D. An atom can gain or lose electrons without changing its charge.

Answer D. When an atom gains or loses electrons that atom becomes electrically charge. If it gains electrons, it will become negatively charged. If it loses electrons, it will become positive.

33. Which of the following sub-atomic particles are located in the nucleus of an atom?

- A. Protons and Neutrons
- B. Electrons and Neutrons
- C. Protons and Electrons
- D. Protons, Neutrons and Electrons

Answer A. Protons and Neutrons are located in the nucleus of the atom. The electrons are the ones revolving around the nucleus.

34. All of the following are charged particles, except

- A. Proton
- B. Alpha particle
- C. Molecule
- D. Ion

Answer C. A molecule is described as uncharged particles. Proton is +1 while alpha particles is +2. Ion is a charged particle because it is an atom that gains or loses electrons, in which case, it will have a net positive or negative charge.

35. Which of the following statements is not true of Dalton's Atomic Theory?

- A. All atoms of an element are identical
- B. A molecule is the smallest particle of an element
- C. Atoms combined in a fixed ratio to form a molecule
- D. Elements are composed of indivisible particles called atom

Answer B. A molecule is not the smallest particle of matter or an element. The atom is regarded as the smallest, indivisible particle of matter. Although, today, an atom is not considered indivisible but rather consisting of sub-particles: proton, neutron and electron.

36. Uranium 238 has an atomic number of 92. It can be inferred that this atom has _____.

- A. 92 protons, 146 electrons, and 92 neutrons
- B. 146 protons, 92 electrons, and 92 neutrons
- C. 92 protons, 92 electrons, and 146 neutrons
- D. 146 protons, 146 electrons, and 238 neutrons

Answer C. For a stable atom like uranium 238, the atomic number is equal to the number of protons, the number of protons is the same as the number of electrons, and the atomic weight is the sum of the number of protons and neutrons.

17. What is the maximum number of electrons that can be accommodated in $n = 4$?

- A. 2
- B. 8
- C. 16
- D. 32

Answer D. The maximum number of electrons that a certain orbit can accommodate is expressed in the formula $N = 2n^2$, where N is the number of electrons and n is the number of orbit. If $n = 4$, then, $N = 2 \times 4^2 = 2 \times 16 = 32$

38. If element X has an atomic configuration of $1s^2 2s^2 2p^2$, what family in the periodic table does this element belong?

- A. Alkali metal
- B. Halogen
- C. Inert gas
- D. Carbon

Answer D. Since the outer atomic configuration of element X is $2p^2$, it is located in group IVA, the carbon family.

39. Ions are formed by _____

- A. Gaining or losing electrons
- B. Gaining or losing protons
- C. Sharing protons
- D. Sharing electrons

Answer A. Ions are charged atoms or molecules that have gained or lost electrons. Protons are not gained nor lost, because they are stationary in the nucleus. Electrons are the ones that move and have the ability to transfer from one atom to another.

40. Which of the following is an extensive property of matter?

- A. Weight
- B. Melting point
- C. Specific gravity
- D. Color

Answer A. Extensive property depends on the amount of material, while intensive property depends on the quality of the material. Weight is an extensive property of matter because it depends on the amount of force exerted by the earth on that material.

41. A given gas occupies 2000 cc at a pressure of 90 cm of mercury. What volume will the same gas occupy if the pressure becomes $\frac{1}{2}$ of the original pressure?

- A. 1000 cc
- B. 2000 cc
- C. 3000 cc
- D. 4000 cc

Answer D. According to Boyle's law, the volume of the confined gas is inversely proportional to its pressure, thus, $P_1V_1 = P_2V_2$. Solving for V_2 , $V_2 = P_1V_1/P_2 = 90 \text{ cm} \times 2000 \text{ cc} / 45 \text{ cm} = 4000 \text{ cc}$.

42. All of the following are characteristics of metals, except _____.

- A. Good conductor
- B. Brittleness
- C. Malleability
- D. Ductility

Answer C. Brittleness is the ability of a certain material to be broken easily. Obviously, it is not a characteristic of a metal. The three other choices are good conductor, malleability and ductility are characteristics of metals.

43. Which is true of metalloids?

- A. They have properties of both metals and non-metals.
- B. They conduct electricity better than metals.
- C. They conduct heat better than metals.
- D. They conduct heat and electricity less effective than non-metals.

Answer A. Metalloids are elements with properties of both metals and non-metals. Example of this is arsenic.

44. Which of the substances below are compounds and not elements or mixtures?

- I. Ethanol
- II. Neon
- III. Water
- IV. Crude oil

- A. III only
- B. I and IV

- C. I and III
- D. I, III and IV

Answer C. The only compounds in the list are ethanol and water. Neon is an element, while crude oil is a mixture.

45. In the given list of mixtures, the one which is considered homogenous is _____.

- A. Alloy, air, sugar solution
- B. Rainwater, petroleum, gravel and sand
- C. Salt solution, sugar, sea water
- D. Water, air, oil

Answer A. Homogenous mixture is made up of only one phase and the composition is the same throughout the mixture. In the given list, alloy, air and sugar solution are homogenous mixtures.

Although choice B is composed of all mixtures, gravel and sand are heterogenous. Choices C and D have compounds in their list, sugar and water respectively.

46. Which of the following substances listed below is considered a colloid?

- A. Water
- B. Alcohol

- C. Soda
- D. Milk

Answer D. Milk is considered a colloid which is a homogenous mixture made up of particles whose size is between the particles of solution and suspension. Solution is homogenous while suspension is heterogenous.

47. Alcohol is considered as an organic compound and derivative of aliphatic carbons. This compound is characterized by _____.

- A. A carbonyl attached to an ether linkage
- B. A carbonyl bonded to two carbons
- C. A halogen attached to an alkyl group
- D. A hydroxyl bonded to an alkyl group

Answer D. Alcohol is considered as a derivative of aliphatic hydrocarbons. The general formula of alcohol is $R-OH$, which means, combination of alkyl group, R and hydroxyl, OH.

48. In a flame test, what will be the color of the flame if there is a presence of boron in a solution?

- A. Brick red
- B. Royal blue
- C. Yellow gold
- D. Bright green

Answer D. A flame test is a procedure in Chemistry to detect the presence of certain metal ions, based on the emission spectrum of an element. Boron has a characteristic green emission spectrum.

49. Which of the following properties are characteristics of acids?

- A. Turn red litmus paper to blue, have pH less than 7 and are electrical conductors
- B. Turn red litmus paper to blue, are electrical conductors and react with metals
- C. Turn blue litmus paper to red, have pH less than 7 and react with metals
- D. Turn blue litmus paper to red, have pH greater than 7 and electrical conductors

Answer C. The characteristics of acid are: blue litmus paper turns red, pH is less than 7, reacts with metals and good conductor of electricity.

50. What will be the possible result if a base and acid react together?

- A. Salt will be formed
- B. No change in base and acid will happen
- C. The mass of base and acid will decrease
- D. The mass of base will increase while that of acid will decrease

Answer A. Salts are usually neutral. In some cases, salt is formed when base reacts with acid.

51. What is the primary reason why toothpaste helps control tooth decay?

- A. It cools the mouth and prevents germs from multiplying.
- B. It removes bad odor that remains after eating.
- C. It surrounds the teeth with protective substance.
- D. It neutralizes acidic food particles that remain in the teeth.

Answer D. Tooth decay is usually caused by acidic food particles. Generally, base neutralizes acidity, and toothpaste is composed of basic substance.

52. The type of chemical reaction which occurs when only one product is formed from two or more reactants, is called _____.

A. Composition	C. Decomposition
B. Single replacement	D. Double replacement

Answer A. When two or more reactants combine to form one product, the type of chemical reaction is called composition. Example: $2H_2 + O_2 = 2H_2O$

53. Which of the following particles are involved in the formation of compounds?

A. Protons	C. Electrons
B. Neutrons	D. Photons

Answer C. Electrons are the only ones which are capable of sharing with other atoms to form compounds.

54. The chemical reaction that releases heat energy during the process is called _____.

A. Endothermic reaction	C. Synthesis reaction
B. Exothermic reaction	D. Decomposition reaction

Answer B. In exothermic reaction, heat energy is released. In endothermic reaction, heat energy is absorbed by the body during the process.

55. Roger has a 2-meter roll of aluminum foil labelled 100% made of aluminium. He takes a thumb-size piece of that aluminum foil. Which of the following statements is true about the piece of aluminum foil that Roger took compared with the rest of aluminum in the roll?

A. Their densities and melting point are the same.
 B. Their densities and melting point are different.
 C. Their densities are different but their melting point are the same.
 D. Their densities are the same but their melting point are different.

Answer A. Density and melting point are distinguishing properties of matter. Whether the material is large or small, as long as it is made up of same material, its density and melting point will not change.

56. Which of the following is an example of a chemical change?

A. Melting of candle	C. Dissolving sugar
B. Rusting of iron	D. Boiling of water

Answer B. Chemical change is the change in appearance as well as the composition of the material or substance. Rusting of iron is a chemical change because in rusting, the iron combines with oxygen to form a different substance, ferric oxide.

57. Paint applied to an iron surface prevents the iron from rusting, because _____.

- A. It prevents nitrogen from coming in contact with iron.
- B. It reacts chemically with iron.
- C. It makes the surface of the iron smoother, thus, preventing rust.
- D. It prevents oxygen and moisture from coming in contact with iron.

Answer D. When oxygen comes in contact with iron, it forms a new substance known as rust. Paint is applied to an iron surface so that oxygen will not get in contact with iron, preventing the iron from rusting.

58. Which two processes are reversible and occur at the same temperature?

- I. Freezing
- II. Condensation
- III. Melting
- IV. Sublimation

- A. I and II
- B. I and III

- C. II and III
- D. III and IV

Answer B. Freezing and melting are reversible processes. In freezing, liquid is changed to solid by releasing heat, while in melting, solid is changed to liquid by absorbing heat. The temperature is the same and does not change during the process.

59. The radioisotope which is now used in cancer radiation treatment is _____.

- A. Cobalt 60
- B. Uranium 238

- C. Sodium 24
- D. Iodine 131

Answer A. Cobalt 60 is a radioactive isotope of cobalt which has an intense gamma-ray activity and is now used in radio-therapy and cancer radiation treatment.

60. Which radioactive radiation coming from external source is most dangerous to human body?

- A. X-rays
- B. Alpha radiation

- C. Beta radiation
- D. Gamma radiation

Answer D. Gamma radiation is an electromagnetic radiation emitted by radioactive decay and having energies which are even higher than x-rays. It is also highly penetrating, that's why it is most dangerous to human body.

61. The science that deals with the physical aspects of the earth is called _____.

- A. Earth Science
- B. Physical Science

- C. Geology
- D. Petrology

Answer A. Earth Science is one of the branches of Physical Science. It deals with the physical aspects of the earth, and not the biological concepts.

62. The history of the earth has been divided by scientists into eras. Which of the following shows the arrangement of these eras into the proper sequence from the oldest to the most recent?

- A. Precambrian, Paleozoic, Mesozoic, Cenozoic
- B. Cenozoic, Mesozoic, Paleozoic, Precambrian
- C. Precambrian, Mesozoic, Cenozoic, Paleozoic
- D. Paleozoic, Cenozoic, Precambrian, Mesozoic

Answer A. In geologic timetable, it was theorized that the earth was formed out of nothing. The Precambrian era described the earth to be void, no water, no air, nothing existed. It was followed by Paleozoic era where unicellular organisms started to exist; Mesozoic era and its Jurassic period, and the latest which is Cenozoic era.

63. All the planets in the solar system _____.

- A. have the same temperature
- B. can maintain life
- C. revolve around the sun
- D. are exactly of same size

Answer C. The general characteristic of a planet is that it revolves around the sun. It is not considered a planet if it does not revolve around the sun.

64. Johannes Kepler developed three basic laws of planetary motion. Which of the following is not one of these laws?

- A. Law of Elliptical Orbits
- B. Law of Equal Areas
- C. Law of Radial Velocities
- D. Law of Universal Gravitation

Answer D. Law of Universal Gravitation is not one of the basic laws of planetary motion. This law was formulated by Sir Isaac Newton.

65. Which scientist paved the way for modern exploration by advancing the theory that planets revolve around the sun?

- A. Ptolemy
- B. Aristotle
- C. Copernicus
- D. Harvey

Answer C. Copernicus was the one who theorized that the sun is the center of the solar system and that all the planets revolve around the sun.

66. The star, Alpha Centauri, which is considered as the closest star to our solar system is about 4.7 light years away from the earth. If the star emits a beam of light, how long will it take the light to reach the earth?

- A. 1 year
- B. 4.7 years
- C. 47 years
- D. 4700 years

Answer B. A light year is the distance travelled by light in one year. So, if a beam of light is emitted from the said star, that beam of light will reach the earth after 4.7 light years.

67. Which of the following is the main cause of seasonal changes?

- A. The distance of the earth from the sun
- B. The rotation of the sun about the earth
- C. The revolution of the earth around the sun
- D. The tilting of the earth on its axis

Answer D. The earth is tilted around 23 degrees with the vertical and as it revolves around the sun, the tilt does not change its direction. For this reason, there are times where one hemisphere receives greater light than the other causing seasonal changes.

68. The vernal equinox is the time at which the sun passes through this point, about March 21, marking the beginning of what season in the northern hemisphere?

- A. Summer
- B. Autumn
- C. Winter
- D. Spring

Answer D. The vernal equinox is the time when both the northern and southern hemisphere receive the same amount of sunlight. This time occurs every March 21. When the sun passes through this point, it marks the beginning of spring in the northern hemisphere.

69. The earth's rotation on its axis gave rise to the occurrence of day and night. What would have happened if the earth's axis is not tilted?

- A. Some places will just have day time while others only have night time.
- B. Some places will have very long nights; others will have short days.
- C. The days and nights throughout the globe will always be of equal length.
- D. The length of day and night will just be the same as when the axis is tilted.

Answer C. If the earth's axis is not tilted, all the parts of the earth will be receiving the same amount of light and so the days and nights throughout the globe will always be of equal length.

70. When the arctic circle is experiencing 24 hours of darkness, the sun is shining directly over the _____.

- A. Equator
- B. Tropic of Cancer
- C. Tropic of Capricorn
- D. Antarctic Circle

Answer C. When the earth is tilted in such a way that the northern hemisphere is receiving lesser amount of light, it is winter in the northern hemisphere. If the sun is shining directly over the tropic of Capricorn, then the arctic circle is experiencing 24 hours of darkness. On the other hand, if the sun is shining directly over the tropic of Cancer, the arctic circle is experiencing 24 hours of daytime.

71. Assuming that the revolution of the earth speeds up such that its revolution takes place every $365 \frac{1}{8}$ days, every how many years will a leap year occur?
- A. 3 years
 - B. 4 years
 - C. 6 years
 - D. 8 years

Answer D. Leap year will occur every 8 years. If the earth's revolution is $365 \frac{1}{8}$ days, it will take 8 years to make an additional 1 year out of $\frac{1}{8}$ day, that is, $\frac{1}{8} \times 8 = 1$.

72. What term is given to the surface of the earth between the tropic of cancer and the equator?
- A. Great circle
 - B. Meridian
 - C. Zone
 - D. Solstice

Answer C. By definition, zone is an area or region distinguished from adjacent parts by some distinctive feature. Since the tropic of cancer and the equator are two adjacent parts, the region between the two great circles is the zone. There are five zones of the surface of the earth: torrid zone, the north and south temperate zones and the north and south frigid zones.

73. The time that it takes for the earth to complete a 60° rotation is
- A. 1 hr
 - B. 4 hrs
 - C. 12 hrs
 - D. 24 hrs

Answer B. Every 15° movement is equivalent to one hour. So, 60° is equivalent to 4 hours. Each 15° is one time zone, which is one hour. 360° rotation is equal to 24 time zones.

74. What phase of the moon is observed when it cannot be seen at all because it passes directly between the earth and the sun?
- A. New moon
 - B. First quarter
 - C. Full moon
 - D. Last quarter

Answer A. During New Moon phase, the moon is between the earth and the sun, blocking the light from the sun; thus, during this phase, the moon is non-observable.

75. During solar eclipse, _____
- A. The earth is between the moon and the sun
 - B. The moon is between the earth and the sun
 - C. The sun is between the earth and the moon
 - D. The earth is behind the sun

Answer B. Solar eclipse occurs when the moon is between the earth and the sun. During this event, the moon blocks the light coming from the sun, casting shadow on earth. Lunar eclipse occurs when the earth is between the moon and the sun.

76. Two minerals are both green and shiny. Which of the following methods can be used to identify whether they are the same or different?

- A. Identify the general characteristics
- B. Find out which one is organic
- C. Determine the visible characteristics
- D. Compute and compare their densities

Answer D. Density is a distinguishing property of a mineral, that is, each mineral has its own density. By comparing the densities of the two minerals, one can tell that the two are the same or different.

77. Which of the following is not a mineral?

- A. Pearl, because it is solid and organic
- B. Topaz, because it is a gem and inorganic
- C. Quartz, because it is silicious and inorganic
- D. Gold, because it is a metal-ore

Answer A. One of the general characteristics of a mineral is that it should be inorganic. That is, the mineral is not derived from living things. Pearl is not a mineral because it is organic. It is derived from oyster.

78. Mineral B can scratch D. Mineral A can scratch D but A can be scratched by C. D can be scratched by C but B can scratch C. Which of the following is the correct order from hard to soft?

- A. ABCD
- B. BACD

- C. BCAD
- D. DACB

Answer C. Hardness is the ability of a mineral to scratch another mineral. If mineral X can scratch mineral Y, then, X is harder than Y. If mineral X can be scratched by Y, mineral Y is harder.

79. The mass of gemstones is usually measured in terms of metric carat. This measure of mass is equivalent to _____.

- A. 2000 mg
- B. 1000 mg

- C. 200 mg
- D. 100 mg

Answer C. One metric carat is equal to 200 grams. This conversion factor was adopted at the 4th General Conference on Weight and Measures.

80. While digging the ground, a man found a well-preserved fossil of an organism in a broken rock. This rock is most likely _____.

- A. Igneous
- B. Sedimentary

- C. Metamorphic
- D. Volcanic tuff

Answer B. Sedimentary rocks are formed out of sediments in water. The first organism is aquatic in nature. Thus, there is a great possibility that the fossil is embedded in sedimentary rock.

86. Petroleum is formed by the _____.

- A. Decay of microorganisms under heat and pressure
- B. Heat from radioactive substances found in the mantle
- C. Volcanic eruptions that burn living matters
- D. Steam trapped underneath a geologic column

Answer A. Petroleum, or commonly called crude oil, is a natural flammable liquid hydrocarbon mixture found beneath the earth's surface in between layers of rocks. It was formed from the decay of remains of organisms, including microorganisms, that lived millions of years ago, which were subjected to extreme heat and pressure.

87. Farmers add commercial fertilizers to the soil to _____.

- A. Speed up the decay of bacteria
- B. Add mineral content
- C. Activate the growth of earthworm
- D. Enhance the growth of the roots

Answer B. Commercial fertilizers mostly contain minerals such as calcium and potassium. The common use of commercial fertilizer is to add minerals to soil.

88. All of the following are examples of soil conservation, except _____.

- A. Contour flowing
- B. Overgrazing
- C. Crop rotation
- D. Terracing

Answer B. In overgrazing, the grass, including the roots are eaten by the animals feeding on the grass. So, when there are no more roots to hold the soil, the soil becomes prone to erosion.

89. Which of the following reasons is not true as to why terraces are built in Mountain Province?

- A. To accommodate excess run of water
- B. To provide space for planting rice
- C. To make the place beautiful
- D. To provide steps for the natives to go uphill

Answer C. Terraces are built in Mountain Province because of choices A, B and D. They did not dream making the terraces beautiful.

90. The prevailing wind systems experienced in the Philippines are _____.

- A. Northwest and Southeast Monsoon
- B. Northeast and Southwest Monsoon
- C. East and West Monsoon
- D. North and South Monsoon

Answer B. The prevailing wind systems in the Philippines are Northeast Monsoon or "amihan" and the Southwest Monsoon or "habagat".

91. A group of researchers formulated a hypothesis that anthurium flower will grow larger if fish bone is added to soil. They made two experimental plots of anthurium, one with fish bone and the other without fish bone. Which data should they be collected to support this hypothesis?

A. Size of anthurium flower in both plots
B. Length of growing time in both plots
C. Height of anthurium plants in both plants
D. Amount of fish bone in both plants

Answer A. It is stated in the hypothesis that "anthurium flower will grow larger if fish bone is added to soil". So, the size of the anthurium should be the data to be collected to support the hypothesis.

92. In an experiment, a biologist noticed that bacteria growing on a plate of agar did not grow next to a mold that was growing on the same plate. So, he wrote in his report, "The mold may be producing a substance that kills bacteria". This statement best described as _____.

A. An observation
B. A hypothesis
C. A generalization
D. A conclusion

Answer B. It is a hypothesis. The words "may be" signifies that the statement is an inference or a scientific guess, so much so that there is a basis in formulating such hypothesis, that is, the bacteria died because of the presence of the mold.

93. Which of the following parts of the microscope does not belong to the group?

A. Arm
B. Condenser
C. Body tube
D. Stage

Answer B. Condenser is the only illuminating part among the choices. The other three; arm, body tube, and the stage are mechanical parts of the microscope.

94. Which of the following structures are found in plant cells but not in animal cells?

1. Cell wall
2. Centrioles
3. Lysosomes
4. Chloroplasts
A. 1 and 4
B. 2 and 3
C. 3 and 4
D. 2 and 4

Answer A. Animal cells have no cell wall and chloroplasts; plant cells have these structures. Chloroplasts are present in the plant cells because they use these as catalysts in photosynthesis.

95. Prokaryotic cells are different from eukaryotic cells because prokaryotes have _____.

A. Small ribosomes used for protein synthesis
B. Nucleic acids found inside a nucleus

- C. Mitochondria that participates in cellular respiration
- D. Genome that is located in the nucleoid

Answer A. Prokaryotic cells or prokaryotes have no distinct nucleus compared to eukaryotic cells. Thus, prokaryotes do not have nucleic acids, mitochondria, and genome. They only have small ribosomes for protein synthesis.

96. When a plant cell is placed in a hypertonic solution, _____.

- A. The plant cell will expand
- B. The plant cell will rupture
- C. The plant cell will shrink
- D. There will be no effect on the plant cell

Answer C. In hypertonic solution, the concentration of solute is higher and the concentration of water is lower outside the cell. The water then moves out of the cell causing the cell to shrink.

97. One of the bases for classifying plants based on phylogenetic consideration is _____.

- A. Chemical composition
- B. Physiological traits
- C. Structural similarities
- D. Evolutionary sequence

Answer D. Phylogenetic involves the origin and evolutionary development of plants.

98. Which of the following organisms is included in the kingdom plantae?

- A. Magnoliidae
- B. Fungi
- C. Protozoa
- D. Prokaryotes

Answer A. Magnoliidae is one of the evergreen trees; thus, it belongs to kingdom plantae. Fungi is under kingdom fungi, and protozoa belongs to kingdom protista while prokaryotes are under kingdom monera.

99. The molecule that allows plants to capture energy from the sun is the _____.

- A. ATP
- B. Carbon dioxide
- C. Chlorophyll
- D. Sugar

Answer C. Chlorophyll is the link between living things and sunlight. It is the pigment that acts as catalyst in photosynthesis.

100. Which of the following microorganisms is considered non-pathogenic?

- A. Plasmodium falcifarum
- B. Lactobacilli
- C. Influenza sp.
- D. Yersinia pestis

Answer B. Lactobacilli is non-pathogenic since it does not carry diseases. It is considered as useful bacteria because it helps in the digestion and assimilation of food in the intestines.

101. Which of the following protists can cause dysentery?

- A. Aedis egypti
- B. Entamoeba histolyca
- C. Gorilla beringei
- D. Streptococcus

Answer B. Of the given choices, Entamoeba histolyca is the only one that can cause dysentery.

102. Amoeba, a single-celled organism, has thin structures essential for cytokinesis, movement, and food getting. These structures are called _____.

- A. Microfilaments
- B. Pseudopodia
- C. Macrofilaments
- D. Analogous structures

Answer B. Pseudopodia is derived from the two words, "pseudo" which means false, and "podia" meaning feet. These feet are used by amoeba in mobility, food getting and other cytokinesis activities of this organism.

103. Which of the following organisms is classified as fungus?

- A. Amoeba
- B. Yeast
- C. Bacteria
- D. Magnolia

Answer B. Yeast is classified under kingdom fungi. Amoeba is under kingdom protista, bacteria is under kingdom monera, while magnolia belongs to kingdom plantae.

104. Spiders can be distinguished from insects by the fact that spiders have _____.

- A. Hard outer covering
- B. Four pairs of legs
- C. Large abdomens
- D. Biting mouth parts

Answer B. Spider is one of the organisms that belongs to class Arachnida of the phylum Arthropoda. One of the general characteristics of arachnids is that they have four pairs of legs. Insects have only three pairs of legs.

105. What organ systems are involved when a chicken flees upon sensing the presence of a snake?

- 1. Skeletal
- 2. Nervous

- 3. Muscular
- 4. Digestive

- A. 1 and 2
- B. 2 and 3

- C. 1, 2 and 3
- D. 1, 2, 3 and 4

Answer C. The chicken flees because it is afraid of the snake. It therefore involves the nervous system. In fleeing, the chicken uses its movement, so it involves the skeletal and muscular systems.

106. Removal of wastes is essential in maintaining body functions. Which group of systems helps the body remove wastes?

- A. Digestive, Respiratory and Integumentary
- B. Digestive, Circulatory and Excretory
- C. Digestive, Respiratory and Nervous
- D. Digestive, Muscular and Integumentary

Answer A. The digestive system removes waste after digestion of food, the respiratory system releases carbon dioxide while integumentary which involves the skin excretes sweat through the pores of the skin. Circulatory, nervous, and muscular systems do not remove wastes.

107. Patients suffering from malfunctioning of kidney are now assisted by a machine that serves as artificial kidney called _____.

- A. Homeostatic machine
- B. Pacemaker
- C. Hemoscope
- D. Hemodialyzer

Answer D. Hemodialyzer is derived from the root words, "hemo" which means blood and "dialize" which means to filter. This machine helps in filtering blood which the kidney cannot function anymore due to illness.

108. If carbohydrates are considered hydrophilic organic compounds, lipids, on the other hand are considered _____.

- A. Hydrophobic
- B. Biomolecules
- C. Saccharides
- D. Peptides

Answer A. Hydrophilic organic compounds are capable of dissolving in water, while hydrophobic organic compounds like lipids are not capable of dissolving in water.

109. Polysaccharides, Triglycerides, Polypeptides, and Nucleic Acids are classified as _____.

- A. Macromolecular
- B. Carbohydrates
- C. Phospholids
- D. Acids and Bases

Answer A. Macromolecular or biomolecules are large organic molecules that are normally present as essential components of living organisms. There are four types of biomolecules: carbohydrates (example is saccharides or sugar); proteins (example is polypeptides); lipids or fats (example is triglycerides); and the nucleic acids (example is the DNA).

110. Sugar compounds such as monosaccharides, disaccharides and polysaccharides are also classified as _____.

- A. Carbohydrates
- B. Lipids
- C. Proteins
- D. Nucleic acids

Answer A. Saccharides are sugar, and they are classified as carbohydrates.

1. What hormones in plants are responsible for promoting axillary budding and apical dominance?

- A. Auxins and Gibberellins
- B. Cytokinins and Gibberellins
- C. Auxins and Abscisates
- D. Cytokinins and Auxins

Answer A. Auxins are hormones in plants that initiate and develop roots, while gibberellins initiate stem elongation, germination, and flowering.

12. In order to determine the cause of an infectious disease, a diagnostic test that allows microorganisms to multiply in a medium is performed in a controlled laboratory condition. This test is called _____.

- A. Complete blood count
- B. Urinalysis
- C. Microbial culture
- D. In vitro fertilization

Answer C. In microbial culture, microorganisms to be tested for infectious disease are allowed to multiply in a medium placed in a petri dish. The medium is usually made up of a gel.

13. The two chemical factors that can cause decomposition are _____.

- A. Death and cellular respiration
- B. Autolysis and photosynthesis
- C. Autolysis and putrefaction
- D. Putrefaction and respiration

Answer C. Both autolysis and putrefaction result in decomposition. Autolysis is the spontaneous disintegration of cells, while putrefaction is the decomposition of organic matter.

14. How will genetic drift affect the population of rabbits living in a stable habitat with no changes in the environment?

- A. It will increase the population of the rabbits.
- B. It will decrease the genetic variation of the rabbits.
- C. It will develop the dominance and superior traits of the rabbits.
- D. It will enhance the appearance of new traits of the rabbits.

Answer C. The genetic drift will promote the dominance and superior traits of animals such as rabbits.

15. A woman has normal ovaries and uterus but cannot bear a child because of defective fallopian tube. Which technological process can help her bear a child?

- A. Hysterectomy
- B. In vivo fertilization
- C. Tubal ligation
- D. In vitro fertilization

Answer D. In vitro fertilization occurs outside the living organism. The term "in vitro" is derived from Latin words, "in glass", that is why in vitro fertilization is usually done in a test tube. On the other hand, in vivo fertilization occurs inside the living organism.

116. Which of the following pairs of body parts are considered homologous?

- A. Human Arms and Seal's flippers
- B. Appendix and Tail bone
- C. Lungs and Gills
- D. Ectoderm and Endoderm

Answer A. Homologous are pairs of body parts that are corresponding in structure and evolutionary origin while analogous are similar in function but not in evolutionary origin. Choice A is homologous, B is vestigial, C is analogous and D is embryonic.

117. A person with diabetes has insulin deficiency. What organ secretes insulin?

- A. Liver
- B. Kidney
- C. Pancreas
- D. Thyroid

Answer C. The Isles of Langerhans, which is in the pancreas, secretes insulin that controls the sugar content of the body.

118. The vermiform appendix, though a part of the large intestine, has no digestive function. It is therefore called _____.

- A. A heterogeneous organ
- B. A vestigial structure
- C. An addendum organ
- D. An analogous structure

Answer B. Appendix is a vermiform (worm-like) structure located at the tip of the large intestine. It is considered as vestigial structure because it is non-functional part of digestive system.

119. The protein shell of a virus is called _____.

- A. Nucleolus
- B. Capsid
- C. Nucleus
- D. Nucleic acid

Answer B. Capsid is the protein shell of a virus. This is usually a coiled structure composed of proteins that encloses the nucleic acid of a virus.

120. The segmented worm that can be used to facilitate anticoagulation is the _____.

- A. Earthworm
- B. Leech
- C. Flatworm
- D. Remora

Answer B. Leech is an aquatic blood-sucking annelid worm of the class Hirudinea. One species of this worm was formerly used by the physicians to bleed their patients.

121. Which of the following is not considered as useful function of bacteria?

- A. They act as decomposers of dead organisms.
- B. They can synthesize enzymes to make vitamins.
- C. They can clean up oil spill by digesting hydrocarbons.
- D. They can be pathogenic.

Answer D. All the choices are useful functions of bacteria except choice D because pathogenic bacteria cause diseases.

122. What stage in chicken pox is considered very contagious?

- A. When wounds are peeling off
- B. When wounds have swollen
- C. When all wounds have disappeared
- D. During the first week

Answer D. Viral diseases, which include chicken pox, are considered very contagious during the first week because it is the incubation period where a person is capable of transmitting a pathogen of a particular disease.

123. Some genetic disorders are caused by the damage to DNA which cannot be repaired. This shows _____.

- A. Mutation
- B. Pathology
- C. Embryology
- D. Speciation

Answer A. Mutation is a heritable alteration of genes or chromosomes of an organism. When this happens, the genetic traits of the offspring become different from the parent, a phenotypic manifestation of a sudden change in the genetic material of a cell.

124. What term is used to describe an organism which depends its subsistence on others?

- A. Prey
- B. Parasite
- C. Medium
- D. Host

Answer B. In parasitism, the organism that benefits from the other organism is called parasite. It is generally smaller than the one which is inhibited is called, the host.

125. Of the following organisms listed below, which one will most likely survive in the tundra biome?

- A. Frog
- B. Turtle
- C. Rice bird
- D. Walrus

Answer D. The tundra biome, which is located in the polar regions, is characterized by extreme cold temperature. The walrus is well adapted in the arctic region because of its thick skin.

126. What term is given to the state of inactivity of an organism such as a frog when passing winter or cold season?

- A. Aestivation
- B. Hibernation
- C. Dormancy
- D. Cold-blooded Response

Answer B Hibernation is the state of inactivity of an organism when passing winter or cold season. It is during this time when the organism "sleeps" mainly because of the cold climate. Aestivation, on the other hand, is the state of inactivity of organism when passing summer or warm season.

127. A duckbilled platypus shows the characteristic of a mammal and not a bird because _____.
- A. It eats other animals
 - B. It feeds its young milk
 - C. It makes a nest and lay eggs
 - D. It has webbed feet

Answer B. Since a duckbilled platypus is a mammal, it possesses mammary glands and so, it feeds its young milk.

128. Organisms, like ants, behave under specific conditions such as storing food during summer, defending themselves by stinging enemies or even invading other ant colonies. The term used to describe this behaviour of the organism is called _____.
- A. Ecological habitat
 - B. Ecological defeat
 - C. Ecological succession
 - D. Ecological niche

Answer D. Ecological niche is the behaviour of organism or group of organisms living under specific environmental condition. It is how the organisms live and not where the organisms live.

129. Which is true about dominant genes?
- A. They prevent the expression of other genes in the same locus
 - B. They should be paired with recessive genes
 - C. They cannot be expressed if heterozygous
 - D. They are aggressive and express the more beneficial traits

Answer D. Dominant genes produce the same phenotypic effect whether paired with an identical or a dissimilar gene. They are mostly aggressive and express the more beneficial traits. On the other hand, recessive genes do not produce a phenotypic effect when heterozygous with dominant genes.

130. Hemophilia is a common sex-linked disorder that is x-linked recessive trait. If a man is hemophiliac, this means that _____.
- A. He got the disease from his mother.
 - B. He will have children with haemophilia.
 - C. All his brothers and sisters will be hemophiliac.
 - D. None of his brothers and sisters will be hemophiliac.

Answer A. Hemophilia is an inherited bleeding disorders that impair blood clotting. It is caused by a defect in genes that are located on the X chromosome, making haemophilia an X-linked recessive disease. Males inherit an X chromosome from their mother and a Y chromosome from their father. For this reason, man who is haemophiliac inherited the disease from his mother.



131. Soil is considered an ecosystem. Which of the following supports this statement?

- A. Soil differs in color, odor, and water holding capacity
- B. It consists of living organisms and abiotic components.
- C. It is a mixture of minerals, water, and gases.
- D. It is the habitat of microorganisms.

Answer B. Soil is considered an ecosystem because it consists of biotic components which are the living things and the abiotic components which are the non living.

132. Why are green plants considered as autotrophs?

- A. They have green pigments that capture light.
- B. They can build simple inorganic substances into complex organic compounds.
- C. They can absorb water through the roots.
- D. They depend on other sources of food.

Answer B. "Autotrophs" mean self nourishment or can make their own food. Plants are autotrophs because they can make their own food through photosynthesis.

133. Of the following organisms listed below, which one is considered heterotrophic?

- A. Bacteria
- B. Moss
- C. Algae
- D. Grasshopper

Answer D. Heterotrophic is a consumer, meaning it consumes the producers. Grasshopper is a consumer because it feeds on the grass.

134. An ecosystem that suffered ecological imbalance can recover naturally through _____.

- A. Ecological succession
- B. Ecological evolution
- C. Ecological niche
- D. Ecological rebirth

Answer A. Ecological succession is the change from one community to another community due to change in environmental condition. An example is the change from pond community to terrestrial community. Succession is a gradual change and happens at a longer period of time.

135. Which of the following shows the transfer of food between all life forms?

- A. Nutrition cycle
- B. Food web
- C. Biogeochemical cycle
- D. Hydrologic pathway

Answer B. There are two kinds of food transfer, food chain and food web. Food chain is the transfer of food in a single path while the transfer of food in a food web branches in different directions like a web.



136. A simple food chain consists of the following: The cob worm that feeds on the corn is eaten by the small bird and this bird is then eaten by the hawk. What will happen to the organisms involved in the food chain if the worm is killed by pesticide?

- A. The bird will thrive.
- B. The corn will decrease.
- C. The hawk will die.
- D. The worm will still reproduce.

Answer C. The hawk will die. The bird will die because there is no worm to be eaten, so, eventually the hawk will die also since there will be no bird to be eaten by the hawk.

137. One environmental activity that may reduce water pollution is the _____.

- A. Construction of flood control in low-lying areas
- B. Treatment of waste water before discharging
- C. Use of detergents to clean the water
- D. Desalination of sea water to make it potable

Answer B. Treatment of waste water before discharging is used today to control water pollution. This is a requirement today in establishments which are using water in their operations.

138. Global warming is an environmental problem being brought about by activities of man that change the quality of air. This may result to the following, except _____.

- A. Extreme rise in temperature due to release of trapped methane gas
- B. Having dry season to most parts of the earth through the years
- C. Spread of tropical diseases due to constant migration of insects
- D. Rising of sea levels and flooding of coastal areas

Answer C. Spread of tropical diseases due to migration of insects is not caused by global warming.

139. Of the following listed below, which one is considered as the most disadvantageous source of energy since it gives detrimental effect on the population's food supply?

- A. Solar
- B. Geothermal
- C. Wind
- D. Hydroelectric plant

Answer B. Geothermal power plant affects the vegetation in the nearby area, that's why the food supply is also affected.

140. The major concern about global warming arises from increased concentration of _____.

- A. Greenhouse gases
- B. Acid rain
- C. Photochemical smog
- D. Aerosols

Answer A. Greenhouse gases is composed of air pollutants such as carbon dioxide and sulphur oxides. Greater concentration of these pollutants will raise the problem of global warming.

**Natural Science**

141. Which of the following does not contribute to greenhouse effect?

- A. Use of detergent
- B. Forest fires
- C. Burning of waste
- D. Deforestation

Answer A. The use of detergents affect the water quality of a certain place, and it does not affect the greenhouse effect. Greenhouse effect is mainly caused by poor air quality.

142. Intensive agricultural and industrial practices may result to eutrophication. Which of the following is the possible consequence of this environmental activity?

- A. The biochemical oxygen demand (BOD) will decrease.
- B. There will be a significant decrease in the water harvest by fishermen.
- C. The water will be less turbid and odor-free.
- D. The toxicity of water will be lessened.

Answer B. Eutrophication exists when the bodies of water are over enriched by nutrients such as nitrogen and phosphorous which generally cause water quality empalment. The effect of this is algal bloom which results to water pollution. Because of this, many aquatic produce will die and so, water harvest will decrease.

143. The main aim of Brundtland Commission is the development that meets the needs of the present without compromising the ability of the future generation to meet their own needs. This development is classified as _____.

- A. Sustainable development
- B. Environmental development
- C. Uncompromising development
- D. Ecological development

Answer A. The Brundtland Commission was named after Norway's former Prime Minister, Gro Harlem Brundtland who was assigned as the first commissioner. The commission aims for sustainable development in the present without compromising the future needs.

144. The "red tide" phenomenon causes what particular marine organism to be poisonous for human consumption?

- A. Arthropods
- B. Mollusks
- C. Earthworms
- D. Algae

Answer B. Mollusks such as shellfish (tahong) are usually affected by red tide phenomenon. These organisms after being affected by dinoflagellates, which contain toxins become poisonous for human consumption.

145. Air, when mixed with sulphur oxide coming from motor vehicles and industrial plants, falls to earth giving detrimental effects to living and non-living things. This is known as _____.

- A. Acid rain
- B. Greenhouse rain
- C. Fall out
- D. El nino

Answer A. Acid rain is composed of water droplets that are acidic due to atmospheric pollution. This occurs when sulphur oxides combine with water to form sulphuric acid. Since this is acid, this gives detrimental effects not only to living things but non-living as well.

146. Which environmental law requires industries to install anti-pollution devices and bans the use of incinerators?

- A. Philippine Clean Air Act
- B. Environmental Security Act
- C. Anti-pollution Act
- D. Greenpeace Act

Answer A. The Philippine Clean Air Act of 1999 simply known as Clean Air Act, adopts and promulgates policies, regulations, etc., in order to maintain good quality of air that protects human health and welfare. This act requires industries to install pollution control devices and bans the use of incinerators.

147. The El Nino phenomenon, which grabbed control of the world's weather machine, is described by meteorologists as _____.

- A. Landslides, flash floods, warm air currents
- B. Heavy downpours, tropical storms, strong winds
- C. Lack of rains, droughts, crop failure
- D. Monsoons, easterly winds, low pressure area

Answer C. The El Nino phenomenon usually occurs during dry season where the temperature is warm. Hence, it is characterized by lack of rains, and bringing droughts and crop failure.

148. The Kyoto Protocol requires to reduce greenhouse effect in the environment. This practice advocates _____.

- A. Environmental protection
- B. Energy conservation
- C. Ecological destruction
- D. Sustainable development

Answer A. The advocacy of Kyoto Protocol is primarily the environmental protection.

149. Social and environmental injustices are interconnected, as exemplified by which of the following activities?

- A. Extinction of rare animal species
- B. Temperature warming and rise of sea level
- C. Toxic wastes from industrialized countries are disposed to poorer countries
- D. Desertification of verdant regions

Answer C. Truly, social and environmental injustices are interconnected, and this usually happens globally. Most of the problems encountered is the disposal of wastes. An example of this is the issue of the garbage from Canada, which is dumped in the Philippines.

150 At present, scientists report about the formation of holes in the ozone layer of the atmosphere. This dangerous trend is being blamed on which of the following wrong practices?

- A. The use of ozone itself on the surface of the earth
- B. The excessive use of aerosols
- C. Removal of the protective cover like the green plants
- D. The use of soil fertilizers on earth

Answer B. The main cause of the formation of holes in the ozone layer is the release of chlorofluorocarbons (CFC) in the atmosphere, and aerosols are examples of fluorocarbons.

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KOMUNIKASYON SA AKADEMIKONG FILIPINO**KOMUNIKASYON****Depinisyon ng Komunikasyon**

- Ito ay ang akto ng pagpapahayag ng ideya sa pamamagitan ng pasalita o pasulat na paraan (Webster, 1974).
- Ito ay ang proseso ng pagpapadala at pagtanggap ng mensahe sa pamamagitan ng simbolikong cues na maaring verbal o di-verbal (Bernales, et al., 2003).
- Ito ay ang pagpapalit-usap para kapwa makinabang ng kaalaman ang isa't isa nang di-nagke'tamangan (Arrogante, 2003).

Uri ng Komunikasyon

- Komunikasyong Verbal - ang uri ng komunikasyon na ginagamitan ng wika, pasalita man o pasulat
- Komunikasyong Di-Verbal - ang uri ng komunikasyon na hindi ginagamitan ng wika, kundi ginagamitan ito ng senyas, ekspresyon ng mukha, simbolo at iba pa

Antas ng Komunikasyon

- A. Komunikasyong Intrapersonal - antas ng komunikasyon na pansarili at sa isang indibidwal lamang.
Hal.: Repleksyon o pag - ilalp sa iyong mga nagawang kasalanan
- B. Komunikasyong Interpersonal - ang antas ng komunikasyon na nagagawa ng dalawang tao o higit pa.
Hal.: Interaksyon ng isang estudyante at ng isang guro
- C. Komunikasyong Pampubliko - ang tawag sa antas ng komunikasyon sa pagitan ng isang tao at sa isang malaking grupo ng mga tao.
Hal.: Pangangampanya ng isang kandidato sa harap ng mga mamamayan
- D. Komunikasyong Pang-Midya - antas ng komunikasyon na kung saan ang pagtanggap o ang pagpapadala ng mensahe ay ginagamitan ng kagamitang pang-midya.
Hal.: Komunikasyon na nagaganap gamit ang radyo, telebisyon, o pahayagan

WIKA**Depinisyon ng Wika**

- Ito ay isang masistemang balangkas ng sinasalitang tunog na pinipili at isinasanyos sa paraang arbitrario upang magamit ng mga taong kabilang sa isang kultura (Gleason, 1988)
- Ito ay ang pangunahin at pinaka-elaboreyt na anyo ng simbolikong gawaing pantao (Hill, 1976).
- Ito ay isang sistema ng komunikasyon sa pagitan ng mga tao sa pamamagitan ng mga pasulat o pasalitang simbolo (Webster, 1974).

Katangian ng Wika

- Ito ay isang masistemang balangkas.
- Ito ay sinasalitang tunog.
- Ito ay pinipili at isinasanyos.

- Ito ay arbitraryo.
- Ito ay ginagamit.
- Ito ay nakabatay sa kultura.
- Ito ay nagbabago.

Teorya ng Pinagmulan ng Wika

- Teoryang Bow-wow - Ang wika ay maaaring nagmula sa panggagaya ng tao sa mga tunog sa kalikasan.
- Teoryang Pooh-pooh - Ang wika ay nailikha bunga ng mga masidhing damdamin ng tao gaya ng sakit, tuwa, sarap, kalungkutan, lakot, pagkabigla at iba pa.
- Teoryang Yo-he-ho - Ang wika ay bunga dumano ng pwersang pisikal ng tao.
- Teoryang Ta-ra-ra-boom-de-ay - Ang wika ay nag-ugat sa mga tunog na nililikha ng mga tao sa kanilang mga ritwal.
- Teoryang Ta-ta - Ang wika ay nagmula sa panggagaya ng dila sa kumpas o galaw ng kamay ng tao na kanyang ginagawa sa bawat partikular na okasyon.
- Teoryang Ding-dong - Ang wika ay nagmula sa panggagaya ng tao sa mga tunog na nailikha ng mga bagay-bagay sa paligid, hindi lamang sa kalikasan kung di maging sa mga bagay na likha ng tao.
- Biblikal na Pananaw - Ang wika ay kaloob ng Diyos at ang pagkakaiba ng wika ay isinasalaysay sa kwento ni Babel at sa kanyang Toro.

Kahalagahan ng Wika

- Ito ay isang instrumento ng komunikasyon.
- Ito ay nag-iingat at nagpapalaganap ng kaalaman.
- Ito ay nagbubuklod ng bansa.
- Ito ay lumilinaang ng malikhaing pag-iisip.

Tungkulin ng Wika

- **Interaksyonal** - nakapagpapapanatili/ nakapagpapatatag ng relasyong sosyal
- **Instrumental** - tumutugon sa mga pangangailangan
- **Regulatori** - kumokontrol at gumagabay sa kilos/asal ng iba
- **Personal** - nakapagpapahayag ng sariling damdamin o opinyon
- **Imahinatibo** - nakapagpapahayag ng imahinasyon sa malikhaing paraan
- **Heuristik** - naghahanap ng impormasyon
- **Impormatib** - nagbibigay ng impormasyon

Antas ng Wika

1. **Pormal** - Ito ang mga salitang istandard dahil kinikilala, tinatanggap, at ginagamit ng higit na nakararami.
 - a. **Pambansa** - Ito ang wikang kadalasang ginagamit ng pamahalaan, aklat pangwika, aklat pambalarila at paaralan. (Hal. Ama)
 - b. **Pampanitikan o Panretonka** - Ito ang mga salitang matatayog, malalim, makulay, masining at ginagamit ng mga manunulat sa kanilang akdang pampanitikan. (Hal. Haligi ng Tahanan)
2. **Impormal** - Ito ang mga salitang karaniwan, pang-araw-araw, palasak, at kadalasang ginagamit sa pakikipag-usap at pakikipagtatalsasan sa mga kakilala at kaibigan.
 - a. **Lalawiganin** - Ito ang mga bokabularyong dayalektal (Hal. Amahan)

- b. Kolokyal - Ito ay mga pang-araw-araw na salita na ginagamit sa mga pagkakataong impormal (Hal. Tatay - Tay)
- c. Balbal - Ito ay ang pinakadinamiko at pinakamababang antas ng wika na kadalasang maririnig sa mga usapang kalye. (Hal. Erpat)

Barayti ng Wika

- Dayalekto - Ito ang barayti ng wikang nalilikha ng dimensyong heograpiko. Ito ang wikang ginagamit sa isang partikular na rehlyon, lalawigan o pook, malaki man o maliit.
Hal. Aba, ang ganda! (Maynila)
Aba, ang ganda nhl (Balangas)
- Sosyolek - Ito ang barayti ng wikang nabubuo ayon sa umiikang sosyal dahil sa ito ay nakabalay sa mga pangkat lipunan.
Hal. Wiz ko feel ang mga hombre dilech, day! (Gay lingo o Wika ng mga Bakla)
Wow pare, ang tindi ng tama ko! Heaven! (Wika ng mga Adik)
- Jargon - Ito ang mga tanging bokabularyo ng isang partikular na pangkat ng gawain.
Hal. Tsok, lesson plan, klas (Tilser)
Appeal, justice, court, hearing (Abogado)
- Idyolek - Ito ay tumutukoy sa personal at kanl-kanlyang paraan ng paggamit ng wika.
Hal. Magandang gabi, bayan! (Noli de Castro)
Halika, kaibigan! Usap tayo. (Boy Abunda)

Ang Kasaysayan ng Wikang Filipino

- 1935 Saligang Batas, Artikulo XIV, Seksyon 3 (Taong 1935)
- Taong 1935 nang unang naisip ang konsepto ng paglikha ng isang Wikang Pambansa. Sa 1935 Saligang Batas ng Pilipinas, itinatag na ang Kongreso ay gagawa ng mga hakbang tungo sa paglikha ng isang Wikang Pambansa na batay sa isa sa mga umiiral na katutubong wika.
- Batas Komonwelt Big. 184 (Taong 1936)
- Taong 1936, pinagtibay ng Balasang Pambansa ang Batas Komonwelt Big. 184 na lumikha ng isang Surian ng Wikang Pambansa na siyang pipili ng katutubong wika na siyang magiging batayan ng wikang pambansa.
- Kautusang Tagapagpaganap Big. 134 (Disyembre 30, 1937)
- Alinsunod sa Batas Komonwelt Big. 184 sa bisa ng Saligang Batas 1935, ipinahayag ng Pangulong Quezon na ang Wikang Pambansa ay ibabata sa Tagalog.
- Kautusang Tagapagpaganap Big. 263 (Hunyo 19, 1940)
- Pagpapalimbag ng isang diksyunaryo at isang gramatika ng Wikang Pambansa at itinakdang mula sa Hunyo 19, 1940, ay pasisimulan nang ituro ang Wikang Pambansa ng Pilipinas sa lahat ng paaralang-pampubliko at pribado sa buong bansa.
- Proklamasyon Big. 12 (Marso 26, 1954)
- Ang petsa ng pagdiriwang ng Linggo ng Wika ay Marso 29-Abril 4 na isasagawa taun-taon bilang paggunita sa kaarawan ni Francisco Balagtas.

Filipino

- **Proklamasyon Blg. 186** (Setyembre 23, 1955)
 - Ang paglilipat ng petsa ng pagdiriwang ng Linggo ng Wika. Ito ay gagawin na mula Agosto 13-19, bilang paggunita sa kaarawan ni Manuel L. Quezon.
- **Kautusang Pangkagawaran Blg. 7** (Agosto 13, 1959)
 - Ang Pambansang Wika ay tatawaging Pilipino.
- **1973 Saligang Batas, Artikulo XV, Seksyon 3** (Taong 1973)
 - Sinasabi dito na ang 1973 Saligang-Batas ay dapat na ipahayag sa Ingles at Pilipino, ang dapat na mga Wikang Opisyal. Isa pa, ang Pambansang Asemblea ay dapat gumawa ng mga hakbang luno sa nagpapaunlad at pormal na adapsyon ng panlahat na Wikang Pambansa na makikilalang Filipino.
- **1987 Saligang Batas, Artikulo XIV, Seksyon 6-9** (Pebrero 2, 1987)
 - Mula sa Pilipino, ang Pambansang Wika ay tatawagin ng Filipino. Kasama ng Filipino, ang wikang Ingles ay ang siyang dalawang opisyal na wika ng bansa.
- **Kautusang Pangkagawaran Blg. 81** (Agosto 6, 1987)
 - Ang Alpabetong Filipino ay binubuo ng 28 letra.
A B C D E F G H I J K L M N N̄ N̄̄ O P Q R S T U V W X Y Z
- **Proklamasyon Blg. 104** (Hulyo 1997)
 - Ang buwan ng Agosto taun-taon ay magiging Buwan ng Wikang Filipino.

Ang Alpabetong Filipino: Pahapyaw na Kasaysayan

1. Baybayin - Ito ang pinakaunang alpabeto na ginamit ng sinaunang Pilipino. Ito ay binubuo ng labimpitong titik: tatlong patinig at labing-apat na katinig.
2. Alpabetong Romano - Kasabay ng pagdating ng Kastila ay ang pagpalit ng lumang alibata ng alpabetong Romano. Ito ay binubuo ng tatlumpong titik at tinawag nang pa-Kastila.
3. Abecedario - Pinalitan ng pangalan ang alpabetong Romano, at ito ay tinawag ng abecedario.
4. Abakada - Taong 1940, binalangkas ni Lope K. Santos ang bagong alpabeto na nakilala sa tawag na Abakada dahil sa unang apat na titik niyon. Ang orihinal na Abakada ay binubuo ng dalawampung titik: labintimang katinig at limang patinig. Pagdating ng 1971, dinagdagan ang abakada ng labing-isang titik kaya ito ay binubuo na ng tatlumpong isang titik.
5. Ang Alpabetong Filipino - Noong Agosto 6, 1987, ipinasa ng Linangan ng mga Wika sa Pilipinas ang Kautusang Pangkagawaran Blg. 81 na may pamagat na "Ang Alpabeto at Patnubay sa Ispelang ng Wikang Filipino." Simula noon hanggang sa kasalukuyan, angating alpabeto ay nagkaroon na ng dalawampu't walong titik na tinatawag nang pa-Ingles maliban sa N̄.

Ang Pagkakaliba ng Tagalog, Pilipino at Filipino

Tagalog

- Ito ang tawag sa wika na ginagamit sa Bulacan, Batangas, Rizal, Laguna, Quezon, Cavite, Mindoro, Marinduque, Ilang parte ng Nueva Ecija, Puerto Princesa, at Metro Manila.

- Nang ideklara ni Pangulong Quezon ang Kautusang Tagapagpaganap Blg. 134 sa taong 1937, lumalabas na ang Wikang Pambansa ay dapat na nakabatay sa Tagalog.
- Pero hindi sinasabi sa kautusang ito na ang wikang pambansa ay tatawaging Tagalog.

Filipino

- Taong 1959, ayon sa Kautusang Pangkagawaran Blg. 7, binigyang pangalan ang wikang pambansa na nakasaad sa Kautusang Tagapagpaganap Blg. 134 sa taong 1937, at ito ay tinawag na Pilipino.
- Kung gayon, wala pang wikang Pilipino na nakilala bago 1959. Ang wikang Pilipino ay nakabatay pa rin sa wikang Tagalog.
- Mula noong 1959 hanggang 1987, itinuro ang Pilipino sa lahat ng eskwelahang publiko at pribado.
- Ngunit, ang Pilipino bilang isang wika ay hindi na umiral sa kasalukuyan sapagkat ito ay pinalitan na ng wikang Filipino mula noong 1987.
- Sa kasalukuyan, ang salitang Pilipino ay tumutukoy nalamang sa mga taong naninirahan sa Pilipinas at hindi na sa isang wikang pambansa.

Filipino

- Sa pamamagitan ng 1987 Saligang Batas, naitatag ang Filipino bilang wikang pambansa na siyang pumalit sa naunang pangalan na Pilipino.
- Kung ang neo'y wikang pambansa na Pilipino ay nakabatay lamang sa Tagalog, ang wikang pambansa naman na Filipino ay nakabatay na sa lahat ng wika sa Pilipinas, kasama na ang Ingles at Kastila.
- Mula 20 letra ng Tagalog, naging 28 letra ang Filipino. Idinagdag ang 8 letra na c, f, j, ñ, q, v, x at z.
- Mula 1987 hanggang sa kasalukuyan, ang wikang Filipino na ang siyang opisyal na wikang pambansa ng Pilipinas.

Mga Sangay ng Linggwistiks

- PONOLOHIYA** - pag-aaral ng makabuluhang tunog ng isang wika.
- MORPOLOHIYA** - pag-aaral ng pinakamaliit na yunit ng isang salita na nagtataglay ng kahulugan
- SINTAKS** - pag-aaral ng pag-uugnay-ugnay ng mga salita upang makabuo ng pangungusap at pag-uugnay-ugnay ng mga pangungusap sa pagbuo ng diskurso

Ang Ponoheyang Filipino

- Ponema ang tawag sa isang makabuluhang tunog ng isang wika.
- Ang ponolohiya ay tinatawag din na "palatunugan."
- May Dalawang Uri ng Ponema: Ponemang Segmental at Ponemang Suprasegmental
- Ponemang Segmental: Diptonggo, Klaster, Pares Minimal, Ponemang Malayang Nagpapalitan
- Diptonggo ang tawag sa alinmang patinig na sinusundan ng /w/ o /y/ sa loob ng isang pantig. Halimbawa: kahoy (ka-hoy), giliw (gi-liw), bahay (ba-hay)
- Ponemang Suprasegmental : Din, Tono o Intonasyon, Antala o Hinto

Filipino

- **Klaster** ang tawag sa nlinmang dalawang magkasunod na magkalbang katinig sa isang pantig.
Halimbawa: klaster (klas-ter), pluma (plu-ma), lroso (tro-so)
- **Pares Minimal** ang tawag sa pares ng salita na magkaiba ang kahulugan ngunit magkapareho ang baybay maliban sa isang titik.
Halimbawa: bata at pata, buhay at bahay, lolo at lola
- **Ponemang Malayang Nagpapalit** ang tawag sa pares ng salita na magkapareho ang kahulugan ngunit magkapareho ang baybay maliban sa isang titik.
Halimbawa: lalaki at lalake, marami at madami, marumi at madumi

Ang Morpolohiyang Filipino

- **Morpema** ang tawag sa pinakamaliit na yunit ng isang salita na nagtataglay ng kahulugan.
- Ang morpolohiya ay tinatawag din na "palabuuan."
- Pagbabagong Morpoponemiko
 - Asimilasyon:** pambata = pang- + bata
 - Pagpapalit ng Ponema:** marami = ma- + dami
 - Pagpapalit-din:** balTAan = baLiita + -an
 - Metatesis:** niligaw = in- + ligaw
 - Pagkakaltas:** lakpan = takip + -an
 - Gradasyon:** regalo = regalo + -han
 - Reduksyon:** teka = hintay + ka

Sintaks

- Ang sintaks ay ang makaagham na pag-aaral at pagsusuri ng mga pangungusap.
- **Dalawang Ayo ng Pangungusap**
 - Karaniwang Ayo** - nagsisimula sa panaguri at nagtatapos sa simuno.
Halimbawa: Nagsinungaling si Janeth Napulesya.
 - Di-Karaniwang Ayo** - nagsisimula sa simuno at nagtatapos sa panaguri.
Halimbawa: Si Janeth Napulesya ay nagsinungaling.
- May dalawang klasipikasyon ang pangungusap: Uri ng Pangungusap Ayon sa Gamit at Uri ng Pangungusap Ayon sa Kayarian
- **Uri ng Pangungusap Ayon sa Gamit**
 - Paturol o Pasalaysay** - nagsasalaysay ng isang katotohanan o pangyayari at nagtatapos sa tuldok.
Halimbawa: Si Jose Rizal ay ang ating pambansang bayani.
 - Patanong** - nagtatanong o nag-uusisa at nagtatapos sa tandang pananong.
Halimbawa: Natutulog ba ang Diyos?
 - Pautos o Pakiusap** - nag-uutos o nakikiusap at nagtatapos sa tuldok o sa tandang pananong.
Halimbawa: Dalhin mo ang mga gamit sa kwarto.
 - Padamdang** - nagsasaad ng matinding damdamin at gumagamit ng tandang padamdang.
Halimbawa: Nakul Nabasag ang mamahaling pinggan.

- Pangungusap Ayon sa Kayarlan

- a. **Payak** - binubuo ng isang sugnay na makapag-iisa.
- may apat ilang kayarian: payak na simuno at payak na panaguri, payak na simuno at tambalang panaguri, tambalang simuno at payak na panaguri, tambalang simuno at tambalang panaguri.

Halimbawa: Si Nicole ay mayumi.

- b. **Tambalan** - binubuo ng dalawang sugnay na makapag-iisa.
- ginagamitan ito ng pangatnig na panimbang: at, ngunit, o.
- ang sugnay na kasunod ng pangatnig na panimbang ay isang sugnay na makapag-iisa.

Halimbawa: Si Monica ay matapat, ngunit si Adrian ay taksil.

- c. **Hugnayan** - binubuo ng isang sugnay na makapag-iisa at isang sugnay na di makapag-iisa.
- ginagamitan ito ng pangatnig na pantulong: dahil, kung, nang.
- ang sugnay na kasunod ng pangatnig na pantulong ay isang sugnay na di makapag-iisa.

Halimbawa: Masaya si Edil Mae dahil siya ay may bagong laruan.

- d. **Langkapan** - binubuo ng dalawang sugnay na makapag-iisa at isang sugnay na di makapag-iisa.
- ginagamitan ito ng mga pangatnig na panimbang at pangatnig na pantulong.

Halimbawa: Nang siya ay naging isang sirena, siya ay nakagat ng jelly fish at siya ay na-ospital.

Mga Bahagi ng Pananalita sa Filipino

A. Pangngalan

- mga salita na nagsasaad ng pangalan ng tao, bagay, pook, hayop, pangyayari, damdamin, kaisipan o ideya.

Uri ng Pangngalan

1. **Pambalana** - tumutukoy sa pangkalahatan o karaniwang pangalan ng tao, hayop at bagay at isinusulat sa maikling titik ang unang letra ng salita

Hal.: bansa, presidente, bayani

2. **Pantangi** - nagsasaad ng tanging pangalan ng tao, hayop at bagay at isinusulat sa malaking titik ang unang letra ng salita

Hal.: Pilipinas, Corazon Aquino, Jose Rizal

B. Panghalip

- mga salita na humahalili o pumapalit sa mga pangngalan

Hal.: ako, ikaw, siya

Fillipino

C. Pang-uri

- mga salitang naglalarawan ng pangngalan o panghalip

Mga Uri ng Pang-uri

1. **Pang-uring Panlarawan** – nagsasabi tungkol sa anyo, laki, lasa, amoy, hugis at iba pang katangian ng pangngalan at panghalip
Hal.: Matamis ang linda niyang mansanas.
2. **Pang-uring Pamllang** - nagsasaad ng dami o bilang ng pangngalan at panghalip
Hal.: Si Mang Baste ay nag-iisang lumaban sa criminal.
3. **Pang-uring Pantangi** - may anyong pangngalang pantangi na naglalarawan sa pangngalan.
Hal.: Pinakapaborito ko sa lahat ang manggang Cebu.
4. **Pang-uring Paari** - mga salitang paari na tumuturing sa pangngalan
Hal.: Ang mga tradisyon ay bahagi ng ating kultura.

Kayarian ng Pang-uri

1. **Payak** - Binubuo ito ng salitang-ugat lamang.
Hal.: sapat, tangi, dakila
2. **Maylapi** - Binubuo ito ng salitang-ugat at panlapi.
Hal.: masipag, maliyaga, masarap
3. **Inuulit** - Binubuo ito ng pag-uulit sa salitang-ugat o sa pang-uring maylapi.
Hal.: baliw na baliw, magandang maganda, kaakit-akit
4. **Tambalan** - Binubuo ng dalawang magkalibang salita.
Hal.: isip-bata, balat-sibuyas, hampaslupa

Kaantasan ng Pang-uri

1. **Lantay** – Nagbibigay ng simple o payak na paglalarawan sa isang pangngalan o panghalip.
Hal.: Ang prinsesa ay maganda.
2. **Pahambing** – Naghahambing ng katangian ng dalawang pangngalan o panghalip. Ginagamitan ito ng mas, lalo, di hamak, lubha, kaysa at iba pa.
Hal.: Ang prinsesa ay mas maganda kaysa sa reyna.
3. **Pasukdol** – Nagbibigay ng sukdulang paglalarawan o katangiang nakahihigit sa lahat. Ang panghahambing ay higit sa dalawa. Ginagamitan ito ng pinaka, napaka, hari ng, ubod ng, saksakan ng, lubos na at iba pa.
Hal.: Pinakamaganda ang prinsesa sa lahat ng babae sa palasyo.

D. Pandiwa

- mga salita na nagsasaad ng kilos o galaw

Halimbawa: kain, inom, lipad

- Ang Pokus ng Pandiwa

- a. **Pokus sa Tagaganap o Aktor Pukos** - ang paksa ng pangungusap ay ang siyang gumagawa ng kilos na sinasaad ng pandiwa.
Halimbawa: Kumain si Loremed.
- b. **Pokus sa Tagatanggap o Benefaktib Pokus** - ang paksa ng pangungusap ay ang siyang pinaglalaanan ng kilos na sinasaad ng pandiwa.

- Halimbawa: Ipinagluto ng kanyang asawa si Elmerisa.
- c. **Pokus sa Sanhi** - Ang paksa ng pangungusap ay ang siyang dahilan o sanhi kung bakit nagaganap ang kilos na sinasaad ng pandiwa.
Halimbawa: Ikinatuwa ng publiko ang pagkapanalo ni Pacquiao.
- d. **Pokus sa Ganapan o Lokatib Pokus** - Ang paksa ng pangungusap ay ang lugar kung saan nangyayari ang kilos na sinasaad ng pandiwa.
Halimbawa: Pinagpasyalan nila ang dagat sa Marigondon.
- e. **Pokus sa Gamit o Instrumental Pokus** - Ang paksa ng pangungusap ay ang siyang ginagamit na bagay upang maisagawa ang kilos na sinasaad ng pandiwa.
Halimbawa: Ipinanghalo ko ang sandok.

E. Pang-abay

- mga salitang naglalarawan ng pandiwa, pang-uri, o kapwa pang-abay

Hal.: Nagdasal nang matalimtim si Honesto.

F. Pang-ukol

- ay ang bahagi ng pananalitang nag-uugnay sa pangngalan, panghalip, pandiwa at pang-abay na pinag-uukulan ng kilos, gawa, aral, balak o layon

Hal.: laban sa, para sa, tungkol sa

**II. FILIPINO 2: PAGBASA AT PAGSULAT TUNGO SA PANANALIKSIK
PAGBASA**

- Ito ay ang pagkilala at pagkuha ng mga idaya at kaisipan sa mga sagisag na nakalimbag upang mabigkas nang pasalita (Austero, et al., 1999).
- Ayon kay Goodman, Ito ay isang "psycholinguistic guessing game" (Badayos, 2000).
- Ito ay isang bahagi ng pakikipagtalastasan na kahanay ng pakikinig, pagsasalita at pagsulat (Bernales, et al., 2001).

Proseso ng Pagbasa

- a. **Persepsyon** - pagkilala sa mga nakalimbag na simbolo
- b. **Komprehensyon** - pag-unawa sa tekstong binabasa
- c. **Reaksyon** - paghahatol ng kawastuhan, kahusayan, at halaga ng isang teksto
- d. **Asimilasyon** - Integrasyon ng binasang teksto sa mga karanasan ng mambabasa

Mga Uri ng Pagbasa

- **Skimming** - Ito ay isang mabilis sa pagbasa na nakakaya ng isang tao na kung saan ang higitangad na mambabasa ay ang makuha ang buong kaisipan ng isang teksto.
- **Scanning** - Ito ay ang paghahanap ng isang tiyak o partikular na impormasyon sa isang pahina.
- **Tahimik na Pagbasa** - Ito ay isang uri ng pagbasa gamit ang mga mata lamang at walang puwang ang paggamit ng bibig.
- **Pasalitang Pagbasa** - Ito ay isang pagbasa na ginagamitan ng mata at bibig kaya may tunog at pagsasalita
- **Study Speed na Pagbasa** - Ito ang pinakamabagal na pagbasa at ginagamit Ito

Filipino

sa mga mahihirap na seloskyon.

- **Matulin na Pagbasa** - Ito ay isang mabibilis na pagbasa na kung saan binabasa lamang ang mahahalagang bahagi batay sa layunin ng mambabasa.

Teorya sa Pagbasa

- **Teoryang Bottom - Up** - Ayon sa teoryang ito, ang proseso ng pag-unawa ay nagsisimula sa teksto patungo sa mambabasa. Kung gayon, pasiv na partisipant lamang ang mambabasa sapagkat tagatanggap lamang siya ng impormasyon mula sa teksto.
- **Teoryang Top-Down** - Sa teoryang ito, ang proseso ng pag-unawa ay nagsisimula sa mambabasa patungo sa teksto. Nagiging isang aktib na partisipant ang mambabasa dahil sa siya ay may taglay na dating kaalaman at kung gayon nakakabuo ng palagay at hinuha sa kanyang binasa.
- **Teoryang Interaktib** - Sinasabi sa teoryang ito na may dalawang interaksyon na nangyayari sa pagbasa, ang interaksyong awtor-mambabasa at mambabasa-awtor. Sa madaling sabi, ang teoryang ito ay kombinasyon ng teoryang bottom-up at top-down.
- **Teoryang Iskima** - Mahalaga ang ginagampanan ng dating kaalaman ng mambabasa sa kanyang pagbasa. Bago pa man basahin ng isang mambabasa ang isang teksto, siya ay may taglay nang ideya sa nilalaman ng teksto mula sa kanyang iskima sa paksa.

PAGSULAT

- Ito ay ang pagsasalit sa papel o sa anumang kasangkapang magaring magamit na mapagsasalitan ng mga nabuong salita o simbolo at ilustrasyon ng isang tao o mga tao sa layuning mapahayag ang kanyang o kanyang kaisipan (Bernales, et al., 2001).
- Ayon kina Peck at Buckingham, ito ay isang ekstensyon ng wika at karanasang natamo ng isang tao mula sa kanyang pakikinig, pagsasalita at pagbabasa (Radillo, 1998).
- Ito ay isang komprehensibo na kakayahang naglalaman ng wastong gamit, talasalitaan, pagbubuo ng kaisipan, retorika at iba pang mga element (Xing at Jin, 1989).

Mga Hakbang sa Pagsulat

- Pre-writing** - pagpapalano at paghahanda sa pagsulat
- Actual Writing** - pagsulat ng burador o draft
- Rewriting** - pag-reedit at pagrerebisa ng burador o draft

Mga Uri ng Pagsulat

- **Akademik** - Halos lahat ng pagsusulat sa paaralan ay maluuuri na akademik kagaya ng sanaysay, pamanahong papel, tesis o disertasyon at lab report.
- **Teknikal** - Ito ay isang espesyalizadong uri ng pagsulat na tumutugon o nagbibigay-solusyon sa isang komplikadong suliranin tulad ng isang feasibility study.
- **Jornalistik** - Ito ay isang pampamahayag na uri ng pagsulat katulad ng pagsulat ng balita.
- **Referensyal** - Ang uri ng pagsulat na may layuning mag-rekomenda ng iba pang sors o referens sa isang paksa katulad ng bibliografi.

- **Profesional** - Ito ay isang uri ng pagsulat na nakatuon o eksklusiv sa isang tiyak na profesyon. Halimbawa nito ay ang "medical report" ng mga doktor at "legal forms" ng mga abogado.
- **Mallikhalin** - Ito ang masining na uri ng pagsulat at ito ay mayaman sa layutay, idyoma at simbolismo kagaya ng pagsulat ng tula, nobela at maikling katha.

PANANALIKSIK

Depinisyon ng Pananaliksik

- Ito ay isang maingat, kritikal, at disiplinadong "inquiry" sa pamamagitan ng iba't-ibang teknik at paraan batay sa kalikasan at kalagayan ng natukoy na suliranin tungo sa klarifikasyon at/o resolusyon nito (Good, 1963)
- Ito ay isang sistematikong paghahanap sa mga mahahalagang informasyon hinggil sa isang tiyak na paksa o suliranin (Aquino, 1974).
- Ito ay isang pagtatangka upang makakuha ng mga solusyon sa mga suliranin (Trece, et al., 1973).

Ang Pamanahong Papel

Ang pamanahong papel ay isang uri ng papel-pampananaliksik na karaniwang ipinagagawa sa mga estudyante sa kolehiyo bilang isa sa mga pangangailangan sa isang larangang akademiko. Ito ay binubuo ng mga sumusunod na bahagi:

- a. Mga Pahinang Preliminari o Front Matters
- b. Kabanata I: Ang Suliranin at Kaligiran Nito
- c. Kabanata II: Mga Kaugnay na Pag-aaral at Literatura
- d. Kabanata III: Disenyo at Paraan ng Pananaliksik
- e. Kabanata IV: Presentasyon at Interpretasyon ng mga Datos
- f. Kabanata V: Lagom, Kongklusyon at Rekomendasyon
- g. Mga Panghuling Pahina

FILIPINO 3: RETORIKA O MASINING NA PAGPAPAHAYAG

Depinisyon ng Retorika

- Ayon kay Aristotle, ito ay ang fakulti ng pagtuklas ng lahat ng aveylabel na paraan ng panghikayat sa anumang partikular na kaso (Bernales, et al., 2006).
- Para kay Cicero, ito ay isang pagpapahayag na dinisenyo upang makapanghikayat (Bernales, et al., 2006).
- Ayon kay Quintilian, ito ay ang sining ng mahusay na pagsasalita (Bernales, et al., 2006).

Wastong Gamit ng mga Salita

1. Nang at Ng

Nang

- a. Ginagamit ang nang bilang pangatnig sa mga hugnayang pangungusap; kalumbas ng "when" sa Ingles.

Hal.: Tulog na ang mga anak nang dumating ang kanilang ina.

- b. Ginagamit ang nang sa gitna ng dalawang salitang-ugat na inuulit.

Filipino

Hal.: tapon nang tapon

c. Ang nang ay nagmula sa na at inaangkupan ng ng at inilalagay sa pagitan ng pandiwa at ng pang-abay

Hal.: Nag-aral nang mabuti si Juan.

Ng

a. Ginagamit ang ng bilang pananda sa tuwirang layon ng pandiwang palipat.

Hal.: Nagtanim ng palay si Maria na isang magsasaka.

b. Ginagamit ang ng bilang pananda ng aktor o tagaganap ng pandiwa.

Hal.: Tinulungan ng dalaga ang kanyang lola sa pagtawid.

c. Ginagamit ang panandang ng kano nagsasaad ng pagmamay-ari ng isang bagay o katangian.

Hal.: Ang boses ng bayan ang siyang dapat na mananalig.

2. May at Mayroon**May**

a. Ginagamit ang may kapag sinusundan ng pangngalan.

Hal.: May kasalanang ginawa sina Nicole at Adrian kagabi.

b. Ginagamit ang may kapag sinusundan ng pandiwa.

Hal.: May tumawa dahil sa nasabing balita.

c. Ginagamit ang may kapag sinusundan ng pang-uri.

Hal.: May magandang karanasan si Maya tungkol sa pag-ibig.

d. Ginagamit ang may kapag sinusundan ng panghalip na panao sa kaukulang paari.

Hal.: Ang mga anak ni Mang Juan ay may kani-kanila nang pamilya.

Mayroon

a. Ginagamit ang mayroon kapag may napapasingit na kataga (kagaya ng po, pa, din at rin) sa salitang sinusundan nito.

Hal.: Mayroon po kaming ipagtatapat sa inyo. (ang katagang ginamit sa pangungusap na ito ay "po")

b. Ginagamit ang mayroon bilang panagot sa isang tanong.

Hal.: May pera ka ba? — Mayroon.

c. Ginagamit ang mayroon kung nangangahulugan ng pagka-maykaya o mayaman.

Hal.: Ang mga Garcia ay mayroon sa probinsya ng Cebu.

3. Kung at Kong**Kung**

a. Ginagamit ang kung bilang isang pangatnig sa mga hugnayang pangungusap; katumbas ng if sa Ingles.

Hal.: Kung may problema ka, puntahan mo lang ako.

Kong

a. Ang kong ay nanggaling sa panghalip na panaong ko at inaangkupan lamang ng ng.

Hal.: Ang tangi kong hangad ngayong taon ay ang makapasa sa L.E.T.

4. Din/Daw at Rin/Raw**Din/Daw**

a. Ginagamit ang din/daw kung ang salitang sinusundan ay nagtatapos sa katinig maliban sa w at y.

Hal.: Magpapatingin daw siya sa doktor ngayon.

Rin/Raw

a. Ginagamit ang rin/rawkung ang salitang sinusundan ay nagtatapos sa patinig at sa malapating na w at y.

Hal.: May handa raw layo sa darating na kaurawan ni tatay.

5. Sina at Sila**Sina**

a. Ang sina ay panandang pangkayarian sa pangngalan.

Hal.: Sina Fe at Maria ay pupunta sa Baguio.

Sila

a. Ang sila ay ginagamit bilang isang panghalip panao; katumbas ng they sa Ingles.

Hal.: Sila ay pupunta sa Baguio.

Ang Mga Idyoma

Ang mga idyoma ay mga di-tuwiran o di-tahasang pagpapahayag na may kahulugang patalinhaga. Ito ay di-literat kung kaya nangangailangan ng konotativo at malalim na pagpapakahulugan.

Mga Halimbawa:

putok sa buho	-	anak sa pagkadalaga
mababaw ang luha	-	madaling umiyak
naglutubid ng buhangin	-	nagsisinungaling
magbatak ng buto	-	magtrabaho
nakahiga sa salapi	-	mayaman
magdildil ng asin	-	maghirap
balat-kalabaw	-	hindi marunong mahiya
magmahabang dulang	-	magpapakasal, mag-asawa
nagpuputok ang butse	-	galit na galit
hilong talilong	-	litung-lito

Mga Uri ng Tayutay

1. **Alliterasyon** - pag-uulit ng mga tunog-katinig sa inisyal na bahagi ng salita
Hal.: Makikita sa mga mala ni Maria ang mga masasayang nangyari sa kaniya kasama si Marco. (makikita, mga, mala, Maria, masasayang, Marco)
2. **Konsonans** - pag-uulit ng mga tunog-katinig sa final naman na bahagi ng salita
Hal.: Ang aking pagmamahal para kay Rosal ay lalong tumatagatag liabang tumatagal. (pagmamahal, Rosal, tumatagatag)
3. **Asonans** - pag-uulit ng mga tunog-patinig sa nalinang bahagi ng salita

Filipino

Hal.: Ang aking alagang aso ay agad kong pinaliguan pagdating ko sa amin.
(ang. aking, alagang, aso, agad, amin)

4. **Onomatopiya** - Sa pamamagitan ng tunog o himig ng salita ay nagagawang maihatid ang kahulugan nito.
Ang dagundong ng kulog ang siyang gumising sa bata.
5. **Anafora** - pag-uulit sa unang bahagi ng pahayag o ng isang taludtod
Hal.: Ikaw ang aking pangarap.
Ikaw ang bigay ng maykapal.
Ikaw ang lahat sa akin.
6. **Eplora** - pag-uulit sa huling bahagi ng pahayag o ng isang taludtod
Hal.: Ang Konstitusyon ay para sa mamamayan,
Gawa ng mamamayan.
At mula sa mamamayan.
7. **Anadiplosis** - pag-uulit sa una at huling bahagi ng pahayag o ng isang taludtod
Hal.: Ang mahal ko ay tanging ikaw.
Ikaw na nagbibigay ng ilaw
Ilaw sa gabi na koy dilim
Dilim man o liwanag, ikaw ay mahal pa rin
8. **Pagtutulad o Simili** - Isang di tuwirang paghahambing ng magkaibang bagay, tao o pangyayari pagkat gumagamit ng mga pariralang tulad ng, kawangis ng, para ng, at gaya ng.
Hal.: Tulad ng isang ibon, ang tao ay namamalay.
9. **Pagwawangis o Metafor** - isang di tuwirang paghahambing ng magkaibang bagay, tao o pangyayari pagkat HINDI NA gumagamit ng mga nabanggit na parirala sa itaas.
Hal.: Ang kanyang buhay ay isang bukas na aklat.
10. **Pagbibigay** - katauhan o Personifikasyon - Inaaring tao ang mga bagay na walang buhay sa pamamagitan ng pagkakapit sa mga ito ng mga gawi o kilos ng tao.
Hal.: Nararamdaman ko siya sa pamamagitan ng haplos ng hangin.
11. **Pagmamalabis o Iperboli** - Lagpas sa katotohanan o eksaherado ang mga pahayag kung pakasusuriin.
Hal.: Gutom na gutom si Juan na kaya niyang kumain ng isang kalabaw.
12. **Pagpapalit - tawag o Metonimi** - ang pagpapalit ng katawagan o ngalan sa bagay na tinutukoy.
Hal.: Ang palasyo ay nag-angunsa na walang pasok bukas. (Palasyo - Presidente ng Pilipinas)
13. **Pagpapalit - saklaw o Sinekdoki** - ang pagbabanggit ng bahagi bilang pagtukoy sa kabuuan.
Hal.: Apat na mga mata ang patuloy na tumititig sa kanya.
14. **Paglumanay o Eupemismo** - tumutukoy sa paggamit ng mga salitang magpapabawas sa lindi ng kahulugan ng orihinal na salita
Hal.: Ang kanyang kasintahan ay isang babaeng mababa ang lipad. (Mababa ang lipad - Prostitiyut)
15. **Tanong Retorikal** - isang tanong na hindi naman talaga kailangan ng sagot kundi ang layunin ay maikintal sa isipan ng nakikinig ang mensahe
Hal.: Natutulog ba ang Diyos?
16. **Pagsusukdol o Klaimaks** - paghahanay ng mga pangyayaring may papataas na tinig, sitwasyon o antas
Hal.: Mablis na humupa ang hangin, napawi ang malakas na ulan, muling sumilay ang liwanag ng araw na nagbabadya ng panibagong pagkakataon para muling bumangon sa buhay!
17. **Antiklaymaks** - paghahanay ng mga pangyayaring may papababa na tinig, sitwasyon o antas

Hal.: Noon, ang bulwagang iyon ay puno ng mga nagkakagulong tagahanga, hanggang sa unti-unting nababawasan ang mga manonood, padalang nang padalang ang mga pumalakpak at ngayo'y maging mga bulong ay waring sigaw sa kanyang pandinig.

18. **Pagtatambis o Oksimoron** - Ito ay ang paglalahad ng mga bagay na magkasalungat upang higit na mapatingkad ang bisa ng pagpapahayag. Ito ay kadalasang mahaba.
Hal.: Ang buhay sa mundo ay tunay na kakalawa: may lungkot at may ligaya, may dilim at may liwanag, may tawa at may luha, may hirap at may ginhawa, may dusa at may pag-asal
19. **Pagsalungat o Epigram** - kahawig ng pagtatambis kaya nga lamang ay maikli at matalinhaga
Hal.: Natalo siya upang muling manalo.
20. **Pag-uyam o Ironiya** - isang pagpapanayag na may layuning mangutya ngunit itinatago sa paraang waring nagbibigay-puri
Hal.: Siya ay may magandang mukha na kung saan tanging ina niya lang ang humahanga.
21. **Paralelismo** - isang pagpapahayag na may halos lisang istruktura
Hal.: Iyan ang disiplinang militar: sama-samang lulusob sa mga kaaway, sama-sama rin kaming mamamatay o sama-sama rin kaming magtatagumpay.
22. **Pagtawag** - isang pahayag na nakikipag-usap sa isang karaniwang begay na wari nakikipag-usap sa isang tao.
Hal.: Pag-ibig, Oh kay lupit mo!
23. **Pagtanggi** - gumagamit ng panangging HINDI upang malpahayag ang makabuluhang pagsang-ayon
Hal.: Si Lucas ay hindi sinungaling, hindi lamang niya kaya ang magsabi ng totoo.
24. **Pagdaramdam** - nagsasaad ng hindi pangkaraniwang damdamin
Hal.: Nakakaawa ang sinapit niya. Dahil sa pagtataksil niya ay iniwan siya ng kanyang asawa.

ANG PANITIKAN NG PILIPINAS

Depinisyon ng Panitikan

- Ito ay ang katipunan ng mga akdang nasusulat na makilala sa pamamagitan ng malikhaing pagpapahayag, aestetikong anyo, pandaigdalang kaisipan, at kawalang-maliw (Webster, 1974).
- Ito ay katumbas ng "literatura" sa wikang Kastila at "literature" naman sa wikang Ingles. Ito ay mula sa salitang Latin na "litera" na ang ibig sabihin ay "letra" o "litik" (Mateo, 1996).
- Ito ay isang salamin ng lahi (Bisa, 1987).

Anyo ng Panitikan

- **Anyong Tuluyan** - ang anyo ng panitikan napatalata o ang karaniwang takbo ng pangungusap at gumagamit ng payak at direktang paglalahad ng kaisipan.
- **Anyong Patula** - ang anyo ng panitikan na pataludtod, may sukat at tugma o malayang taludturan at gumagamit ng masining at matalinhagang

Mga Uri ng Anyong Tuluyan

- **NOBELA** - mahabang salaysay na nahahati sa mga kabanata
- **MAIKLING KWENTO** - maikling katha na nagsasalaysay ng pang-araw-araw na

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- buhay na may ilang tauhan lang, pangyayari at may isang kakintalan
- **DULA** - sinasadula at tinatanghal sa tanghalan
- **ALAMAT** - nagsasalaysay sa pinagmulan ng isang bagay
- **PARABULA** - katha mula sa Bibliya
- **PABULA** - kwentong may aral at hayop ang pangunahing tauhan
- **TALAMBUHAY** - akda sa kasaysayan ng buhay ng isang tao
- **SANAYSAY** - akdang tumatalakay sa isang paksa at naglalayong maglahad ng opinyon o pananaw
- **TALUMPATI** - binibigkas sa harap ng madla
- **BALITA** - naglalahad sa mga pang-araw-araw na mga pangyayari sa lipunan, pamahalaan, industriya at iba pang paksang nagaganap sa buong bansa
- **ANEKDOTA** - kwento na ang pangyayari ay hango sa tunay na karanasan, nakawiwili at kapupulutan ng aral
- **EDITORIAL** - isang sanaysay na naglalahad ng kuru-kuro o opinyon ng isang editor
- **KASAYSAYAN** - Ito'y tala o mga nakasulat tungkol sa mga pangyayari ng nakaraan.
- **MITOLOHIYA** - kwento hinggil sa pinagmulan ng sansinukuban, diyos, at diyosa at iba pang mga mahiwagang nilikha
- **ULAT** - nasusulat bunga ng isinasagawang pananaliksik, pagsusuri, pag-aaral at iba pa

Mga Uri ng Anyong Patul...

- **TULANG LIRIKO** - tulang naglalahad ng mga masidhing damdamin, imahinasyon at karanasan ng tao at kadalasang inaawit
- **TULANG PASALAYSAY** - nagsasalaysay ng mga pangyayari sa paraang pataludtod
- **TULANG PADULA** - tulang sinadyang isulat upang itanghal sa entablado
- **TULANG PATNIGAN** - tula ng paglatalo, pangangatwiran at tagisan ng talino

IBA'T IBANG URI NG TULANG LIRIKO

- **PASTORAL** - naglalarawan ng tunay na buhay sa kabundukan
- **DALIT** - awit na pumupuri sa Diyos o sa Mahal na Birhen
- **PASYON** - aklat na naglalarawan ng pasakit at pagdurusa ni Hesuskristo
- **ODA** - nagpapahayag ng papuri at masiglang damdamin tungkol sa isang paksa
- **ELEHIYA** - madamdaming tula tungkol sa patay
- **SONETO** - tulang binubuo ng 14 na taludtod
- **KANTA** - madamdamin na tulang liriko na ang karaniwang pinapaksa ay tungkol sa pag-ibig, pag-asa at kaligayahan
- **PANUBONG** - mahabang tula ng pagpaparangal o paghahandog sa isang taong nagdaraos ng kaarawan o kapistahan

IBA'T IBANG URI NG TULANG PASALAYSAY

- **EPIKO** - mahabang tula tungkol sa magiting na pakikipagsapalaran at kabayanihan ng isang taong may pambihirang katangian
- **AWIT** - tulang maromansa kung saan nakaharap sa mga pakikipagsapalaran ang mga tauhan at hango sa tunay na buhay
- **KORIDO** - tulang maromansa kung saan ang mga tauhan ay may kakayahang supernatural

IBA'T IBANG URI NG TULANG PADULA

- **ZARZUELA** - dulang musikal na karaniwang binubuo ng tallong akto tungkol sa pag-ibig, kasakiman at poot
- **MORO-MORO** - nagpapakita ng hidwaan at labanan ng Kristyano at di-Kristyano
- **SENAKULO** - pagtatanghal tungkol sa paghihirap at kamatayan ni Hesus
- **TIBAG** - ang paghahanap nina Reyna Elena at Constantino sa krus na pinagpakuan ni Hesus
- **PANUNULUYAN** - nagpapakita ng paghahanap ng matutuluyan nina Maria at Jose para doon isilang ang sanggol na si Jesus

IBA'T IBANG URI NG TULANG PATNIGAN

- **DUPLO** - paligsahan sa pangangatwiran na kadalasang masaksihan sa paglalamay sa palay
- **BALAGTASAN** - tagisao ng talino sa pamamagitan ng katwiran sa pamamaraang patula
- **KARAGATAN** - dula tungkol sa isang prinsesa na sadyang humulog sa isang singsing sa dagat. Kung sinuman ang talaking makakakuha ng singsing ay pakakasalan niya

IMPLUWENSYA NG PANITIKAN SA DAIGDIG

- Aklat ng mga Araw - batayan ng pananampalataya ng mga Intsik
- Aklat ng Mga Palay - naglalarawan ng mga kullo ni Osiris at tumatalakay sa mitolohiyang Ehipto
- Awit ni Rolando - Isinasalaysay dito ang gintong panahon ng Kristyanismo sa Pransya
- Bibliya - ang pinakabatayan ng pananampalatayang Kristyano
- Canterbury Tales - naglalarawan ng pananampalataya at pag-uugali ng mga Ingles noong unang panahon
- Divina Comedia - nagpapahayag ng pananampalataya at pag-uugali ng mga Italyano noong unang panahon
- El Cid Campeador - naglalarawan sa kalangian ng mga Kastila at kasaysayan ng Espanya
- Iliad at Odyssey - nagsasalaysay sa mitolohiya at alamat ng Gresya
- Isang Libo at Isang Gabi - Naglalarawan ito sa pamahalaan at lipunan ng Arabya at Persya
- Koran - batayan ng pananampalataya ng mga Muslim
- Mahabharata - pinakamahabang epiko ng buong daigdig na naglalarawan sa kasaysayan ng pananampalataya sa India
- Uncle Tom's Cabin - nagbukas sa mga mata ng Amerikano sa kaapihan ng mga lahing itim at naging simula ng pagloganap ng demokrasya sa buong daigdig

IBA'T IBANG YUGTO NG PANITIKAN SA PILIPINAS

- Panitikan bago dumating ang mga Kastila
- Panitikan sa panahon ng mga Kastila
- Panitikan sa panahon ng Propaganda at Himagsikan
- Panitikan sa panahon ng mga Amerikano
- Panitikan sa panahon ng mga Hapones

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**PANITIKAN SA PANAHON NG REPUBLIKA HANGGANG SA KASALUKUYAN
PANITIKAN BAGO DUMATING ANG MGA KASTILA**

- Baybayin ang tawag sa unang sistema ng pagsulat ng mga sinaunang Pilipino.
- Mga Awiting Bayan:
 1. An-anoy - inaawit habang ang mga magsasaka ay gumagawa ng pilapil sa kanilang bukirin
 2. Ayoweng - inaawit sa pagkabyaw ng tubo
 3. Daeleng - awit tungkol sa mga pista pagdiriwang
 4. Dalit o Himno - awit sa pagpuri o pagsamba. Ito ang isa sa pinakamatandang awit ng pagdalangin sa Birhen.
 5. Danyo - awit sa pagsasamba o pananampalataya
 6. Dione - awit sa panligaw o kasalan
 7. Dung-aw - awit na nagpapahayag ng kalungkutan at pagdurusa
 8. Hillraw at Panambat - mga awit sa pag-inuman
 9. Indonatin at Kutang- mga awiting panlansangan. Awit ng lasing na naglalayong magpalawa, magpasaring o manukso.
 10. Kalusan - isang uri ng "ludawit" (awit sa lunday)
 11. Kumintang - sa simula ito'y awit sa pakikidigma (battle songs) subalit sa paglakad ng panahon, ang diwang ito'y nawala at sa halip ay naging syaw at awit sa pag-ibig.
 12. Kundiman at Ballaw - awit ng pag-ibig
 13. Maluway - awit sa sama-samang paggawa
 14. Mayeka - isang awit na panggabi ng mga Igorote
 15. Oyayi - awit sa pagpaptulog ng bata o paghehele sa sanggol
 16. Panlian - inaawit sa pagkuha ng bahay-pukyutan
 17. Papag - inaawit sa tuwing may bayuhan ng palay
 18. Salagintok - awit sa pakikipagkaibigan
 19. Sambotani - awit sa tagumpay
 20. Soliranin - awit sa paggaod ng bangka
 21. Umbay - awit sa paglilibing
- **Epiko Bago Dumating ang Mga Kastila:**
 - EPIKO NG IFUGAO**
 - ALIM** - kahawig ng Ramayana ng India. Nagpakasal ang magkapatid na sina Bagan at Wigan.
 - HUDHUD** - pakikipagsapalaran ni Aliguyan ng Gomhandam
 - EPIKO NG MUSLIM**
 - BIDASARI** - pinagmalupitan ang dalagang si Bidasari; patay kung araw at buhay kong gabi.
 - BANTUGAN** - isang prinsipal na ubod ng tapang at lakas. Namatay siya at muling nabuhay.
 - INDARAPATRA AT SULAYMAN** - itinuturing na alamat ng Mindanao. Namatay si Sulayman at muling nabuhay.
 - EPIKO NG TAGALOG**
 - KUMINTANG** - kasaysayan ng pagsusugo ng Haring Soledad sa kanyang tatlong anak na sina Bagtas, Mandukit, at Dikyaw.

EPIKO NG MGA BISAYA

HINILAWOD - kasaysayan ng pag-libigan ng mga Bathala ng mga taga-Iloilo, Antique at Aklan. Pinakamahaba at pinakamatandang epiko ng Panay.

LAGDA - kalipunan ng mga kautusan ng pamahalaan.

MARAGTAS - kasaysayan ng mga nagsitakas na sampung datung Malay dahil sa kalupitan ni Sultang Makatunaw ng Borneo.

EPIKO NG BIKOLANO

IBALON AT ASLON - hulwaran ng mabuting pamumuhay ng mga taga-Bikol.

EPIKO NG ILOKANO

BIAG NI LAM-ANG - Buha, ni Lam-Ang. Ito ay isang akda ni Pedro Bukaneg. Ito ang pinakasikat na epiko ng mga Ilokano.

PANITIKAN SA PANAHOON NG MGA KASTILA

- **Mga Akdang Panrelihiyon at Pangkagandahang Asal:**
 - a. **Ang Doctrina Cristiana** - ang kauna-unahang aklat na nallimbag sa Pilipinas
 - b. **Nuestra Señora Del Rosario** - ang ikalawang aklat na nallimbag sa Pilipinas at naglalaman ng talambuhay ng mga santo
 - c. **Ang Barlaan at Josaphat** - ang kauna-unahang nobela na nallimbag sa Pilipinas
 - d. **Ang Urbana at Feliza** - aklat na sinulat ni Padre Modesto de Castro at ito ay naglalaman ng mga pangaral tungkol sa kagandahang asal at wastong pag-uugali.
- **Mga Anyo ng Dula sa Panahon ng Kastila:**
 - a. **Karagatan** - isang tulang patnigan at hango sa isang alamat ng isang dalagang naghulog ng singsing sa dagat at ang sinumang binatang makakakita nito ay siyang pakakasalan ng dalaga
 - b. **Duplo** - isinasagawa sa patulang paglatalo at higit na pinahalagahan ang kahusayan sa paghabi ng mga taludtod. Ito ay ginagawa sa bakuran ng namatayan
 - c. **Juego de Prenda** - karaniwan ding ginagawa ito sa lamay ng palay na kung saan ang mga babaeng kasali sa larong ito ay binabansagan ng pangalan ng mga bulaklak, punongkahoy naman sa katalakihan. Sinisimulan ito sa paghahanap ng hari ng kanyang alagang ibon
 - d. **Moro-moro** - dula tungkol sa paglalaban ng mga Muslim at mga Kristyano
 - e. **Zarsuela** - dulang musikal na karaniwang binubuo ng tatlong akto tungkol sa pag-ibig, kasakiman, at poot
- **Mga Tulang Maromansa:**
 - a. **AWIT** - tulang maromansa kung saan nakaharap sa mga pakikipagsapalaran ang mga tauhan at hango sa tunay na buhay. Halimbawa: Florante at Laura, Bahay ni Segismundo
 - b. **ILORIDO** - tulang maromansa kung saan ang mga tauhan ay may kakayahang supernatural. Halimbawa: Ibong Adarna, Don Juan Tenoso, Bernardo Carpio

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PANITIKAN SA PANAHOON NG PROPAGANDA AT HIMAGSIKAN

- **Tatsulok ng Kilusang Propaganda:**
 - a. **Jose Rizal** - ang "Dakilang Manunulat" ng Kilusang Propaganda. Iilan sa kanyang mga akda: Sa Aking Mga Kabata, Noli Me Tangere, El Filibusterismo at Mi Ultimo Adios.
 - b. **Marcelo H. Del Pilar** - ang "Dakilang Political Analyst" ng Kilusang Propaganda. Iilan sa kanyang mga akda: Calingat Cayo, Dasalan at Tocsohan at Sagot ng Espanya sa Hibik ng Pilipinas.
 - c. **Graciano Lopez Jaena** - ang "Dakilang Mananalumpati" ng Kilusang Propaganda. Iilan sa kanyang mga akda: Fray Botod, El Bandolerismo sa Pilipinas at Ang Anak ng Prayle
- **Manunulat sa Panahon ng Himagsikan:**
 - a. **Andres Bonifacio** - ang "Ama ng Demokrasyang Pilipino". Iilan sa kanyang mga akda: Huling Paalam, Katapusang Hibik ng Pilipinas at Pag-ibig sa Tinubuang Lupa.
 - b. **Emilio Jacinto** - ang "Utak ng Katipunan". Iilan sa kanyang mga akda: Ang Kartilya ng Katipunan, A La Patria at Liwanag at Dilim.
 - c. **Apolinario Mabini** - ang "Utak ng Himagsikan". Iilan sa kanyang mga akda: Ang Himagsikang Pilipino, El Liberal at El Verdadero Decalogo.

PANITIKAN SA PANAHOON NG MGA AMERIKANO

- **Iilan sa Mga Kilalang Manunulat**
 - a. **Jose Corazon de Jesus** - ang Makata ng Pag-ibig at Hari ng Balagtas sa kanyang panahon. Mga Akda: Isang Punongkahoy, Manggagawa, Bayan Ko.
 - b. **Alejandro Abadilla** - kilala dahil sa kanyang tulang "Ako ang Daigdig"
 - c. **Amado Hernandez** - ang Makata ng Manggagawa. Mga Akda: Isang Dipang Langit, Bayang Malaya, Ibong Mandaragit at Luha ng Buwaya.
 - d. **Ildefonso Santos** - napabantog dahil sa kanyang tulang Ang Guryon. Iba pang mga akda: Gabi, Tatlong Inakay at Sa Tabi ng Dagat.
 - e. **Deogracias Rosario** - ang Ama ng Maikling Kwento sa Pilipinas. Mga akda: Dahil sa Pag-ibig, Walang Panginoon, Ang Geisha, Dalawang Larawan at Bulaklak ng Inyong Panahon.
 - f. **Severino Reyes** - ang Ama ng Zarsuelang Tagalog. Mga Akda: Walang Sugat, Mga Bihag ni Cupido, Huling Patil, Halik ng Isang Patay at Kalupi.
 - g. **Lope K. Santos** - ang Ama ng Balarila. Mga akda: Banaag at Sikat, Pangginggera

PANITIKAN SA PANAHOON NG MGA HAPONES

- **Iilan sa Mga Kilalang Manunulat**
 - a. **Liwayway Arceo** - Mga akda: Uhaw ang Tigang na Lupa, Canal dela Reina at Ang Mag-anak na Cruz.
 - b. **Jullan Balmaceda** - Mga Akda: Sino Ba Kayo, Dahil sa Anak, Ang Palabas ni Suwan
 - c. **Francisco Rodrigo** - Kilalang Akda: Sa Pula, Sa Puti
 - d. **Narciso Reyes** - Akda: Lupang Tinubuan
 - e. **N.V.M. Gonzalez** - Akda: Lunsod, Nayon at Dagat-dagatan

PANITIKAN SA PANAHOON NG REPUBLIKA HANGGANG SA KASALUKUYAN

- Iilan sa Mga Kilalang Manunulat
 - a. Genoveva Edroza Matute - Mga Akda: Ako'y Isang Tinig, Kwento ni Mabuti
 - b. Teodoro Agoncillo - Akda: Maikling Kwentong Tagalog
 - c. Elpidio Kapulong - Akda: Planeta, Buwan at Mga Bituin
 - d. Amado Hernandez - Akda: Luha ng Buwaya
 - e. Liwayway Arceo - Akda: Nagbabagang Paraiso
 - f. Dominador Mirasol - Akda: Mga Halik sa Alabok
 - g. Lualhati Bautista - Mga Akda: Bata, Bata, Paano ka Ginawa?, Gapo, Sakada, Dekada '70 at Bulaklak ng City Jail

MGA SAGISAG PANULAT NG MGA PILIPINONG MANUNULAT

- Emilio Aguinaldo - Rosalia Magdalo, Magdalo
- Virgilio Almarino - Rio Alma
- Cecilio Apostol - Calulo, Calipso at Calypso
- Francisco Baltazar - Balagtas
- Andres Bonifacio - Agapito Bagumbayan, Maypagasa, Magdiwang
- Felipe Calderon - Simoun, Elias
- José Corazón de Jesús - Huseng Batute, Pusong Hapis, Luksang Paruparo
- Jose dela Cruz - Huseng Sisiw
- Epifanio Delos Santos - G. Solon
- Nestor Vicente Madali Gonzalez - N.V.M. Gonzalez
- Marcelo H. Del Pilar - Plaridel, Dolores Manapat, Piping Dilat, Siling Labuyo, Kupang, Hailalaga, Patos, Carmelo, D.A. Murgas, L.O. Crame, D.M. Calero, Hilario, M. Dali.
- Severino Reyes - Lola Basyang
- Fernando Ma. Guerrero - Fluvio Gil, Florisel
- Amado Hernandez - Amante Ernani, Herininia de la Riva, Julio Abril
- Emilio Jacinto - Dimas-Ilaw, Pingkian
- Nick Joaquin - Quijano de Manila
- Graciano Lopez Jaena - Bolivar, Diego Laura
- Gen. Antonio Luna - Taga-Ilog
- Juan Luna - J.B., Buan
- Apollinario Mabini - Bini, Paralitico, Katabay
- Jose Palma - Ana-haw, Esteban Estebanes, Gan Hanlik
- Jose Maria Panganiban - Jomapa, J.M.P.
- Pascual H. Poblete - Anak-Bayan
- Mariano Ponce - Nanding, Tikbalang, Kalipulako
- Dr. José Protasio Rizal Mercado y Alonso Realonda - José Rizal, Dimas-alang, Laong-Laan, Agno, Calambaño
- Lope K. Santos - Anak-Bayan, Doctor Lukas, Lakandalita
- Dr. Pio Valenzuela - Madlang-Away
- Jose Garcia Villa - Doveglion

Multiple choice

1. Ito ay isang sangay ng linggwistika na sumasaklaw sa pag-aaral ng kayarian o istruktura ng salita.

A. Ortograpiya
B. Morpolohiya
C. Semantika
D. Sintaks

Sagot: B-Morpolohiya. Ang ortograpiya ay nag-aaral sa wastong baybay ng mga salita. Ang semantika ay nag-aaral naman sa kahulugan ng salita samantalang ang sintaks ay nakapokus sa pag-aaral tungkol sa wastong pag-uugnay at pagsasaayos ng mga salita para makabuo ng isang pangungusap.

2. "Lumilindol." Anong uri ito ng pangungusap na walang liyak na paksa?

A. Temporal
B. Eksistensyal
C. Penomenal
D. Modal

Sagot: C-Ang penomenal na pangungusap ay nagpapakita ng pangyayaring pangkalikasan o pangkapaligiran.

3. Ibigay ang panlapi na ginamit sa mga sumusunod na salita: kaligayahan, pagmamahalan, at pagkatiwalaan.

A. hulapi
B. tambalan
C. kabilaan
D. laguhan

Sagot: C-Kabilaan ang tawag sa dalawang panlapi na kinakabit sa isang salitang-ugat. Ang kabilaan ay binubuo ng tatlong uri. Ito ay maaaring unlapi at gitlapi (isinulat), unlapi at hulapi (kaligayahan), gitlapi at hulapi (sinamahan).

4. Anong bantas ang siyang inilalagay sa pagitan ng unlaping "ika" at "tambalang"?

A. tuldok
B. panaklong
C. kuwit
D. gitling

Sagot: D-Maliban sa pagitan ng unlaping "ika" at "tambalang", ang gitling ay inilalagay sa pagitan ng panlaping nagtatapos sa katinig at salitang -ugat na nagsisimula sa patinig (lag-araw, pag-unlad), at sa pagitan ng panlapi at pangngalang panlangi (maka-Diyos, taga-Marikina).

5. Ayon kay Gleason, ang wika ay pinagkakasunduan ng isang lahi at kaya naman ay naunawaan ng lahat ng kasapi ng lahi.

A. masistema
B. dinamiko
C. likas
D. arbitraryo

Sagot: D-Ang wika ay arbitraryo dahil ito ay pinagkakasunduan ng isang lahi.

6. Isang akda ni Padre Modesto de Castro na binubuo ng palitan ng liham ng dalawang magkapatid.

A. Urbana at Feliza
B. Barlaan at Josaphat
C. Dasalan at Tocsohan
D. Indarapatra at Sulayman

Sagot: A-Ang Urbana at Feliza ay isang akda ni Padre Modesto de Castro na naglalaman ng mga pangaral tungkol sa kagandahang asal at dapat ugaliin sa iba't-ibang pagkakataon.

7. Anong teorya ng wika ang nagsasabing ang wika ay nailikha bunga ng masidhing damdamin ng tao?

A. Bow-wow
B. Pooh-pooh
C. Ding-dong
D. Yoheho

Sagot: B-Ang Teoryang Pooh-pooh ay ang teorya ng wika na nagsasabing ang wika ay nailikha bunga ng masidhing damdamin ng tao gaya ng sakit, luwa, sarap, kalungkutan, takot, pagkabigla at iba pa.

8. "Nahulog ang batal" Anong uri ng pangungusap ayon sa kayarian ang pangungusap na ito?

A. padamdang
B. pasalaysay
C. payak
D. tambalan

Sagot: C-Ang payak na pangungusap ay isang uri ng pangungusap ayon sa kayarian na binubuo ng isang sugnay na makapag-lisa. Ang tambalan naman ay binubuo ng dalawang sugnay na makapag-lisa. Ang pasalaysay at padamdang ay uri ng pangungusap ayon sa gamit.

9. Sa ponemang segmental, ano ang tinataglay ng mga salitang galaw, baliw, lamay, at kahoy?

A. ponema
B. klaster
C. diptonggo
D. pares minimal

Sagot: C- Upang magkaroon ng diptonggo ang isang salita, hindi lang dapat magkasunod ang isang patinig at ang malapalinig na w o y, kung hindi, dapat din na ang dalawa ay magkasama sa lisang pantig (halimbawa: ga-law, ba-liw, la-may, ka-hoy).

10. Sa kasaysayan ng ating panlikan, ang kinikilalang Gintong Panahon ng Panlikan ng Pilipinas ay ang panahon ng _____.

A. Amerikano
B. Hapones
C. Kastila
D. Kontemporaryo

Sagot: B-Pinagbawal ng pamahalaang Hapon ang mga Pilipino sa pagsulat ng anumang akda sa Ingles kaya umusbong ang mga akdang naisulat sa wikang Filipino.

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11. Alin sa mga sumusunod ay sadyang isinulat upang ibigkas sa harap ng madla?

- A. anekdota
- B. pabula
- C. talambuhay
- D. talumpati

Sagot: D-Ang talumpati ay isang halimbawa ng anyong tuluyan na binibigkas sa harap ng madla.

12. Kararating lang ni tatay mula sa trabaho. Ano ang aspekto ng pandiwa sa pangungusap?

- A. pangnagdaan
- B. pangkasalukuyan
- C. panghinaharap
- D. katalapos

Sagot: D-Katalapos ang aspekto ng isang pandiwa kapag ang kilos ay bago lang natapos. Ito ay karaniwang nagsisimula sa unlaping ka- at inuulit ang unang pantig ng salitang-ugat.

13. Anong titulo sa panitikan ng Pilipinas ang ibinigay sa manunulat ng akdang pinamagatang "Isang Dipang Langit"?

- A. Makata ng Pag-ibig
- B. Makata ng Manggagawa
- C. Makata ng Masa
- D. Ama ng Wikang Pambansa

Sagot: B-Ang may-akda ng "Isang Dipang Langit" na si Amado V. Hernandez ay tinaguriang ang "Makata ng Manggagawa". Minsan din siyang naging lider ng isang labor union at pinagtanggol niya ang karapatan ng mga manggagawa.

14. Siya ay kilala bilang ang "dakilang mananalumpati" ng kilusang propaganda.

- A. Graciano Lopez Jaena
- B. Jose Rizal
- C. Marcelo H. Del Pilar
- D. Gregorio Del Pilar

Sagot: A-Ang Triumvirate ng Kilusang Propaganda ay sina Graciano Lopez Jaena, Jose Rizal, at Marcelo H. Del Pilar. At ang tinagurian na "dakilang mananalumpati" ay si Graciano Lopez Jaena.

15. Sa pananaliksik, saang kabanata matatagpuan ang mga kaugnay na pag-aaral at literatura?

- A. Kabanata I
- B. Kabanata II
- C. Kabanata III
- D. Kabanata IV

Sagot: B-May limang kabanata ang isang pananaliksik. Ito ay ang mga sumusunod: Kabanata I (Ang Suliranin at ang Kaligiran Nito), Kabanata II (Mga Kaugnay na Pag-aaral at Literatura), Kabanata III (Disenyo at Paraan ng Pananaliksik), Kabanata IV (Presentasyon at Interpretasyon ng mga Datos), at Kabanata V (Lagom, Kongklusyon at Rekomendasyon).

16. Isang mahabang tulang pasalaysay tungkol sa kabayanihan ng pangunahing tauhan na may pambihirang katangian.

A. epiko
B. pabula
C. parabula
D. dalit

Sagot: A-Epiko. Ang pabula ay tungkol sa kwento na ang nagsisiganap ay mga hayop, parabula ay mula sa Bibliya at ang dalit ay awit na pumupuri sa Diyos.

17. Alin sa mga sumusunod ang HINDI epiko ng Mindanao?

A. Bantugan
B. Didasari
C. Maragtas
D. Indarapatra at Sulayman

Sagot: C-Ang Maragtas ay epiko ng mga Bisaya.

18. Sino ang may-akda ng Fray Botod?

A. Jose Garcia Villa
B. Graciano Lopez Jaena
C. Marcelo del Pilar
D. Jose Rizal

Sagot: B-Ang akdang "Fray Botod" ay isang akdang lumutulgisa sa kamangmangan at pagmamalabis ng mga prayle. Ito ay sinulat ni Graciano Lopez Jaena.

19. Siya ay isang Pilipinong manunulat na tanyag sa kanyang sagisag-panulat na Dimas-ilaw.

A. Jose dela Cruz
B. Antonio Luna
C. Jose Corazon de Jesus
D. Emilio Jacinto

Sagot: D-Si Emilio Jacinto ay si "Dimas-ilaw" at siya ang kanang kamay ni Andres Bonifacio. Si Jose dela Cruz ay Huseng Sisiw, Antonio Luna ay Taga-Ilog, at Jose Corazon de Jesus ay Huseng Batule.

20. Ito ay isang tulang maromansa na kung saan nakaharap sa mga pakikipagsapalaran ang mga tauhan at hango sa tunay na buhay.

A. Moro-moro
B. Epiko
C. Awit
D. Korido

Sagot: C-May dalawang tulang maromansa sa Panitikang Filipino. Ito ay ang awit at korido. Ang awit ay makatotohanan dahil ito ay hango sa tunay na buhay (halimbawa: Florante at Laura). Ang korido ay di-makatotohanan dahil ang mga tauhan ay may kakayahang supernatural (halimbawa: Ibong Adarna).

21. Ang salitang "parak" ay nasa anong antas ng wika?

A. Jargon
B. Pidgin
C. Kolokyal
D. Balbal

Sagot: D-Kapag sinabing balbal, ito ay mga salita na ginagamit sa usapang kalye, at ito ang pinakadinamiko at pinakamababang antas ng wika. Sa balbal, kadalasan

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ay pinapalitan ang mga salita ng iba pang salita (pulis-parak) o di kaya ay binaliktad at dinagdagan ang baybay ng salita (hiya-dyahi).

22. Alin sa mga sumusunod na pangngalan ang HINDI tahas?

- A. pagkain
- B. gamot
- C. lapis
- D. pag-asa

Sagot: D-Ang "pag-asa" ay isang pangngalang basal (pangngalang di-kongkreto, hindi nahihipo at hindi nakikita). Samantala, ang "pagkain", "gamot", at "lapis" ay mga pangngalang tahas (pangngalang kongkreto, materyal, nahihipo, at nakikita).

23. Alin sa mga sumusunod ay isang pangngalang palansak?

- A. pag-ibig
- B. Jose
- C. bahay-kubo
- D. buwig

Sagot: D-Ang pangngalang palansak ay isang pangngalan na tumutukoy sa grupo o pangkat ng isang uri ng tao, hayop o bagay (halimbawa: buwig, tropa, lahi).

24. Ang "pangarap" ay isang pangngalang _____.

- A. pantangi
- B. palansak
- C. tahas
- D. basal

Sagot: D-Ang "pangarap" ay isang pangngalang basal (pangngalang di-kongkreto, hindi nahihipo at hindi nakikita).

25. Hangerin nilong makapagbigay nang wasto at epektibong pakikipag-ugnayan gamit ang mga sagisag pangwika?

- A. talastasan
- B. bokabularyo
- C. talasanggunian
- D. linggwistika

Sagot: A-Ang "talastasan" o kilala rin sa tawag na "komunikasyon" ay isang proseso ng pagpapalitan ng kasisipan gamit ang mga simbolong pangwika at may layuning makapagbigay ng matagumpay na pakikipag-ugnayan at pagkakaintindihan.

26. Nasa anong antas ng wika kabilang ang mga salitang tulad ng NASAN at PA'NO?

- A. Pabalbal
- B. Kolokyal
- C. Panretorika
- D. Pampanitikan

Sagot: B-Ang kolokyal ay binubuo ng mga salitang impormal na pinaikli katulad ng talay-tay, mayroon-meron, saan-san.

27. Anong antas ng wika ang salitang DYAHÍ?

- A. Jargon
- B. Pidgin
- C. Kolokyal
- D. Balbal

Sagot: D-Kapag sinabing balbal, Ito ay mga salita na ginagamit sa usapang kalye, pinakadinamiko at pinakamababa na antas ng wika. Sa balbal, kadalasan ay pinapalitan ang mga salita ng iba pang salita (pulis-parak) o di kaya ay binaliklad at dinagdagan ang baybay ng salita (hiya-dyahi).

28. Ang salitang "bana" ay halimbawa ng anong antas ng wika?

- A. Sosyolek
- B. Pabalbal
- C. Lalawiganin
- D. Idyolek

Sagot: C-Ang lalawiganin o dayalekto ay isang antas ng wika na nakabatay sa lugar o rehiyon (dimensyong heograpiko). Ang "bana" ay isang salitang Bisaya na ang ibig sabihin ay "asawang lalaki".

29. Anong uri ng pangungusap ang "Magandang araw po"?

- A. Temporal
- B. Pormulasyong Panlipunan
- C. Sambitla
- D. Pamanahon

Sagot: B-Ang "Pormulasyong Panlipunan" ay isang uri ng pangungusap na walang tiyak na paksa na tumutukoy sa ekspresyon o pagbati na ginagamit sa pakikipagkapwa katulad ng "Magandang Umaga!", "Maraming Salamat!" at "Walang Anuman."

30. Ito ay isang uri ng morpema ayon sa kahulugan na may kahulugan sa ganang sarili. Ito ay nakakatayo ng mag-isa sapagkat may angkin itong kahulugan na hindi na nangangailangan ng iba pang morpema.

- A. Morpemang Leksikal
- B. Di-malayang Morpema
- C. Malayang Morpema
- D. Morpemang Pangkayarian

Sagot: A-Ang "Morpemang Leksikal" ay tinalawag ding morpemang pangnilalaman. Ito ay nakakatayo nang mag-isa sapagkat may angkin itong kahulugan na hindi na nangangailangan ng iba pang morpema. Ang mga pangngalan, panghalip, pandiwa, pang-uri at pang-abay ay mga morpemang leksikal.

31. Anong sangay ng linggwistika ang nakatuon sa tamang pagsasaayos ng mga salita para makabuo ng isang pangungusap na nagsasaad ng buong diwa?

- A. Ortograpiya
- B. Morpolohiya
- C. Semantika
- D. Sintaks

Sagot: D-Sintaks. Ang ortograpiya ay nag-aaral sa wastong baybay ng mga salita. Ang semantika ay nag-aaral naman sa kahulugan ng salita samantala ang morpolohiya ay sumasaklaw sa pag-aaral ng kayarian o istruktura ng salita.

Ilipino

32. "Maraming Salamat." Ito ay anong uri ng pangungusap?

- | | |
|----------------------------|--------------|
| A. Temporal | C. Sambitla |
| B. Pormulasyong Panlipunan | D. Pamanahon |

Sagot: B-Ang "Pormulasyong Panlipunan" ay isang uri ng pangungusap na walang tiyak na paksa na tumutukoy sa ekspresyon o pagbati na ginagamit sa pakikipagkapwa katulad ng "Magandang Umaga!", "Maraming Salamat!" at "Walang Anuman."

33. Isang dulong pampanitikan na kilala sa katawagan na reader-response theory.

- | | |
|------------------|------------------|
| A. Impresyonisya | C. antropolohiya |
| B. patalambuhay | D. pansikolohiya |

Sagot: A-Ang Impresyonista ay isang dulong pampanitikan na kung saan ang pagpapakahulugan sa isang tekstong binasa ay nakapokus sa sariling panlasa ng mambabasa at kilala rin sa katawagan na reader-response theory.

34. Ibigay ang panaguri sa pangungusap na nasa ibaba.

"Ang gatas ay mayaman sa sustansya."

- | | |
|--------------|------------|
| A. sustansya | C. gatas |
| B. ang | D. mayaman |

Sagot: D-May dalawang bahagi ang isang pangungusap: simuno at panaguri. Ang simuno o paksa ay siyang pinag-uusapan sa pangungusap, samantalang ang panaguri ay ang nagsasabi kung ano ang ginagawa o ano tungkol ang simuno. Sa pangungusap sa itaas, "ang gatas" ang simuno at "mayaman" ang panaguri.

35. Tukuyin kung anong uri ng panlapi ang matatagpuan sa mga sumusunod na salita: kabutihan, pag-ipunan, magsuyuan.

- | | |
|-------------|-------------|
| A. hulapi | C. kabilaan |
| B. tambalan | D. laguhan |

Sagot: C-Kabilaan ang tawag sa dalawang panlapi na kinakabit sa isang salitang-ugat. Ang kabilaan ay binubuo ng tatlong uri. Ito ay maaaring unlapi at gillapi (isinulat), unlapi at hulapi (kaligayahan), gillapi at hulapi (sinamahan).

36. Alin sa mga sumusunod ang tumutukoy sa isang talumpati na kung saan maagang ipinaalam ang mga kalahok tungkol sa ano ang paksa ng talumpati?

- | | |
|-----------------------|--------------|
| A. may kahandaan | C. impromptu |
| B. biglaang talumpati | D. di-handa |

Sagot: A-Sa talumpating "may kahandaan", ang talumpati ay talumpati na binubuo ng isang paksa lamang at maagang ipinaalam ang mga kalahok tungkol sa nasabing paksa.

37. Anong bahagi ng pananalita na nagsasaad ng kilos, galaw, at pangyayari?

- A. pangngalan
- B. panghalip
- C. pang-uri
- D. pandiwa

Sagot: D-Pandiwa. Ang pangngalan ay nagsasaad ng pangalan ng tao, bagay, hayop o lugar. Ang panghalip naman ay siyang humahalili sa pangngalan at ang pang-uri ay naglalarawan ng pangngalan at panghalip.

38. Si Ana ay mabagal na naglakad papunta sa altar. Anong bahagi ng pananalita ang sinalangguhitang salita?

- A. pang-ukol
- B. pang-abay
- C. pang-uri
- D. pandiwa

Sagot: B-Pang-abay. Ang pang-abay ay naglalarawan ng pandiwa, pang-uri at kapwa pang-abay. Sa pangungusap sa itaas, ang salitang "mabagal" ay naglalarawan sa salitang "naglakad" na isang pandiwa. Ginamit ang salitang "mabagal" bilang isang pang-abay na pamaraan.

39. Si Fe ay masaya sa kanyang kaarawan. Ang salitang "masaya" ay isang _____.

- A. pang-ukol
- B. pang-abay
- C. pang-uri
- D. pandiwa

Sagot: C-Pang-uri. Ang pang-uri ay naglalarawan ng pangngalan at panghalip. Sa pangungusap sa itaas, ang salitang "masaya" ay naglalarawan kay "Fe" na isang pangngalan.

40. Si Fe ay masayang naghihintay sa kanyang ina. Paano ginamit ang salitang sinalangguhitang pangungusap?

- A. pang-ukol
- B. pang-abay
- C. pang-uri
- D. pandiwa

Sagot: B-Pang-abay. Sa pangungusap sa itaas, ang naglalarawan ng salitang "masayang" ay ang salitang "naghihintay" hindi si "Fe". At ang salitang "naghihintay" ay isang pandiwa, kaya ang tamang sagot ay pang-abay.

41. "Tanghali na." Anong uri ito ng pangungusap na walang tiyak na paksa?

- A. Temporal
- B. Eksistensyal
- C. Penomenal
- D. Modal

Sagot: A-Ang temporal na pangungusap ay nagpapakita ng oras o araw.

Filipino

42. Anong sangay ng linggwistika ang nakatuon sa pag-aaral ng kahulugan ng tunog o ponema?

- A. Ortograpiya
- B. Morpolohiya
- C. Semantika
- D. Ponolohiya

Sagot: D-Ponolohiya. Ang ortograpiya ay nag-aaral sa wastong baybay ng mga salita. Ang semantika ay nag-aaral naman sa kahulugan ng salita samantalang ang morpolohiya ay sumasaklaw sa pag-aaral ng kayarian o istruktura ng salita.

43. Sa pananaliksik, saang kabanata matatagpuan ang tungkol sa suliranin at ang kaligiran nito?

- A. Kabanata I
- B. Kabanata II
- C. Kabanata III
- D. Kabanata IV

Sagot: A-May limang kabanata ang isang pananaliksik. Ito ay ang mga sumusunod: Kabanata I (Ang Suliranin at ang Kaligiran Nito), Kabanata II (Mga Kaugnay na Pag-aaral at Literatura), Kabanata III (Disenyo at Paraan ng Pananaliksik), Kabanata IV (Presentasyon at Interpretasyon ng mga Datos) at Kabanata V (Lagom, Kongklusyon, at Rekomendasyon).

44. Aling salita ang may diptonggo?

- A. awal
- B. sibuyas
- C. bahay
- D. lahat ng nabanggit

Sagot: C- Upang magkaroon ng diptonggo ang isang salita, hindi lang dapat magkasunod ang isang patinig at ang malapatinig na w o y, kung hindi, dapat din na ang dalawa ay magkasama sa isang pantig (halimbawa: ga-law, ba-liw, la-may, ka-hoy).

45. Siya ay kilala sa kanyang sagisag-panulat na Huseng Sisiw.

- A. Julian Cruz Balmaceda
- B. Jose Corazon de Jesus
- C. Florentino Collantes
- D. Jose dela Cruz

Sagot: D-Si Jose Dela Cruz ang guro ni Francisco Balagtas. Tinawag siyang Huseng Sisiw dahil sa tuwing may magpapagawa sa kanya ng tula, ang hihingin niya na kabayaran ay hindi pera kundi isang sisiw.

46. Ano ang pamagat ng ating pambansang awit?

- A. Bayang Magiliw
- B. Perlas ng Silangan
- C. Alab ng Puso
- D. Lupang Hinirang

Sagot: D- "Lupang Hinirang" ang pamagat ng ating pambansang awit at hindi "Bayang Magiliw". Si Jose Palma ang sumulat sa titik nito at nilapatan ng musika ni Julian Felipe.

47. Nagpupuyos sa galit ang kanyang ina nang malaman nito ang pagbubulakbol sa klase.

- | | |
|--------------------------|----------------------------|
| A. inis na inis sa galit | C. na-ngingitngit sa galit |
| B. nagbubuhos ng galit | D. nakikimkim ng galit |

Sagot: C- Ang nagpupuyos sa galit ay isang idyoma na ang ibig sabihin ay na-ngingitngit sa galit.

48. "Kumakain ng prutas si Jherame". Ibigay ang pokus ng pandiwa na nasa pangungusap.

- | | |
|------------|----------------|
| A. ganapan | C. tagaganap |
| B. sanhi | D. tagatanggap |

Sagot: C- Ang paksa na si "Jherame" ang siyang gumaganap sa kilos na "kumakain" kaya ang pokus ng pandiwa ay pokus sa tagaganap o aktor.

49. Hayun ang mga bata na masayang naglalaro. Ang salitang hayun ay isang _____.

- | | |
|----------------|---------------|
| A. pang-abay | C. pangngalan |
| B. pang-angkop | D. panghalip |

Sagot: D- Ang hayun ay isang panghalip pamatlig. Ita pang halimbawa ng panghalip pamatlig ay ang mga sumusunod : doon, diyan, at dito.

50. "Pinagbakasyunan nina Jhera at Tonton ang Camotes". Ano ang pokus ng pandiwa na nasa pangungusap?

- | | |
|------------|----------------|
| A. ganapan | C. tagaganap |
| B. sanhi | D. tagatanggap |

Sagot: A- Ang paksa na "ang Camotes" ay ang lugar kung saan naganap ang kilos na "pinagbakasyunan" kaya ang pokus ng pandiwa ay pokus sa ganapan.

51. Alin sa mga salita sa ibaba ang nasa kayariang KPPKKPKPKP?

- | | |
|---------------|---------------|
| A. kasaysayan | C. katapangan |
| B. heograpiya | D. klastering |

Sagot: B-Ang heograpiya ay nasa kayariang KPPKKPKPKP. Sa pagtukoy ng kayarian, ginagamit ang simbolong P para sa patinig at K para sa katinig.

52. "Ipinagluto ng kanyang asawa si Jestoni". Ano ang pokus ng pandiwa na nasa pangungusap?

- | | |
|------------|----------------|
| A. ganapan | C. tagaganap |
| B. sanhi | D. tagatanggap |

Sagot: D-Ang paksa na "si Jestoni" ay ang tagatanggap ng kilos na "ipinagluto" kaya ang pokus ng pandiwa ay pokus sa tagatanggap.

Filipino

53. Sino ang may-akda ng Dasalan at Tocsohan?

- A. Graciano Lopez Jaena
- B. N.V.M. Gonzalez
- C. Andres Bonifacio
- D. Marcelo H. del Pilar

Sagot: D-Ang Dasalan at Tocsohan ay akda ni del Pilar. Ito ay kahawig ng mga dasal na ilinuturo ng mga prayle sa mga Pilipino. Sa dasal na ito, lantaran niyang tinutuligsa ang mga kasalanan ng mga prayle noon.

54. Isang tanyag na Pilipinong manunulat na may sagisag-panulat na "Agap-Ito Bagumbayan".

- A. Graciano Lopez Jaena
- B. N.V.M. Gonzalez
- C. Andres Bonifacio
- D. Marcelo H. del Pilar

Sagot: C-Ilan sa kilalang sagisag-panulat ni Bonifacio ay ang mga ss.: May Pag-asa, Magdiwang at Agap-Ito Bagumbayan.

55. Isang awiting bayan na tungkol sa pakikipagkaibigan.

- A. sambotani
- B. salagintok
- C. daeleng
- D. oyayi

Sagot: B-Ang sambotani ay awit sa tagumpay, ang daeleng ay sa pista, at oyayi ay sa pagpapatulog ng bata.

56. Sino si Dolores Manapat?

- A. Graciano Lopez Jaena
- B. N.V.M. Gonzalez
- C. Andres Bonifacio
- D. Marcelo H. del Pilar

Sagot: D-Ilan sa mga sagisag panulat ni del Pilar ay ang mga ss.: Plaridel, Piping Dilat at Dolores Manapat.

57. Isang tulang maromansa na kung saan ang mga tauhan ay may kapangyarihang supernatural at kaya naman ito ay hindi kapani-paniwala.

- A. oda
- B. korido
- C. soneto
- D. elehiya

Sagot: B - Ang oda, soneto at elehiya ay hindi tulang maromansa. Ang mga ito ay mga tulang liriko.

58. Ito ay isang tulang patnigan at hango sa isang alamat ng isang dalagang naghulog ng singsing sa dagat at ang sinumang binatang makakakita nito ay siyang pakakasalan ng dalaga.

- | | |
|--------------|------------|
| A. panubong | C. elehiya |
| B. karagatan | D. oda |

Sagot: B- Ang oda, panubong at elehiya ay hindi tulang patnigan. Ang mga ito ay mga tulang liriko.

59. Sino ang may akda ng awiting ang "Bayan Ko"?

- | | |
|--------------------------|--------------------|
| A. Constancio De Guzman | C. Dolores Manapat |
| B. Jose Corazon de Jesus | D. Jomapa |

Sagot: B-Ang "Bayan Ko" ay sinulat ni Jose Corazon de Jesus at nilapatan ng musika ni Constancio De Guzman.

60. Isang manunulat sa panahon ng Amerikano na naging lanyag sa kanyang tulang "Ang Guryon".

- | | |
|---------------------|-----------------------|
| A. Ildefonso Santos | C. Alejandro Abadilla |
| B. Amado Hernandez | D. Teodor Gener |

Sagot: A-Si Ildefonso Santos ay napabantog dahil sa kanyang tulang "Ang Guryon" at ang kanyang naging sagisag panulat ay "Ilaw Silangan".

61. Siya ang "Ama ng Maikling Kwento sa Pilipinas".

- | |
|-----------------------|
| A. Jose Garcia Villa |
| B. Deogracias Rosario |
| C. Aurelio Tolentino |
| D. Zulueta de Costa |

Sagot: B-Si Deogracias Rosario ay ang "Ama ng Maikling Kwento sa Pilipinas" at lilan sa kanyang mga akda ay ang mga ss. : Walang Panginoon, Ang Geisha, at Dahil sa Pag-ibig.

62. Isang dula na sumikat nang humina ang zarzuela sa Pilipinas. Ito ay tinatawag ding stage show sa Ingles.

- | |
|--------------|
| A. bodabil |
| B. duplo |
| C. karagatan |
| D. korido |

Sagot: A-Taong 1916 ipinakilala ang bodabil sa Pilipinas. Ito ay isang uri ng dula na pinaghalu-halong awitan, sayawan, drama, at katalawanan.

Filipino

63. Ito ay isang epiko tungkol sa kasaysayan ng pag-ilibigan ng mga bathala mula sa Iloilo, Antique at Aklan.

- A. Hinilawod
- B. Biag-ni-Lam-ang
- C. Bidasari
- D. Maragtas

Sagot: A- Ang Hinilawod ang tinuturing na pinakamahaba at pinakamatandang epiko ng Panay.

64. Siya ay kilala bilang ang "dakilang manunulat" ng kilusang propaganda.

- A. Graciano Lopez Jaena
- B. Jose Rizal
- C. Marcelo H. Del Pilar
- D. Gregorio Del Pilar

Sagot: B- Ang Triumvirate ng Kilusang Propaganda ay sina Graciano Lopez Jaena, Jose Rizal, at Marcelo H. Del Pilar, at ang tinaguriang ang "dakilang manunulat" ay si Jose Rizal.

65. Alin sa mga sumusunod na titik ng Alpabetong Filipino ay isang hiram?

- A. y
- B. b
- C. c
- D. ng

Sagot: C- Ang titik na c ay mula sa Alpabetong Ingles.

66. Anong teorya ng wika ang nagsasabing ang wika ay nailikha bunga ng masidhing damdamin ng tao?

- A. Bow-wow
- B. Pooch-pooch
- C. Ding-dong
- D. Yeh-heh

Sagot: B- Ang Teoryang Pooch-pooch ay ang teorya ng wika na nagsasabing ang wika ay nailikha bunga ng masidhing damdamin ng tao gaya ng sakil, tuwa, sarap, kalungkutan, takot, pagkabigla at iba pa.

67. "Katoliko ba ang Santo Papa?"

Ang pahayag sa itaas ay halimbawa ng anong uri ng tayutay?

- A. Pagdaramdam
- B. Tanong Retorikal
- C. Pagtanggil
- D. Pagsalungat

Sagot: B- Ang Tanong Retorikal ay isang tanong na may pangunahing layunin na makuha ang alensyon ng mambabasa sa pamamagitan ng pagbibigay ng tanong na hindi naman kailangang bigyan ng kasagutan.

68. Bakit Tagalog ang siyang napiling batayan ng kauna-unahang wikang pambansa sa Pilipinas?

- A. Dahil sa ito ang ginagamit ng mga taga-Manila kung saan naman malatagpuan ang kabisera ng Pilipinas.
- B. Dahil sa ito ay binubuo ng mga kaakit-akit na mga salita at bokabularyo
- C. Dahil sa ito ay tinatanggap at ginagamit na ng mas nakararaming Pilipino.
- D. A at C

Sagot: C- Maliban sa napatunayan na ang wikang Tagalog ay ang may pinakamaunlad na kayarian, mekanismo, at literatura, pinili ang Tagalog bilang batayan ng kauna-unahang wikang pambansa dahil sa ito ay tinatanggap at ginagamit na ng mas nakararaming Pilipino

69. Ilan lahat ang hiram na titik ng Alpabetong Filipino?

- | | |
|------|------|
| A. 6 | C. 8 |
| B. 7 | D. 9 |

Sagot: C- Walo ang hiram na titik ng Alpabetong Filipino mula sa mga banyagang Alpabeto: c, f, j, ñ, q, v, x at z.

70. Ano ang naging pangalan ng wikang pambansa noong 1959?

- | | |
|-------------|--------------------|
| A. Pilipino | C. Tagalog |
| B. Filipino | D. Wikang Pambansa |

Sagot: A-Pinaikli ang pangalan ng naunang wikang pambansa noong taong 1939, at ito ay naging Pilipino sa taong 1959.

71. Ito ay isang awiting bayan na ginamit sa pagpapatulog ng bata.

- | | |
|----------|--------------|
| A. diona | C. soliranin |
| B. oyayl | D. umbay |

Sagot: B-Ang diona ay para sa panliligaw, stet para sa paggaod ng bangka at umbay para sa paglilibing.

72. "Ang palasyo ay nag-anunsyo na walang pasok bukas."

Ano ang layutay ang ginamit sa pahayag sa itaas?

- A. Pagpapalit-saklaw
- B. Pagpapalit-tawag
- C. Paglanggi
- D. Pagsalungat

Sagot: B-Ang pagpapalit-tawag ay ang paggamit ng isang pangalan upang tukuyin ang isa pang pangalan na pinapalitan nito. Sa itaas, ang palasyo ay ginamit na tumutukoy naman sa presidente ng Pilipinas.

Filipino

73. "Apat na mga mata ang tumitilig sa kanya."

Ano ang tayutay ang ginamit sa pahayag na ito?

- | | |
|----------------------|-------------------------|
| A. Pagpapalit-saklaw | C. Pagtangi Pagsalungat |
| B. Pagpapalit-lawag | D. Pagsalungat |

Sagot: A-Ang pagpapalit-saklaw ay ang pagbanggit ng isang bahagi bilang paglukoy sa kabuuan. Sa itaas, ang apat na mata ay ginamit na tumutukoy naman sa dalawang tao.

74. "San ha niyo nakatira?" Ano ang antas ng wikang nakasalungguit?

- | | |
|-------------|----------------|
| A. kolokyal | C. balbal |
| B. pambansa | D. lalawiganin |

Sagot: A- Ang kolokyal ay isang halimbawa ng impormal na antas ng wika na kung saan pinapaikli ang isang salita mula sa orihinal na salita.

75. Ang "Maupay na Agal" ng mga taga-Samar ay halimbawa ng anong antas ng wika?

- | | |
|-------------|----------------|
| A. kolokyal | C. balbal |
| B. pambansa | D. lalawiganin |

Sagot: D-Waray-waray ang lawag sa dayalekto sa mga taga-Samar. Ang ibang katawagan ng dayalekto ay lalawiganin. Ang dayalekto o lalawiganin ay isang barayti ng wika na tumutukoy sa isang wika na sinasalita sa isang pook, rehiyon o lugar.

76. Ang "Hindi po namin kayo tatantanan" at "Dahil hindi natutulog ang balita 24 oras" ay mga tanyag na pahayag ni Mike Enriquez sa telebisyon. Sa anong barayti ng wika ito naiuri?

- | | |
|--------------|-------------|
| A. Jargon | C. Sosyolek |
| B. Dayalekto | D. Idyolek |

Sagot: D-Ang Idyolek ay isang barayti ng wika na kung saan iba ang wika ng bawat isa dahil ang bawat tao ay may kanya-kanyang personal na paraan sa paggamit nito.

77. "Neneng ang pangalan ng aking ermal." Ano ang antas ng wikang nakasalungguit?

- | | |
|-------------|----------------|
| A. kolokyal | C. balbal |
| B. pambansa | D. lalawiganin |

Sagot: C- Ang balbal ang pinakamababang antas ng wika. Ito ay kadalasang ginagamit sa usapang kalye.

78. Ano ang tamang spelling ng salitang barbershop sa Filipino?

- | | |
|---------------|----------------|
| A. barbersyap | C. barbershap |
| B. barbershop | D. barbersiyap |

Sagot: B- Ang mga salitang may digrapong sh ay pinananatili na lamang sa orihinal na anyo.

79. Isang kwento hinggil sa pinagmulan ng sansinukuban at kalipunan ng iba't-ibang paniniwala sa mga diyos at diyosa.

A. pabula
B. parabula
C. mitolohiya
D. anekdota

Sagot: C- Ang pabula ay tungkol sa kwento na ang nagsisiganap ay mga hayop, parabula ay mula sa Bibliya, at ang anekdota ay mga pangyayari na hango sa tunay na buhay, karanasan at kapupulutan ng aral.

80. Ito ay isang mahabang tulang pasalaysay tungkol sa kabayanihan ng pangunahing tauhan na may pambihirang katangian.

A. epiko
B. pabula
C. parabula
D. dalit

Sagot: A- Ang pabula ay tungkol sa kwento na ang nagsisiganap ay mga hayop, parabula ay mula sa Bibliya, at ang dalit ay awit na pumupuri sa Diyos.

81. Alin sa apat na uri ng akdang pampanitikan na patula ay tungkol sa pangangatwiran at tagisan ng talino?

A. tulang pasalaysay
B. tulang patnigan
C. tulang padula
D. tulang liriko

Sagot: B- Ang tulang patnigan ay isang uri ng pagtatalong patula na itinatanghal ng mga natutunggaling makata.

82. "Ikaw ang aking mahal". Ibigay ang ayos ng pangungusap na ito.

A. payak
B. tambalan
C. karaniban
D. di-karaniwan

Sagot: D- Di-karaniwan ang ayos ng isang pangungusap kung nauna ang simuno kaysa sa panaguri.

83. Ang may-akda ng "Kahapon, Ngayon at Bukas" ay si _____.

A. Aurelio Tolentino
B. Juan Abad
C. Alejandro Abadilla
D. Severino Reyes

Sagot: A- Ang "Kahapon, Ngayon at Bukas" ay isang akda ni Aurelio Tolentino. Ito ay sumikat sa taong 1903 at tungkol sa pagtutulligsa ng mga Amerikano.

Filipino

84. Siya ang sumulat ng dulang ang "Tanikalang Ginto" na kung saan inakulat ng mga alagad ng batas ang Batangas habang itinatanghal ang dulang ito at dinakip ang may-akda.

A. Aurelio Tolentino
B. Juan Abad
C. Alejandro Abadilla
D. Severino Reyes

Sagot: B-Ang "Tanikalang Ginto" ay isa sa mga unang dulang itinatanghal sa panahon ng mga Amerikano na umuusig sa mga Amerikano at sedisyoso.

85. Ang mga sumusunod ay mga nobela ni Lualhati Bautista MALIBAN sa _____.

A. Dekada '70
B. Satanas sa Lupa
C. Gapo
D. Bulaklak ng City Jail

Sagot: B-Ang Satanas sa Lupa ay akda ni Celso Al Carunungan.

86. Isang balikan at kilalang feministang manunulat na kung saan ang kanyang akda ay nakapokus sa mga kababaihan. Siya ang may-akda ng "Bata, Bata, Paano ka Ginawa?"

A. Estrella Alfon
B. Lualhati Bautista
C. Fausto Galsuran
D. Gervacio Santiago

Sagot: B-Ilan sa mga kilalang akda ni Lualhati Bautista ay ang mga sumusunod: Dekada '70, "Nena" at "Bata, Bata, Paano ka Ginawa?".

87. Ito ay isang kwento tungkol sa pagpapadala ng prinsipal sa kanyang kapatid na lalaki upang patayin ang mga halimaw sa kabilang bundok. Ito ang tinaguriang alamat ng Mindanao.

A. Alim
B. Hudhud
C. Indarapatra at Sulayman
D. wala sa nabanggit

Sagot: C-Ang Alim at Hudhud ay epiko ng mga Ifugao.

88. Handa ng lumisan ang taong "amoy lupa" nang malaman niyang nasa maayos na nakalagayan ang mga anak nito. Ang pariralang "amoy lupa" ay nagsasaad ng anong antas ng wika?

A. kolokyal
B. pambansa
C. balbal
D. pampanitikan

Sagot: D-Ang amoy lupa ay isang idyoma na ang ibig-sabihin ay malanda na at malapit nang matatay. Ang isang idyoma ay nasa antas pampanitikan.

89. Ito ay isang epiko ng mga bisaya na tungkol sa kalipunan ng mga kautusan ng pamahalaan. Gaya ng "Kodigo ni Kalantiyaw" ng tribu ng Aklan.

- A. Bantugan
- B. Lagda
- C. Bidasari
- D. Darangan

Sagot: B-Ang Bantugan, Bidasari, at Darangan ay mga epiko ng mga Muslim.

90. Siya ang may-akda ng dula ang "Dalagang Bukid".

- A. Hermogenes Ilagan
- B. N.V.M. Gonzalez
- C. Alejandro Abadilla
- D. Patricio Mariano

Sagot: A-Si Hermogenes Ilagan ang may-akda ng dula ang "Dalagang Bukid" na siyang naging batayan ng awiting panzarsuela.

91. Sino ang may-akda ng Fray Botod?

- A. Jose Garcia Villa
- B. Graciano Lopez Jaena
- C. Marcelo del Pilar
- D. Jose Rizal

Sagot: B-Ang akdang "Fray Botod" ay isang akdang tumutuligsa sa kamangmangan at pagmamalabis ng mga praylo. Ito ay sinulat ni Graciano Lopez Jaena.

92. Ito ay isang awiting bayan na tungkol sa paglilibing.

- A. Umbay
- B. kundiman
- C. sambotani
- D. soliranin

Sagot: A-Ang kundiman ay tungkol sa pag-ibig, sambotani sa tagumpay, at soliranin sa paggaod ng Bangka.

93. Alin sa mga sumusunod ay HINDI tulang pasalaysay?

- A. Moro-moro
- B. Epiko
- C. Awit
- D. Korido

Sagot: A-Ang moro-moro ay isang tulang padula na nagpapakita ng hidwaan ng mga Kristyano at ng mga Di-Kristyano.

94. Alin sa mga lula sa ibaba ang isang tulang liriko?

- A. Panunuluyan
- B. Duplo
- C. Pastoral
- D. Balagtasan

Sagot: C-Ang pastoral ay isang tulang liriko na naglalayong ilarawan o ipahayag ang tunay na buhay sa kabundukan.

Fillipino

95. Ito ay isang tulang maromansa na kung saan nakaharap sa mga pakikipagsapalaran ang mga tauhan at hango sa tunay na buhay.

- | | |
|---------|------------|
| A. oda | C. soneto |
| B. awit | D. elehiya |

Sagot: B-Ang oda, soneto at elehiya ay hindi tulang maromansa. Ang mga ito ay mga tulang liriko.

96. Alin sa mga sumusunod ang HINDI epiko ng Bisaya?

- | | |
|-------------|--------------|
| A. Lagda | C. Bidasari |
| B. Maragtas | D. Hinilawod |

Sagot: C-Ang Bidasari ay epiko ng mga Muslim.

97. Ang tinaguriang pinakasikat na epiko ng mga Ilokano ay ang _____.

- | | |
|--------------------|--------------------|
| A. Ibalon at Aslon | C. Hinilawod |
| B. Bantugan | D. Biag ni Lam-ang |

Sagot: D-Ang Biag ni Lam-ang ay isang tanyag na epiko ng mga Ilokano bago pa man dumaling ang mga Kastila, subalit nasulat lamang ito noong dakong dantaong labimpito. Ito ay akda ni Pedro Bukaneg.

98. Isang epiko na tungkol sa mga bathalang Ifugao ni Punholdayan at Makanungan. Tinutukoy rito ang pagpapakasal ng magkapatid na Bugan at Wigan.

- | | |
|-----------|------------------|
| A. Haraya | C. Hari sa Bukid |
| B. Alim | D. Lagda |

Sagot: B-Ang Haraya, Hari sa Bukid, at Lagda ay mga epiko ng mga Bisaya.

99. Alin sa mga sumusunod ay isang epiko ng mga Ifugao?

- | | |
|--------------------|--------------------|
| A. Ibalon at Aslon | C. Biag ni Lam-ang |
| B. Hudhud | D. Haraya |

Sagot: B-Ang Hudhud ay isang epiko ng mga Ifugao na tungkol sa mga buhay at pakikipagsapalaran ng mga dakilang bayani ng Ifugao na ang tanging bida ay si Aliguyan.

100. Isang manunulat sa wikang Kastila na may segisag panulat na Balikuling at nahirang na Makatang Laureado.

- | | |
|--------------------|-----------------------|
| A. Jesus Balmori | C. Alejandro Abadilla |
| B. N.V.M. Gonzalez | D. Zulueta de Costa |

Sagot: A-Si Jesus Balmori ay nahirang na Makatang Laureado dahil tinato niya si Manuel Bernabe sa balagtasang wikang Kastila.

101. Ang kauna-unahang nobelang sinulat ng isang Pilipino gamit ang wikang Ingles ay ang _____.

- | | |
|------------------------|-----------------------|
| A. The Wound and Stars | C. Like the Molave |
| B. A Child of Sorrow | D. A Vision of Beauty |

Sagot: B-Ang "A Child Of Sorrow" ay isang akda ni Zoilo Galang.

102. Siya ay isang Cebuana na ipinalalagay na pinakapangunahing manunulat na babae sa Ingles bago makadigma.

- | | |
|--------------------|--------------------|
| A. Estrella Alfon | C. Dolores Manapat |
| B. N.V.M. Gonzalez | D. Imapa |

Sagot: A-Si Estrella Alfon ay ipinanganak sa San Nicolas, Cebu at nagwagi ng mga karangalan mula sa maikling kwento at dula sa Free Press, Carlos Palanca Award, at iba pa.

103. Sino ang may-akda ng maikling kwento na "Kwento ni Mabuti"?

- | | |
|-----------------------|----------------------|
| A. Estrella Alfon | C. Genoveva Matute |
| B. Deogracias Rosario | D. Lualhati Bautista |

Sagot: C-Si Genoveva Matute ay ang may-akda ng "Kwento ni Mabuti". Ito ay tungkol sa buhay ng isang karaniwang guro na may suliranin sa buhay, ngunit ito ay hindi naging sagabal sa kanyang tapat na paglilingkod.

104. Aling salita ang may klaster?

- | | |
|------------|------------|
| A. bulsa | C. trabaho |
| B. bistado | D. kahoy |

Sagot: C-Upang magkaroon ng klaster ang isang salita, hindi lang dapat magkasunod ang dalawa o higit pa na katinig, kung hindi, dapat din na nasa isang pantig ang nasabing magkasunod na katinig.

105. "Ikaw ay may pusong bato". Ihigay ang ayos ng pangungusap na ito.

- | | |
|-------------|-----------------|
| A. payak | C. karaniwan |
| B. tambalan | D. di-karaniwan |

Sagot: D-Di-karaniwan ang ayos ng isang pangungusap kung nauna ang simuno kaysa sa panaguri.

106. Aling salita ang may diplonggo?

- | | |
|----------|----------|
| A. buwal | C. patay |
| B. bayan | D. twas |

Sagot: C-Upang magkaroon ng diplonggo ang isang salita, hindi lang dapat magkasunod ang isang patinig at ang malapatinig na w o y, kung hindi, dapat din na ang dalawa ay nasa isang pantig.

Filipino

107. Siya ang may-akda ng maikling kwento na "Uhaw ang Tigang na Lupa".

- | | |
|----------------------|--------------------|
| A. Liwayway Arceo | C. Genoveva Malute |
| B. Lualhati Bautista | D. Estrella Alfon |

Sagot: A- Si Liwayway Arceo ang may-akda ng kwentong "Uhaw ang Tigang na Lupa". Ito ay tungkol sa kwento ng mga taong uhaw sa pag-ibig at silang lahat ay maihahambing sa tigang na lupa na nauuhaw sa patak ng ulan.

108. Ikinalulungkot ko ang mga kalamidad na dumating sa Pilipinas sa taong ito. Ang kaganapan ng pandiwang nakasalangguhit sa pangungusap ay _____.

- | | |
|------------|----------------|
| A. ganapan | C. tagaganap |
| B. sanhi | D. tagatanggap |

Sagot: B-Ang paksa na "ang mga kalamidad na dumating sa Pilipinas" ay siyang dahilan o sanhi kung bakit nangyayari ang pandiwa na "ikinalulungkot".

109. Si Janeth Napolesya ay naglulubid ng buhangin. Ang pariralang "naglulubid ng buhangin" ay nagsasaad ng anong antas ng wika?

- | | |
|-------------|-----------------|
| A. kolokyal | C. balbal |
| B. pambansa | D. pampanitikan |

Sagot: D-Ang naglulubid ng buhangin ay isang idyoma na ang ibig sabihin ay nagsisinungaling. Ang isang idyoma ay nasa antas pampanitikan.

110. Ang Alibata ay hango sa alpabetong Arabo na "alif-ba-ia". Ito ay may 17 titik: 3 patinig at 14 na katnig. Ano ang ibang tawag sa alibata?

- | | |
|--------------|---------------|
| A. Baybayin | C. Diona |
| B. Cuneiform | D. Abecedario |

Sagot: A- Ito ang pinakaunang sistema ng pagsulat ng mga katutubong Pilipino

111. Sino si Herninda dela Riva sa tunay na buhay?

- | | |
|--------------------|-----------------------|
| A. Idefonso Santos | C. Alejandro Abadilla |
| B. Amado Hernandez | D. Teodor Gener |

Sagot: B-Ang mga sagisag panulat ni Amado Hernandez ay ang mga sumusunod: Amante Emani, Julio Abril, at Herninda dela Riva.

112. Alin sa mga sumusunod na akda ni Aurelio Tolentino ang siyang naging sanhi ng kanyang pagkakakulong?

- | | |
|-----------------------------|------------------|
| A. Luha ng Tagalog | C. Bagong Kristo |
| B. Kahapon, Ngayon at Bukas | D. Manood Kayo |

Sagot: B-Ang "Kahapon, Ngayon at Bukas" ay isang protesta sa pamamalakad ng mga Amerikano, isang pagbabanta at paghamon ng paglalaban.

113. Siya ang may-akda ng "Ninay".

- | | |
|-------------------|------------------------|
| A. Pedro Paterno | C. Jomapa |
| B. Emilio Jacinto | D. Isabelo delos Reyes |

Sagot: A- Si Pedro Paterno ay ang may-akda ng Ninay. Ang Ninay ay ang kauna-unahang nobelang Panlipunan sa wikang Kastila na sinulat ng isang Pilipino.

114. Alin sa mga sumusunod ay HINDI uri ng pangungusap ayon sa gamit?

- | | |
|--------------|-------------|
| A. padamdang | C. paulos |
| B. langkapan | D. patanong |

Sagot: B-Ang langkapan ay uri ng pangungusap ayon sa kayarian.

115. Sino ang may-akda ng nobelang Banaag at Sikat?

- | | |
|-------------------|--------------------------|
| A. Jose dela Cruz | C. Jose Corazon de Jesus |
| B. Lope K. Santos | D. Emilio Jacinto |

Sagot: B-Si Lope K. Santos ay isang tanyag na Pilipinong manunulat at tinaguriang "Ama ng Balarilang Tagalog".

116. Isang Pilipinong manunulat na may sagisag panulat na Dimas-ilaw.

- | | |
|-------------------|--------------------------|
| A. Jose dela Cruz | C. Jose Corazon de Jesus |
| B. Antonio Luna | D. Emilio Jacinto |

Sagot: D-Si Jose dela Cruz ay Huseng Sisiw, Antonio Luna ay Taga-Ilog, at Jose Corazon de Jesus ay Huseng Batule.

117. Siya ay may sagisag-panulat na Paralitiko at ang tinaguriang "Utak ng Himagsikan".

- | | |
|-------------------|--------------------------|
| A. Emilio Jacinto | C. Jose Corazon de Jesus |
| B. Antonio Luna | D. Apolinario Mabini |

Sagot: D-Si Apolinario Mabini ay tinaguriang "Utak ng Himagsikan". Siya ang kanang kamay ni Emilio Aguinaldo.

118. Ito ay isang satirikong bersyon ni Del Pilar sa akdang sinulat ni Padre Jose Rodriguez na may ganito ring pamagat.

- | |
|---|
| A. Calingat Cayo |
| B. Dasalan at Tocsohan |
| C. Fray Botod |
| D. Sagot ng Espanya sa Hibik ng Pilipinas |

Sagot: A-Ang Calingat Cayo ay isang akdang isinulat ni Del Pilar na nagtatanggol sa Noli Me Tangere ni Rizal. Ito ay sagot sa Calingat Cayo ni Padre Jose Rodriguez.

Filipino

119. "Meron akong nalalaman. 'Di ko sasabihin sa iyo." Nasa anong antas ng wika ang mga salitang nakasalungguhit?

- | | |
|-------------|-----------------|
| A. kolokyal | C. pampanitikan |
| B. balbal | D. talawiganin |

Sagot: A- Ang kolokyal ay isang halimbawa ng impormal na antas ng wika na kung saan pinapaikli ang isang salita mula sa orihinal na salita.

120. Ano ang tamang salin sa idyomang "You are the apple of my eye"?

- | | |
|-----------------------------|-----------------------------|
| A. masayahin ka pala | C. katuwa-tuwa ka |
| B. ikaw ay mahalaga sa akin | D. mansanas ang paborito ko |

Sagot: B-Ang idyomang "you are the apple of my eye" ay nangangahulugang ikaw ay mahalaga sa akin.

121. Sa ponemang segmental, ano ang tinataglay ng mga salitang sabaw, giliw, damay, reyna?

- | | |
|------------|------------------|
| A. ponema | C. diptonggo |
| B. klaster | D. pares minimal |

Sagot: C-Upang magkaroon ng diptonggo ang isang salita, hindi lang dapat magkasunod ang isang patinig at ang malapatinig na w o y, kung hindi, dapat din na ang dalawa ay magkasama sa isang pantig (halimbawa: ga-law, ba-liw, la-may, ka-hoy).

122. Ano ang tawag sa uri ng wika na nailikha sa pamamagitan ng pagpapaikli o pagsasama-sama ng mga salitang impormal at binigyan ng buong kahulugan?

- | | |
|-------------|-----------------|
| A. kolokyal | C. panlalawigan |
| B. lokal | D. pampanitikan |

Sagot: A-Ang kolokyal ay binubuo ng mga salitang impormal na pinaikli (halimbawa: tatay-tay, mayroon-meron, saan-saan) o pagsasama-sama ng mga salitang impormal at binigyan ng buong kahulugan (halimbawa: tapa+sinangag+ilog=tapsilog, maghintay+ka=teka).

123. Ano ang tawag sa bantas na sinisimbolo ng sunod-sunod na tatlong tuldok para ipakita na may mga bahaging hindi na sinipi sa isang talata?

- | |
|-------------|
| A. synopsis |
| B. Ellipsis |
| C. sintesis |
| D. abstrak |

Sagot: B-Ang ellipsis ay isang bantas na binubuo ng sunod-sunod na tatlong tuldok para ipakita na may mga bahaging hindi na sinipi sa isang talata.

124. "May pag-asa." Anong uri ito ng pangungusap na walang tiyak na paksa?

- A. Temporal
- B. Eksistensyal
- C. Penomenal
- D. Modal

Sagot: B-Ang eksistensyal na pangungusap ay nagpapakita ng pagka-mayroon o pagkawala.

125. Sa pananaliksik, saang kabanata matatagpuan ang presentasyon at Interpretasyon ng mga datos?

- A. Kabanata V
- B. Kabanata IV
- C. Kabanata III
- D. Kabanata II

Sagot: B-May limang kabanata ang isang pananaliksik. Ito ay ang mga sumusunod: Kabanata I (Ang Suliranin at ang Kaligiran Nito), Kabanata II (Mga Kaugnay na Pag-aaral at Literatura), Kabanata III (Disenyo at Paraan ng Pananaliksik), Kabanata IV (Presentasyon at Interpretasyon ng mga Datos) at Kabanata V (Lagom, Kongklusyon, at Rekomendasyon).

126. Ang mga sumusunod ay mga mahahalagang salik sa pagtatalumpati MALIBAN sa

- A. okasyon
- B. pagyayabang
- C. tagapakinig
- D. paksa

Sagot: B-Ang pagyayabang ay hindi isa sa mahalagang salik sa pagtatalumpati. Ito ay isang maling ugali na dapat iwasan ng isang mahusay na mananalumpati.

127. Ito ay tumutukoy sa proseso ng pagpapalitan ng kaisipan, opinyon o salaysay gamit ang mga simbolo o sagisag.

- A. pakikinig
- B. pagtuklas
- C. talastasan
- D. paglalahad

Sagot: C-Ang "talastasan" o kilala rin sa tawag na "komunikasyon" ay isang proseso ng pagpapalitan ng kaisipan gamit ang mga simbolong pangwika at may hangaring makapagbigay ng matagumpay na pakikipag-ugnayan at pagkakaintindihan.

128. Anong bahagi ng pahayagan ang nagpapakita ng opinyon ng buong pahayagan hinggil sa isang napapanahong balita?

- A. editorial
- B. kolum
- C. pahayag ng tagapayo
- D. abstrak

Sagot: A-Ang editorial ay kilala rin sa tawag na pangulong tudling. Ito ang bahagi ng pahayagan na nagpapakita ng opinyon ng buong pahayagan hinggil sa isang napapanahong balita. Samantala, ang kolum naman ay ang bahagi ng pahayagan na nagpapakita ng opinyon ng isang manunulat o kolumnista hinggil sa isang isyu o balita.

Filipino

129. Isang Pilipinong manunulat na tinaguriang "Utak ng Katipunan" si _____.

- A. Jose dela Cruz
- B. Anotonio Luna
- C. Jose Corazon de Jesus
- D. Emilio Jacinto

Sagot: D-Si Emilio Jacinto ang "Utak ng Katipunan" at siya ang kanang kamay ni Andres Bonifacio. Siya rin ang gumawa ng Kartilya ng Katipunan.

130. Anong bantas ang siyang ginagamit sa paghihiwalay ng magkakasunod na salita at lipon ng mga salitang magkaka-uri?

- A. kuwit
- B. tuldok-kuwit
- C. gilling
- D. tutuldok

Sagot: A-Ang kuwit ang siyang tamang bantas na gagamitin paghihiwalay ng magkakasunod na salita at lipon ng mga salitang magkaka-uri. Ginagamit din ang kuwit sa pagsusulat ng potsa at address.

131. Ano ang wastong pagpapakahulugan sa idyomang "My bank account is in the red"?

- | | |
|------------------------|------------------------|
| A. nakapa-lipon | C. nanakawan nang pera |
| B. malapit nang maubos | D. bale-wala |

Sagot: B. Ang idyomang "My bank account is in the red" ay nangangahulugan na malapit nang maubos ang pera.

132. Sa panitikan, ano ang tawag sa bilang ng pantig sa bawat linya o taludtod ng isang tula?

- | | |
|------------|---------------|
| A. sukat | C. talinghaga |
| B. saknong | D. tugma |

Sagot: A-Sukat. Ang saknong ay tumutukoy sa bawat grupo ng mga taludtod sa isang tula. Ang talinghaga naman ay ang malalim na pagpapakahulugan at pag-uunawa sa mensahe ng tula, at ang tugma ay ang pagkakapareho ng tunog sa huling pantig sa bawat huling salita ng isang tula.

133. Ito ay tumutukoy sa instrumento ng komunikasyon na siyang ginagamit sa pakikipagtalastasan, ugnayan, at pagpapalitan ng kaisipan.

- | | |
|----------|----------------|
| A. tunog | C. bokabularyo |
| B. wika | D. sining |

Sagot: B-Ang wika ay isang masistemang balangkas ng sinasalitang tunog na pinipili at isinasagay sa paraang arbitrario at nakabatay sa isang kultura. Ito ang instrumento ng komunikasyon.

134. Ito ay isang barayti ng wika na tumutukoy sa wikang nalilikha batay sa dimensyong heograpiko.

- A. Etnolek
- B. Ekolek
- C. Sosyolek
- D. Dayalekto

Sagot: D- Ang dayalekto o lalawiganin ay ang wikang ginagamit sa isang partikular na rehiyon, lalawigan o pook, malaki man o maliit. Ito ay nalilikha batay sa dimensyong heograpiko. Halimbawa ng mga dayalekto ay Hiligaynon (sa Iloilo), Sinugbuanong Binisaya (sa Cebu), Waray (sa Samar).

135. Sa komunikasyon na pasulat, alin sa mga sumusunod ang nararapat na isaalang-alang?

- A. Lakas ng boses
- B. Maayos na pagpapalugit
- C. Maliksing mga mata
- D. Pagkibit ng balikat

Sagot: B- Ang maayos na pagpapalugit o proper margin ay mahalagang salik sa komunikasyon na pasulat.

136. Ano ang tamang pagpapakahulugan sa idyomang "The present problem is only a storm in a teacup"?

- A. bale-wala
- B. buong puso
- C. may gallit
- D. matagumpay

Sagot: A- Ang idyomang "The present problem is only a storm in a teacup" ay nangangahulugan na ang kasalukuyang problema ay hindi naman talaga tunay na problema o bale-wala.

137. Anong sangay ng linggwistika na sumasaklaw sa pag-aaral ng wastong baybay ng mga salita?

- A. Ortograpiya
- B. Morpolohiya
- C. Semantika
- D. Sintaks

Sagot: A- Ortograpiya. Ang morpolohiya ay nag-aaral sa istruktura ng salita. Ang semantika ay nag-aaral naman sa kahulugan ng salita samantala ang sintaks ay nakapokus sa pag-aaral tungkol sa wastong pag-uugnay at pagsasaayos ng mga salita para makabuo ng isang pangungusap.

138. Anong pagbabagong morpoponemiko ang ginamit sa mga sumusunod na salita: NIYAKAP, NILIGAW, NILIPAD?

- A. Reduksyon
- B. Metatesis
- C. Pagpapalit
- D. Asimilasyon

Sagot: B- Ang metatesis ang pagpapalitan ng posisyon ng mga ponema o tunog sa isang salitang nilalapat. Halimbawa: in+yaya=niyaya (ang unlaping in ay naging ni).

Filipino

139. Isa sa pinakatanyag na Pilipinong manunulat sa Ingles at kilala sa kanyang sagisag panulat na "Dovegllion".

- | | |
|----------------------|-----------------------|
| A. Jose Garcia Villa | C. Alejandro Abadilla |
| B. N.V.M. Gonzalez | D. Zulueta de Costa |

Sagot: A-Si Jose Garcia Villa ay kilala sa paggamit ng mga matatalim na mga salita. Ang kanyang sagisag panulat na "Dovegllion" ay mula sa mga salitang dove, eagle, at lion.

140. Ibigay ang pagbabagong morpomonemiko ang nangyari sa mga sumusunod na salita: TAKPAN, DALHAN, BUKSAN.

- | | |
|----------------|----------------|
| A. Pagkakaltas | C. Pagpapalit |
| B. Metatesis | D. Asimilasyon |

Sagot: A-Ang pagkakaltas ng ponema ay nangyayari kung ang huling ponemang patinig ng salitang-ugat ay nawawala kapag nilalagyan ng hulapi. Halimbawa: takip-takpan (nawala ang "i"), dala-dalhan (nawala ang "a"), bukas-buksan (nawala ang "a").

141. Ano ang panlapi na ginamit sa mga sumusunod na salita: PINAGSUMIKAPAN, SANSINUKUBAN?

- | | |
|-----------|------------|
| A. hulapi | C. kabilan |
| B. lamban | D. laguhan |

Sagot: D-Laguhan ang tawag sa isang panlapi na binubuo ng tatlong panlapi na kinakabit sa isang salitang-ugat. Ito ay may unlapi, gitlapi at hulapi. Halimbawa: pinagsumikapan, sansinukuban.

142. Alin sa mga sumusunod na salita ang HINDI kabilang sa pangkat?

- | | |
|------------|------------------|
| A. klaster | C. diptonggo |
| B. diin | D. pares minimal |

Sagot: B.-Ang "diin" ay isang ponemang suprasegmental. Samantala, ang klaster, diptonggo at pares minimal ay mga ponemang segmental.

143. Ibigay ang uri ng tayutay ang pinapakita sa pahayag na: "Pag-ibig, huwag mo akong talikuran".

- | |
|-----------------|
| A. Pagmamalabis |
| B. Pagtawag |
| C. Palit-lawag |
| D. Palit-saklaw |

Sagot: B-Ang pagtawag o apostrophe ay ang pakikipag-usap sa isang bagay na wala namang kakayahang magbigay ng sagot.

144. Ang tono, diin, at antala ay mga halimbawa ng _____.

- A. ponemang segmental
- B. morpemang segmental
- C. ponemang suprasegmental
- D. morpemang suprasegmental

Sagot: C-Ang mga ponemang suprasegmental ay ang mga ponemang hindi makikita sa alpabeto dahil hindi sila sinisimboluhan ng mga titik. Halimbawa ng ponemang suprasegmental ay ang tono, diin, antala at haba.

145. Tila imposible na magkatotoo ang iyong mga pangarap. Nasa anong uri ng pang-abay ang bahaging may malalaking titik?

- A. Pang-abay na Pamaraan
- B. Pang-abay na Panlunan
- C. Pang-abay na Pang-agam
- D. Pang-abay na Kondisyunal

Sagot: C-Ang pang-abay na pang-agam ay nagbabadya ng di-katiyakan sa pagganap sa kilos ng pandiwa. Halimbawa sa mga ito ay ang mga salitang marahil, parang, tila, siguro, baka.

146. Ano ang uri ng tayutay ang ginamit sa pahayag na nasa ibaba?

"Kapalaran, kumampl ka naman sa akin!"

- A. Pagmamalabis
- B. Pagtawag
- C. Palit-tawag
- D. Palit-saklaw

Sagot: B-Ang pagtawag o apostrophe ay ang pakikipag-usap sa isang bagay na wala namang kakayahang magbigay ng sagot.

147. KUNG matapos mo ito nang maaga, may premyo ka mula sa akin. Nasa anong uri ng pang-abay ang bahaging may malalaking titik?

- A. Pang-abay na Pamaraan
- B. Pang-abay na Panlunan
- C. Pang-abay na Pang-agam
- D. Pang-abay na Kondisyunal

Sagot: D-Ang pang-abay na kondisyunal ay ginagamit sa pagsasaad ng kondisyon para maganap ang kilos. Halimbawa sa mga ito ay ang mga salitang kung, kapag, pag, pagka.

Filipino



148. Ang mga salitang "dukha, daga, pasa" ay mga halimbawa ng mga salitang binibigkas ng ____.

- A. Malumi
- B. Maragsa
- C. Malumanay
- D. Mabilis

Sagot: B- Ang mga salitang maragsa ay binibigkas nang tuluy-tuloy na tulad ng mga salitang binibigkas nang mabilis subalit ito ay may implit o pasarang tunog sa hulihan. Tulad ng malumi, ito ay palagiang nagtatapos sa tunog na patinig. Ginagamit dito ang tuldik na pakupya (A) na inilalagay sa ibabaw ng huling patinig ng salita.

149. Anim na malalaking mangga ang ibinigay niya sa akin. Anong uri ng pang-uring pamilang ang sinalangguhiang salita?

- A. Patakeran
- B. Penunuran
- C. Pamahagi
- D. Pamatlilig

Sagot: A. (Ang tamang sagot ay Option "A") Ang patakeran ay ang karaniwang paraan ng pagbilang (halimbawa: isa, dalawa, tatlo).

Ang panunuran naman ay nagsasaad ng ayos o pagkakasunud-sunod ng mga tao o bagay (halimbawa: una, ikalawa, ikatlo). Ang pamahagi ay nagsasaad ng pagbabahagi ng isang kabuuan (halimbawa: kalahatian, sangkepatan). Samantala, ang pamatlilig ay tumutukoy sa panghalip na ginagamit sa pagturo o pagtukoy sa isang bagay (halimbawa: iyan, iyon, dito).